Cyclone Remal (2024)
Complete WRF Analysis Report

FNL - GFS - CFSv2 - ERA5

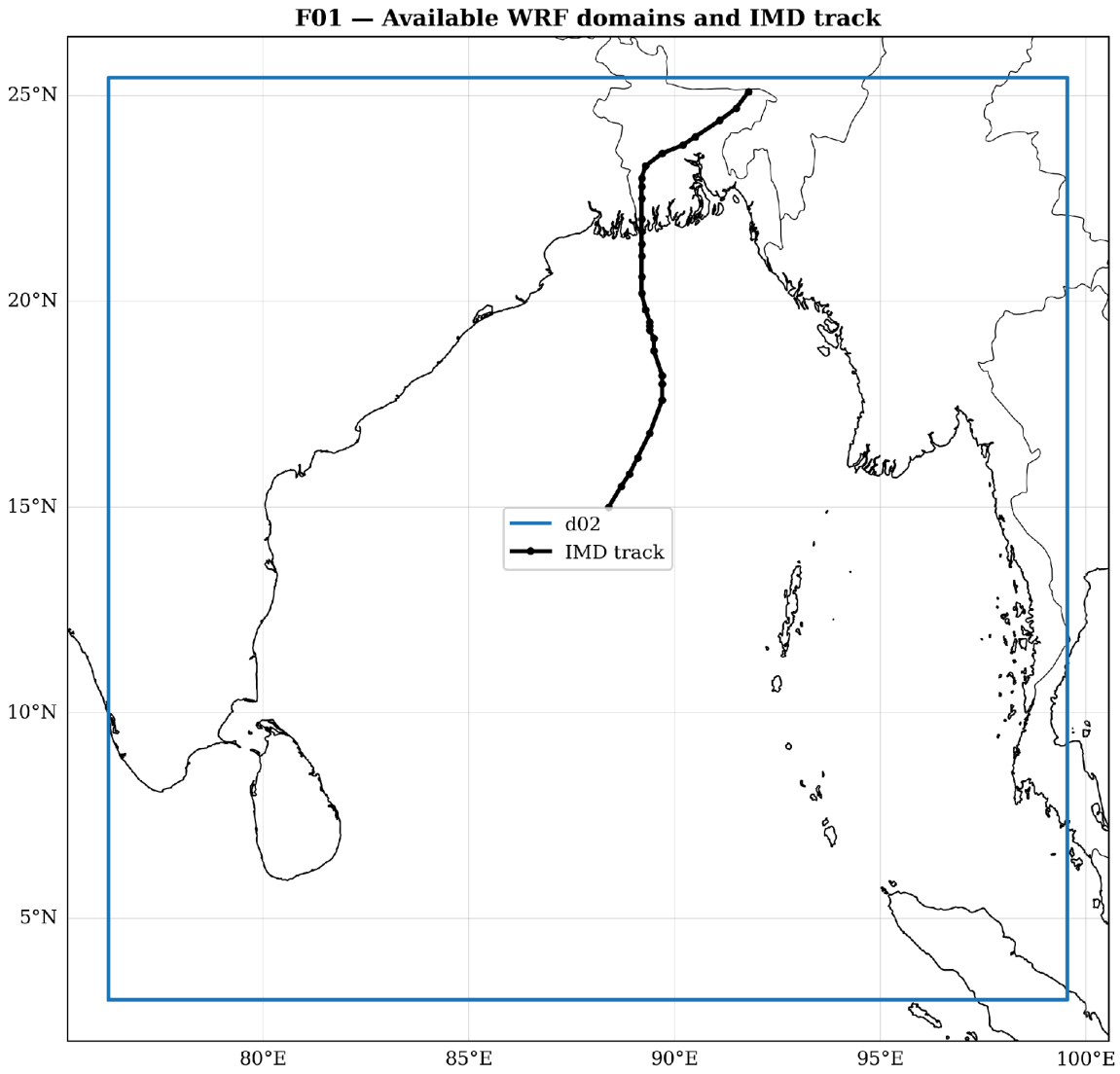
- Primary references: IMD for cyclone track/intensity, IMERG for rainfall, CCMP for ocean-wind patterns.
- Track, translation speed and track-dependent landfall: frozen regular 6-hourly workflow.
- Wind, pressure, peaks, change rates and main PDI: all exact common IMD-WRF timestamps.
- Full-time wind/pressure validation: both ordinary paired-observation scores and time-weighted scores are reported.
- A16: explicit all-match versus regular-6-hour sampling sensitivity.
- PDI: trapezoidal integral of V^3 using wind in m s-1 and actual elapsed seconds; units m^3 s^-2.
- Full-time intensity timestamps used: 30
- Regular 6-hour sensitivity timestamps used: 18
- Generated: 2026-10-20 07:02 UTC

This Word report is assembled directly from the figures and tables saved by the Remal_Analysis.ipynb notebook. NetCDF diagnostic arrays are listed by filename rather than printed cell-by-cell.

A01 - Configuration, inventory and domains

Figures: 1 | CSV tables: 4 | NetCDF extracts: 0

F01_Domain_Map.png



A01_domain_metadata.csv

Column block 1 of 2

domain	source_file	ny	nx	dx_m	dy_m	lat_min
d02	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-24_00_00_00	279	279	9000.0	9000.0	3.021

Column block 2 of 2

domain	lat_max	lon_min	lon_max
d02	25.441	76.248	99.546

Remal_Inventory_Check.csv

model	found	missing	duplicates	unexpected	unparseable_filenames	note
FNL	35	0	0	0	0	
GFS	35	0	0	0	0	
CFSv2	35	0	0	0	0	
ERA5	35	0	0	0	0	

Remal_Model_Configuration.csv

Column block 1 of 3

model	file_checked	MP_PHYSICS	RA_LW_PHYSICS	RA_SW_PHYSICS	SF_SFCLAY_PHYSICS	SF_SURFACE_PHYSICS
FNL	wrfout_d02_2024-05-24_00_00_00	8	1	1	1	2
GFS	wrfout_d02_2024-05-24_00_00_00	8	1	1	1	2
CFSv2	wrfout_d02_2024-05-24_00_00_00	8	1	1	1	2
ERA5	wrfout_d02_2024-05-24_00_00_00	8	1	1	1	2

Column block 2 of 3

model	BL_PBL_PHYSICS	CU_PHYSICS	DX	DY	TIME_STEP	WEST-EAST_GRID_DIMENSION
FNL	1	0	9000.0	9000.0		280
GFS	1	0	9000.0	9000.0		280
CFSv2	1	0	9000.0	9000.0		280
ERA5	1	0	9000.0	9000.0		280

Column block 3 of 3

model	SOUTH-NORTH_GRID_DIMENSION
FNL	280
GFS	280
CFSv2	280
ERA5	280

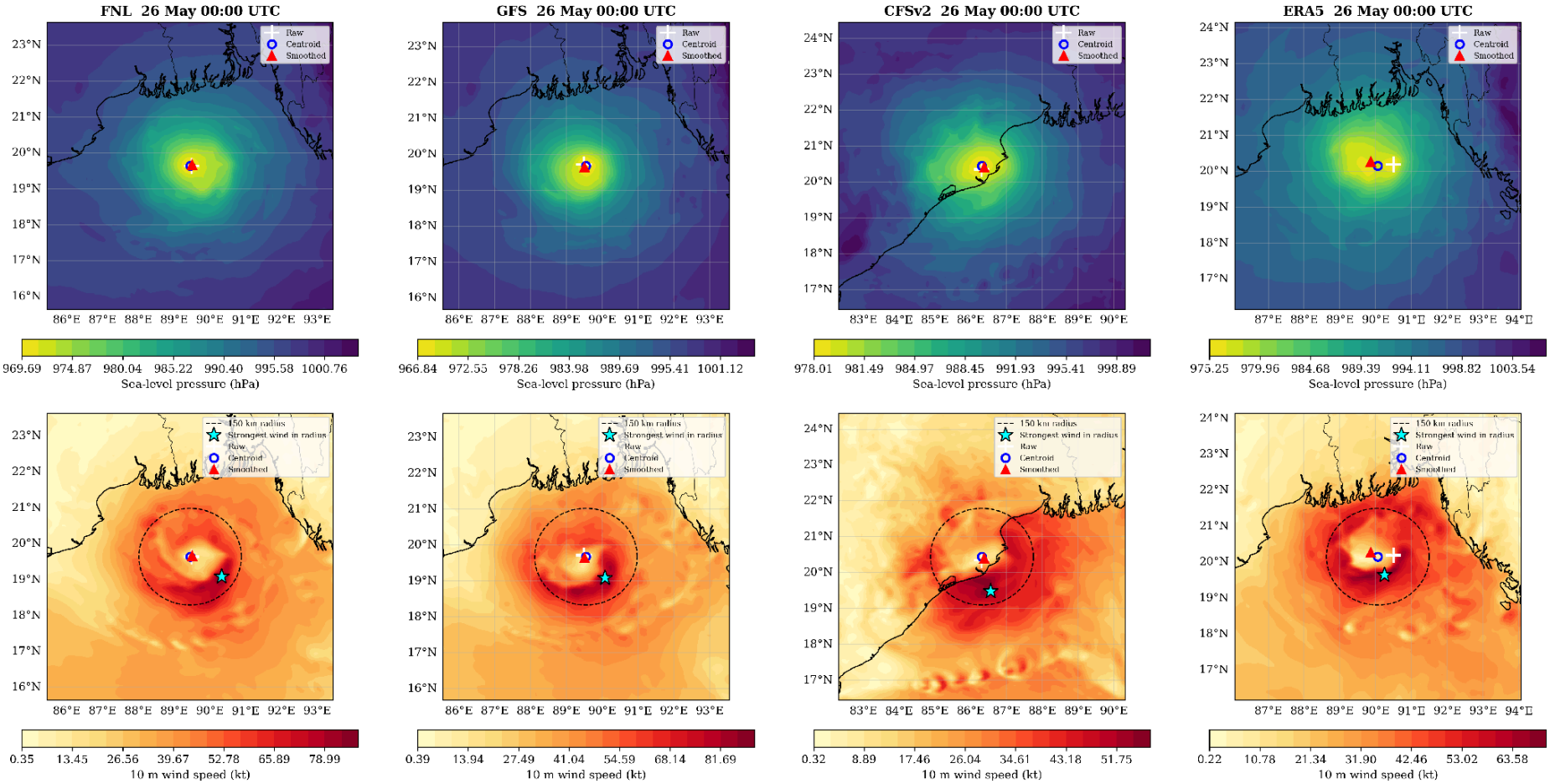
Remal_Timestamp_Differences.csv

model_a	model_b	only_in_a	only_in_b	times_only_in_a	times_only_in_b
FNL	GFS	0	0		
FNL	CFSv2	0	0		
FNL	ERA5	0	0		
GFS	CFSv2	0	0		
GFS	ERA5	0	0		
CFSv2	ERA5	0	0		

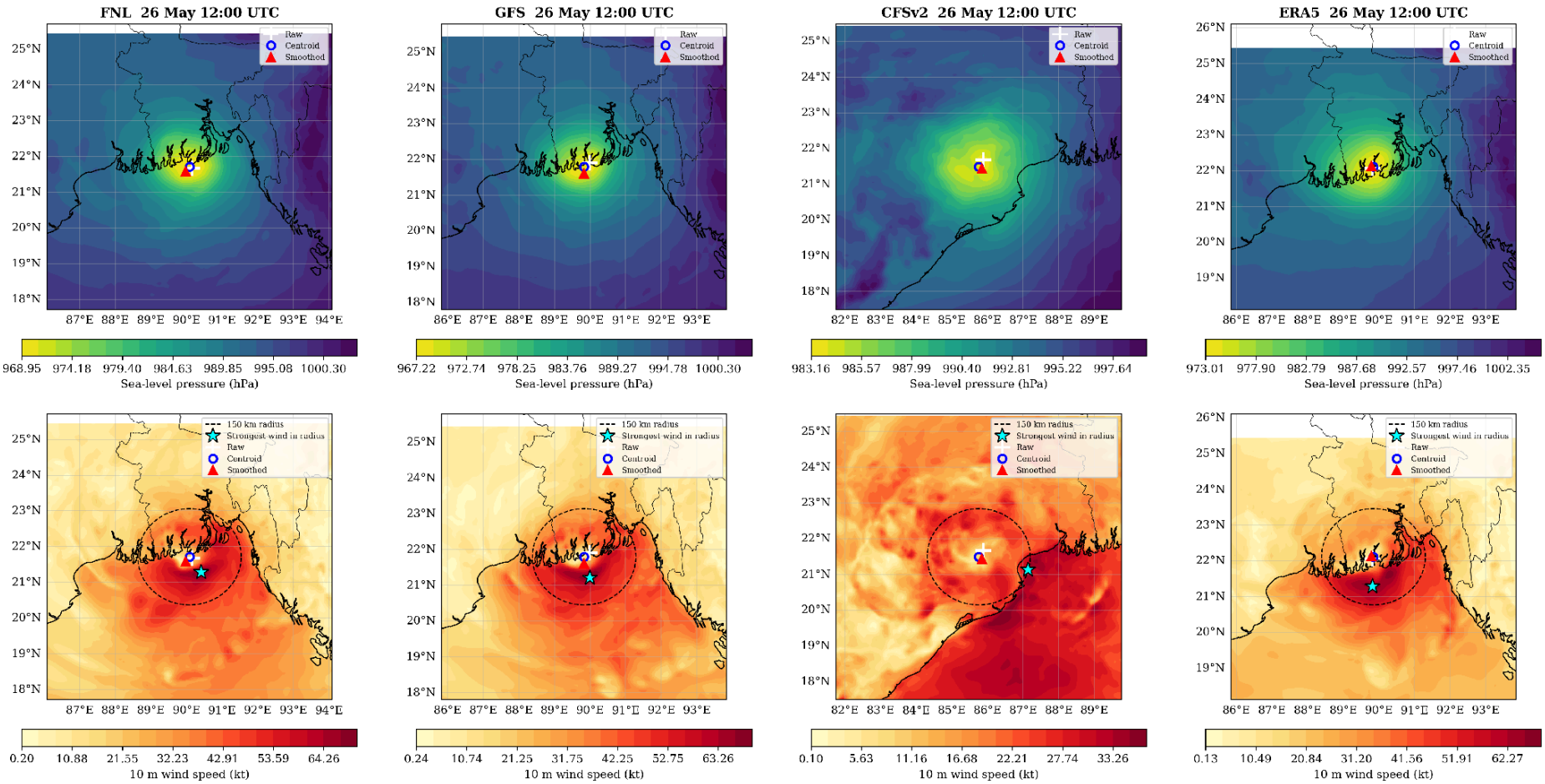
A02 - Tracking sensitivity and diagnostics

Figures: 3 | CSV tables: 22 | NetCDF extracts: 0

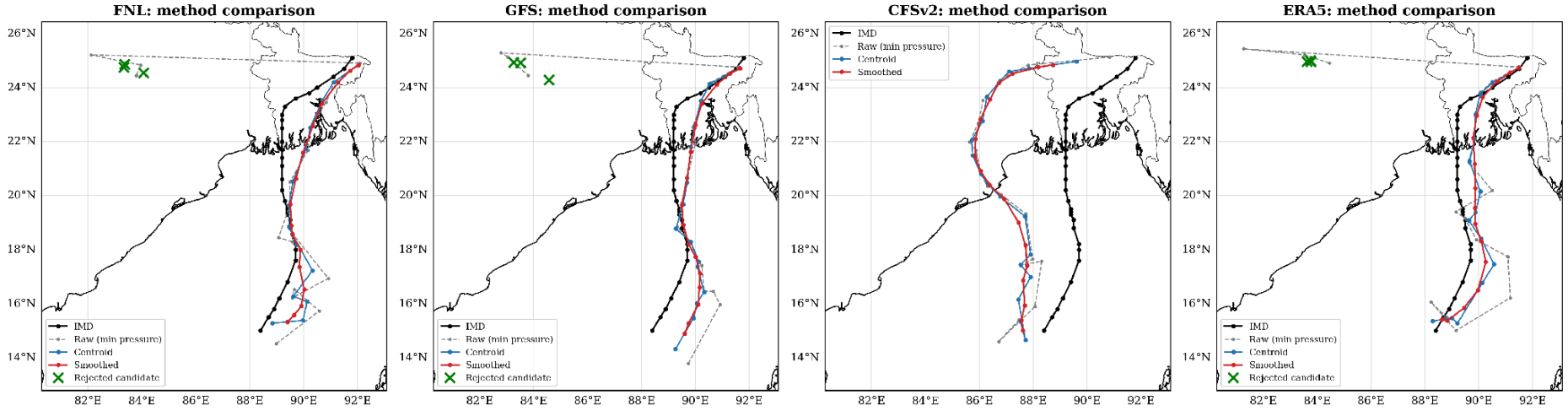
Remal_A1_Selected_Maps_20240526_00.png



Remal_A1_Selected_Maps_20240526_12.png



Remal_A1_Tracking_Method.png



Remal_A1_Method_Errors.csv

model	method	n_matched	mean_error_km	rmse_km	max_error_km
FNL	Raw	18	233.028	351.154	858.273
FNL	Centroid	18	108.58	129.16	274.288
FNL	Smoothed	18	102.128	123.464	249.855
GFS	Raw	18	236.727	353.027	818.15
GFS	Centroid	18	102.524	114.856	233.763
GFS	Smoothed	18	99.46	110.864	205.546
CFSv2	Raw	18	259.501	275.173	369.403
CFSv2	Centroid	18	263.795	275.368	361.563
CFSv2	Smoothed	18	261.708	273.257	349.622
ERA5	Raw	18	239.961	358.899	945.095
ERA5	Centroid	18	100.974	115.084	221.574
ERA5	Smoothed	18	95.789	109.202	203.785

Remal_A1_Rejected_Candidates.csv

Column block 1 of 2

model	time_utc	status	candidate_latitude	candidate_longitude	candidate_pressure_hpa	movement_km
FNL	2024-05-27 18:00:00+00:00	REJECTED	24.748	83.335	990.801	878.199
FNL	2024-05-28 00:00:00+00:00	REJECTED	24.871	83.363	991.189	874.819
FNL	2024-05-28 06:00:00+00:00	REJECTED	24.546	84.06	991.695	806.314
GFS	2024-05-27 18:00:00+00:00	REJECTED	24.927	83.261	990.562	847.5
GFS	2024-05-28 00:00:00+00:00	REJECTED	24.921	83.519	990.793	821.481
GFS	2024-05-28 06:00:00+00:00	REJECTED	24.286	84.568	991.955	718.64
ERA5	2024-05-27 18:00:00+00:00	REJECTED	24.964	83.63	992.55	793.205

ERA5	2024-05-28 00:00:00+00:00	REJECTED	24.997	83.761	992.897	779.98
ERA5	2024-05-28 06:00:00+00:00	REJECTED	24.96	83.775	992.971	778.557

Column block 2 of 2

model	pressure_change_hpa
FNL	5.612
FNL	6.0
FNL	6.507
GFS	6.381
GFS	6.612
GFS	7.774
ERA5	6.826
ERA5	7.172
ERA5	7.247

Remal_CFSv2_6hr_Centroid.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24T00:00:00Z	15.34	87.526	7.162	3.685	994.543	23.25
2024-05-24T06:00:00Z	14.652	87.715	18.452	9.492	993.517	40.736
2024-05-24T12:00:00Z	16.156	87.448	6.666	3.429	991.532	38.887
2024-05-24T18:00:00Z	16.983	87.904	9.079	4.671	990.357	47.449
2024-05-25T00:00:00Z	17.443	87.535	4.023	2.07	985.717	51.246
2024-05-25T06:00:00Z	17.813	87.904	14.121	7.265	982.594	64.74
2024-05-25T12:00:00Z	19.227	87.703	13.93	7.166	979.397	57.689
2024-05-25T18:00:00Z	19.97	86.767	4.82	2.48	977.909	59.111
2024-05-26T00:00:00Z	20.445	86.314	6.378	3.281	978.015	57.469
2024-05-26T06:00:00Z	20.796	86.065	6.837	3.517	982.21	50.082
2024-05-26T12:00:00Z	21.493	85.758	8.725	4.489	983.162	33.871
2024-05-26T18:00:00Z	22.043	85.716	7.263	3.737	986.179	30.56
2024-05-27T00:00:00Z	22.765	86.122	5.103	2.625	985.759	33.901
2024-05-27T06:00:00Z	23.663	86.288	8.509	4.377	986.709	34.226
2024-05-27T12:00:00Z	24.252	86.774	5.065	2.606	985.529	32.76
2024-05-27T18:00:00Z	24.597	87.108	3.294	1.695	987.062	30.908
2024-05-28T00:00:00Z	24.713	87.847	7.075	3.64	987.913	28.454
2024-05-28T06:00:00Z	24.969	89.611	11.513	5.923	991.277	31.141

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km
2024-05-24T00:00:00Z	11.961	148.879
2024-05-24T06:00:00Z	20.957	106.879
2024-05-24T12:00:00Z	20.005	140.605
2024-05-24T18:00:00Z	24.41	148.656
2024-05-25T00:00:00Z	26.363	91.857
2024-05-25T06:00:00Z	33.305	45.121
2024-05-25T12:00:00Z	29.678	109.752
2024-05-25T18:00:00Z	30.409	100.693
2024-05-26T00:00:00Z	29.565	110.147
2024-05-26T06:00:00Z	25.765	125.207
2024-05-26T12:00:00Z	17.425	149.032

2024-05-26T18:00:00Z	15.721	114.247
2024-05-27T00:00:00Z	17.44	129.598
2024-05-27T06:00:00Z	17.608	122.732
2024-05-27T12:00:00Z	16.853	97.479
2024-05-27T18:00:00Z	15.901	140.759
2024-05-28T00:00:00Z	14.638	108.745
2024-05-28T06:00:00Z	16.02	147.629

Remal_CFSv2_6hr_Diagnostics.csv

time_utc	status	candidate_latitude	candidate_longitude	candidate_pressure_hpa	movement_km	pressure_change_hpa
2024-05-24T00:00:00Z	FIRST	15.34	87.526	994.543		
2024-05-24T06:00:00Z	ACCEPTED	14.652	87.715	993.517	79.223	1.026
2024-05-24T12:00:00Z	ACCEPTED	16.156	87.448	991.532	169.643	1.985
2024-05-24T18:00:00Z	ACCEPTED	16.983	87.904	990.357	104.091	1.175
2024-05-25T00:00:00Z	ACCEPTED	17.443	87.535	985.717	64.417	4.64
2024-05-25T06:00:00Z	ACCEPTED	17.813	87.904	982.594	56.778	3.123
2024-05-25T12:00:00Z	ACCEPTED	19.227	87.703	979.397	158.628	3.197
2024-05-25T18:00:00Z	ACCEPTED	19.97	86.767	977.909	128.157	1.488
2024-05-26T00:00:00Z	ACCEPTED	20.445	86.314	978.015	70.971	0.106
2024-05-26T06:00:00Z	ACCEPTED	20.796	86.065	982.21	46.813	4.195
2024-05-26T12:00:00Z	ACCEPTED	21.493	85.758	983.162	83.843	0.952
2024-05-26T18:00:00Z	ACCEPTED	22.043	85.716	986.179	61.259	3.017
2024-05-27T00:00:00Z	ACCEPTED	22.765	86.122	985.759	90.574	0.42
2024-05-27T06:00:00Z	ACCEPTED	23.663	86.288	986.709	101.205	0.95
2024-05-27T12:00:00Z	ACCEPTED	24.252	86.774	985.529	82.018	1.18
2024-05-27T18:00:00Z	ACCEPTED	24.597	87.108	987.062	51.183	1.534
2024-05-28T00:00:00Z	ACCEPTED	24.713	87.847	987.913	75.792	0.851
2024-05-28T06:00:00Z	ACCEPTED	24.969	89.611	991.277	180.271	3.364

Remal_CFSv2_6hr_Raw.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24T00:00:00Z	15.404	87.478	8.48	4.363	994.543	23.573
2024-05-24T06:00:00Z	14.595	86.724	40.736	20.957	993.517	40.736
2024-05-24T12:00:00Z	15.889	88.065	14.256	7.334	991.532	38.887
2024-05-24T18:00:00Z	17.574	88.316	9.369	4.82	990.357	39.461
2024-05-25T00:00:00Z	17.414	87.562	4.023	2.07	985.717	51.246
2024-05-25T06:00:00Z	17.654	87.981	27.623	14.211	982.594	64.74
2024-05-25T12:00:00Z	19.323	87.73	17.19	8.843	979.397	57.689
2024-05-25T18:00:00Z	20.033	86.808	10.695	5.502	977.909	59.111
2024-05-26T00:00:00Z	20.348	86.305	5.962	3.067	978.015	57.469
2024-05-26T06:00:00Z	20.818	86.053	6.837	3.517	982.21	50.082
2024-05-26T12:00:00Z	21.678	85.886	8.756	4.504	983.162	35.329
2024-05-26T18:00:00Z	21.989	85.634	5.096	2.622	986.179	30.56
2024-05-27T00:00:00Z	22.686	85.97	3.191	1.642	985.759	33.901
2024-05-27T06:00:00Z	23.534	86.137	5.73	2.948	986.709	34.226
2024-05-27T12:00:00Z	24.148	86.724	2.917	1.501	985.529	32.76
2024-05-27T18:00:00Z	24.453	87.059	4.866	2.503	987.062	30.908
2024-05-28T00:00:00Z	24.834	87.813	4.306	2.215	987.913	28.454
2024-05-28T06:00:00Z	25.138	90.83	30.87	15.881	991.277	32.316

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km
2024-05-24T00:00:00Z	12.127	149.11
2024-05-24T06:00:00Z	20.957	0.0
2024-05-24T12:00:00Z	20.005	99.213
2024-05-24T18:00:00Z	20.301	64.007
2024-05-25T00:00:00Z	26.363	87.745
2024-05-25T06:00:00Z	33.305	26.658
2024-05-25T12:00:00Z	29.678	119.926
2024-05-25T18:00:00Z	30.409	106.827
2024-05-26T00:00:00Z	29.565	99.897
2024-05-26T06:00:00Z	25.765	127.282
2024-05-26T12:00:00Z	18.175	145.361
2024-05-26T18:00:00Z	15.721	117.108
2024-05-27T00:00:00Z	17.44	132.903
2024-05-27T06:00:00Z	17.608	136.053
2024-05-27T12:00:00Z	16.853	102.117
2024-05-27T18:00:00Z	15.901	135.952
2024-05-28T00:00:00Z	14.638	101.586
2024-05-28T06:00:00Z	16.625	53.951

Remal_CFSv2_6hr_Smoothed.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24T00:00:00Z	14.996	87.621	7.162	3.685	994.543	23.25
2024-05-24T06:00:00Z	15.383	87.563	18.452	9.492	993.517	40.736
2024-05-24T12:00:00Z	15.93	87.689	6.666	3.429	991.532	38.887
2024-05-24T18:00:00Z	16.861	87.629	9.079	4.671	990.357	47.449
2024-05-25T00:00:00Z	17.413	87.781	4.023	2.07	985.717	51.246
2024-05-25T06:00:00Z	18.161	87.714	14.121	7.265	982.594	64.74
2024-05-25T12:00:00Z	19.003	87.458	13.93	7.166	979.397	57.689
2024-05-25T18:00:00Z	19.88	86.928	4.82	2.48	977.909	59.111
2024-05-26T00:00:00Z	20.403	86.382	6.378	3.281	978.015	57.469
2024-05-26T06:00:00Z	20.911	86.045	6.837	3.517	982.21	50.082
2024-05-26T12:00:00Z	21.444	85.846	8.725	4.489	983.162	33.871
2024-05-26T18:00:00Z	22.1	85.865	7.263	3.737	986.179	30.56
2024-05-27T00:00:00Z	22.824	86.042	5.103	2.625	985.759	33.901
2024-05-27T06:00:00Z	23.56	86.395	8.509	4.377	986.709	34.226
2024-05-27T12:00:00Z	24.17	86.723	5.065	2.606	985.529	32.76
2024-05-27T18:00:00Z	24.52	87.243	3.294	1.695	987.062	30.908
2024-05-28T00:00:00Z	24.76	88.189	7.075	3.64	987.913	28.454
2024-05-28T06:00:00Z	24.841	88.729	11.513	5.923	991.277	31.141

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km
2024-05-24T00:00:00Z	11.961	148.879
2024-05-24T06:00:00Z	20.957	106.879
2024-05-24T12:00:00Z	20.005	140.605
2024-05-24T18:00:00Z	24.41	148.656
2024-05-25T00:00:00Z	26.363	91.857
2024-05-25T06:00:00Z	33.305	45.121

2024-05-25T12:00:00Z	29.678	109.752
2024-05-25T18:00:00Z	30.409	100.693
2024-05-26T00:00:00Z	29.565	110.147
2024-05-26T06:00:00Z	25.765	125.207
2024-05-26T12:00:00Z	17.425	149.032
2024-05-26T18:00:00Z	15.721	114.247
2024-05-27T00:00:00Z	17.44	129.598
2024-05-27T06:00:00Z	17.608	122.732
2024-05-27T12:00:00Z	16.853	97.479
2024-05-27T18:00:00Z	15.901	140.759
2024-05-28T00:00:00Z	14.638	108.745
2024-05-28T06:00:00Z	16.02	147.629

Remal_CFSv2_6hr_Smoothed_Flagged.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24 00:00:00+00:00	14.996	87.621	7.162	3.685	994.543	23.25
2024-05-24 06:00:00+00:00	15.383	87.563	18.452	9.492	993.517	40.736
2024-05-24 12:00:00+00:00	15.93	87.689	6.666	3.429	991.532	38.887
2024-05-24 18:00:00+00:00	16.861	87.629	9.079	4.671	990.357	47.449
2024-05-25 00:00:00+00:00	17.413	87.781	4.023	2.07	985.717	51.246
2024-05-25 06:00:00+00:00	18.161	87.714	14.121	7.265	982.594	64.74
2024-05-25 12:00:00+00:00	19.003	87.458	13.93	7.166	979.397	57.689
2024-05-25 18:00:00+00:00	19.88	86.928	4.82	2.48	977.909	59.111
2024-05-26 00:00:00+00:00	20.403	86.382	6.378	3.281	978.015	57.469
2024-05-26 06:00:00+00:00	20.911	86.045	6.837	3.517	982.21	50.082
2024-05-26 12:00:00+00:00	21.444	85.846	8.725	4.489	983.162	33.871
2024-05-26 18:00:00+00:00	22.1	85.865	7.263	3.737	986.179	30.56
2024-05-27 00:00:00+00:00	22.824	86.042	5.103	2.625	985.759	33.901
2024-05-27 06:00:00+00:00	23.56	86.395	8.509	4.377	986.709	34.226
2024-05-27 12:00:00+00:00	24.17	86.723	5.065	2.606	985.529	32.76
2024-05-27 18:00:00+00:00	24.52	87.243	3.294	1.695	987.062	30.908
2024-05-28 00:00:00+00:00	24.76	88.189	7.075	3.64	987.913	28.454
2024-05-28 06:00:00+00:00	24.841	88.729	11.513	5.923	991.277	31.141

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km	carried_forward	smoothing_uses_carried_forward
2024-05-24 00:00:00+00:00	11.961	148.879	False	False
2024-05-24 06:00:00+00:00	20.957	106.879	False	False
2024-05-24 12:00:00+00:00	20.005	140.605	False	False
2024-05-24 18:00:00+00:00	24.41	148.656	False	False
2024-05-25 00:00:00+00:00	26.363	91.857	False	False
2024-05-25 06:00:00+00:00	33.305	45.121	False	False
2024-05-25 12:00:00+00:00	29.678	109.752	False	False
2024-05-25 18:00:00+00:00	30.409	100.693	False	False
2024-05-26 00:00:00+00:00	29.565	110.147	False	False
2024-05-26 06:00:00+00:00	25.765	125.207	False	False
2024-05-26 12:00:00+00:00	17.425	149.032	False	False
2024-05-26 18:00:00+00:00	15.721	114.247	False	False
2024-05-27 00:00:00+00:00	17.44	129.598	False	False
2024-05-27 06:00:00+00:00	17.608	122.732	False	False
2024-05-27 12:00:00+00:00	16.853	97.479	False	False
2024-05-27 18:00:00+00:00	15.901	140.759	False	False

2024-05-28 00:00:00+00:00	14.638	108.745	False	False
2024-05-28 06:00:00+00:00	16.02	147.629	False	False

Remal_ERA5_6hr_Centroid.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24T00:00:00Z	15.353	88.286	6.786	3.491	997.081	18.807
2024-05-24T06:00:00Z	15.465	88.999	9.911	5.099	996.421	30.99
2024-05-24T12:00:00Z	15.271	89.208	11.541	5.937	992.596	43.835
2024-05-24T18:00:00Z	16.768	90.135	10.362	5.331	993.724	39.048
2024-05-25T00:00:00Z	17.455	90.567	17.427	8.965	991.339	43.243
2024-05-25T06:00:00Z	18.407	90.086	12.142	6.246	988.791	47.751
2024-05-25T12:00:00Z	19.076	89.63	9.418	4.845	985.755	44.784
2024-05-25T18:00:00Z	19.381	89.882	17.011	8.751	983.638	51.944
2024-05-26T00:00:00Z	20.15	90.059	8.197	4.217	975.25	70.618
2024-05-26T06:00:00Z	21.282	89.651	8.396	4.319	972.373	69.171
2024-05-26T12:00:00Z	22.113	89.846	1.686	0.867	973.012	69.174
2024-05-26T18:00:00Z	23.008	89.875	5.1	2.624	976.344	52.848
2024-05-27T00:00:00Z	23.774	90.061	2.518	1.296	980.161	52.887
2024-05-27T06:00:00Z	24.173	90.498	7.026	3.614	983.073	36.86
2024-05-27T12:00:00Z	24.743	91.489	6.778	3.487	985.724	36.286
2024-05-27T18:00:00Z	24.743	91.489	11.339	5.833	985.724	21.662
2024-05-28T00:00:00Z	24.743	91.489	7.015	3.609	985.724	16.586
2024-05-28T06:00:00Z	24.743	91.489	1.743	0.897	985.724	13.98

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km
2024-05-24T00:00:00Z	9.675	149.765
2024-05-24T06:00:00Z	15.943	120.709
2024-05-24T12:00:00Z	22.551	38.422
2024-05-24T18:00:00Z	20.088	147.916
2024-05-25T00:00:00Z	22.246	127.212
2024-05-25T06:00:00Z	24.565	146.929
2024-05-25T12:00:00Z	23.039	122.001
2024-05-25T18:00:00Z	26.722	145.374
2024-05-26T00:00:00Z	36.329	60.035
2024-05-26T06:00:00Z	35.585	70.32
2024-05-26T12:00:00Z	35.586	91.829
2024-05-26T18:00:00Z	27.188	139.351
2024-05-27T00:00:00Z	27.208	116.395
2024-05-27T06:00:00Z	18.963	44.67
2024-05-27T12:00:00Z	18.667	47.408
2024-05-27T18:00:00Z	11.144	81.98
2024-05-28T00:00:00Z	8.533	121.903
2024-05-28T06:00:00Z	7.192	135.35

Remal_ERA5_6hr_Diagnostics.csv

time_utc	status	candidate_latitude	candidate_longitude	candidate_pressure_hpa	movement_km	pressure_change_hpa
2024-05-24T00:00:00Z	FIRST	15.353	88.286	997.081		
2024-05-24T06:00:00Z	ACCEPTED	15.465	88.999	996.421	77.412	0.661

2024-05-24T12:00:00Z	ACCEPTED	15.271	89.208	992.596	31.141	3.824
2024-05-24T18:00:00Z	ACCEPTED	16.768	90.135	993.724	193.635	1.128
2024-05-25T00:00:00Z	ACCEPTED	17.455	90.567	991.339	89.187	2.385
2024-05-25T06:00:00Z	ACCEPTED	18.407	90.086	988.791	117.417	2.548
2024-05-25T12:00:00Z	ACCEPTED	19.076	89.63	985.755	88.542	3.036
2024-05-25T18:00:00Z	ACCEPTED	19.381	89.882	983.638	42.967	2.116
2024-05-26T00:00:00Z	ACCEPTED	20.15	90.059	975.25	87.561	8.388
2024-05-26T06:00:00Z	ACCEPTED	21.282	89.651	972.373	132.818	2.877
2024-05-26T12:00:00Z	ACCEPTED	22.113	89.846	973.012	94.609	0.639
2024-05-26T18:00:00Z	ACCEPTED	23.008	89.875	976.344	99.513	3.331
2024-05-27T00:00:00Z	ACCEPTED	23.774	90.061	980.161	87.28	3.817
2024-05-27T06:00:00Z	ACCEPTED	24.173	90.498	983.073	62.823	2.912
2024-05-27T12:00:00Z	ACCEPTED	24.743	91.489	985.724	118.624	2.652
2024-05-27T18:00:00Z	REJECTED	24.964	83.63	992.55	793.205	6.826
2024-05-28T00:00:00Z	REJECTED	24.997	83.761	992.897	779.98	7.172
2024-05-28T06:00:00Z	REJECTED	24.96	83.775	992.971	778.557	7.247

Remal_ERA5_6hr_Raw.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24T00:00:00Z	15.485	88.819	3.263	1.679	997.081	20.15
2024-05-24T06:00:00Z	16.05	88.232	12.826	6.598	996.421	30.99
2024-05-24T12:00:00Z	15.0	89.154	31.47	16.189	992.596	43.835
2024-05-24T18:00:00Z	16.211	91.166	22.642	11.648	993.724	39.864
2024-05-25T00:00:00Z	17.734	91.082	22.01	11.323	991.339	42.565
2024-05-25T06:00:00Z	18.371	89.909	5.055	2.6	988.791	47.751
2024-05-25T12:00:00Z	19.165	89.489	5.206	2.678	985.755	44.784
2024-05-25T18:00:00Z	19.402	89.154	29.3	15.073	983.638	48.576
2024-05-26T00:00:00Z	20.19	90.495	34.07	17.527	975.25	70.618
2024-05-26T06:00:00Z	21.21	89.657	12.844	6.607	972.373	69.171
2024-05-26T12:00:00Z	22.067	89.825	5.943	3.057	973.012	69.174
2024-05-26T18:00:00Z	23.073	89.909	6.756	3.476	976.344	52.848
2024-05-27T00:00:00Z	23.841	90.076	5.225	2.688	980.161	52.887
2024-05-27T06:00:00Z	24.224	90.495	3.443	1.771	983.073	36.86
2024-05-27T12:00:00Z	24.758	91.501	6.778	3.487	985.724	36.286
2024-05-27T18:00:00Z	25.441	81.276	5.831	3.0	992.55	9.052
2024-05-28T00:00:00Z	25.214	83.539	4.414	2.271	992.897	15.331
2024-05-28T06:00:00Z	24.91	84.461	9.648	4.963	992.971	14.075

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km
2024-05-24T00:00:00Z	10.366	149.082
2024-05-24T06:00:00Z	15.943	96.648
2024-05-24T12:00:00Z	22.551	18.003
2024-05-24T18:00:00Z	20.508	54.496
2024-05-25T00:00:00Z	21.897	137.916
2024-05-25T06:00:00Z	24.565	149.746
2024-05-25T12:00:00Z	23.039	124.183
2024-05-25T18:00:00Z	24.989	143.7
2024-05-26T00:00:00Z	36.329	66.727
2024-05-26T06:00:00Z	35.585	66.215
2024-05-26T12:00:00Z	35.586	86.599
2024-05-26T18:00:00Z	27.188	146.703

2024-05-27T00:00:00Z	27.208	110.904
2024-05-27T06:00:00Z	18.963	49.629
2024-05-27T12:00:00Z	18.667	49.418
2024-05-27T18:00:00Z	4.657	72.549
2024-05-28T00:00:00Z	7.887	121.895
2024-05-28T06:00:00Z	7.241	144.676

Remal_ERA5_6hr_Smoothed.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24T00:00:00Z	15.409	88.643	6.786	3.491	997.081	18.807
2024-05-24T06:00:00Z	15.363	88.831	9.911	5.099	996.421	30.99
2024-05-24T12:00:00Z	15.835	89.447	11.541	5.937	992.596	43.835
2024-05-24T18:00:00Z	16.498	89.97	10.362	5.331	993.724	39.048
2024-05-25T00:00:00Z	17.543	90.262	17.427	8.965	991.339	43.243
2024-05-25T06:00:00Z	18.313	90.094	12.142	6.246	988.791	47.751
2024-05-25T12:00:00Z	18.955	89.866	9.418	4.845	985.755	44.784
2024-05-25T18:00:00Z	19.536	89.857	17.011	8.751	983.638	51.944
2024-05-26T00:00:00Z	20.271	89.864	8.197	4.217	975.25	70.618
2024-05-26T06:00:00Z	21.182	89.852	8.396	4.319	972.373	69.171
2024-05-26T12:00:00Z	22.134	89.79	1.686	0.867	973.012	69.174
2024-05-26T18:00:00Z	22.965	89.927	5.1	2.624	976.344	52.848
2024-05-27T00:00:00Z	23.652	90.145	2.518	1.296	980.161	52.887
2024-05-27T06:00:00Z	24.23	90.683	7.026	3.614	983.073	36.86
2024-05-27T12:00:00Z	24.553	91.159	6.778	3.487	985.724	36.286
2024-05-27T18:00:00Z	24.743	91.489	11.339	5.833	985.724	21.662
2024-05-28T00:00:00Z	24.743	91.489	7.015	3.609	985.724	16.586
2024-05-28T06:00:00Z	24.743	91.489	1.743	0.897	985.724	13.98

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km
2024-05-24T00:00:00Z	9.675	149.765
2024-05-24T06:00:00Z	15.943	120.709
2024-05-24T12:00:00Z	22.551	38.422
2024-05-24T18:00:00Z	20.088	147.916
2024-05-25T00:00:00Z	22.246	127.212
2024-05-25T06:00:00Z	24.565	146.929
2024-05-25T12:00:00Z	23.039	122.001
2024-05-25T18:00:00Z	26.722	145.374
2024-05-26T00:00:00Z	36.329	60.035
2024-05-26T06:00:00Z	35.585	70.32
2024-05-26T12:00:00Z	35.586	91.829
2024-05-26T18:00:00Z	27.188	139.351
2024-05-27T00:00:00Z	27.208	116.395
2024-05-27T06:00:00Z	18.963	44.67
2024-05-27T12:00:00Z	18.667	47.408
2024-05-27T18:00:00Z	11.144	81.98
2024-05-28T00:00:00Z	8.533	121.903
2024-05-28T06:00:00Z	7.192	135.35

Remal_ERA5_6hr_Smoothed_Flagged.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24 00:00:00+00:00	15.409	88.643	6.786	3.491	997.081	18.807
2024-05-24 06:00:00+00:00	15.363	88.831	9.911	5.099	996.421	30.99
2024-05-24 12:00:00+00:00	15.835	89.447	11.541	5.937	992.596	43.835
2024-05-24 18:00:00+00:00	16.498	89.97	10.362	5.331	993.724	39.048
2024-05-25 00:00:00+00:00	17.543	90.262	17.427	8.965	991.339	43.243
2024-05-25 06:00:00+00:00	18.313	90.094	12.142	6.246	988.791	47.751
2024-05-25 12:00:00+00:00	18.955	89.866	9.418	4.845	985.755	44.784
2024-05-25 18:00:00+00:00	19.536	89.857	17.011	8.751	983.638	51.944
2024-05-26 00:00:00+00:00	20.271	89.864	8.197	4.217	975.25	70.618
2024-05-26 06:00:00+00:00	21.182	89.852	8.396	4.319	972.373	69.171
2024-05-26 12:00:00+00:00	22.134	89.79	1.686	0.867	973.012	69.174
2024-05-26 18:00:00+00:00	22.965	89.927	5.1	2.624	976.344	52.848
2024-05-27 00:00:00+00:00	23.652	90.145	2.518	1.296	980.161	52.887
2024-05-27 06:00:00+00:00	24.23	90.683	7.026	3.614	983.073	36.86
2024-05-27 12:00:00+00:00	24.553	91.159	6.778	3.487	985.724	36.286
2024-05-27 18:00:00+00:00	24.743	91.489	11.339	5.833	985.724	21.662
2024-05-28 00:00:00+00:00	24.743	91.489	7.015	3.609	985.724	16.586
2024-05-28 06:00:00+00:00	24.743	91.489	1.743	0.897	985.724	13.98

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km	carried_forward	smoothing_uses_carried_forward
2024-05-24 00:00:00+00:00	9.675	149.765	False	False
2024-05-24 06:00:00+00:00	15.943	120.709	False	False
2024-05-24 12:00:00+00:00	22.551	38.422	False	False
2024-05-24 18:00:00+00:00	20.088	147.916	False	False
2024-05-25 00:00:00+00:00	22.246	127.212	False	False
2024-05-25 06:00:00+00:00	24.565	146.929	False	False
2024-05-25 12:00:00+00:00	23.039	122.001	False	False
2024-05-25 18:00:00+00:00	26.722	145.374	False	False
2024-05-26 00:00:00+00:00	36.329	60.035	False	False
2024-05-26 06:00:00+00:00	35.585	70.32	False	False
2024-05-26 12:00:00+00:00	35.586	91.829	False	False
2024-05-26 18:00:00+00:00	27.188	139.351	False	False
2024-05-27 00:00:00+00:00	27.208	116.395	False	False
2024-05-27 06:00:00+00:00	18.963	44.67	False	False
2024-05-27 12:00:00+00:00	18.667	47.408	False	True
2024-05-27 18:00:00+00:00	11.144	81.98	True	True
2024-05-28 00:00:00+00:00	8.533	121.903	True	True
2024-05-28 06:00:00+00:00	7.192	135.35	True	True

Remal_FNL_6hr_Centroid.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24T00:00:00Z	15.271	88.835	6.832	3.514	996.495	27.534
2024-05-24T06:00:00Z	15.377	89.978	1.131	0.582	995.149	37.592
2024-05-24T12:00:00Z	16.07	90.13	10.485	5.394	992.54	35.911
2024-05-24T18:00:00Z	16.271	89.618	7.4	3.807	991.876	45.595
2024-05-25T00:00:00Z	17.226	90.325	1.746	0.898	990.428	44.169
2024-05-25T06:00:00Z	18.59	89.573	34.842	17.924	983.934	56.344

2024-05-25T12:00:00Z	18.177	89.73	8.837	4.546	981.328	56.782
2024-05-25T18:00:00Z	18.854	89.459	20.205	10.394	979.248	67.184
2024-05-26T00:00:00Z	19.639	89.448	10.45	5.376	969.687	87.732
2024-05-26T06:00:00Z	20.522	89.601	7.243	3.726	969.25	83.407
2024-05-26T12:00:00Z	21.715	90.086	17.551	9.029	968.954	71.381
2024-05-26T18:00:00Z	22.518	90.244	4.887	2.514	973.926	65.896
2024-05-27T00:00:00Z	23.528	90.696	4.887	2.514	977.931	50.216
2024-05-27T06:00:00Z	24.197	91.109	7.493	3.855	982.86	36.774
2024-05-27T12:00:00Z	24.844	92.036	14.636	7.529	985.189	33.182
2024-05-27T18:00:00Z	24.844	92.036	11.259	5.792	985.189	29.309
2024-05-28T00:00:00Z	24.844	92.036	4.657	2.396	985.189	19.763
2024-05-28T06:00:00Z	24.844	92.036	7.857	4.042	985.189	20.726

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km
2024-05-24T00:00:00Z	14.165	149.555
2024-05-24T06:00:00Z	19.339	127.21
2024-05-24T12:00:00Z	18.474	146.483
2024-05-24T18:00:00Z	23.456	116.62
2024-05-25T00:00:00Z	22.723	139.093
2024-05-25T06:00:00Z	28.986	14.136
2024-05-25T12:00:00Z	29.211	74.966
2024-05-25T18:00:00Z	34.563	94.754
2024-05-26T00:00:00Z	45.133	110.946
2024-05-26T06:00:00Z	42.908	98.637
2024-05-26T12:00:00Z	36.721	58.178
2024-05-26T18:00:00Z	33.9	102.523
2024-05-27T00:00:00Z	25.833	83.324
2024-05-27T06:00:00Z	18.918	61.054
2024-05-27T12:00:00Z	17.07	37.624
2024-05-27T18:00:00Z	15.078	121.321
2024-05-28T00:00:00Z	10.167	109.538
2024-05-28T06:00:00Z	10.662	116.382

Remal_FNL_6hr_Diagnostics.csv

time_utc	status	candidate_latitude	candidate_longitude	candidate_pressure_hpa	movement_km	pressure_change_hpa
2024-05-24T00:00:00Z	FIRST	15.271	88.835	996.495		
2024-05-24T06:00:00Z	ACCEPTED	15.377	89.978	995.149	123.175	1.345
2024-05-24T12:00:00Z	ACCEPTED	16.07	90.13	992.54	78.788	2.609
2024-05-24T18:00:00Z	ACCEPTED	16.271	89.618	991.876	59.039	0.664
2024-05-25T00:00:00Z	ACCEPTED	17.226	90.325	990.428	130.22	1.448
2024-05-25T06:00:00Z	ACCEPTED	18.59	89.573	983.934	171.237	6.495
2024-05-25T12:00:00Z	ACCEPTED	18.177	89.73	981.328	48.837	2.606
2024-05-25T18:00:00Z	ACCEPTED	18.854	89.459	979.248	80.617	2.08
2024-05-26T00:00:00Z	ACCEPTED	19.639	89.448	969.687	87.246	9.561
2024-05-26T06:00:00Z	ACCEPTED	20.522	89.601	969.25	99.52	0.437
2024-05-26T12:00:00Z	ACCEPTED	21.715	90.086	968.954	141.802	0.296
2024-05-26T18:00:00Z	ACCEPTED	22.518	90.244	973.926	90.861	4.972
2024-05-27T00:00:00Z	ACCEPTED	23.528	90.696	977.931	121.365	4.005
2024-05-27T06:00:00Z	ACCEPTED	24.197	91.109	982.86	85.53	4.929
2024-05-27T12:00:00Z	ACCEPTED	24.844	92.036	985.189	118.148	2.329
2024-05-27T18:00:00Z	REJECTED	24.748	83.335	990.801	878.199	5.612
2024-05-28T00:00:00Z	REJECTED	24.871	83.363	991.189	874.819	6.0
2024-05-28T06:00:00Z	REJECTED	24.546	84.06	991.695	806.314	6.507

Remal_FNL_6hr_Raw.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24T00:00:00Z	14.514	88.987	11.598	5.967	996.495	29.685
2024-05-24T06:00:00Z	15.727	90.579	17.639	9.074	995.149	37.592
2024-05-24T12:00:00Z	16.532	89.657	10.946	5.631	992.54	34.986
2024-05-24T18:00:00Z	16.211	89.573	8.67	4.46	991.876	45.595
2024-05-25T00:00:00Z	16.934	90.914	35.042	18.027	990.428	44.169
2024-05-25T06:00:00Z	18.61	89.573	34.842	17.924	983.934	56.344
2024-05-25T12:00:00Z	18.292	89.573	5.075	2.611	981.328	56.782
2024-05-25T18:00:00Z	18.451	89.07	21.642	11.133	979.248	67.184
2024-05-26T00:00:00Z	19.639	89.489	10.45	5.376	969.687	87.732
2024-05-26T06:00:00Z	20.505	89.489	1.105	0.569	969.25	83.407
2024-05-26T12:00:00Z	21.678	90.16	20.349	10.469	968.954	71.381
2024-05-26T18:00:00Z	22.532	90.244	4.887	2.514	973.926	65.896
2024-05-27T00:00:00Z	23.458	90.83	9.164	4.714	977.931	50.216
2024-05-27T06:00:00Z	24.224	91.166	10.351	5.325	982.86	36.774
2024-05-27T12:00:00Z	24.91	92.088	7.846	4.036	985.189	33.182
2024-05-27T18:00:00Z	25.214	82.114	6.405	3.295	990.801	12.64
2024-05-28T00:00:00Z	24.834	83.958	10.287	5.292	991.189	19.293
2024-05-28T06:00:00Z	24.453	83.791	10.903	5.609	991.695	15.422

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km
2024-05-24T00:00:00Z	15.271	108.492
2024-05-24T06:00:00Z	19.339	122.315
2024-05-24T12:00:00Z	17.998	145.338
2024-05-24T18:00:00Z	23.456	116.838
2024-05-25T00:00:00Z	22.723	100.687
2024-05-25T06:00:00Z	28.986	12.487
2024-05-25T12:00:00Z	29.211	95.837
2024-05-25T18:00:00Z	34.563	45.126
2024-05-26T00:00:00Z	45.133	107.318
2024-05-26T06:00:00Z	42.908	106.735
2024-05-26T12:00:00Z	36.721	50.563
2024-05-26T18:00:00Z	33.9	103.995
2024-05-27T00:00:00Z	25.833	86.926
2024-05-27T06:00:00Z	18.918	61.999
2024-05-27T12:00:00Z	17.07	43.162
2024-05-27T18:00:00Z	6.503	54.07
2024-05-28T00:00:00Z	9.925	146.127
2024-05-28T06:00:00Z	7.934	145.219

Remal_FNL_6hr_Smoothed.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24T00:00:00Z	15.324	89.406	6.832	3.514	996.495	27.534
2024-05-24T06:00:00Z	15.572	89.648	1.131	0.582	995.149	37.592

2024-05-24T12:00:00Z	15.906	89.909	10.485	5.394	992.54	35.911
2024-05-24T18:00:00Z	16.522	90.025	7.4	3.807	991.876	45.595
2024-05-25T00:00:00Z	17.362	89.839	1.746	0.898	990.428	44.169
2024-05-25T06:00:00Z	17.997	89.876	34.842	17.924	983.934	56.344
2024-05-25T12:00:00Z	18.54	89.587	8.837	4.546	981.328	56.782
2024-05-25T18:00:00Z	18.89	89.546	20.205	10.394	979.248	67.184
2024-05-26T00:00:00Z	19.672	89.503	10.45	5.376	969.687	87.732
2024-05-26T06:00:00Z	20.625	89.712	7.243	3.726	969.25	83.407
2024-05-26T12:00:00Z	21.585	89.977	17.551	9.029	968.954	71.381
2024-05-26T18:00:00Z	22.587	90.342	4.887	2.514	973.926	65.896
2024-05-27T00:00:00Z	23.415	90.683	4.887	2.514	977.931	50.216
2024-05-27T06:00:00Z	24.19	91.28	7.493	3.855	982.86	36.774
2024-05-27T12:00:00Z	24.629	91.727	14.636	7.529	985.189	33.182
2024-05-27T18:00:00Z	24.844	92.036	11.259	5.792	985.189	29.309
2024-05-28T00:00:00Z	24.844	92.036	4.657	2.396	985.189	19.763
2024-05-28T06:00:00Z	24.844	92.036	7.857	4.042	985.189	20.726

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km
2024-05-24T00:00:00Z	14.165	149.555
2024-05-24T06:00:00Z	19.339	127.21
2024-05-24T12:00:00Z	18.474	146.483
2024-05-24T18:00:00Z	23.456	116.62
2024-05-25T00:00:00Z	22.723	139.093
2024-05-25T06:00:00Z	28.986	14.136
2024-05-25T12:00:00Z	29.211	74.966
2024-05-25T18:00:00Z	34.563	94.754
2024-05-26T00:00:00Z	45.133	110.946
2024-05-26T06:00:00Z	42.908	98.637
2024-05-26T12:00:00Z	36.721	58.178
2024-05-26T18:00:00Z	33.9	102.523
2024-05-27T00:00:00Z	25.833	83.324
2024-05-27T06:00:00Z	18.918	61.054
2024-05-27T12:00:00Z	17.07	37.624
2024-05-27T18:00:00Z	15.078	121.321
2024-05-28T00:00:00Z	10.167	109.538
2024-05-28T06:00:00Z	10.662	116.382

Remal_FNL_6hr_Smoothed_Flagged.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24 00:00:00+00:00	15.324	89.406	6.832	3.514	996.495	27.534
2024-05-24 06:00:00+00:00	15.572	89.648	1.131	0.582	995.149	37.592
2024-05-24 12:00:00+00:00	15.906	89.909	10.485	5.394	992.54	35.911
2024-05-24 18:00:00+00:00	16.522	90.025	7.4	3.807	991.876	45.595
2024-05-25 00:00:00+00:00	17.362	89.839	1.746	0.898	990.428	44.169
2024-05-25 06:00:00+00:00	17.997	89.876	34.842	17.924	983.934	56.344
2024-05-25 12:00:00+00:00	18.54	89.587	8.837	4.546	981.328	56.782
2024-05-25 18:00:00+00:00	18.89	89.546	20.205	10.394	979.248	67.184
2024-05-26 00:00:00+00:00	19.672	89.503	10.45	5.376	969.687	87.732
2024-05-26 06:00:00+00:00	20.625	89.712	7.243	3.726	969.25	83.407
2024-05-26 12:00:00+00:00	21.585	89.977	17.551	9.029	968.954	71.381
2024-05-26 18:00:00+00:00	22.587	90.342	4.887	2.514	973.926	65.896

2024-05-27 00:00:00+00:00	23.415	90.683	4.887	2.514	977.931	50.216
2024-05-27 06:00:00+00:00	24.19	91.28	7.493	3.855	982.86	36.774
2024-05-27 12:00:00+00:00	24.629	91.727	14.636	7.529	985.189	33.182
2024-05-27 18:00:00+00:00	24.844	92.036	11.259	5.792	985.189	29.309
2024-05-28 00:00:00+00:00	24.844	92.036	4.657	2.396	985.189	19.763
2024-05-28 06:00:00+00:00	24.844	92.036	7.857	4.042	985.189	20.726

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km	carried_forward	smoothing_uses_carried_forward
2024-05-24 00:00:00+00:00	14.165	149.555	False	False
2024-05-24 06:00:00+00:00	19.339	127.21	False	False
2024-05-24 12:00:00+00:00	18.474	146.483	False	False
2024-05-24 18:00:00+00:00	23.456	116.62	False	False
2024-05-25 00:00:00+00:00	22.723	139.093	False	False
2024-05-25 06:00:00+00:00	28.986	14.136	False	False
2024-05-25 12:00:00+00:00	29.211	74.966	False	False
2024-05-25 18:00:00+00:00	34.563	94.754	False	False
2024-05-26 00:00:00+00:00	45.133	110.946	False	False
2024-05-26 06:00:00+00:00	42.908	98.637	False	False
2024-05-26 12:00:00+00:00	36.721	58.178	False	False
2024-05-26 18:00:00+00:00	33.9	102.523	False	False
2024-05-27 00:00:00+00:00	25.833	83.324	False	False
2024-05-27 06:00:00+00:00	18.918	61.054	False	False
2024-05-27 12:00:00+00:00	17.07	37.624	False	True
2024-05-27 18:00:00+00:00	15.078	121.321	True	True
2024-05-28 00:00:00+00:00	10.167	109.538	True	True
2024-05-28 06:00:00+00:00	10.662	116.382	True	True

Remal_GFS_6hr_Centroid.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24T00:00:00Z	14.312	89.261	18.84	9.692	994.595	35.808
2024-05-24T06:00:00Z	15.463	89.933	6.433	3.309	994.715	38.007
2024-05-24T12:00:00Z	16.009	90.053	9.664	4.972	992.595	37.204
2024-05-24T18:00:00Z	16.429	90.328	7.187	3.697	989.55	47.484
2024-05-25T00:00:00Z	17.376	90.075	2.915	1.499	986.911	42.445
2024-05-25T06:00:00Z	17.548	90.124	17.104	8.799	982.054	51.848
2024-05-25T12:00:00Z	18.289	89.835	3.093	1.591	979.108	52.856
2024-05-25T18:00:00Z	18.774	89.278	2.716	1.397	974.391	72.186
2024-05-26T00:00:00Z	19.678	89.549	6.005	3.089	966.838	90.721
2024-05-26T06:00:00Z	20.485	89.69	5.61	2.886	968.131	77.003
2024-05-26T12:00:00Z	21.798	89.834	4.189	2.155	967.224	70.257
2024-05-26T18:00:00Z	22.571	89.97	7.783	4.004	973.094	65.484
2024-05-27T00:00:00Z	23.512	90.203	7.707	3.965	977.045	48.002
2024-05-27T06:00:00Z	24.144	90.549	10.103	5.198	982.225	39.422
2024-05-27T12:00:00Z	24.711	91.656	12.658	6.512	984.181	35.19
2024-05-27T18:00:00Z	24.711	91.656	11.796	6.068	984.181	28.139
2024-05-28T00:00:00Z	24.711	91.656	7.995	4.113	984.181	17.169
2024-05-28T06:00:00Z	24.711	91.656	5.25	2.701	984.181	17.491

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km
2024-05-24T00:00:00Z	18.421	75.88
2024-05-24T06:00:00Z	19.552	127.097
2024-05-24T12:00:00Z	19.14	145.714
2024-05-24T18:00:00Z	24.428	148.829
2024-05-25T00:00:00Z	21.836	125.906
2024-05-25T06:00:00Z	26.673	78.083
2024-05-25T12:00:00Z	27.192	74.402
2024-05-25T18:00:00Z	37.135	86.174
2024-05-26T00:00:00Z	46.671	86.054
2024-05-26T06:00:00Z	39.614	71.132
2024-05-26T12:00:00Z	36.143	67.504
2024-05-26T18:00:00Z	33.688	90.747
2024-05-27T00:00:00Z	24.695	119.288
2024-05-27T06:00:00Z	20.281	80.564
2024-05-27T12:00:00Z	18.103	67.42
2024-05-27T18:00:00Z	14.476	148.855
2024-05-28T00:00:00Z	8.832	132.327
2024-05-28T06:00:00Z	8.998	83.182

Remal_GFS_6hr_Diagnostics.csv

time_utc	status	candidate_latitude	candidate_longitude	candidate_pressure_hpa	movement_km	pressure_change_hpa
2024-05-24T00:00:00Z	FIRST	14.312	89.261	994.595		
2024-05-24T06:00:00Z	ACCEPTED	15.463	89.933	994.715	146.934	0.121
2024-05-24T12:00:00Z	ACCEPTED	16.009	90.053	992.595	62.049	2.12
2024-05-24T18:00:00Z	ACCEPTED	16.429	90.328	989.55	55.171	3.045
2024-05-25T00:00:00Z	ACCEPTED	17.376	90.075	986.911	108.676	2.638
2024-05-25T06:00:00Z	ACCEPTED	17.548	90.124	982.054	19.853	4.858
2024-05-25T12:00:00Z	ACCEPTED	18.289	89.835	979.108	87.894	2.946
2024-05-25T18:00:00Z	ACCEPTED	18.774	89.278	974.391	79.686	4.717
2024-05-26T00:00:00Z	ACCEPTED	19.678	89.549	966.838	104.548	7.553
2024-05-26T06:00:00Z	ACCEPTED	20.485	89.69	968.131	90.944	1.293
2024-05-26T12:00:00Z	ACCEPTED	21.798	89.834	967.224	146.763	0.907
2024-05-26T18:00:00Z	ACCEPTED	22.571	89.97	973.094	87.081	5.87
2024-05-27T00:00:00Z	ACCEPTED	23.512	90.203	977.045	107.242	3.952
2024-05-27T06:00:00Z	ACCEPTED	24.144	90.549	982.225	78.609	5.18
2024-05-27T12:00:00Z	ACCEPTED	24.711	91.656	984.181	128.605	1.956
2024-05-27T18:00:00Z	REJECTED	24.927	83.261	990.562	847.5	6.381
2024-05-28T00:00:00Z	REJECTED	24.921	83.519	990.793	821.481	6.612
2024-05-28T06:00:00Z	REJECTED	24.286	84.568	991.955	718.64	7.774

Remal_GFS_6hr_Raw.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24T00:00:00Z	13.782	89.741	31.521	16.216	994.595	35.808
2024-05-24T06:00:00Z	15.969	90.914	26.987	13.883	994.715	38.007
2024-05-24T12:00:00Z	16.452	90.663	25.981	13.366	992.595	37.684
2024-05-24T18:00:00Z	16.452	90.328	7.187	3.697	989.55	47.484
2024-05-25T00:00:00Z	17.414	90.244	9.316	4.792	986.911	42.445
2024-05-25T06:00:00Z	17.574	90.076	5.722	2.944	982.054	51.848
2024-05-25T12:00:00Z	18.292	89.825	3.093	1.591	979.108	52.856
2024-05-25T18:00:00Z	18.848	89.322	9.763	5.023	974.391	72.186
2024-05-26T00:00:00Z	19.718	89.489	9.61	4.944	966.838	90.721

2024-05-26T06:00:00Z	20.426	89.657	6.159	3.169	968.131	77.003
2024-05-26T12:00:00Z	21.911	89.992	17.142	8.818	967.224	70.257
2024-05-26T18:00:00Z	22.532	89.909	9.38	4.826	973.094	65.484
2024-05-27T00:00:00Z	23.534	90.244	7.707	3.965	977.045	48.002
2024-05-27T06:00:00Z	24.148	90.495	4.579	2.356	982.225	39.422
2024-05-27T12:00:00Z	24.758	91.585	3.849	1.98	984.181	35.19
2024-05-27T18:00:00Z	25.29	82.785	3.084	1.587	990.562	14.364
2024-05-28T00:00:00Z	25.138	83.036	8.046	4.139	990.793	13.281
2024-05-28T06:00:00Z	24.453	83.791	11.683	6.01	991.955	16.222

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km
2024-05-24T00:00:00Z	18.421	20.241
2024-05-24T06:00:00Z	19.552	8.959
2024-05-24T12:00:00Z	19.386	99.965
2024-05-24T18:00:00Z	24.428	147.723
2024-05-25T00:00:00Z	21.836	111.56
2024-05-25T06:00:00Z	26.673	83.904
2024-05-25T12:00:00Z	27.192	74.975
2024-05-25T18:00:00Z	37.135	95.595
2024-05-26T00:00:00Z	46.671	93.436
2024-05-26T06:00:00Z	39.614	70.479
2024-05-26T12:00:00Z	36.143	78.002
2024-05-26T18:00:00Z	33.688	86.747
2024-05-27T00:00:00Z	24.695	114.606
2024-05-27T06:00:00Z	20.281	84.003
2024-05-27T12:00:00Z	18.103	66.214
2024-05-27T18:00:00Z	7.389	83.594
2024-05-28T00:00:00Z	6.832	86.992
2024-05-28T06:00:00Z	8.345	108.962

Remal_GFS_6hr_Smoothed.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24T00:00:00Z	14.888	89.597	18.84	9.692	994.595	35.808
2024-05-24T06:00:00Z	15.261	89.749	6.433	3.309	994.715	38.007
2024-05-24T12:00:00Z	15.967	90.105	9.664	4.972	992.595	37.204
2024-05-24T18:00:00Z	16.605	90.152	7.187	3.697	989.55	47.484
2024-05-25T00:00:00Z	17.118	90.175	2.915	1.499	986.911	42.445
2024-05-25T06:00:00Z	17.738	90.011	17.104	8.799	982.054	51.848
2024-05-25T12:00:00Z	18.204	89.746	3.093	1.591	979.108	52.856
2024-05-25T18:00:00Z	18.914	89.554	2.716	1.397	974.391	72.186
2024-05-26T00:00:00Z	19.646	89.506	6.005	3.089	966.838	90.721
2024-05-26T06:00:00Z	20.654	89.691	5.61	2.886	968.131	77.003
2024-05-26T12:00:00Z	21.618	89.831	4.189	2.155	967.224	70.257
2024-05-26T18:00:00Z	22.627	90.002	7.783	4.004	973.094	65.484
2024-05-27T00:00:00Z	23.409	90.241	7.707	3.965	977.045	48.002
2024-05-27T06:00:00Z	24.122	90.803	10.103	5.198	982.225	39.422
2024-05-27T12:00:00Z	24.522	91.287	12.658	6.512	984.181	35.19
2024-05-27T18:00:00Z	24.711	91.656	11.796	6.068	984.181	28.139
2024-05-28T00:00:00Z	24.711	91.656	7.995	4.113	984.181	17.169
2024-05-28T06:00:00Z	24.711	91.656	5.25	2.701	984.181	17.491

Column block 2 of 2

time_utc	max_wind_ms	max_wind_distance_km
2024-05-24T00:00:00Z	18.421	75.88
2024-05-24T06:00:00Z	19.552	127.097
2024-05-24T12:00:00Z	19.14	145.714
2024-05-24T18:00:00Z	24.428	148.829
2024-05-25T00:00:00Z	21.836	125.906
2024-05-25T06:00:00Z	26.673	78.083
2024-05-25T12:00:00Z	27.192	74.402
2024-05-25T18:00:00Z	37.135	86.174
2024-05-26T00:00:00Z	46.671	86.054
2024-05-26T06:00:00Z	39.614	71.132
2024-05-26T12:00:00Z	36.143	67.504
2024-05-26T18:00:00Z	33.688	90.747
2024-05-27T00:00:00Z	24.695	119.288
2024-05-27T06:00:00Z	20.281	80.564
2024-05-27T12:00:00Z	18.103	67.42
2024-05-27T18:00:00Z	14.476	148.855
2024-05-28T00:00:00Z	8.832	132.327
2024-05-28T06:00:00Z	8.998	83.182

Remal_GFS_6hr_Smoothed_Flagged.csv

Column block 1 of 2

time_utc	latitude	longitude	wind_speed_kt	wind_speed_ms	central_pressure_hpa	max_wind_kt
2024-05-24 00:00:00+00:00	14.888	89.597	18.84	9.692	994.595	35.808
2024-05-24 06:00:00+00:00	15.261	89.749	6.433	3.309	994.715	38.007
2024-05-24 12:00:00+00:00	15.967	90.105	9.664	4.972	992.595	37.204
2024-05-24 18:00:00+00:00	16.605	90.152	7.187	3.697	989.55	47.484
2024-05-25 00:00:00+00:00	17.118	90.175	2.915	1.499	986.911	42.445
2024-05-25 06:00:00+00:00	17.738	90.011	17.104	8.799	982.054	51.848
2024-05-25 12:00:00+00:00	18.204	89.746	3.093	1.591	979.108	52.856
2024-05-25 18:00:00+00:00	18.914	89.554	2.716	1.397	974.391	72.186
2024-05-26 00:00:00+00:00	19.646	89.506	6.005	3.089	966.838	90.721
2024-05-26 06:00:00+00:00	20.654	89.691	5.61	2.886	968.131	77.003
2024-05-26 12:00:00+00:00	21.618	89.831	4.189	2.155	967.224	70.257
2024-05-26 18:00:00+00:00	22.627	90.002	7.783	4.004	973.094	65.484
2024-05-27 00:00:00+00:00	23.409	90.241	7.707	3.965	977.045	48.002
2024-05-27 06:00:00+00:00	24.122	90.803	10.103	5.198	982.225	39.422
2024-05-27 12:00:00+00:00	24.522	91.287	12.658	6.512	984.181	35.19
2024-05-27 18:00:00+00:00	24.711	91.656	11.796	6.068	984.181	28.139
2024-05-28 00:00:00+00:00	24.711	91.656	7.995	4.113	984.181	17.169
2024-05-28 06:00:00+00:00	24.711	91.656	5.25	2.701	984.181	17.491

Column block 2 of 2

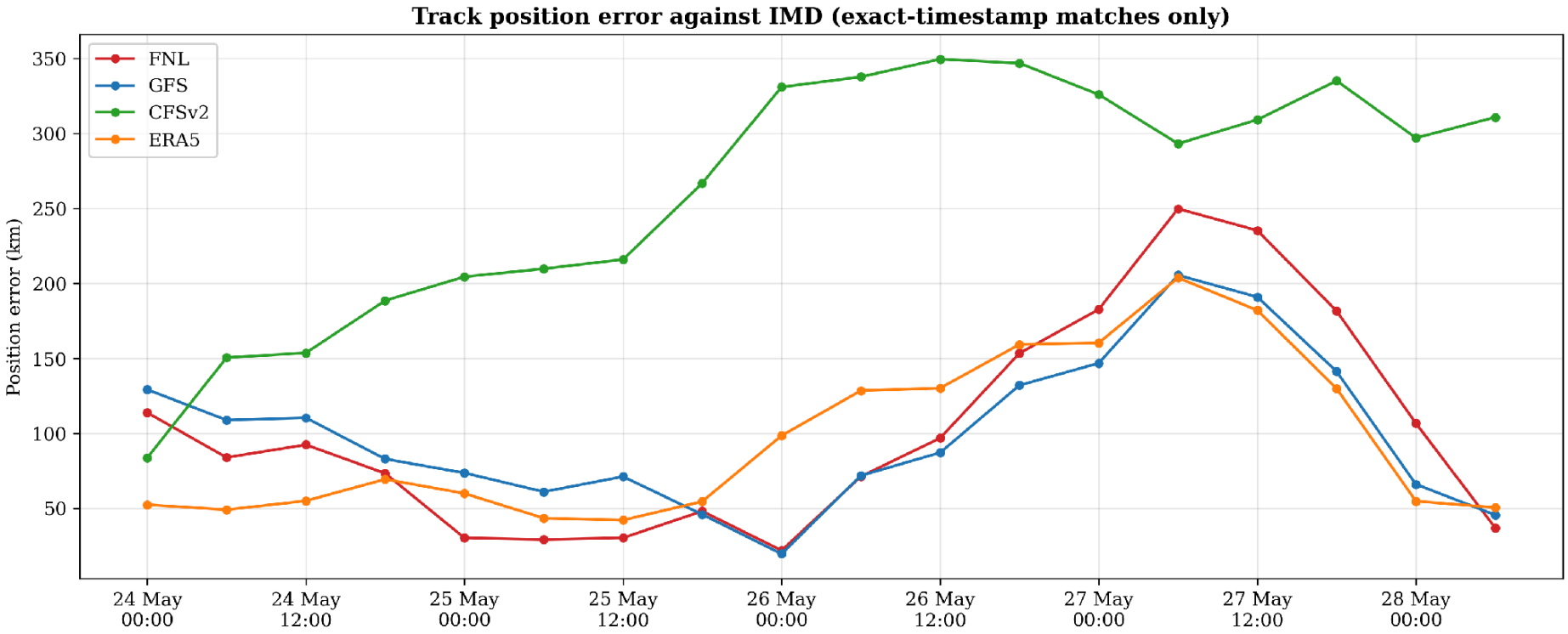
time_utc	max_wind_ms	max_wind_distance_km	carried_forward	smoothing_uses_carried_forward
2024-05-24 00:00:00+00:00	18.421	75.88	False	False
2024-05-24 06:00:00+00:00	19.552	127.097	False	False
2024-05-24 12:00:00+00:00	19.14	145.714	False	False
2024-05-24 18:00:00+00:00	24.428	148.829	False	False
2024-05-25 00:00:00+00:00	21.836	125.906	False	False
2024-05-25 06:00:00+00:00	26.673	78.083	False	False

2024-05-25 12:00:00+00:00	27.192	74.402	False	False
2024-05-25 18:00:00+00:00	37.135	86.174	False	False
2024-05-26 00:00:00+00:00	46.671	86.054	False	False
2024-05-26 06:00:00+00:00	39.614	71.132	False	False
2024-05-26 12:00:00+00:00	36.143	67.504	False	False
2024-05-26 18:00:00+00:00	33.688	90.747	False	False
2024-05-27 00:00:00+00:00	24.695	119.288	False	False
2024-05-27 06:00:00+00:00	20.281	80.564	False	False
2024-05-27 12:00:00+00:00	18.103	67.42	False	True
2024-05-27 18:00:00+00:00	14.476	148.855	True	True
2024-05-28 00:00:00+00:00	8.832	132.327	True	True
2024-05-28 06:00:00+00:00	8.998	83.182	True	True

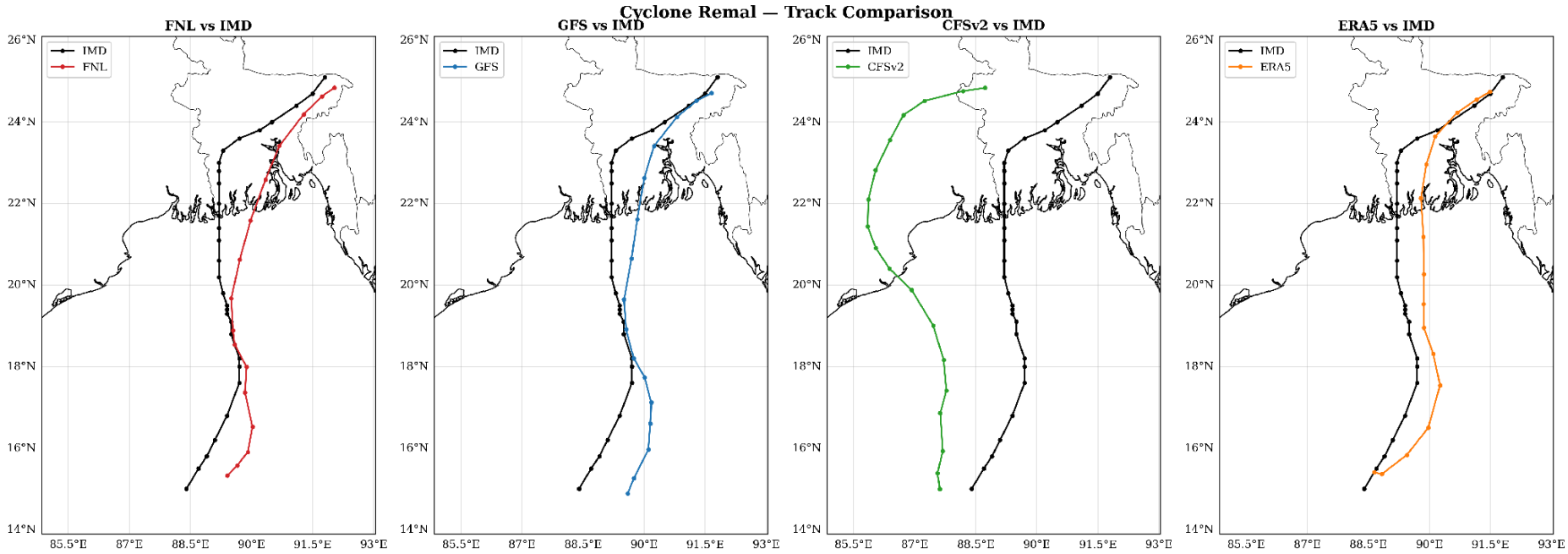
A03 - Track validation against IMD

Figures: 2 | CSV tables: 9 | NetCDF extracts: 0

Remal_A2_Position_Error_Series.png



Remal_A2_Track_Maps.png



Remal_A2_Track_Stats.csv

model	n_matched	mean_error_km	track_rmse_km	max_error_km	max_error_time	n_carried_forward_in_sample
FNL	18	102.128	123.464	249.855	2024-05-27 06:00:00+00:00	3
GFS	18	99.46	110.864	205.546	2024-05-27 06:00:00+00:00	3
CFSv2	18	261.708	273.257	349.622	2024-05-26 12:00:00+00:00	0
ERA5	18	95.789	109.202	203.785	2024-05-27 06:00:00+00:00	3

Remal_CFSv2_vs_IMD.csv

Column block 1 of 3

time_utc	lat_imd	lon_imd	wind_imd	pres_imd	lat_model	lon_model
2024-05-24 00:00:00+00:00	15.0	88.4	25	996	14.996	87.621
2024-05-24 06:00:00+00:00	15.8	88.9	25	996	15.383	87.563
2024-05-24 12:00:00+00:00	16.2	89.1	25	995	15.93	87.689
2024-05-24 18:00:00+00:00	16.8	89.4	25	995	16.861	87.629
2024-05-25 00:00:00+00:00	17.6	89.7	30	993	17.413	87.781
2024-05-25 06:00:00+00:00	18.2	89.7	30	991	18.161	87.714
2024-05-25 12:00:00+00:00	18.8	89.5	35	990	19.003	87.458
2024-05-25 18:00:00+00:00	19.3	89.4	40	988	19.88	86.928
2024-05-26 00:00:00+00:00	19.5	89.4	50	984	20.403	86.382
2024-05-26 06:00:00+00:00	20.2	89.2	50	983	20.911	86.045
2024-05-26 12:00:00+00:00	21.1	89.2	60	978	21.444	85.846
2024-05-26 18:00:00+00:00	21.7	89.2	60	978	22.1	85.865
2024-05-27 00:00:00+00:00	22.5	89.2	45	982	22.824	86.042
2024-05-27 06:00:00+00:00	23.0	89.2	35	985	23.56	86.395

2024-05-27 12:00:00+00:00	23.6	89.7	35	987	24.17	86.723
2024-05-27 18:00:00+00:00	24.0	90.5	30	990	24.52	87.243
2024-05-28 00:00:00+00:00	24.4	91.1	20	991	24.76	88.189
2024-05-28 06:00:00+00:00	25.1	91.8	20	993	24.841	88.729

Column block 2 of 3

time_utc	wind_model	wind_centre_model	pres_model	carried_forward	track_error_km	wind_bias_kt
2024-05-24 00:00:00+00:00	23.25	7.162	994.543	False	83.707	-1.75
2024-05-24 06:00:00+00:00	40.736	18.452	993.517	False	150.505	15.736
2024-05-24 12:00:00+00:00	38.887	6.666	991.532	False	153.679	13.887
2024-05-24 18:00:00+00:00	47.449	9.079	990.357	False	188.576	22.449
2024-05-25 00:00:00+00:00	51.246	4.023	985.717	False	204.515	21.246
2024-05-25 06:00:00+00:00	64.74	14.121	982.594	False	209.823	34.74
2024-05-25 12:00:00+00:00	57.689	13.93	979.397	False	215.969	22.689
2024-05-25 18:00:00+00:00	59.111	4.82	977.909	False	266.881	19.111
2024-05-26 00:00:00+00:00	57.469	6.378	978.015	False	331.047	7.469
2024-05-26 06:00:00+00:00	50.082	6.837	982.21	False	337.826	0.082
2024-05-26 12:00:00+00:00	33.871	8.725	983.162	False	349.622	-26.129
2024-05-26 18:00:00+00:00	30.56	7.263	986.179	False	346.921	-29.44
2024-05-27 00:00:00+00:00	33.901	5.103	985.759	False	326.045	-11.099
2024-05-27 06:00:00+00:00	34.226	8.509	986.709	False	293.236	-0.774
2024-05-27 12:00:00+00:00	32.76	5.065	985.529	False	309.229	-2.24
2024-05-27 18:00:00+00:00	30.908	3.294	987.062	False	335.206	0.908
2024-05-28 00:00:00+00:00	28.454	7.075	987.913	False	297.079	8.454
2024-05-28 06:00:00+00:00	31.141	11.513	991.277	False	310.878	11.141

Column block 3 of 3

time_utc	wind_centre_bias_kt	pres_bias_hpa
2024-05-24 00:00:00+00:00	-17.838	-1.457
2024-05-24 06:00:00+00:00	-6.548	-2.483
2024-05-24 12:00:00+00:00	-18.334	-3.468
2024-05-24 18:00:00+00:00	-15.921	-4.643
2024-05-25 00:00:00+00:00	-25.977	-7.283
2024-05-25 06:00:00+00:00	-15.879	-8.406
2024-05-25 12:00:00+00:00	-21.07	-10.603
2024-05-25 18:00:00+00:00	-35.18	-10.091
2024-05-26 00:00:00+00:00	-43.622	-5.985
2024-05-26 06:00:00+00:00	-43.163	-0.79
2024-05-26 12:00:00+00:00	-51.275	5.162
2024-05-26 18:00:00+00:00	-52.737	8.179
2024-05-27 00:00:00+00:00	-39.897	3.759
2024-05-27 06:00:00+00:00	-26.491	1.709
2024-05-27 12:00:00+00:00	-29.935	-1.471
2024-05-27 18:00:00+00:00	-26.706	-2.938
2024-05-28 00:00:00+00:00	-12.925	-3.087
2024-05-28 06:00:00+00:00	-8.487	-1.723

Remal_CFSv2_vs_IMD_excluding_carried_forward.csv

Column block 1 of 3

time_utc	lat_imd	lon_imd	wind_imd	pres_imd	lat_model	lon_model

2024-05-24 00:00:00+00:00	15.0	88.4	25	996	14.996	87.621
2024-05-24 06:00:00+00:00	15.8	88.9	25	996	15.383	87.563
2024-05-24 12:00:00+00:00	16.2	89.1	25	995	15.93	87.689
2024-05-24 18:00:00+00:00	16.8	89.4	25	995	16.861	87.629
2024-05-25 00:00:00+00:00	17.6	89.7	30	993	17.413	87.781
2024-05-25 06:00:00+00:00	18.2	89.7	30	991	18.161	87.714
2024-05-25 12:00:00+00:00	18.8	89.5	35	990	19.003	87.458
2024-05-25 18:00:00+00:00	19.3	89.4	40	988	19.88	86.928
2024-05-26 00:00:00+00:00	19.5	89.4	50	984	20.403	86.382
2024-05-26 06:00:00+00:00	20.2	89.2	50	983	20.911	86.045
2024-05-26 12:00:00+00:00	21.1	89.2	60	978	21.444	85.846
2024-05-26 18:00:00+00:00	21.7	89.2	60	978	22.1	85.865
2024-05-27 00:00:00+00:00	22.5	89.2	45	982	22.824	86.042
2024-05-27 06:00:00+00:00	23.0	89.2	35	985	23.56	86.395
2024-05-27 12:00:00+00:00	23.6	89.7	35	987	24.17	86.723
2024-05-27 18:00:00+00:00	24.0	90.5	30	990	24.52	87.243
2024-05-28 00:00:00+00:00	24.4	91.1	20	991	24.76	88.189
2024-05-28 06:00:00+00:00	25.1	91.8	20	993	24.841	88.729

Column block 2 of 3

time_utc	wind_model	wind_centre_model	pres_model	carried_forward	track_error_km	wind_bias_kt
2024-05-24 00:00:00+00:00	23.25	7.162	994.543	False	83.707	-1.75
2024-05-24 06:00:00+00:00	40.736	18.452	993.517	False	150.505	15.736
2024-05-24 12:00:00+00:00	38.887	6.666	991.532	False	153.679	13.887
2024-05-24 18:00:00+00:00	47.449	9.079	990.357	False	188.576	22.449
2024-05-25 00:00:00+00:00	51.246	4.023	985.717	False	204.515	21.246
2024-05-25 06:00:00+00:00	64.74	14.121	982.594	False	209.823	34.74
2024-05-25 12:00:00+00:00	57.689	13.93	979.397	False	215.969	22.689
2024-05-25 18:00:00+00:00	59.111	4.82	977.909	False	266.881	19.111
2024-05-26 00:00:00+00:00	57.469	6.378	978.015	False	331.047	7.469
2024-05-26 06:00:00+00:00	50.082	6.837	982.21	False	337.826	0.082
2024-05-26 12:00:00+00:00	33.871	8.725	983.162	False	349.622	-26.129
2024-05-26 18:00:00+00:00	30.56	7.263	986.179	False	346.921	-29.44
2024-05-27 00:00:00+00:00	33.901	5.103	985.759	False	326.045	-11.099
2024-05-27 06:00:00+00:00	34.226	8.509	986.709	False	293.236	-0.774
2024-05-27 12:00:00+00:00	32.76	5.065	985.529	False	309.229	-2.24
2024-05-27 18:00:00+00:00	30.908	3.294	987.062	False	335.206	0.908
2024-05-28 00:00:00+00:00	28.454	7.075	987.913	False	297.079	8.454
2024-05-28 06:00:00+00:00	31.141	11.513	991.277	False	310.878	11.141

Column block 3 of 3

time_utc	wind_centre_bias_kt	pres_bias_hpa
2024-05-24 00:00:00+00:00	-17.838	-1.457
2024-05-24 06:00:00+00:00	-6.548	-2.483
2024-05-24 12:00:00+00:00	-18.334	-3.468
2024-05-24 18:00:00+00:00	-15.921	-4.643
2024-05-25 00:00:00+00:00	-25.977	-7.283
2024-05-25 06:00:00+00:00	-15.879	-8.406
2024-05-25 12:00:00+00:00	-21.07	-10.603
2024-05-25 18:00:00+00:00	-35.18	-10.091
2024-05-26 00:00:00+00:00	-43.622	-5.985
2024-05-26 06:00:00+00:00	-43.163	-0.79
2024-05-26 12:00:00+00:00	-51.275	5.162
2024-05-26 18:00:00+00:00	-52.737	8.179
2024-05-27 00:00:00+00:00	-39.897	3.759

2024-05-27 06:00:00+00:00	-26.491	1.709
2024-05-27 12:00:00+00:00	-29.935	-1.471
2024-05-27 18:00:00+00:00	-26.706	-2.938
2024-05-28 00:00:00+00:00	-12.925	-3.087
2024-05-28 06:00:00+00:00	-8.487	-1.723

Remal_ERA5_vs_IMD.csv

Column block 1 of 3

time_utc	lat_imd	lon_imd	wind_imd	pres_imd	lat_model	lon_model
2024-05-24 00:00:00+00:00	15.0	88.4	25	996	15.409	88.643
2024-05-24 06:00:00+00:00	15.8	88.9	25	996	15.363	88.831
2024-05-24 12:00:00+00:00	16.2	89.1	25	995	15.835	89.447
2024-05-24 18:00:00+00:00	16.8	89.4	25	995	16.498	89.97
2024-05-25 00:00:00+00:00	17.6	89.7	30	993	17.543	90.262
2024-05-25 06:00:00+00:00	18.2	89.7	30	991	18.313	90.094
2024-05-25 12:00:00+00:00	18.8	89.5	35	990	18.955	89.866
2024-05-25 18:00:00+00:00	19.3	89.4	40	988	19.536	89.857
2024-05-26 00:00:00+00:00	19.5	89.4	50	984	20.271	89.864
2024-05-26 06:00:00+00:00	20.2	89.2	50	983	21.182	89.852
2024-05-26 12:00:00+00:00	21.1	89.2	60	978	22.134	89.79
2024-05-26 18:00:00+00:00	21.7	89.2	60	978	22.965	89.927
2024-05-27 00:00:00+00:00	22.5	89.2	45	982	23.652	90.145
2024-05-27 06:00:00+00:00	23.0	89.2	35	985	24.23	90.683
2024-05-27 12:00:00+00:00	23.6	89.7	35	987	24.553	91.159
2024-05-27 18:00:00+00:00	24.0	90.5	30	990	24.743	91.489
2024-05-28 00:00:00+00:00	24.4	91.1	20	991	24.743	91.489
2024-05-28 06:00:00+00:00	25.1	91.8	20	993	24.743	91.489

Column block 2 of 3

time_utc	wind_model	wind_centre_model	pres_model	carried_forward	track_error_km	wind_bias_kt
2024-05-24 00:00:00+00:00	18.807	6.786	997.081	False	52.433	-6.193
2024-05-24 06:00:00+00:00	30.99	9.911	996.421	False	49.113	5.99
2024-05-24 12:00:00+00:00	43.835	11.541	992.596	False	55.013	18.835
2024-05-24 18:00:00+00:00	39.048	10.362	993.724	False	69.373	14.048
2024-05-25 00:00:00+00:00	43.243	17.427	991.339	False	59.947	13.243
2024-05-25 06:00:00+00:00	47.751	12.142	988.791	False	43.461	17.751
2024-05-25 12:00:00+00:00	44.784	9.418	985.755	False	42.139	9.784
2024-05-25 18:00:00+00:00	51.944	17.011	983.638	False	54.613	11.944
2024-05-26 00:00:00+00:00	70.618	8.197	975.25	False	98.503	20.618
2024-05-26 06:00:00+00:00	69.171	8.396	972.373	False	128.527	19.171
2024-05-26 12:00:00+00:00	69.174	1.686	973.012	False	130.208	9.174
2024-05-26 18:00:00+00:00	52.848	5.1	976.344	False	159.307	-7.152
2024-05-27 00:00:00+00:00	52.887	2.518	980.161	False	160.421	7.887
2024-05-27 06:00:00+00:00	36.86	7.026	983.073	False	203.785	1.86
2024-05-27 12:00:00+00:00	36.286	6.778	985.724	False	182.116	1.286
2024-05-27 18:00:00+00:00	21.662	11.339	985.724	True	129.864	-8.338
2024-05-28 00:00:00+00:00	16.586	7.015	985.724	True	54.813	-3.414
2024-05-28 06:00:00+00:00	13.98	1.743	985.724	True	50.564	-6.02

Column block 3 of 3

time_utc	wind_centre_bias_kt	pres_bias_hpa
2024-05-24 00:00:00+00:00	-18.214	1.081
2024-05-24 06:00:00+00:00	-15.089	0.421
2024-05-24 12:00:00+00:00	-13.459	-2.404
2024-05-24 18:00:00+00:00	-14.638	-1.276
2024-05-25 00:00:00+00:00	-12.573	-1.661
2024-05-25 06:00:00+00:00	-17.858	-2.209
2024-05-25 12:00:00+00:00	-25.582	-4.245
2024-05-25 18:00:00+00:00	-22.989	-4.362
2024-05-26 00:00:00+00:00	-41.803	-8.75
2024-05-26 06:00:00+00:00	-41.604	-10.627
2024-05-26 12:00:00+00:00	-58.314	-4.988
2024-05-26 18:00:00+00:00	-54.9	-1.656
2024-05-27 00:00:00+00:00	-42.482	-1.839
2024-05-27 06:00:00+00:00	-27.974	-1.927
2024-05-27 12:00:00+00:00	-28.222	-1.276
2024-05-27 18:00:00+00:00	-18.661	-4.276
2024-05-28 00:00:00+00:00	-12.985	-5.276
2024-05-28 06:00:00+00:00	-18.257	-7.276

Remal_ERA5_vs_IMD_excluding_carried_forward.csv

Column block 1 of 3

time_utc	lat_imd	lon_imd	wind_imd	pres_imd	lat_model	lon_model
2024-05-24 00:00:00+00:00	15.0	88.4	25	996	15.409	88.643
2024-05-24 06:00:00+00:00	15.8	88.9	25	996	15.363	88.831
2024-05-24 12:00:00+00:00	16.2	89.1	25	995	15.835	89.447
2024-05-24 18:00:00+00:00	16.8	89.4	25	995	16.498	89.97
2024-05-25 00:00:00+00:00	17.6	89.7	30	993	17.543	90.262
2024-05-25 06:00:00+00:00	18.2	89.7	30	991	18.313	90.094
2024-05-25 12:00:00+00:00	18.8	89.5	35	990	18.955	89.866
2024-05-25 18:00:00+00:00	19.3	89.4	40	988	19.536	89.857
2024-05-26 00:00:00+00:00	19.5	89.4	50	984	20.271	89.864
2024-05-26 06:00:00+00:00	20.2	89.2	50	983	21.182	89.852
2024-05-26 12:00:00+00:00	21.1	89.2	60	978	22.134	89.79
2024-05-26 18:00:00+00:00	21.7	89.2	60	978	22.965	89.927
2024-05-27 00:00:00+00:00	22.5	89.2	45	982	23.652	90.145
2024-05-27 06:00:00+00:00	23.0	89.2	35	985	24.23	90.683
2024-05-27 12:00:00+00:00	23.6	89.7	35	987	24.553	91.159

Column block 2 of 3

time_utc	wind_model	wind_centre_model	pres_model	carried_forward	track_error_km	wind_bias_kt
2024-05-24 00:00:00+00:00	18.807	6.786	997.081	False	52.433	-6.193
2024-05-24 06:00:00+00:00	30.99	9.911	996.421	False	49.113	5.99
2024-05-24 12:00:00+00:00	43.835	11.541	992.596	False	55.013	18.835
2024-05-24 18:00:00+00:00	39.048	10.362	993.724	False	69.373	14.048
2024-05-25 00:00:00+00:00	43.243	17.427	991.339	False	59.947	13.243
2024-05-25 06:00:00+00:00	47.751	12.142	988.791	False	43.461	17.751
2024-05-25 12:00:00+00:00	44.784	9.418	985.755	False	42.139	9.784
2024-05-25 18:00:00+00:00	51.944	17.011	983.638	False	54.613	11.944
2024-05-26 00:00:00+00:00	70.618	8.197	975.25	False	98.503	20.618
2024-05-26 06:00:00+00:00	69.171	8.396	972.373	False	128.527	19.171
2024-05-26 12:00:00+00:00	69.174	1.686	973.012	False	130.208	9.174
2024-05-26 18:00:00+00:00	52.848	5.1	976.344	False	159.307	-7.152

2024-05-27 00:00:00+00:00	52.887	2.518	980.161	False	160.421	7.887
2024-05-27 06:00:00+00:00	36.86	7.026	983.073	False	203.785	1.86
2024-05-27 12:00:00+00:00	36.286	6.778	985.724	False	182.116	1.286

Column block 3 of 3

time_utc	wind_centre_bias_kt	pres_bias_hpa
2024-05-24 00:00:00+00:00	-18.214	1.081
2024-05-24 06:00:00+00:00	-15.089	0.421
2024-05-24 12:00:00+00:00	-13.459	-2.404
2024-05-24 18:00:00+00:00	-14.638	-1.276
2024-05-25 00:00:00+00:00	-12.573	-1.661
2024-05-25 06:00:00+00:00	-17.858	-2.209
2024-05-25 12:00:00+00:00	-25.582	-4.245
2024-05-25 18:00:00+00:00	-22.989	-4.362
2024-05-26 00:00:00+00:00	-41.803	-8.75
2024-05-26 06:00:00+00:00	-41.604	-10.627
2024-05-26 12:00:00+00:00	-58.314	-4.988
2024-05-26 18:00:00+00:00	-54.9	-1.656
2024-05-27 00:00:00+00:00	-42.482	-1.839
2024-05-27 06:00:00+00:00	-27.974	-1.927
2024-05-27 12:00:00+00:00	-28.222	-1.276

Remal_FNL_vs_IMD.csv

Column block 1 of 3

time_utc	lat_imd	lon_imd	wind_imd	pres_imd	lat_model	lon_model
2024-05-24 00:00:00+00:00	15.0	88.4	25	996	15.324	89.406
2024-05-24 06:00:00+00:00	15.8	88.9	25	996	15.572	89.648
2024-05-24 12:00:00+00:00	16.2	89.1	25	995	15.906	89.909
2024-05-24 18:00:00+00:00	16.8	89.4	25	995	16.522	90.025
2024-05-25 00:00:00+00:00	17.6	89.7	30	993	17.362	89.839
2024-05-25 06:00:00+00:00	18.2	89.7	30	991	17.997	89.876
2024-05-25 12:00:00+00:00	18.8	89.5	35	990	18.54	89.587
2024-05-25 18:00:00+00:00	19.3	89.4	40	988	18.89	89.546
2024-05-26 00:00:00+00:00	19.5	89.4	50	984	19.672	89.503
2024-05-26 06:00:00+00:00	20.2	89.2	50	983	20.625	89.712
2024-05-26 12:00:00+00:00	21.1	89.2	60	978	21.585	89.977
2024-05-26 18:00:00+00:00	21.7	89.2	60	978	22.587	90.342
2024-05-27 00:00:00+00:00	22.5	89.2	45	982	23.415	90.683
2024-05-27 06:00:00+00:00	23.0	89.2	35	985	24.19	91.28
2024-05-27 12:00:00+00:00	23.6	89.7	35	987	24.629	91.727
2024-05-27 18:00:00+00:00	24.0	90.5	30	990	24.844	92.036
2024-05-28 00:00:00+00:00	24.4	91.1	20	991	24.844	92.036
2024-05-28 06:00:00+00:00	25.1	91.8	20	993	24.844	92.036

Column block 2 of 3

time_utc	wind_model	wind_centre_model	pres_model	carried_forward	track_error_km	wind_bias_kt
2024-05-24 00:00:00+00:00	27.534	6.832	996.495	False	113.823	2.534
2024-05-24 06:00:00+00:00	37.592	1.131	995.149	False	83.934	12.592
2024-05-24 12:00:00+00:00	35.911	10.485	992.54	False	92.418	10.911
2024-05-24 18:00:00+00:00	45.595	7.4	991.876	False	73.367	20.595

2024-05-25 00:00:00+00:00	44.169	1.746	990.428	False	30.283	14.169
2024-05-25 06:00:00+00:00	56.344	34.842	983.934	False	29.234	26.344
2024-05-25 12:00:00+00:00	56.782	8.837	981.328	False	30.325	21.782
2024-05-25 18:00:00+00:00	67.184	20.205	979.248	False	48.098	27.184
2024-05-26 00:00:00+00:00	87.732	10.45	969.687	False	21.918	37.732
2024-05-26 06:00:00+00:00	83.407	7.243	969.25	False	71.261	33.407
2024-05-26 12:00:00+00:00	71.381	17.551	968.954	False	96.878	11.381
2024-05-26 18:00:00+00:00	65.896	4.887	973.926	False	153.475	5.896
2024-05-27 00:00:00+00:00	50.216	4.887	977.931	False	182.73	5.216
2024-05-27 06:00:00+00:00	36.774	7.493	982.86	False	249.855	1.774
2024-05-27 12:00:00+00:00	33.182	14.636	985.189	False	235.359	-1.818
2024-05-27 18:00:00+00:00	29.309	11.259	985.189	True	181.618	-0.691
2024-05-28 00:00:00+00:00	19.763	4.657	985.189	True	106.701	-0.237
2024-05-28 06:00:00+00:00	20.726	7.857	985.189	True	37.024	0.726

Column block 3 of 3

time_utc	wind_centre_bias_kt	pres_bias_hpa
2024-05-24 00:00:00+00:00	-18.168	0.495
2024-05-24 06:00:00+00:00	-23.869	-0.851
2024-05-24 12:00:00+00:00	-14.515	-2.46
2024-05-24 18:00:00+00:00	-17.6	-3.124
2024-05-25 00:00:00+00:00	-28.254	-2.572
2024-05-25 06:00:00+00:00	4.842	-7.066
2024-05-25 12:00:00+00:00	-26.163	-8.672
2024-05-25 18:00:00+00:00	-19.795	-8.752
2024-05-26 00:00:00+00:00	-39.55	-14.313
2024-05-26 06:00:00+00:00	-42.757	-13.75
2024-05-26 12:00:00+00:00	-42.449	-9.046
2024-05-26 18:00:00+00:00	-55.113	-4.074
2024-05-27 00:00:00+00:00	-40.113	-4.069
2024-05-27 06:00:00+00:00	-27.507	-2.14
2024-05-27 12:00:00+00:00	-20.364	-1.811
2024-05-27 18:00:00+00:00	-18.741	-4.811
2024-05-28 00:00:00+00:00	-15.343	-5.811
2024-05-28 06:00:00+00:00	-12.143	-7.811

Remal_FNL_vs_IMD_excluding_carried_forward.csv

Column block 1 of 3

time_utc	lat_imd	lon_imd	wind_imd	pres_imd	lat_model	lon_model
2024-05-24 00:00:00+00:00	15.0	88.4	25	996	15.324	89.406
2024-05-24 06:00:00+00:00	15.8	88.9	25	996	15.572	89.648
2024-05-24 12:00:00+00:00	16.2	89.1	25	995	15.906	89.909
2024-05-24 18:00:00+00:00	16.8	89.4	25	995	16.522	90.025
2024-05-25 00:00:00+00:00	17.6	89.7	30	993	17.362	89.839
2024-05-25 06:00:00+00:00	18.2	89.7	30	991	17.997	89.876
2024-05-25 12:00:00+00:00	18.8	89.5	35	990	18.54	89.587
2024-05-25 18:00:00+00:00	19.3	89.4	40	988	18.89	89.546
2024-05-26 00:00:00+00:00	19.5	89.4	50	984	19.672	89.503
2024-05-26 06:00:00+00:00	20.2	89.2	50	983	20.625	89.712
2024-05-26 12:00:00+00:00	21.1	89.2	60	978	21.585	89.977
2024-05-26 18:00:00+00:00	21.7	89.2	60	978	22.587	90.342
2024-05-27 00:00:00+00:00	22.5	89.2	45	982	23.415	90.683
2024-05-27 06:00:00+00:00	23.0	89.2	35	985	24.19	91.28

2024-05-27 12:00:00+00:00	23.6	89.7	35	987	24.629	91.727

Column block 2 of 3

time_utc	wind_model	wind_centre_model	pres_model	carried_forward	track_error_km	wind_bias_kt
2024-05-24 00:00:00+00:00	27.534	6.832	996.495	False	113.823	2.534
2024-05-24 06:00:00+00:00	37.592	1.131	995.149	False	83.934	12.592
2024-05-24 12:00:00+00:00	35.911	10.485	992.54	False	92.418	10.911
2024-05-24 18:00:00+00:00	45.595	7.4	991.876	False	73.367	20.595
2024-05-25 00:00:00+00:00	44.169	1.746	990.428	False	30.283	14.169
2024-05-25 06:00:00+00:00	56.344	34.842	983.934	False	29.234	26.344
2024-05-25 12:00:00+00:00	56.782	8.837	981.328	False	30.325	21.782
2024-05-25 18:00:00+00:00	67.184	20.205	979.248	False	48.098	27.184
2024-05-26 00:00:00+00:00	87.732	10.45	969.687	False	21.918	37.732
2024-05-26 06:00:00+00:00	83.407	7.243	969.25	False	71.261	33.407
2024-05-26 12:00:00+00:00	71.381	17.551	968.954	False	96.878	11.381
2024-05-26 18:00:00+00:00	65.896	4.887	973.926	False	153.475	5.896
2024-05-27 00:00:00+00:00	50.216	4.887	977.931	False	182.73	5.216
2024-05-27 06:00:00+00:00	36.774	7.493	982.86	False	249.855	1.774
2024-05-27 12:00:00+00:00	33.182	14.636	985.189	False	235.359	-1.818

Column block 3 of 3

time_utc	wind_centre_bias_kt	pres_bias_hpa
2024-05-24 00:00:00+00:00	-18.168	0.495
2024-05-24 06:00:00+00:00	-23.869	-0.851
2024-05-24 12:00:00+00:00	-14.515	-2.46
2024-05-24 18:00:00+00:00	-17.6	-3.124
2024-05-25 00:00:00+00:00	-28.254	-2.572
2024-05-25 06:00:00+00:00	4.842	-7.066
2024-05-25 12:00:00+00:00	-26.163	-8.672
2024-05-25 18:00:00+00:00	-19.795	-8.752
2024-05-26 00:00:00+00:00	-39.55	-14.313
2024-05-26 06:00:00+00:00	-42.757	-13.75
2024-05-26 12:00:00+00:00	-42.449	-9.046
2024-05-26 18:00:00+00:00	-55.113	-4.074
2024-05-27 00:00:00+00:00	-40.113	-4.069
2024-05-27 06:00:00+00:00	-27.507	-2.14
2024-05-27 12:00:00+00:00	-20.364	-1.811

Remal_GFS_vs_IMD.csv

Column block 1 of 3

time_utc	lat_imd	lon_imd	wind_imd	pres_imd	lat_model	lon_model
2024-05-24 00:00:00+00:00	15.0	88.4	25	996	14.888	89.597
2024-05-24 06:00:00+00:00	15.8	88.9	25	996	15.261	89.749
2024-05-24 12:00:00+00:00	16.2	89.1	25	995	15.967	90.105
2024-05-24 18:00:00+00:00	16.8	89.4	25	995	16.605	90.152
2024-05-25 00:00:00+00:00	17.6	89.7	30	993	17.118	90.175
2024-05-25 06:00:00+00:00	18.2	89.7	30	991	17.738	90.011
2024-05-25 12:00:00+00:00	18.8	89.5	35	990	18.204	89.746
2024-05-25 18:00:00+00:00	19.3	89.4	40	988	18.914	89.554
2024-05-26 00:00:00+00:00	19.5	89.4	50	984	19.646	89.506

2024-05-26 06:00:00+00:00	20.2	89.2	50	983	20.654	89.691
2024-05-26 12:00:00+00:00	21.1	89.2	60	978	21.618	89.831
2024-05-26 18:00:00+00:00	21.7	89.2	60	978	22.627	90.002
2024-05-27 00:00:00+00:00	22.5	89.2	45	982	23.409	90.241
2024-05-27 06:00:00+00:00	23.0	89.2	35	985	24.122	90.803
2024-05-27 12:00:00+00:00	23.6	89.7	35	987	24.522	91.287
2024-05-27 18:00:00+00:00	24.0	90.5	30	990	24.711	91.656
2024-05-28 00:00:00+00:00	24.4	91.1	20	991	24.711	91.656
2024-05-28 06:00:00+00:00	25.1	91.8	20	993	24.711	91.656

Column block 2 of 3

time_utc	wind_model	wind_centre_model	pres_model	carried_forward	track_error_km	wind_bias_kt
2024-05-24 00:00:00+00:00	35.808	18.84	994.595	False	129.226	10.808
2024-05-24 06:00:00+00:00	38.007	6.433	994.715	False	108.9	13.007
2024-05-24 12:00:00+00:00	37.204	9.664	992.595	False	110.409	12.204
2024-05-24 18:00:00+00:00	47.484	7.187	989.55	False	82.944	22.484
2024-05-25 00:00:00+00:00	42.445	2.915	986.911	False	73.633	12.445
2024-05-25 06:00:00+00:00	51.848	17.104	982.054	False	61.032	21.848
2024-05-25 12:00:00+00:00	52.856	3.093	979.108	False	71.185	17.856
2024-05-25 18:00:00+00:00	72.186	2.716	974.391	False	45.894	32.186
2024-05-26 00:00:00+00:00	90.721	6.005	966.838	False	19.624	40.721
2024-05-26 06:00:00+00:00	77.003	5.61	968.131	False	71.867	27.003
2024-05-26 12:00:00+00:00	70.257	4.189	967.224	False	87.147	10.257
2024-05-26 18:00:00+00:00	65.484	7.783	973.094	False	132.099	5.484
2024-05-27 00:00:00+00:00	48.002	7.707	977.045	False	146.85	3.002
2024-05-27 06:00:00+00:00	39.422	10.103	982.225	False	205.546	4.422
2024-05-27 12:00:00+00:00	35.19	12.658	984.181	False	190.981	0.19
2024-05-27 18:00:00+00:00	28.139	11.796	984.181	True	141.295	-1.861
2024-05-28 00:00:00+00:00	17.169	7.995	984.181	True	66.021	-2.831
2024-05-28 06:00:00+00:00	17.491	5.25	984.181	True	45.636	-2.509

Column block 3 of 3

time_utc	wind_centre_bias_kt	pres_bias_hpa
2024-05-24 00:00:00+00:00	-6.16	-1.405
2024-05-24 06:00:00+00:00	-18.567	-1.285
2024-05-24 12:00:00+00:00	-15.336	-2.405
2024-05-24 18:00:00+00:00	-17.813	-5.45
2024-05-25 00:00:00+00:00	-27.085	-6.089
2024-05-25 06:00:00+00:00	-12.896	-8.946
2024-05-25 12:00:00+00:00	-31.907	-10.892
2024-05-25 18:00:00+00:00	-37.284	-13.609
2024-05-26 00:00:00+00:00	-43.995	-17.162
2024-05-26 06:00:00+00:00	-44.39	-14.869
2024-05-26 12:00:00+00:00	-55.811	-10.776
2024-05-26 18:00:00+00:00	-52.217	-4.906
2024-05-27 00:00:00+00:00	-37.293	-4.955
2024-05-27 06:00:00+00:00	-24.897	-2.775
2024-05-27 12:00:00+00:00	-22.342	-2.819
2024-05-27 18:00:00+00:00	-18.204	-5.819
2024-05-28 00:00:00+00:00	-12.005	-6.819
2024-05-28 06:00:00+00:00	-14.75	-8.819

Remal_GFS_vs_IMD_excluding_carried_forward.csv

Column block 1 of 3

time_utc	lat_imd	lon_imd	wind_imd	pres_imd	lat_model	lon_model
2024-05-24 00:00:00+00:00	15.0	88.4	25	996	14.888	89.597
2024-05-24 06:00:00+00:00	15.8	88.9	25	996	15.261	89.749
2024-05-24 12:00:00+00:00	16.2	89.1	25	995	15.967	90.105
2024-05-24 18:00:00+00:00	16.8	89.4	25	995	16.605	90.152
2024-05-25 00:00:00+00:00	17.6	89.7	30	993	17.118	90.175
2024-05-25 06:00:00+00:00	18.2	89.7	30	991	17.738	90.011
2024-05-25 12:00:00+00:00	18.8	89.5	35	990	18.204	89.746
2024-05-25 18:00:00+00:00	19.3	89.4	40	988	18.914	89.554
2024-05-26 00:00:00+00:00	19.5	89.4	50	984	19.646	89.506
2024-05-26 06:00:00+00:00	20.2	89.2	50	983	20.654	89.691
2024-05-26 12:00:00+00:00	21.1	89.2	60	978	21.618	89.831
2024-05-26 18:00:00+00:00	21.7	89.2	60	978	22.627	90.002
2024-05-27 00:00:00+00:00	22.5	89.2	45	982	23.409	90.241
2024-05-27 06:00:00+00:00	23.0	89.2	35	985	24.122	90.803
2024-05-27 12:00:00+00:00	23.6	89.7	35	987	24.522	91.287

Column block 2 of 3

time_utc	wind_model	wind_centre_model	pres_model	carried_forward	track_error_km	wind_bias_kt
2024-05-24 00:00:00+00:00	35.808	18.84	994.595	False	129.226	10.808
2024-05-24 06:00:00+00:00	38.007	6.433	994.715	False	108.9	13.007
2024-05-24 12:00:00+00:00	37.204	9.664	992.595	False	110.409	12.204
2024-05-24 18:00:00+00:00	47.484	7.187	989.55	False	82.944	22.484
2024-05-25 00:00:00+00:00	42.445	2.915	986.911	False	73.633	12.445
2024-05-25 06:00:00+00:00	51.848	17.104	982.054	False	61.032	21.848
2024-05-25 12:00:00+00:00	52.856	3.093	979.108	False	71.185	17.856
2024-05-25 18:00:00+00:00	72.186	2.716	974.391	False	45.894	32.186
2024-05-26 00:00:00+00:00	90.721	6.005	966.838	False	19.624	40.721
2024-05-26 06:00:00+00:00	77.003	5.61	968.131	False	71.867	27.003
2024-05-26 12:00:00+00:00	70.257	4.189	967.224	False	87.147	10.257
2024-05-26 18:00:00+00:00	65.484	7.783	973.094	False	132.099	5.484
2024-05-27 00:00:00+00:00	48.002	7.707	977.045	False	146.85	3.002
2024-05-27 06:00:00+00:00	39.422	10.103	982.225	False	205.546	4.422
2024-05-27 12:00:00+00:00	35.19	12.658	984.181	False	190.981	0.19

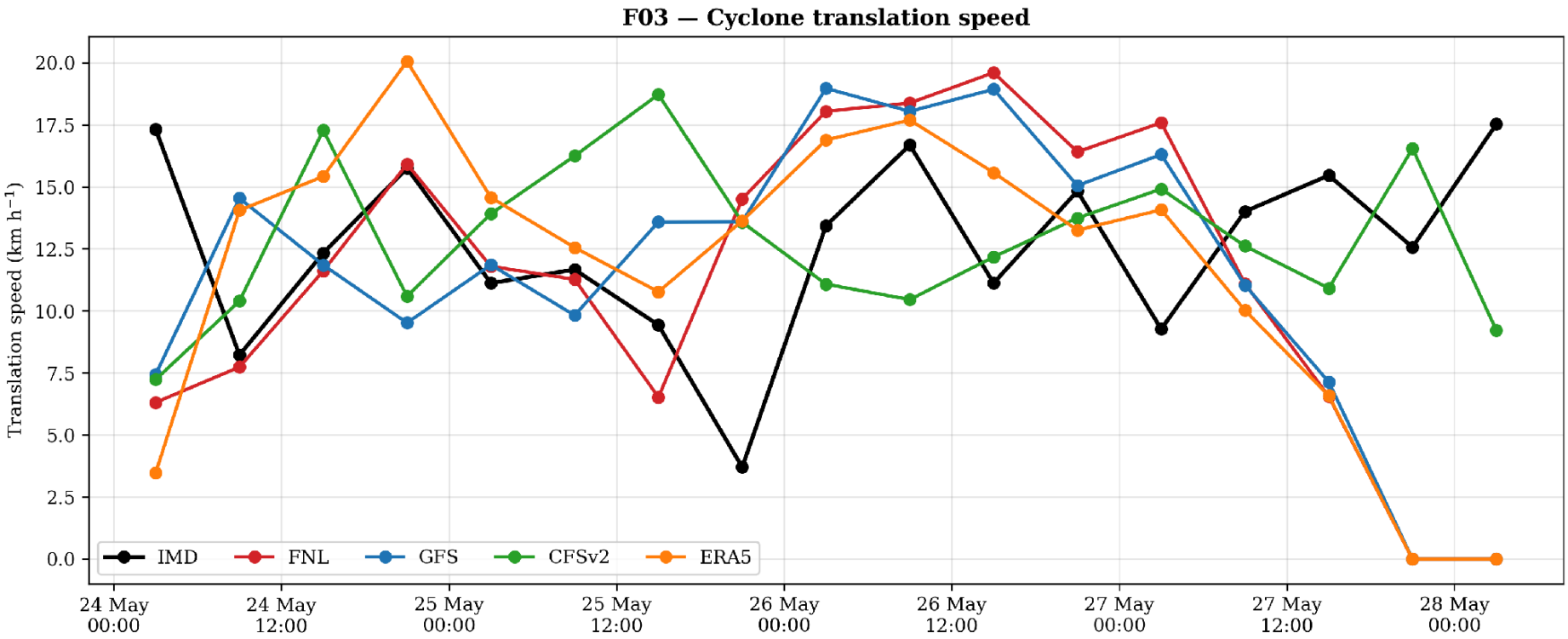
Column block 3 of 3

time_utc	wind_centre_bias_kt	pres_bias_hpa
2024-05-24 00:00:00+00:00	-6.16	-1.405
2024-05-24 06:00:00+00:00	-18.567	-1.285
2024-05-24 12:00:00+00:00	-15.336	-2.405
2024-05-24 18:00:00+00:00	-17.813	-5.45
2024-05-25 00:00:00+00:00	-27.085	-6.089
2024-05-25 06:00:00+00:00	-12.896	-8.946
2024-05-25 12:00:00+00:00	-31.907	-10.892
2024-05-25 18:00:00+00:00	-37.284	-13.609
2024-05-26 00:00:00+00:00	-43.995	-17.162
2024-05-26 06:00:00+00:00	-44.39	-14.869
2024-05-26 12:00:00+00:00	-55.811	-10.776
2024-05-26 18:00:00+00:00	-52.217	-4.906
2024-05-27 00:00:00+00:00	-37.293	-4.955
2024-05-27 06:00:00+00:00	-24.897	-2.775
2024-05-27 12:00:00+00:00	-22.342	-2.819

A04 - Translation speed

Figures: 1 | CSV tables: 3 | NetCDF extracts: 0

F03_Translation_Speed.png



A04_translation_speed_main.csv

Column block 1 of 2

source	start_time_utc	end_time_utc	mid_time_utc	elapsed_hours	distance_km	translation_speed_kmh_model
FNL	2024-05-24 00:00:00+00:00	2024-05-24 06:00:00+00:00	2024-05-24 03:00:00+00:00	6.0	37.869	6.312
FNL	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	2024-05-24 09:00:00+00:00	6.0	46.433	7.739
FNL	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	2024-05-24 15:00:00+00:00	6.0	69.662	11.61
FNL	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	2024-05-24 21:00:00+00:00	6.0	95.465	15.911
FNL	2024-05-25 00:00:00+00:00	2024-05-25 06:00:00+00:00	2024-05-25 03:00:00+00:00	6.0	70.753	11.792
FNL	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	2024-05-25 09:00:00+00:00	6.0	67.618	11.27
FNL	2024-05-25 12:00:00+00:00	2024-05-25 18:00:00+00:00	2024-05-25 15:00:00+00:00	6.0	39.137	6.523
FNL	2024-05-25 18:00:00+00:00	2024-05-26 00:00:00+00:00	2024-05-25 21:00:00+00:00	6.0	87.058	14.51

FNL	2024-05-26 00:00:00+00:00	2024-05-26 06:00:00+00:00	2024-05-26 03:00:00+00:00	6.0	108.236	18.039
FNL	2024-05-26 06:00:00+00:00	2024-05-26 12:00:00+00:00	2024-05-26 09:00:00+00:00	6.0	110.226	18.371
FNL	2024-05-26 12:00:00+00:00	2024-05-26 18:00:00+00:00	2024-05-26 15:00:00+00:00	6.0	117.567	19.595
FNL	2024-05-26 18:00:00+00:00	2024-05-27 00:00:00+00:00	2024-05-26 21:00:00+00:00	6.0	98.426	16.404
FNL	2024-05-27 00:00:00+00:00	2024-05-27 06:00:00+00:00	2024-05-27 03:00:00+00:00	6.0	105.474	17.579
FNL	2024-05-27 06:00:00+00:00	2024-05-27 12:00:00+00:00	2024-05-27 09:00:00+00:00	6.0	66.539	11.09
FNL	2024-05-27 12:00:00+00:00	2024-05-27 18:00:00+00:00	2024-05-27 15:00:00+00:00	6.0	39.341	6.557
FNL	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	2024-05-27 21:00:00+00:00	6.0	0.0	0.0
FNL	2024-05-28 00:00:00+00:00	2024-05-28 06:00:00+00:00	2024-05-28 03:00:00+00:00	6.0	0.0	0.0
GFS	2024-05-24 00:00:00+00:00	2024-05-24 06:00:00+00:00	2024-05-24 03:00:00+00:00	6.0	44.645	7.441
GFS	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	2024-05-24 09:00:00+00:00	6.0	87.206	14.534
GFS	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	2024-05-24 15:00:00+00:00	6.0	71.087	11.848
GFS	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	2024-05-24 21:00:00+00:00	6.0	57.108	9.518
GFS	2024-05-25 00:00:00+00:00	2024-05-25 06:00:00+00:00	2024-05-25 03:00:00+00:00	6.0	71.116	11.853
GFS	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	2024-05-25 09:00:00+00:00	6.0	58.931	9.822
GFS	2024-05-25 12:00:00+00:00	2024-05-25 18:00:00+00:00	2024-05-25 15:00:00+00:00	6.0	81.498	13.583
GFS	2024-05-25 18:00:00+00:00	2024-05-26 00:00:00+00:00	2024-05-25 21:00:00+00:00	6.0	81.556	13.593
GFS	2024-05-26 00:00:00+00:00	2024-05-26 06:00:00+00:00	2024-05-26 03:00:00+00:00	6.0	113.771	18.962
GFS	2024-05-26 06:00:00+00:00	2024-05-26 12:00:00+00:00	2024-05-26 09:00:00+00:00	6.0	108.213	18.035
GFS	2024-05-26 12:00:00+00:00	2024-05-26 18:00:00+00:00	2024-05-26 15:00:00+00:00	6.0	113.543	18.924
GFS	2024-05-26 18:00:00+00:00	2024-05-27 00:00:00+00:00	2024-05-26 21:00:00+00:00	6.0	90.286	15.048
GFS	2024-05-27 00:00:00+00:00	2024-05-27 06:00:00+00:00	2024-05-27 03:00:00+00:00	6.0	97.78	16.297
GFS	2024-05-27 06:00:00+00:00	2024-05-27 12:00:00+00:00	2024-05-27 09:00:00+00:00	6.0	66.223	11.037
GFS	2024-05-27 12:00:00+00:00	2024-05-27 18:00:00+00:00	2024-05-27 15:00:00+00:00	6.0	42.82	7.137
GFS	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	2024-05-27 21:00:00+00:00	6.0	0.0	0.0
GFS	2024-05-28 00:00:00+00:00	2024-05-28 06:00:00+00:00	2024-05-28 03:00:00+00:00	6.0	0.0	0.0
CFSv2	2024-05-24 00:00:00+00:00	2024-05-24 06:00:00+00:00	2024-05-24 03:00:00+00:00	6.0	43.416	7.236
CFSv2	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	2024-05-24 09:00:00+00:00	6.0	62.378	10.396
CFSv2	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	2024-05-24 15:00:00+00:00	6.0	103.654	17.276
CFSv2	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	2024-05-24 21:00:00+00:00	6.0	63.526	10.588
CFSv2	2024-05-25 00:00:00+00:00	2024-05-25 06:00:00+00:00	2024-05-25 03:00:00+00:00	6.0	83.459	13.91
CFSv2	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	2024-05-25 09:00:00+00:00	6.0	97.449	16.241
CFSv2	2024-05-25 12:00:00+00:00	2024-05-25 18:00:00+00:00	2024-05-25 15:00:00+00:00	6.0	112.278	18.713
CFSv2	2024-05-25 18:00:00+00:00	2024-05-26 00:00:00+00:00	2024-05-25 21:00:00+00:00	6.0	81.427	13.571
CFSv2	2024-05-26 00:00:00+00:00	2024-05-26 06:00:00+00:00	2024-05-26 03:00:00+00:00	6.0	66.452	11.075
CFSv2	2024-05-26 06:00:00+00:00	2024-05-26 12:00:00+00:00	2024-05-26 09:00:00+00:00	6.0	62.724	10.454
CFSv2	2024-05-26 12:00:00+00:00	2024-05-26 18:00:00+00:00	2024-05-26 15:00:00+00:00	6.0	73.03	12.172
CFSv2	2024-05-26 18:00:00+00:00	2024-05-27 00:00:00+00:00	2024-05-26 21:00:00+00:00	6.0	82.438	13.74
CFSv2	2024-05-27 00:00:00+00:00	2024-05-27 06:00:00+00:00	2024-05-27 03:00:00+00:00	6.0	89.458	14.91
CFSv2	2024-05-27 06:00:00+00:00	2024-05-27 12:00:00+00:00	2024-05-27 09:00:00+00:00	6.0	75.666	12.611
CFSv2	2024-05-27 12:00:00+00:00	2024-05-27 18:00:00+00:00	2024-05-27 15:00:00+00:00	6.0	65.475	10.913
CFSv2	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	2024-05-27 21:00:00+00:00	6.0	99.224	16.537
CFSv2	2024-05-28 00:00:00+00:00	2024-05-28 06:00:00+00:00	2024-05-28 03:00:00+00:00	6.0	55.289	9.215
ERA5	2024-05-24 00:00:00+00:00	2024-05-24 06:00:00+00:00	2024-05-24 03:00:00+00:00	6.0	20.852	3.475
ERA5	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	2024-05-24 09:00:00+00:00	6.0	84.286	14.048
ERA5	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	2024-05-24 15:00:00+00:00	6.0	92.497	15.416
ERA5	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	2024-05-24 21:00:00+00:00	6.0	120.315	20.052
ERA5	2024-05-25 00:00:00+00:00	2024-05-25 06:00:00+00:00	2024-05-25 03:00:00+00:00	6.0	87.393	14.565
ERA5	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	2024-05-25 09:00:00+00:00	6.0	75.308	12.551
ERA5	2024-05-25 12:00:00+00:00	2024-05-25 18:00:00+00:00	2024-05-25 15:00:00+00:00	6.0	64.615	10.769
ERA5	2024-05-25 18:00:00+00:00	2024-05-26 00:00:00+00:00	2024-05-25 21:00:00+00:00	6.0	81.761	13.627
ERA5	2024-05-26 00:00:00+00:00	2024-05-26 06:00:00+00:00	2024-05-26 03:00:00+00:00	6.0	101.29	16.882
ERA5	2024-05-26 06:00:00+00:00	2024-05-26 12:00:00+00:00	2024-05-26 09:00:00+00:00	6.0	106.11	17.685
ERA5	2024-05-26 12:00:00+00:00	2024-05-26 18:00:00+00:00	2024-05-26 15:00:00+00:00	6.0	93.425	15.571
ERA5	2024-05-26 18:00:00+00:00	2024-05-27 00:00:00+00:00	2024-05-26 21:00:00+00:00	6.0	79.511	13.252
ERA5	2024-05-27 00:00:00+00:00	2024-05-27 06:00:00+00:00	2024-05-27 03:00:00+00:00	6.0	84.431	14.072
ERA5	2024-05-27 06:00:00+00:00	2024-05-27 12:00:00+00:00	2024-05-27 09:00:00+00:00	6.0	60.126	10.021
ERA5	2024-05-27 12:00:00+00:00	2024-05-27 18:00:00+00:00	2024-05-27 15:00:00+00:00	6.0	39.499	6.583
ERA5	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	2024-05-27 21:00:00+00:00	6.0	0.0	0.0
ERA5	2024-05-28 00:00:00+00:00	2024-05-28 06:00:00+00:00	2024-05-28 03:00:00+00:00	6.0	0.0	0.0

Column block 2 of 2

source	translation_speed_kmh_imd	model	speed_bias_kmh
FNL	17.309	FNL	-10.998
FNL	8.225	FNL	-0.486
FNL	12.331	FNL	-0.721
FNL	15.749	FNL	0.162
FNL	11.119	FNL	0.673
FNL	11.662	FNL	-0.392
FNL	9.43	FNL	-2.907
FNL	3.706	FNL	10.803
FNL	13.433	FNL	4.606
FNL	16.679	FNL	1.692
FNL	11.119	FNL	8.475
FNL	14.826	FNL	1.578
FNL	9.266	FNL	8.313
FNL	14.003	FNL	-2.913
FNL	15.458	FNL	-8.902
FNL	12.563	FNL	-12.563
FNL	17.524	FNL	-17.524
GFS	17.309	GFS	-9.869
GFS	8.225	GFS	6.31
GFS	12.331	GFS	-0.483
GFS	15.749	GFS	-6.231
GFS	11.119	GFS	0.733
GFS	11.662	GFS	-1.84
GFS	9.43	GFS	4.153
GFS	3.706	GFS	9.886
GFS	13.433	GFS	5.529
GFS	16.679	GFS	1.356
GFS	11.119	GFS	7.804
GFS	14.826	GFS	0.222
GFS	9.266	GFS	7.03
GFS	14.003	GFS	-2.965
GFS	15.458	GFS	-8.322
GFS	12.563	GFS	-12.563
GFS	17.524	GFS	-17.524
CFSv2	17.309	CFSv2	-10.073
CFSv2	8.225	CFSv2	2.172
CFSv2	12.331	CFSv2	4.944
CFSv2	15.749	CFSv2	-5.161
CFSv2	11.119	CFSv2	2.79
CFSv2	11.662	CFSv2	4.58
CFSv2	9.43	CFSv2	9.283
CFSv2	3.706	CFSv2	9.865
CFSv2	13.433	CFSv2	-2.358
CFSv2	16.679	CFSv2	-6.225
CFSv2	11.119	CFSv2	1.052
CFSv2	14.826	CFSv2	-1.086
CFSv2	9.266	CFSv2	5.643
CFSv2	14.003	CFSv2	-1.392
CFSv2	15.458	CFSv2	-4.546
CFSv2	12.563	CFSv2	3.975
CFSv2	17.524	CFSv2	-8.309
ERA5	17.309	ERA5	-13.834
ERA5	8.225	ERA5	5.823
ERA5	12.331	ERA5	3.085
ERA5	15.749	ERA5	4.304
ERA5	11.119	ERA5	3.446
ERA5	11.662	ERA5	0.89

ERA5	9.43	ERA5	1.339
ERA5	3.706	ERA5	9.92
ERA5	13.433	ERA5	3.449
ERA5	16.679	ERA5	1.006
ERA5	11.119	ERA5	4.451
ERA5	14.826	ERA5	-1.574
ERA5	9.266	ERA5	4.806
ERA5	14.003	ERA5	-3.982
ERA5	15.458	ERA5	-8.875
ERA5	12.563	ERA5	-12.563
ERA5	17.524	ERA5	-17.524

A04_translation_speed_unsmoothed_sensitivity.csv

source	start_time_utc	end_time_utc	mid_time_utc	elapsed_hours	distance_km	translation_speed_kmh
FNL_unsmoothed_centroid	2024-05-24 00:00:00+00:00	2024-05-24 06:00:00+00:00	2024-05-24 03:00:00+00:00	6.0	123.175	20.529
FNL_unsmoothed_centroid	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	2024-05-24 09:00:00+00:00	6.0	78.788	13.131
FNL_unsmoothed_centroid	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	2024-05-24 15:00:00+00:00	6.0	59.039	9.84
FNL_unsmoothed_centroid	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	2024-05-24 21:00:00+00:00	6.0	130.22	21.703
FNL_unsmoothed_centroid	2024-05-25 00:00:00+00:00	2024-05-25 06:00:00+00:00	2024-05-25 03:00:00+00:00	6.0	171.237	28.539
FNL_unsmoothed_centroid	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	2024-05-25 09:00:00+00:00	6.0	48.837	8.14
FNL_unsmoothed_centroid	2024-05-25 12:00:00+00:00	2024-05-25 18:00:00+00:00	2024-05-25 15:00:00+00:00	6.0	80.617	13.436
FNL_unsmoothed_centroid	2024-05-25 18:00:00+00:00	2024-05-26 00:00:00+00:00	2024-05-25 21:00:00+00:00	6.0	87.246	14.541
FNL_unsmoothed_centroid	2024-05-26 00:00:00+00:00	2024-05-26 06:00:00+00:00	2024-05-26 03:00:00+00:00	6.0	99.52	16.587
FNL_unsmoothed_centroid	2024-05-26 06:00:00+00:00	2024-05-26 12:00:00+00:00	2024-05-26 09:00:00+00:00	6.0	141.802	23.634
FNL_unsmoothed_centroid	2024-05-26 12:00:00+00:00	2024-05-26 18:00:00+00:00	2024-05-26 15:00:00+00:00	6.0	90.861	15.143
FNL_unsmoothed_centroid	2024-05-26 18:00:00+00:00	2024-05-27 00:00:00+00:00	2024-05-26 21:00:00+00:00	6.0	121.365	20.228
FNL_unsmoothed_centroid	2024-05-27 00:00:00+00:00	2024-05-27 06:00:00+00:00	2024-05-27 03:00:00+00:00	6.0	85.53	14.255
FNL_unsmoothed_centroid	2024-05-27 06:00:00+00:00	2024-05-27 12:00:00+00:00	2024-05-27 09:00:00+00:00	6.0	118.148	19.691
FNL_unsmoothed_centroid	2024-05-27 12:00:00+00:00	2024-05-27 18:00:00+00:00	2024-05-27 15:00:00+00:00	6.0	0.0	0.0
FNL_unsmoothed_centroid	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	2024-05-27 21:00:00+00:00	6.0	0.0	0.0
FNL_unsmoothed_centroid	2024-05-28 00:00:00+00:00	2024-05-28 06:00:00+00:00	2024-05-28 03:00:00+00:00	6.0	0.0	0.0
GFS_unsmoothed_centroid	2024-05-24 00:00:00+00:00	2024-05-24 06:00:00+00:00	2024-05-24 03:00:00+00:00	6.0	146.934	24.489
GFS_unsmoothed_centroid	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	2024-05-24 09:00:00+00:00	6.0	62.049	10.342
GFS_unsmoothed_centroid	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	2024-05-24 15:00:00+00:00	6.0	55.171	9.195
GFS_unsmoothed_centroid	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	2024-05-24 21:00:00+00:00	6.0	108.676	18.113
GFS_unsmoothed_centroid	2024-05-25 00:00:00+00:00	2024-05-25 06:00:00+00:00	2024-05-25 03:00:00+00:00	6.0	19.853	3.309
GFS_unsmoothed_centroid	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	2024-05-25 09:00:00+00:00	6.0	87.894	14.649
GFS_unsmoothed_centroid	2024-05-25 12:00:00+00:00	2024-05-25 18:00:00+00:00	2024-05-25 15:00:00+00:00	6.0	79.686	13.281
GFS_unsmoothed_centroid	2024-05-25 18:00:00+00:00	2024-05-26 00:00:00+00:00	2024-05-25 21:00:00+00:00	6.0	104.548	17.425
GFS_unsmoothed_centroid	2024-05-26 00:00:00+00:00	2024-05-26 06:00:00+00:00	2024-05-26 03:00:00+00:00	6.0	90.944	15.157
GFS_unsmoothed_centroid	2024-05-26 06:00:00+00:00	2024-05-26 12:00:00+00:00	2024-05-26 09:00:00+00:00	6.0	146.763	24.46
GFS_unsmoothed_centroid	2024-05-26 12:00:00+00:00	2024-05-26 18:00:00+00:00	2024-05-26 15:00:00+00:00	6.0	87.081	14.513
GFS_unsmoothed_centroid	2024-05-26 18:00:00+00:00	2024-05-27 00:00:00+00:00	2024-05-26 21:00:00+00:00	6.0	107.242	17.874
GFS_unsmoothed_centroid	2024-05-27 00:00:00+00:00	2024-05-27 06:00:00+00:00	2024-05-27 03:00:00+00:00	6.0	78.609	13.102
GFS_unsmoothed_centroid	2024-05-27 06:00:00+00:00	2024-05-27 12:00:00+00:00	2024-05-27 09:00:00+00:00	6.0	128.605	21.434
GFS_unsmoothed_centroid	2024-05-27 12:00:00+00:00	2024-05-27 18:00:00+00:00	2024-05-27 15:00:00+00:00	6.0	0.0	0.0
GFS_unsmoothed_centroid	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	2024-05-27 21:00:00+00:00	6.0	0.0	0.0
GFS_unsmoothed_centroid	2024-05-28 00:00:00+00:00	2024-05-28 06:00:00+00:00	2024-05-28 03:00:00+00:00	6.0	0.0	0.0
CFSv2_unsmoothed_centroid	2024-05-24 00:00:00+00:00	2024-05-24 06:00:00+00:00	2024-05-24 03:00:00+00:00	6.0	79.223	13.204
CFSv2_unsmoothed_centroid	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	2024-05-24 09:00:00+00:00	6.0	169.643	28.274
CFSv2_unsmoothed_centroid	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	2024-05-24 15:00:00+00:00	6.0	104.091	17.348
CFSv2_unsmoothed_centroid	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	2024-05-24 21:00:00+00:00	6.0	64.417	10.736
CFSv2_unsmoothed_centroid	2024-05-25 00:00:00+00:00	2024-05-25 06:00:00+00:00	2024-05-25 03:00:00+00:00	6.0	56.778	9.463
CFSv2_unsmoothed_centroid	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	2024-05-25 09:00:00+00:00	6.0	158.628	26.438
CFSv2_unsmoothed_centroid	2024-05-25 12:00:00+00:00	2024-05-25 18:00:00+00:00	2024-05-25 15:00:00+00:00	6.0	128.157	21.36
CFSv2_unsmoothed_centroid	2024-05-25 18:00:00+00:00	2024-05-26 00:00:00+00:00	2024-05-25 21:00:00+00:00	6.0	70.971	11.828
CFSv2_unsmoothed_centroid	2024-05-26 00:00:00+00:00	2024-05-26 06:00:00+00:00	2024-05-26 03:00:00+00:00	6.0	46.813	7.802
CFSv2_unsmoothed_centroid	2024-05-26 06:00:00+00:00	2024-05-26 12:00:00+00:00	2024-05-26 09:00:00+00:00	6.0	83.843	13.974

CFSv2_unsmoothed_centroid	2024-05-26 12:00:00+00:00	2024-05-26 18:00:00+00:00	2024-05-26 15:00:00+00:00	6.0	61.259	10.21
CFSv2_unsmoothed_centroid	2024-05-26 18:00:00+00:00	2024-05-27 00:00:00+00:00	2024-05-26 21:00:00+00:00	6.0	90.574	15.096
CFSv2_unsmoothed_centroid	2024-05-27 00:00:00+00:00	2024-05-27 06:00:00+00:00	2024-05-27 03:00:00+00:00	6.0	101.205	16.868
CFSv2_unsmoothed_centroid	2024-05-27 06:00:00+00:00	2024-05-27 12:00:00+00:00	2024-05-27 09:00:00+00:00	6.0	82.018	13.67
CFSv2_unsmoothed_centroid	2024-05-27 12:00:00+00:00	2024-05-27 18:00:00+00:00	2024-05-27 15:00:00+00:00	6.0	51.183	8.53
CFSv2_unsmoothed_centroid	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	2024-05-27 21:00:00+00:00	6.0	75.792	12.632
CFSv2_unsmoothed_centroid	2024-05-28 00:00:00+00:00	2024-05-28 06:00:00+00:00	2024-05-28 03:00:00+00:00	6.0	180.271	30.045
ERA5_unsmoothed_centroid	2024-05-24 00:00:00+00:00	2024-05-24 06:00:00+00:00	2024-05-24 03:00:00+00:00	6.0	77.412	12.902
ERA5_unsmoothed_centroid	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	2024-05-24 09:00:00+00:00	6.0	31.141	5.19
ERA5_unsmoothed_centroid	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	2024-05-24 15:00:00+00:00	6.0	193.635	32.272
ERA5_unsmoothed_centroid	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	2024-05-24 21:00:00+00:00	6.0	89.187	14.865
ERA5_unsmoothed_centroid	2024-05-25 00:00:00+00:00	2024-05-25 06:00:00+00:00	2024-05-25 03:00:00+00:00	6.0	117.417	19.569
ERA5_unsmoothed_centroid	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	2024-05-25 09:00:00+00:00	6.0	88.542	14.757
ERA5_unsmoothed_centroid	2024-05-25 12:00:00+00:00	2024-05-25 18:00:00+00:00	2024-05-25 15:00:00+00:00	6.0	42.967	7.161
ERA5_unsmoothed_centroid	2024-05-25 18:00:00+00:00	2024-05-26 00:00:00+00:00	2024-05-25 21:00:00+00:00	6.0	87.561	14.594
ERA5_unsmoothed_centroid	2024-05-26 00:00:00+00:00	2024-05-26 06:00:00+00:00	2024-05-26 03:00:00+00:00	6.0	132.818	22.136
ERA5_unsmoothed_centroid	2024-05-26 06:00:00+00:00	2024-05-26 12:00:00+00:00	2024-05-26 09:00:00+00:00	6.0	94.609	15.768
ERA5_unsmoothed_centroid	2024-05-26 12:00:00+00:00	2024-05-26 18:00:00+00:00	2024-05-26 15:00:00+00:00	6.0	99.513	16.585
ERA5_unsmoothed_centroid	2024-05-26 18:00:00+00:00	2024-05-27 00:00:00+00:00	2024-05-26 21:00:00+00:00	6.0	87.28	14.547
ERA5_unsmoothed_centroid	2024-05-27 00:00:00+00:00	2024-05-27 06:00:00+00:00	2024-05-27 03:00:00+00:00	6.0	62.823	10.471
ERA5_unsmoothed_centroid	2024-05-27 06:00:00+00:00	2024-05-27 12:00:00+00:00	2024-05-27 09:00:00+00:00	6.0	118.624	19.771
ERA5_unsmoothed_centroid	2024-05-27 12:00:00+00:00	2024-05-27 18:00:00+00:00	2024-05-27 15:00:00+00:00	6.0	0.0	0.0
ERA5_unsmoothed_centroid	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	2024-05-27 21:00:00+00:00	6.0	0.0	0.0
ERA5_unsmoothed_centroid	2024-05-28 00:00:00+00:00	2024-05-28 06:00:00+00:00	2024-05-28 03:00:00+00:00	6.0	0.0	0.0

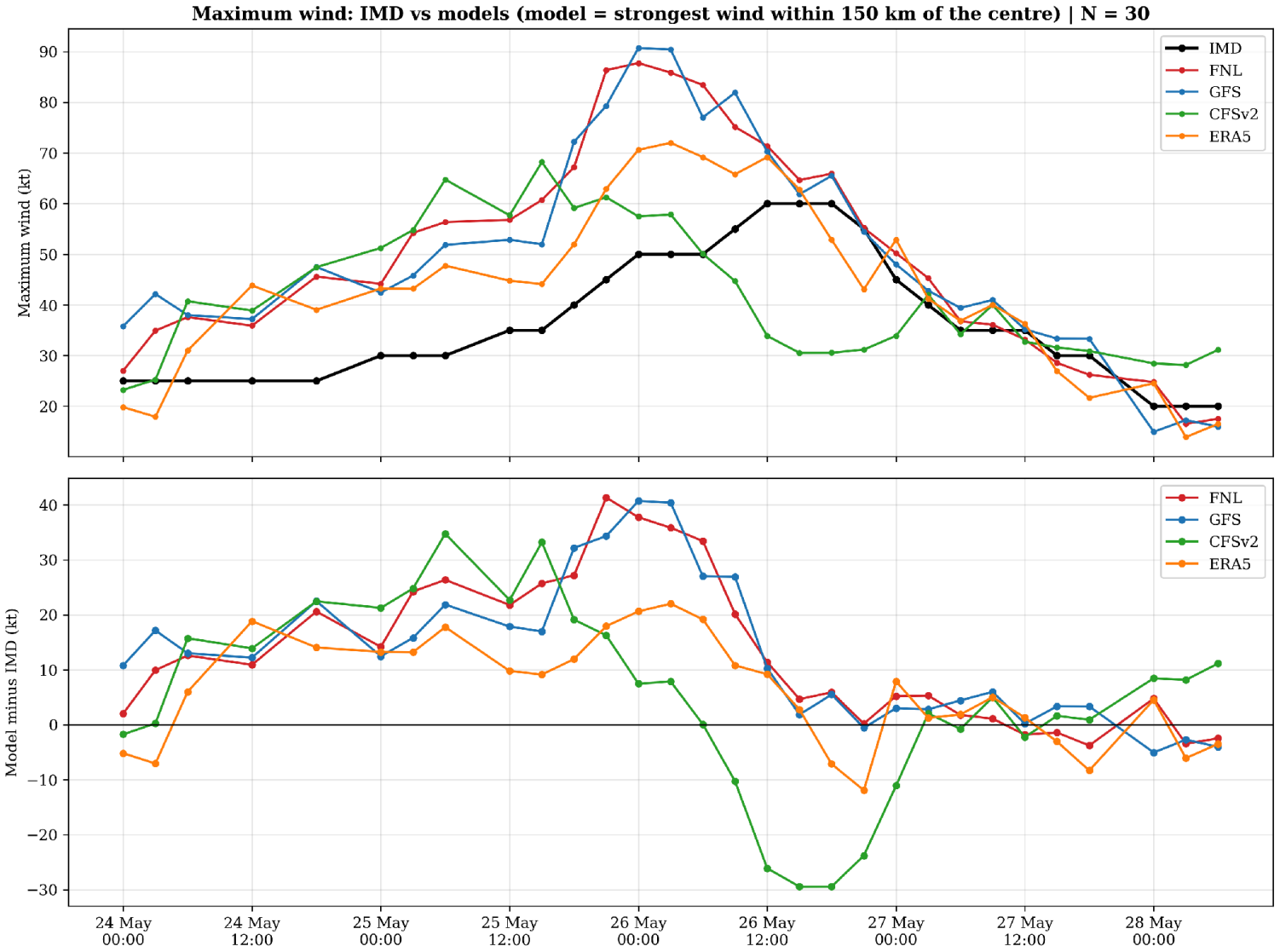
T03_translation_speed_scores.csv

model	n_intervals	bias_kmh	mae_kmh	rmse_kmh	correlation
FNL	17	-1.241	5.512	7.544	-0.179
GFS	17	-0.987	6.048	7.596	-0.358
CFSv2	17	0.303	4.909	5.716	-0.511
ERA5	17	-0.931	5.934	7.608	-0.285

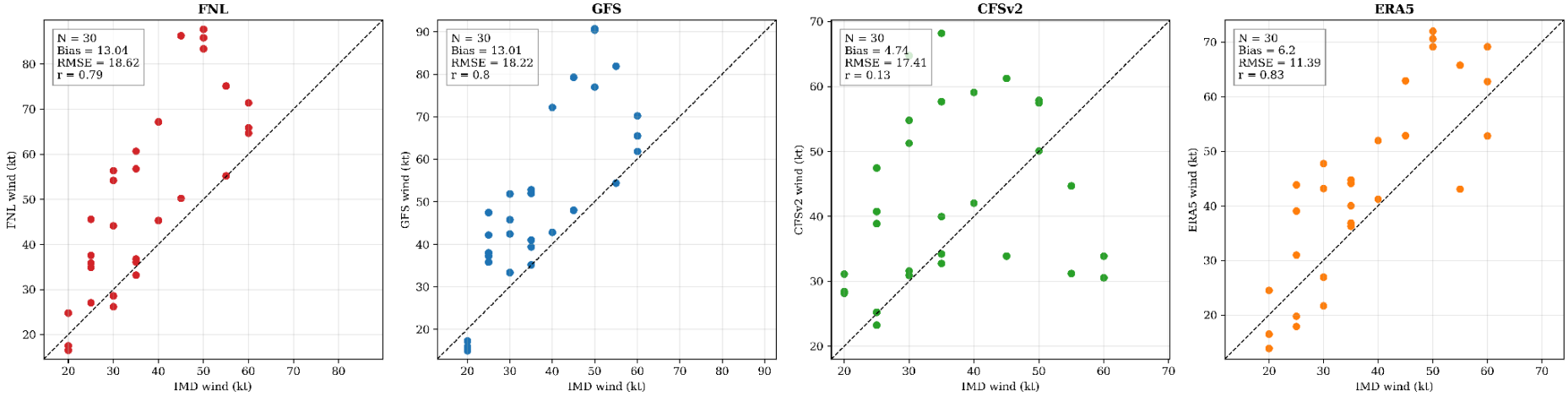
A05 - Maximum 10 m wind validation

Figures: 2 | CSV tables: 11 | NetCDF extracts: 0

F04_Wind_Series_and_Bias.png



S03_Wind_Scatter.png



A05_A06_Intensity_Match_Coverage.csv

model	imd_records_requested	model_exact_matches_before_common_filter	common_matches_used_for_validation
FNL	30	30	30
GFS	30	30	30
CFSv2	30	30	30
ERA5	30	30	30

Remal_CFSv2_All_IMD_Intensity_Extraction.csv

Column block 1 of 3

time_utc	latitude	longitude	central_pressure_hpa	wind_speed_ms	wind_speed_kt	max_wind_ms
2024-05-24 00:00:00+00:00	15.34	87.526	994.543	3.685	7.162	11.961
2024-05-24 03:00:00+00:00	15.39	87.516	997.265	3.443	6.693	12.992
2024-05-24 06:00:00+00:00	14.652	87.715	993.517	9.492	18.452	20.957
2024-05-24 12:00:00+00:00	16.156	87.448	991.532	3.429	6.666	20.005
2024-05-24 18:00:00+00:00	16.983	87.904	990.357	4.671	9.079	24.41
2024-05-25 00:00:00+00:00	17.443	87.535	985.717	2.07	4.023	26.363
2024-05-25 03:00:00+00:00	17.841	87.929	984.994	1.967	3.824	28.187
2024-05-25 06:00:00+00:00	17.813	87.904	982.594	7.265	14.121	33.305
2024-05-25 12:00:00+00:00	19.227	87.703	979.397	7.166	13.93	29.678
2024-05-25 15:00:00+00:00	19.671	87.201	977.775	8.965	17.427	35.088
2024-05-25 18:00:00+00:00	19.97	86.767	977.909	2.48	4.82	30.409
2024-05-25 21:00:00+00:00	20.193	86.529	978.141	2.208	4.293	31.509
2024-05-26 00:00:00+00:00	20.445	86.314	978.015	3.281	6.378	29.565
2024-05-26 03:00:00+00:00	20.629	86.088	980.081	5.132	9.977	29.768
2024-05-26 06:00:00+00:00	20.796	86.065	982.21	3.517	6.837	25.765
2024-05-26 09:00:00+00:00	21.053	85.974	982.732	4.055	7.883	22.997
2024-05-26 12:00:00+00:00	21.493	85.758	983.162	4.489	8.725	17.425
2024-05-26 15:00:00+00:00	21.951	85.603	985.073	1.363	2.649	15.715
2024-05-26 18:00:00+00:00	22.043	85.716	986.179	3.737	7.263	15.721
2024-05-26 21:00:00+00:00	22.365	86.001	986.381	2.821	5.483	16.047

2024-05-27 00:00:00+00:00	22.765	86.122	985.759	2.625	5.103	17.44
2024-05-27 03:00:00+00:00	23.372	86.327	986.189	4.475	8.699	21.624
2024-05-27 06:00:00+00:00	23.663	86.288	986.709	4.377	8.509	17.608
2024-05-27 09:00:00+00:00	23.819	86.447	985.717	4.161	8.088	20.57
2024-05-27 12:00:00+00:00	24.252	86.774	985.529	2.606	5.065	16.853
2024-05-27 15:00:00+00:00	24.639	86.91	986.288	3.242	6.302	16.259
2024-05-27 18:00:00+00:00	24.597	87.108	987.062	1.695	3.294	15.901
2024-05-28 00:00:00+00:00	24.713	87.847	987.913	3.64	7.075	14.638
2024-05-28 03:00:00+00:00	24.829	88.337	990.147	6.139	11.933	14.481
2024-05-28 06:00:00+00:00	24.969	89.611	991.277	5.923	11.513	16.02

Column block 2 of 3

time_utc	max_wind_kt	max_wind_distance_km	centre_method	n_centroid_cells	source_file	guide_latitude
2024-05-24 00:00:00+00:00	23.25	148.879	1.5_hPa_low_SLP_centroid	1506	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-24_00_...	15.34
2024-05-24 03:00:00+00:00	25.254	144.136	1.5_hPa_low_SLP_centroid	1340	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-24_03_...	14.996
2024-05-24 06:00:00+00:00	40.736	106.879	1.5_hPa_low_SLP_centroid	119	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-24_06_...	14.652
2024-05-24 12:00:00+00:00	38.887	140.605	1.5_hPa_low_SLP_centroid	734	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-24_12_...	16.156
2024-05-24 18:00:00+00:00	47.449	148.656	1.5_hPa_low_SLP_centroid	382	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-24_18_...	16.983
2024-05-25 00:00:00+00:00	51.246	91.857	1.5_hPa_low_SLP_centroid	63	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-25_00_...	17.443
2024-05-25 03:00:00+00:00	54.791	111.804	1.5_hPa_low_SLP_centroid	223	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-25_03_...	17.628
2024-05-25 06:00:00+00:00	64.74	45.121	1.5_hPa_low_SLP_centroid	23	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-25_06_...	17.813
2024-05-25 12:00:00+00:00	57.689	109.752	1.5_hPa_low_SLP_centroid	38	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-25_12_...	19.227
2024-05-25 15:00:00+00:00	68.206	144.133	1.5_hPa_low_SLP_centroid	72	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-25_15_...	19.598
2024-05-25 18:00:00+00:00	59.111	100.693	1.5_hPa_low_SLP_centroid	77	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-25_18_...	19.97
2024-05-25 21:00:00+00:00	61.249	96.831	1.5_hPa_low_SLP_centroid	79	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-25_21_...	20.207
2024-05-26 00:00:00+00:00	57.469	110.148	1.5_hPa_low_SLP_centroid	67	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-26_00_...	20.445
2024-05-26 03:00:00+00:00	57.865	141.82	1.5_hPa_low_SLP_centroid	82	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-26_03_...	20.62
2024-05-26 06:00:00+00:00	50.082	125.207	1.5_hPa_low_SLP_centroid	94	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-26_06_...	20.796
2024-05-26 09:00:00+00:00	44.703	121.309	1.5_hPa_low_SLP_centroid	78	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024	21.145

					-05-26_09_...	
2024-05-26 12:00:00+00:00	33.871	149.032	1.5_hPa_low_SLP_centroid	85	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-26_12_...	21.493
2024-05-26 15:00:00+00:00	30.548	136.177	1.5_hPa_low_SLP_centroid	114	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-26_15_...	21.768
2024-05-26 18:00:00+00:00	30.56	114.247	1.5_hPa_low_SLP_centroid	165	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-26_18_...	22.043
2024-05-26 21:00:00+00:00	31.192	149.661	1.5_hPa_low_SLP_centroid	176	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-26_21_...	22.404
2024-05-27 00:00:00+00:00	33.901	129.598	1.5_hPa_low_SLP_centroid	167	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-27_00_...	22.765
2024-05-27 03:00:00+00:00	42.034	132.479	1.5_hPa_low_SLP_centroid	172	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-27_03_...	23.214
2024-05-27 06:00:00+00:00	34.226	122.732	1.5_hPa_low_SLP_centroid	201	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-27_06_...	23.663
2024-05-27 09:00:00+00:00	39.984	98.083	1.5_hPa_low_SLP_centroid	181	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-27_09_...	23.957
2024-05-27 12:00:00+00:00	32.76	97.479	1.5_hPa_low_SLP_centroid	215	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-27_12_...	24.252
2024-05-27 15:00:00+00:00	31.605	89.375	1.5_hPa_low_SLP_centroid	205	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-27_15_...	24.424
2024-05-27 18:00:00+00:00	30.908	140.759	1.5_hPa_low_SLP_centroid	173	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-27_18_...	24.597
2024-05-28 00:00:00+00:00	28.454	108.745	1.5_hPa_low_SLP_centroid	191	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-28_00_...	24.713
2024-05-28 03:00:00+00:00	28.15	149.132	1.5_hPa_low_SLP_centroid	209	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-28_03_...	24.841
2024-05-28 06:00:00+00:00	31.141	147.629	1.5_hPa_low_SLP_centroid	51	/home/ushmo/Desktop/remaL_portfolio/csfv2/wrfout_d02_2024-05-28_06_...	24.969

Column block 3 of 3

time_utc	guide_longitude	search_radius_km	domain_min_slp_hpa	domain_min_outside_search
2024-05-24 00:00:00+00:00	87.526	300.0	994.543	False
2024-05-24 03:00:00+00:00	87.621	300.0	997.265	False
2024-05-24 06:00:00+00:00	87.715	300.0	993.517	False
2024-05-24 12:00:00+00:00	87.448	300.0	991.532	False
2024-05-24 18:00:00+00:00	87.904	300.0	990.357	False
2024-05-25 00:00:00+00:00	87.535	300.0	985.717	False
2024-05-25 03:00:00+00:00	87.72	300.0	984.994	False
2024-05-25 06:00:00+00:00	87.904	300.0	982.594	False
2024-05-25 12:00:00+00:00	87.703	300.0	979.397	False
2024-05-25 15:00:00+00:00	87.235	300.0	977.775	False
2024-05-25 18:00:00+00:00	86.767	300.0	977.909	False
2024-05-25 21:00:00+00:00	86.54	300.0	978.141	False
2024-05-26 00:00:00+00:00	86.314	300.0	978.015	False
2024-05-26 03:00:00+00:00	86.189	300.0	980.081	False

2024-05-26 06:00:00+00:00	86.065	300.0	982.21	False
2024-05-26 09:00:00+00:00	85.911	300.0	982.732	False
2024-05-26 12:00:00+00:00	85.758	300.0	983.162	False
2024-05-26 15:00:00+00:00	85.737	300.0	985.073	False
2024-05-26 18:00:00+00:00	85.716	300.0	986.179	False
2024-05-26 21:00:00+00:00	85.919	300.0	986.381	False
2024-05-27 00:00:00+00:00	86.122	300.0	985.759	False
2024-05-27 03:00:00+00:00	86.205	300.0	986.189	False
2024-05-27 06:00:00+00:00	86.288	300.0	986.709	False
2024-05-27 09:00:00+00:00	86.531	300.0	985.717	False
2024-05-27 12:00:00+00:00	86.774	300.0	985.529	False
2024-05-27 15:00:00+00:00	86.941	300.0	986.288	False
2024-05-27 18:00:00+00:00	87.108	300.0	987.062	False
2024-05-28 00:00:00+00:00	87.847	300.0	987.913	False
2024-05-28 03:00:00+00:00	88.729	300.0	990.147	False
2024-05-28 06:00:00+00:00	89.611	300.0	991.277	False

Remal_CFSv2_All_IMD_Wind_Matches.csv

Column block 1 of 2

time_utc	wind_imd	wind_model	wind_bias_kt	wind_centre_model	wind_centre_bias_kt	max_wind_distance_km
2024-05-24 00:00:00+00:00	25	23.25	-1.75	7.162	-17.838	148.879
2024-05-24 03:00:00+00:00	25	25.254	0.254	6.693	-18.307	144.136
2024-05-24 06:00:00+00:00	25	40.736	15.736	18.452	-6.548	106.879
2024-05-24 12:00:00+00:00	25	38.887	13.887	6.666	-18.334	140.605
2024-05-24 18:00:00+00:00	25	47.449	22.449	9.079	-15.921	148.656
2024-05-25 00:00:00+00:00	30	51.246	21.246	4.023	-25.977	91.857
2024-05-25 03:00:00+00:00	30	54.791	24.791	3.824	-26.176	111.804
2024-05-25 06:00:00+00:00	30	64.74	34.74	14.121	-15.879	45.121
2024-05-25 12:00:00+00:00	35	57.689	22.689	13.93	-21.07	109.752
2024-05-25 15:00:00+00:00	35	68.206	33.206	17.427	-17.573	144.133
2024-05-25 18:00:00+00:00	40	59.111	19.111	4.82	-35.18	100.693
2024-05-25 21:00:00+00:00	45	61.249	16.249	4.293	-40.707	96.831
2024-05-26 00:00:00+00:00	50	57.469	7.469	6.378	-43.622	110.148
2024-05-26 03:00:00+00:00	50	57.865	7.865	9.977	-40.023	141.82
2024-05-26 06:00:00+00:00	50	50.082	0.082	6.837	-43.163	125.207
2024-05-26 09:00:00+00:00	55	44.703	-10.297	7.883	-47.117	121.309
2024-05-26 12:00:00+00:00	60	33.871	-26.129	8.725	-51.275	149.032
2024-05-26 15:00:00+00:00	60	30.548	-29.452	2.649	-57.351	136.177
2024-05-26 18:00:00+00:00	60	30.56	-29.44	7.263	-52.737	114.247
2024-05-26 21:00:00+00:00	55	31.192	-23.808	5.483	-49.517	149.661
2024-05-27 00:00:00+00:00	45	33.901	-11.099	5.103	-39.897	129.598
2024-05-27 03:00:00+00:00	40	42.034	2.034	8.699	-31.301	132.479
2024-05-27 06:00:00+00:00	35	34.226	-0.774	8.509	-26.491	122.732
2024-05-27 09:00:00+00:00	35	39.984	4.984	8.088	-26.912	98.083
2024-05-27 12:00:00+00:00	35	32.76	-2.24	5.065	-29.935	97.479
2024-05-27 15:00:00+00:00	30	31.605	1.605	6.302	-23.698	89.375
2024-05-27 18:00:00+00:00	30	30.908	0.908	3.294	-26.706	140.759
2024-05-28 00:00:00+00:00	20	28.454	8.454	7.075	-12.925	108.745
2024-05-28 03:00:00+00:00	20	28.15	8.15	11.933	-8.067	149.132
2024-05-28 06:00:00+00:00	20	31.141	11.141	11.513	-8.487	147.629

Column block 2 of 2

time_utc	lat_model_intensity	lon_model_intensity	centre_method	n_centroid_cells

2024-05-24 00:00:00+00:00	15.34	87.526	1.5_hPa_low_SLP_centroid	1506
2024-05-24 03:00:00+00:00	15.39	87.516	1.5_hPa_low_SLP_centroid	1340
2024-05-24 06:00:00+00:00	14.652	87.715	1.5_hPa_low_SLP_centroid	119
2024-05-24 12:00:00+00:00	16.156	87.448	1.5_hPa_low_SLP_centroid	734
2024-05-24 18:00:00+00:00	16.983	87.904	1.5_hPa_low_SLP_centroid	382
2024-05-25 00:00:00+00:00	17.443	87.535	1.5_hPa_low_SLP_centroid	63
2024-05-25 03:00:00+00:00	17.841	87.929	1.5_hPa_low_SLP_centroid	223
2024-05-25 06:00:00+00:00	17.813	87.904	1.5_hPa_low_SLP_centroid	23
2024-05-25 12:00:00+00:00	19.227	87.703	1.5_hPa_low_SLP_centroid	38
2024-05-25 15:00:00+00:00	19.671	87.201	1.5_hPa_low_SLP_centroid	72
2024-05-25 18:00:00+00:00	19.97	86.767	1.5_hPa_low_SLP_centroid	77
2024-05-25 21:00:00+00:00	20.193	86.529	1.5_hPa_low_SLP_centroid	79
2024-05-26 00:00:00+00:00	20.445	86.314	1.5_hPa_low_SLP_centroid	67
2024-05-26 03:00:00+00:00	20.629	86.088	1.5_hPa_low_SLP_centroid	82
2024-05-26 06:00:00+00:00	20.796	86.065	1.5_hPa_low_SLP_centroid	94
2024-05-26 09:00:00+00:00	21.053	85.974	1.5_hPa_low_SLP_centroid	78
2024-05-26 12:00:00+00:00	21.493	85.758	1.5_hPa_low_SLP_centroid	85
2024-05-26 15:00:00+00:00	21.951	85.603	1.5_hPa_low_SLP_centroid	114
2024-05-26 18:00:00+00:00	22.043	85.716	1.5_hPa_low_SLP_centroid	165
2024-05-26 21:00:00+00:00	22.365	86.001	1.5_hPa_low_SLP_centroid	176
2024-05-27 00:00:00+00:00	22.765	86.122	1.5_hPa_low_SLP_centroid	167
2024-05-27 03:00:00+00:00	23.372	86.327	1.5_hPa_low_SLP_centroid	172
2024-05-27 06:00:00+00:00	23.663	86.288	1.5_hPa_low_SLP_centroid	201
2024-05-27 09:00:00+00:00	23.819	86.447	1.5_hPa_low_SLP_centroid	181
2024-05-27 12:00:00+00:00	24.252	86.774	1.5_hPa_low_SLP_centroid	215
2024-05-27 15:00:00+00:00	24.639	86.91	1.5_hPa_low_SLP_centroid	205
2024-05-27 18:00:00+00:00	24.597	87.108	1.5_hPa_low_SLP_centroid	173
2024-05-28 00:00:00+00:00	24.713	87.847	1.5_hPa_low_SLP_centroid	191
2024-05-28 03:00:00+00:00	24.829	88.337	1.5_hPa_low_SLP_centroid	209
2024-05-28 06:00:00+00:00	24.969	89.611	1.5_hPa_low_SLP_centroid	51

Remal_ERA5_All_IMD_Intensity_Extraction.csv

Column block 1 of 3

time_utc	latitude	longitude	central_pressure_hpa	wind_speed_ms	wind_speed_kt	max_wind_ms
2024-05-24 00:00:00+00:00	15.29	88.392	997.081	3.33	6.473	10.191
2024-05-24 03:00:00+00:00	15.831	88.395	998.919	3.512	6.828	9.227
2024-05-24 06:00:00+00:00	15.51	88.967	996.421	5.099	9.911	15.943
2024-05-24 12:00:00+00:00	15.271	89.208	992.596	5.937	11.541	22.551
2024-05-24 18:00:00+00:00	16.768	90.135	993.724	5.331	10.362	20.088
2024-05-25 00:00:00+00:00	17.455	90.567	991.339	8.965	17.427	22.246
2024-05-25 03:00:00+00:00	17.964	90.388	990.159	2.572	5.0	22.222
2024-05-25 06:00:00+00:00	18.407	90.086	988.791	6.246	12.142	24.565
2024-05-25 12:00:00+00:00	19.076	89.63	985.755	4.845	9.418	23.039
2024-05-25 15:00:00+00:00	18.989	89.631	984.658	5.295	10.292	22.699
2024-05-25 18:00:00+00:00	19.381	89.882	983.638	8.751	17.011	26.722
2024-05-25 21:00:00+00:00	19.737	89.348	979.713	8.874	17.25	32.374
2024-05-26 00:00:00+00:00	20.15	90.059	975.25	4.217	8.197	36.329
2024-05-26 03:00:00+00:00	20.918	89.622	972.388	6.283	12.213	37.043
2024-05-26 06:00:00+00:00	21.282	89.651	972.373	4.319	8.396	35.585
2024-05-26 09:00:00+00:00	21.569	89.538	972.022	7.45	14.482	33.848
2024-05-26 12:00:00+00:00	22.113	89.846	973.012	0.867	1.686	35.586
2024-05-26 15:00:00+00:00	22.62	89.719	974.579	4.075	7.922	32.297
2024-05-26 18:00:00+00:00	23.008	89.875	976.344	2.624	5.1	27.188
2024-05-26 21:00:00+00:00	23.374	89.933	978.227	1.549	3.011	22.152
2024-05-27 00:00:00+00:00	23.774	90.061	980.161	1.296	2.518	27.208
2024-05-27 03:00:00+00:00	23.919	90.088	982.191	0.965	1.876	21.209

2024-05-27 06:00:00+00:00	24.173	90.498	983.073	3.614	7.026	18.963
2024-05-27 09:00:00+00:00	24.479	90.868	984.295	4.59	8.922	20.586
2024-05-27 12:00:00+00:00	24.743	91.489	985.724	3.487	6.778	18.667
2024-05-27 15:00:00+00:00	25.082	92.277	989.553	10.261	19.945	13.853
2024-05-27 18:00:00+00:00	24.978	92.267	992.635	5.6	10.886	11.144
2024-05-28 00:00:00+00:00	24.313	94.265	995.14	1.396	2.714	12.62
2024-05-28 03:00:00+00:00	24.467	89.575	997.353	4.783	9.297	7.163
2024-05-28 06:00:00+00:00	24.46	89.102	995.971	5.753	11.183	8.493

Column block 2 of 3

time_utc	max_wind_kt	max_wind_distance_km	centre_method	n_centroid_cells	source_file	guide_latitude
2024-05-24 00:00:00+00:00	19.81	147.327	1.5_hPa_low_SLP_centroid	2479	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-24_00_...	15.353
2024-05-24 03:00:00+00:00	17.935	146.966	1.5_hPa_low_SLP_centroid	1999	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-24_03_...	15.409
2024-05-24 06:00:00+00:00	30.99	118.474	1.5_hPa_low_SLP_centroid	1661	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-24_06_...	15.465
2024-05-24 12:00:00+00:00	43.835	38.422	1.5_hPa_low_SLP_centroid	45	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-24_12_...	15.271
2024-05-24 18:00:00+00:00	39.048	147.916	1.5_hPa_low_SLP_centroid	173	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-24_18_...	16.768
2024-05-25 00:00:00+00:00	43.243	127.212	1.5_hPa_low_SLP_centroid	82	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-25_00_...	17.455
2024-05-25 03:00:00+00:00	43.197	134.641	1.5_hPa_low_SLP_centroid	82	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-25_03_...	17.931
2024-05-25 06:00:00+00:00	47.751	146.929	1.5_hPa_low_SLP_centroid	97	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-25_06_...	18.407
2024-05-25 12:00:00+00:00	44.784	122.001	1.5_hPa_low_SLP_centroid	52	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-25_12_...	19.076
2024-05-25 15:00:00+00:00	44.123	146.356	1.5_hPa_low_SLP_centroid	81	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-25_15_...	19.228
2024-05-25 18:00:00+00:00	51.944	145.374	1.5_hPa_low_SLP_centroid	90	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-25_18_...	19.381
2024-05-25 21:00:00+00:00	62.931	145.283	1.5_hPa_low_SLP_centroid	16	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-25_21_...	19.765
2024-05-26 00:00:00+00:00	70.618	60.035	1.5_hPa_low_SLP_centroid	45	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-26_00_...	20.15
2024-05-26 03:00:00+00:00	72.006	94.502	1.5_hPa_low_SLP_centroid	29	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-26_03_...	20.716
2024-05-26 06:00:00+00:00	69.171	70.32	1.5_hPa_low_SLP_centroid	41	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-26_06_...	21.282
2024-05-26 09:00:00+00:00	65.795	88.318	1.5_hPa_low_SLP_centroid	40	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-26_09_...	21.698
2024-05-26 12:00:00+00:00	69.174	91.829	1.5_hPa_low_SLP_centroid	48	/home/ushmo/Desktop/remaL_p	22.113

					ortfolio/era_5/wrfout_d02_2024-05-26_12_...	
2024-05-26 15:00:00+00:00	62.78	137.213	1.5_hPa_low_SLP_centroid	42	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-26_15_...	22.561
2024-05-26 18:00:00+00:00	52.848	139.351	1.5_hPa_low_SLP_centroid	37	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-26_18_...	23.008
2024-05-26 21:00:00+00:00	43.06	149.481	1.5_hPa_low_SLP_centroid	47	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-26_21_...	23.391
2024-05-27 00:00:00+00:00	52.887	116.395	1.5_hPa_low_SLP_centroid	49	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-27_00_...	23.774
2024-05-27 03:00:00+00:00	41.226	55.136	1.5_hPa_low_SLP_centroid	55	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-27_03_...	23.974
2024-05-27 06:00:00+00:00	36.86	44.67	1.5_hPa_low_SLP_centroid	51	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-27_06_...	24.173
2024-05-27 09:00:00+00:00	40.016	32.216	1.5_hPa_low_SLP_centroid	54	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-27_09_...	24.458
2024-05-27 12:00:00+00:00	36.286	47.408	1.5_hPa_low_SLP_centroid	42	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-27_12_...	24.743
2024-05-27 15:00:00+00:00	26.928	19.205	1.5_hPa_low_SLP_centroid	73	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-27_15_...	24.743
2024-05-27 18:00:00+00:00	21.662	73.015	1.5_hPa_low_SLP_centroid	230	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-27_18_...	24.743
2024-05-28 00:00:00+00:00	24.532	78.639	1.5_hPa_low_SLP_centroid	44	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-28_00_...	24.743
2024-05-28 03:00:00+00:00	13.924	47.126	1.5_hPa_low_SLP_centroid	407	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-28_03_...	24.743
2024-05-28 06:00:00+00:00	16.51	72.567	1.5_hPa_low_SLP_centroid	316	/home/ushmo/Desktop/remaL_portfolio/era_5/wrfout_d02_2024-05-28_06_...	24.743

Column block 3 of 3

time_utc	guide_longitude	search_radius_km	domain_min_slp_hpa	domain_min_outside_search
2024-05-24 00:00:00+00:00	88.286	300.0	997.081	False
2024-05-24 03:00:00+00:00	88.643	300.0	998.919	False
2024-05-24 06:00:00+00:00	88.999	300.0	996.421	False
2024-05-24 12:00:00+00:00	89.208	300.0	992.596	False
2024-05-24 18:00:00+00:00	90.135	300.0	993.724	False
2024-05-25 00:00:00+00:00	90.567	300.0	991.339	False
2024-05-25 03:00:00+00:00	90.326	300.0	990.159	False
2024-05-25 06:00:00+00:00	90.086	300.0	988.791	False
2024-05-25 12:00:00+00:00	89.63	300.0	985.755	False
2024-05-25 15:00:00+00:00	89.756	300.0	984.658	False
2024-05-25 18:00:00+00:00	89.882	300.0	983.638	False
2024-05-25 21:00:00+00:00	89.97	300.0	979.713	False
2024-05-26 00:00:00+00:00	90.059	300.0	975.25	False
2024-05-26 03:00:00+00:00	89.855	300.0	972.388	False
2024-05-26 06:00:00+00:00	89.651	300.0	972.373	False
2024-05-26 09:00:00+00:00	89.748	300.0	972.022	False

2024-05-26 12:00:00+00:00	89.846	300.0	973.012	False
2024-05-26 15:00:00+00:00	89.86	300.0	974.579	False
2024-05-26 18:00:00+00:00	89.875	300.0	976.344	False
2024-05-26 21:00:00+00:00	89.968	300.0	978.227	False
2024-05-27 00:00:00+00:00	90.061	300.0	980.161	False
2024-05-27 03:00:00+00:00	90.28	300.0	982.191	False
2024-05-27 06:00:00+00:00	90.498	300.0	983.073	False
2024-05-27 09:00:00+00:00	90.994	300.0	984.295	False
2024-05-27 12:00:00+00:00	91.489	300.0	985.724	False
2024-05-27 15:00:00+00:00	91.489	300.0	989.553	False
2024-05-27 18:00:00+00:00	91.489	300.0	992.55	True
2024-05-28 00:00:00+00:00	91.489	300.0	992.897	True
2024-05-28 03:00:00+00:00	91.489	300.0	992.425	True
2024-05-28 06:00:00+00:00	91.489	300.0	992.971	True

Remal_ERA5_All_IMD_Wind_Matches.csv

Column block 1 of 2

time_utc	wind_imd	wind_model	wind_bias_kt	wind_centre_model	wind_centre_bias_kt	max_wind_distance_km
2024-05-24 00:00:00+00:00	25	19.81	-5.19	6.473	-18.527	147.327
2024-05-24 03:00:00+00:00	25	17.935	-7.065	6.828	-18.172	146.966
2024-05-24 06:00:00+00:00	25	30.99	5.99	9.911	-15.089	118.474
2024-05-24 12:00:00+00:00	25	43.835	18.835	11.541	-13.459	38.422
2024-05-24 18:00:00+00:00	25	39.048	14.048	10.362	-14.638	147.916
2024-05-25 00:00:00+00:00	30	43.243	13.243	17.427	-12.573	127.212
2024-05-25 03:00:00+00:00	30	43.197	13.197	5.0	-25.0	134.641
2024-05-25 06:00:00+00:00	30	47.751	17.751	12.142	-17.858	146.929
2024-05-25 12:00:00+00:00	35	44.784	9.784	9.418	-25.582	122.001
2024-05-25 15:00:00+00:00	35	44.123	9.123	10.292	-24.708	146.356
2024-05-25 18:00:00+00:00	40	51.944	11.944	17.011	-22.989	145.374
2024-05-25 21:00:00+00:00	45	62.931	17.931	17.25	-27.75	145.283
2024-05-26 00:00:00+00:00	50	70.618	20.618	8.197	-41.803	60.035
2024-05-26 03:00:00+00:00	50	72.006	22.006	12.213	-37.787	94.502
2024-05-26 06:00:00+00:00	50	69.171	19.171	8.396	-41.604	70.32
2024-05-26 09:00:00+00:00	55	65.795	10.795	14.482	-40.518	88.318
2024-05-26 12:00:00+00:00	60	69.174	9.174	1.686	-58.314	91.829
2024-05-26 15:00:00+00:00	60	62.78	2.78	7.922	-52.078	137.213
2024-05-26 18:00:00+00:00	60	52.848	-7.152	5.1	-54.9	139.351
2024-05-26 21:00:00+00:00	55	43.06	-11.94	3.011	-51.989	149.481
2024-05-27 00:00:00+00:00	45	52.887	7.887	2.518	-42.482	116.395
2024-05-27 03:00:00+00:00	40	41.226	1.226	1.876	-38.124	55.136
2024-05-27 06:00:00+00:00	35	36.86	1.86	7.026	-27.974	44.67
2024-05-27 09:00:00+00:00	35	40.016	5.016	8.922	-26.078	32.216
2024-05-27 12:00:00+00:00	35	36.286	1.286	6.778	-28.222	47.408
2024-05-27 15:00:00+00:00	30	26.928	-3.072	19.945	-10.055	19.205
2024-05-27 18:00:00+00:00	30	21.662	-8.338	10.886	-19.114	73.015
2024-05-28 00:00:00+00:00	20	24.532	4.532	2.714	-17.286	78.639
2024-05-28 03:00:00+00:00	20	13.924	-6.076	9.297	-10.703	47.126
2024-05-28 06:00:00+00:00	20	16.51	-3.49	11.183	-8.817	72.567

Column block 2 of 2

time_utc	lat_model_intensity	lon_model_intensity	centre_method	n_centroid_cells
2024-05-24 00:00:00+00:00	15.29	88.392	1.5_hPa_low_SLP_centroid	2479
2024-05-24 03:00:00+00:00	15.831	88.395	1.5_hPa_low_SLP_centroid	1999

2024-05-24 06:00:00+00:00	15.51	88.967	1.5_hPa_low_SLP_centroid	1661
2024-05-24 12:00:00+00:00	15.271	89.208	1.5_hPa_low_SLP_centroid	45
2024-05-24 18:00:00+00:00	16.768	90.135	1.5_hPa_low_SLP_centroid	173
2024-05-25 00:00:00+00:00	17.455	90.567	1.5_hPa_low_SLP_centroid	82
2024-05-25 03:00:00+00:00	17.964	90.388	1.5_hPa_low_SLP_centroid	82
2024-05-25 06:00:00+00:00	18.407	90.086	1.5_hPa_low_SLP_centroid	97
2024-05-25 12:00:00+00:00	19.076	89.63	1.5_hPa_low_SLP_centroid	52
2024-05-25 15:00:00+00:00	18.989	89.631	1.5_hPa_low_SLP_centroid	81
2024-05-25 18:00:00+00:00	19.381	89.882	1.5_hPa_low_SLP_centroid	90
2024-05-25 21:00:00+00:00	19.737	89.348	1.5_hPa_low_SLP_centroid	16
2024-05-26 00:00:00+00:00	20.15	90.059	1.5_hPa_low_SLP_centroid	45
2024-05-26 03:00:00+00:00	20.918	89.622	1.5_hPa_low_SLP_centroid	29
2024-05-26 06:00:00+00:00	21.282	89.651	1.5_hPa_low_SLP_centroid	41
2024-05-26 09:00:00+00:00	21.569	89.538	1.5_hPa_low_SLP_centroid	40
2024-05-26 12:00:00+00:00	22.113	89.846	1.5_hPa_low_SLP_centroid	48
2024-05-26 15:00:00+00:00	22.62	89.719	1.5_hPa_low_SLP_centroid	42
2024-05-26 18:00:00+00:00	23.008	89.875	1.5_hPa_low_SLP_centroid	37
2024-05-26 21:00:00+00:00	23.374	89.933	1.5_hPa_low_SLP_centroid	47
2024-05-27 00:00:00+00:00	23.774	90.061	1.5_hPa_low_SLP_centroid	49
2024-05-27 03:00:00+00:00	23.919	90.088	1.5_hPa_low_SLP_centroid	55
2024-05-27 06:00:00+00:00	24.173	90.498	1.5_hPa_low_SLP_centroid	51
2024-05-27 09:00:00+00:00	24.479	90.868	1.5_hPa_low_SLP_centroid	54
2024-05-27 12:00:00+00:00	24.743	91.489	1.5_hPa_low_SLP_centroid	42
2024-05-27 15:00:00+00:00	25.082	92.277	1.5_hPa_low_SLP_centroid	73
2024-05-27 18:00:00+00:00	24.978	92.267	1.5_hPa_low_SLP_centroid	230
2024-05-28 00:00:00+00:00	24.313	94.265	1.5_hPa_low_SLP_centroid	44
2024-05-28 03:00:00+00:00	24.467	89.575	1.5_hPa_low_SLP_centroid	407
2024-05-28 06:00:00+00:00	24.46	89.102	1.5_hPa_low_SLP_centroid	316

Remal_FNL_All_IMD_Intensity_Extraction.csv

Column block 1 of 3

time_utc	latitude	longitude	central_pressure_hpa	wind_speed_ms	wind_speed_kt	max_wind_ms
2024-05-24 00:00:00+00:00	15.29	88.799	996.495	3.989	7.754	13.91
2024-05-24 03:00:00+00:00	14.956	89.836	996.843	7.044	13.693	17.966
2024-05-24 06:00:00+00:00	15.377	89.978	995.149	0.582	1.131	19.339
2024-05-24 12:00:00+00:00	16.07	90.13	992.54	5.394	10.485	18.474
2024-05-24 18:00:00+00:00	16.271	89.618	991.876	3.807	7.4	23.456
2024-05-25 00:00:00+00:00	17.226	90.325	990.428	0.898	1.746	22.723
2024-05-25 03:00:00+00:00	18.106	90.653	987.252	14.671	28.519	27.899
2024-05-25 06:00:00+00:00	18.59	89.573	983.934	17.924	34.842	28.986
2024-05-25 12:00:00+00:00	18.177	89.73	981.328	4.546	8.837	29.211
2024-05-25 15:00:00+00:00	18.408	89.695	980.561	5.03	9.777	31.226
2024-05-25 18:00:00+00:00	18.854	89.459	979.248	10.394	20.205	34.563
2024-05-25 21:00:00+00:00	18.79	89.718	973.506	10.206	19.839	44.415
2024-05-26 00:00:00+00:00	19.639	89.448	969.687	5.376	10.45	45.133
2024-05-26 03:00:00+00:00	20.104	89.828	969.99	7.363	14.312	44.164
2024-05-26 06:00:00+00:00	20.522	89.601	969.25	3.726	7.243	42.908
2024-05-26 09:00:00+00:00	21.174	89.95	969.078	4.503	8.753	38.653
2024-05-26 12:00:00+00:00	21.715	90.086	968.954	9.029	17.551	36.721
2024-05-26 15:00:00+00:00	22.176	89.98	970.574	3.564	6.929	33.26
2024-05-26 18:00:00+00:00	22.518	90.244	973.926	2.514	4.887	33.9
2024-05-26 21:00:00+00:00	23.108	90.418	975.505	3.557	6.914	28.39
2024-05-27 00:00:00+00:00	23.528	90.696	977.931	2.514	4.887	25.833
2024-05-27 03:00:00+00:00	23.877	90.686	979.826	3.924	7.628	23.297
2024-05-27 06:00:00+00:00	24.197	91.109	982.86	3.855	7.493	18.918
2024-05-27 09:00:00+00:00	24.57	91.423	984.196	2.345	4.559	18.57

2024-05-27 12:00:00+00:00	24.844	92.035	985.189	7.529	14.636	17.07
2024-05-27 15:00:00+00:00	24.921	92.529	990.033	9.998	19.435	14.698
2024-05-27 18:00:00+00:00	25.156	93.164	991.512	5.947	11.56	13.484
2024-05-28 00:00:00+00:00	24.826	94.385	994.349	1.005	1.954	12.745
2024-05-28 03:00:00+00:00	24.394	90.264	995.956	5.348	10.395	8.509
2024-05-28 06:00:00+00:00	24.57	90.102	995.656	5.678	11.038	9.012

Column block 2 of 3

time_utc	max_wind_kt	max_wind_distance_km	centre_method	n_centroid_cells	source_file	guide_latitude
2024-05-24 00:00:00+00:00	27.038	149.615	1.5_hPa_low_SLP_centroid	1864	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-24_00_00_00	15.271
2024-05-24 03:00:00+00:00	34.923	89.165	1.5_hPa_low_SLP_centroid	523	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-24_03_00_00	15.324
2024-05-24 06:00:00+00:00	37.592	127.21	1.5_hPa_low_SLP_centroid	583	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-24_06_00_00	15.377
2024-05-24 12:00:00+00:00	35.911	146.483	1.5_hPa_low_SLP_centroid	259	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-24_12_00_00	16.07
2024-05-24 18:00:00+00:00	45.595	116.62	1.5_hPa_low_SLP_centroid	182	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-24_18_00_00	16.271
2024-05-25 00:00:00+00:00	44.169	139.093	1.5_hPa_low_SLP_centroid	441	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-25_00_00_00	17.226
2024-05-25 03:00:00+00:00	54.231	39.586	1.5_hPa_low_SLP_centroid	9	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-25_03_00_00	17.908
2024-05-25 06:00:00+00:00	56.344	14.136	1.5_hPa_low_SLP_centroid	4	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-25_06_00_00	18.59
2024-05-25 12:00:00+00:00	56.782	74.966	1.5_hPa_low_SLP_centroid	63	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-25_12_00_00	18.177
2024-05-25 15:00:00+00:00	60.698	126.794	1.5_hPa_low_SLP_centroid	103	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-25_15_00_00	18.515
2024-05-25 18:00:00+00:00	67.184	94.754	1.5_hPa_low_SLP_centroid	148	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-25_18_00_00	18.854
2024-05-25 21:00:00+00:00	86.335	40.925	1.5_hPa_low_SLP_centroid	18	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-25_21_00_00	19.247
2024-05-26 00:00:00+00:00	87.732	110.946	1.5_hPa_low_SLP_centroid	26	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-26_00_00_00	19.639
2024-05-26 03:00:00+00:00	85.848	71.378	1.5_hPa_low_SLP_centroid	30	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-26_03_00_00	20.081
2024-05-26 06:00:00+00:00	83.407	98.637	1.5_hPa_low_SLP_centroid	54	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-26_06_00_00	20.522
2024-05-26 09:00:00+00:00	75.134	91.983	1.5_hPa_low_SLP_centroid	44	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-26_09_00_00	21.118
2024-05-26 12:00:00+00:00	71.381	58.178	1.5_hPa_low_SLP_centroid	59	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-26_12_00_00	21.715
2024-05-26 15:00:00+00:00	64.652	83.473	1.5_hPa_low_SLP_centroid	49	/home/ushmo/Desktop/remaL_p	22.117

					ortfolio/fnl/wrfout_d02_2024-05-26_15_00_00	
2024-05-26 18:00:00+00:00	65.896	102.523	1.5_hPa_low_SLP_centroid	47	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-26_18_00_00	22.518
2024-05-26 21:00:00+00:00	55.187	137.336	1.5_hPa_low_SLP_centroid	41	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-26_21_00_00	23.023
2024-05-27 00:00:00+00:00	50.216	83.324	1.5_hPa_low_SLP_centroid	46	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-27_00_00_00	23.528
2024-05-27 03:00:00+00:00	45.286	68.569	1.5_hPa_low_SLP_centroid	43	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-27_03_00_00	23.863
2024-05-27 06:00:00+00:00	36.774	61.054	1.5_hPa_low_SLP_centroid	46	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-27_06_00_00	24.197
2024-05-27 09:00:00+00:00	36.096	46.467	1.5_hPa_low_SLP_centroid	43	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-27_09_00_00	24.521
2024-05-27 12:00:00+00:00	33.182	37.624	1.5_hPa_low_SLP_centroid	29	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-27_12_00_00	24.844
2024-05-27 15:00:00+00:00	28.571	111.598	1.5_hPa_low_SLP_centroid	160	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-27_15_00_00	24.844
2024-05-27 18:00:00+00:00	26.211	146.968	1.5_hPa_low_SLP_centroid	63	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-27_18_00_00	24.844
2024-05-28 00:00:00+00:00	24.774	38.78	1.5_hPa_low_SLP_centroid	60	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-28_00_00_00	24.844
2024-05-28 03:00:00+00:00	16.54	126.799	1.5_hPa_low_SLP_centroid	416	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-28_03_00_00	24.844
2024-05-28 06:00:00+00:00	17.518	147.856	1.5_hPa_low_SLP_centroid	344	/home/ushmo/Desktop/remaL_portfolio/fnl/wrfout_d02_2024-05-28_06_00_00	24.844

Column block 3 of 3

time_utc	guide_longitude	search_radius_km	domain_min_slp_hpa	domain_min_outside_search
2024-05-24 00:00:00+00:00	88.835	300.0	996.495	False
2024-05-24 03:00:00+00:00	89.406	300.0	996.843	False
2024-05-24 06:00:00+00:00	89.978	300.0	995.149	False
2024-05-24 12:00:00+00:00	90.13	300.0	992.54	False
2024-05-24 18:00:00+00:00	89.618	300.0	991.876	False
2024-05-25 00:00:00+00:00	90.325	300.0	990.428	False
2024-05-25 03:00:00+00:00	89.949	300.0	987.252	False
2024-05-25 06:00:00+00:00	89.573	300.0	983.934	False
2024-05-25 12:00:00+00:00	89.73	300.0	981.328	False
2024-05-25 15:00:00+00:00	89.595	300.0	980.561	False
2024-05-25 18:00:00+00:00	89.459	300.0	979.248	False
2024-05-25 21:00:00+00:00	89.453	300.0	973.506	False
2024-05-26 00:00:00+00:00	89.448	300.0	969.687	False
2024-05-26 03:00:00+00:00	89.524	300.0	969.99	False
2024-05-26 06:00:00+00:00	89.601	300.0	969.25	False
2024-05-26 09:00:00+00:00	89.844	300.0	969.078	False
2024-05-26 12:00:00+00:00	90.086	300.0	968.954	False
2024-05-26 15:00:00+00:00	90.165	300.0	970.574	False
2024-05-26 18:00:00+00:00	90.244	300.0	973.926	False

2024-05-26 21:00:00+00:00	90.47	300.0	975.505	False
2024-05-27 00:00:00+00:00	90.696	300.0	977.931	False
2024-05-27 03:00:00+00:00	90.902	300.0	979.826	False
2024-05-27 06:00:00+00:00	91.109	300.0	982.86	False
2024-05-27 09:00:00+00:00	91.572	300.0	984.196	False
2024-05-27 12:00:00+00:00	92.035	300.0	985.189	False
2024-05-27 15:00:00+00:00	92.035	300.0	990.033	False
2024-05-27 18:00:00+00:00	92.035	300.0	990.801	True
2024-05-28 00:00:00+00:00	92.035	300.0	991.189	True
2024-05-28 03:00:00+00:00	92.035	300.0	992.711	True
2024-05-28 06:00:00+00:00	92.035	300.0	991.695	True

Remal_FNL_All_IMD_Wind_Matches.csv

Column block 1 of 2

time_utc	wind_imd	wind_model	wind_bias_kt	wind_centre_model	wind_centre_bias_kt	max_wind_distance_km
2024-05-24 00:00:00+00:00	25	27.038	2.038	7.754	-17.246	149.615
2024-05-24 03:00:00+00:00	25	34.923	9.923	13.693	-11.307	89.165
2024-05-24 06:00:00+00:00	25	37.592	12.592	1.131	-23.869	127.21
2024-05-24 12:00:00+00:00	25	35.911	10.911	10.485	-14.515	146.483
2024-05-24 18:00:00+00:00	25	45.595	20.595	7.4	-17.6	116.62
2024-05-25 00:00:00+00:00	30	44.169	14.169	1.746	-28.254	139.093
2024-05-25 03:00:00+00:00	30	54.231	24.231	28.519	-1.481	39.586
2024-05-25 06:00:00+00:00	30	56.344	26.344	34.842	4.842	14.136
2024-05-25 12:00:00+00:00	35	56.782	21.782	8.837	-26.163	74.966
2024-05-25 15:00:00+00:00	35	60.698	25.698	9.777	-25.223	126.794
2024-05-25 18:00:00+00:00	40	67.184	27.184	20.205	-19.795	94.754
2024-05-25 21:00:00+00:00	45	86.335	41.335	19.839	-25.161	40.925
2024-05-26 00:00:00+00:00	50	87.732	37.732	10.45	-39.55	110.946
2024-05-26 03:00:00+00:00	50	85.848	35.848	14.312	-35.688	71.378
2024-05-26 06:00:00+00:00	50	83.407	33.407	7.243	-42.757	98.637
2024-05-26 09:00:00+00:00	55	75.134	20.134	8.753	-46.247	91.983
2024-05-26 12:00:00+00:00	60	71.381	11.381	17.551	-42.449	58.178
2024-05-26 15:00:00+00:00	60	64.652	4.652	6.929	-53.071	83.473
2024-05-26 18:00:00+00:00	60	65.896	5.896	4.887	-55.113	102.523
2024-05-26 21:00:00+00:00	55	55.187	0.187	6.914	-48.086	137.336
2024-05-27 00:00:00+00:00	45	50.216	5.216	4.887	-40.113	83.324
2024-05-27 03:00:00+00:00	40	45.286	5.286	7.628	-32.372	68.569
2024-05-27 06:00:00+00:00	35	36.774	1.774	7.493	-27.507	61.054
2024-05-27 09:00:00+00:00	35	36.096	1.096	4.559	-30.441	46.467
2024-05-27 12:00:00+00:00	35	33.182	-1.818	14.636	-20.364	37.624
2024-05-27 15:00:00+00:00	30	28.571	-1.429	19.435	-10.565	111.598
2024-05-27 18:00:00+00:00	30	26.211	-3.789	11.56	-18.44	146.968
2024-05-28 00:00:00+00:00	20	24.774	4.774	1.954	-18.046	38.78
2024-05-28 03:00:00+00:00	20	16.54	-3.46	10.395	-9.605	126.799
2024-05-28 06:00:00+00:00	20	17.518	-2.482	11.038	-8.962	147.856

Column block 2 of 2

time_utc	lat_model_intensity	lon_model_intensity	centre_method	n_centroid_cells
2024-05-24 00:00:00+00:00	15.29	88.799	1.5_hPa_low_SLP_centroid	1864
2024-05-24 03:00:00+00:00	14.956	89.836	1.5_hPa_low_SLP_centroid	523
2024-05-24 06:00:00+00:00	15.377	89.978	1.5_hPa_low_SLP_centroid	583
2024-05-24 12:00:00+00:00	16.07	90.13	1.5_hPa_low_SLP_centroid	259
2024-05-24 18:00:00+00:00	16.271	89.618	1.5_hPa_low_SLP_centroid	182

2024-05-25 00:00:00+00:00	17.226	90.325	1.5_hPa_low_SLP_centroid	441
2024-05-25 03:00:00+00:00	18.106	90.653	1.5_hPa_low_SLP_centroid	9
2024-05-25 06:00:00+00:00	18.59	89.573	1.5_hPa_low_SLP_centroid	4
2024-05-25 12:00:00+00:00	18.177	89.73	1.5_hPa_low_SLP_centroid	63
2024-05-25 15:00:00+00:00	18.408	89.695	1.5_hPa_low_SLP_centroid	103
2024-05-25 18:00:00+00:00	18.854	89.459	1.5_hPa_low_SLP_centroid	148
2024-05-25 21:00:00+00:00	18.79	89.718	1.5_hPa_low_SLP_centroid	18
2024-05-26 00:00:00+00:00	19.639	89.448	1.5_hPa_low_SLP_centroid	26
2024-05-26 03:00:00+00:00	20.104	89.828	1.5_hPa_low_SLP_centroid	30
2024-05-26 06:00:00+00:00	20.522	89.601	1.5_hPa_low_SLP_centroid	54
2024-05-26 09:00:00+00:00	21.174	89.95	1.5_hPa_low_SLP_centroid	44
2024-05-26 12:00:00+00:00	21.715	90.086	1.5_hPa_low_SLP_centroid	59
2024-05-26 15:00:00+00:00	22.176	89.98	1.5_hPa_low_SLP_centroid	49
2024-05-26 18:00:00+00:00	22.518	90.244	1.5_hPa_low_SLP_centroid	47
2024-05-26 21:00:00+00:00	23.108	90.418	1.5_hPa_low_SLP_centroid	41
2024-05-27 00:00:00+00:00	23.528	90.696	1.5_hPa_low_SLP_centroid	46
2024-05-27 03:00:00+00:00	23.877	90.686	1.5_hPa_low_SLP_centroid	43
2024-05-27 06:00:00+00:00	24.197	91.109	1.5_hPa_low_SLP_centroid	46
2024-05-27 09:00:00+00:00	24.57	91.423	1.5_hPa_low_SLP_centroid	43
2024-05-27 12:00:00+00:00	24.844	92.035	1.5_hPa_low_SLP_centroid	29
2024-05-27 15:00:00+00:00	24.921	92.529	1.5_hPa_low_SLP_centroid	160
2024-05-27 18:00:00+00:00	25.156	93.164	1.5_hPa_low_SLP_centroid	63
2024-05-28 00:00:00+00:00	24.826	94.385	1.5_hPa_low_SLP_centroid	60
2024-05-28 03:00:00+00:00	24.394	90.264	1.5_hPa_low_SLP_centroid	416
2024-05-28 06:00:00+00:00	24.57	90.102	1.5_hPa_low_SLP_centroid	344

Remal_GFS_All_IMD_Intensity_Extraction.csv

Column block 1 of 3

time_utc	latitude	longitude	central_pressure_hpa	wind_speed_ms	wind_speed_kt	max_wind_ms
2024-05-24 00:00:00+00:00	14.312	89.261	994.595	9.692	18.84	18.421
2024-05-24 03:00:00+00:00	15.094	89.525	996.603	3.444	6.694	21.686
2024-05-24 06:00:00+00:00	15.463	89.933	994.715	3.309	6.433	19.552
2024-05-24 12:00:00+00:00	16.009	90.053	992.595	4.972	9.664	19.14
2024-05-24 18:00:00+00:00	16.429	90.328	989.55	3.697	7.187	24.428
2024-05-25 00:00:00+00:00	17.376	90.075	986.911	1.499	2.915	21.836
2024-05-25 03:00:00+00:00	17.113	89.684	985.81	4.32	8.398	23.564
2024-05-25 06:00:00+00:00	17.548	90.124	982.054	8.799	17.104	26.673
2024-05-25 12:00:00+00:00	18.289	89.835	979.108	1.591	3.093	27.192
2024-05-25 15:00:00+00:00	18.848	89.489	976.777	6.09	11.839	26.734
2024-05-25 18:00:00+00:00	18.774	89.278	974.391	1.397	2.716	37.135
2024-05-25 21:00:00+00:00	19.174	89.513	970.307	5.171	10.053	40.801
2024-05-26 00:00:00+00:00	19.678	89.549	966.838	3.089	6.005	46.671
2024-05-26 03:00:00+00:00	19.988	89.625	966.186	1.626	3.161	46.512
2024-05-26 06:00:00+00:00	20.485	89.69	968.131	2.886	5.61	39.614
2024-05-26 09:00:00+00:00	21.092	89.788	966.17	6.371	12.385	42.132
2024-05-26 12:00:00+00:00	21.798	89.834	967.224	2.155	4.189	36.143
2024-05-26 15:00:00+00:00	22.274	89.745	970.569	3.351	6.514	31.818
2024-05-26 18:00:00+00:00	22.571	89.97	973.094	4.004	7.783	33.688
2024-05-26 21:00:00+00:00	23.111	90.094	975.299	2.227	4.33	28.004
2024-05-27 00:00:00+00:00	23.512	90.203	977.045	3.965	7.707	24.695
2024-05-27 03:00:00+00:00	23.811	90.393	980.22	2.576	5.007	22.02
2024-05-27 06:00:00+00:00	24.144	90.549	982.225	5.198	10.103	20.281
2024-05-27 09:00:00+00:00	24.367	91.004	983.106	5.129	9.97	21.087
2024-05-27 12:00:00+00:00	24.711	91.656	984.181	6.512	12.658	18.103
2024-05-27 15:00:00+00:00	24.929	92.343	986.522	8.405	16.338	17.157
2024-05-27 18:00:00+00:00	24.86	92.584	991.335	7.554	14.684	17.144

2024-05-28 00:00:00+00:00	24.56	90.886	994.677	1.659	3.226	7.691
2024-05-28 03:00:00+00:00	24.249	89.965	996.135	4.681	9.099	8.882
2024-05-28 06:00:00+00:00	24.449	89.34	995.502	5.564	10.815	8.214

Column block 2 of 3

time_utc	max_wind_kt	max_wind_distance_km	centre_method	n_centroid_cells	source_file	guide_latitude
2024-05-24 00:00:00+00:00	35.808	75.88	1.5_hPa_low_SLP_centroid	44	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-24_00_00_00	14.312
2024-05-24 03:00:00+00:00	42.155	147.759	1.5_hPa_low_SLP_centroid	520	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-24_03_00_00	14.888
2024-05-24 06:00:00+00:00	38.007	127.097	1.5_hPa_low_SLP_centroid	558	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-24_06_00_00	15.463
2024-05-24 12:00:00+00:00	37.204	145.714	1.5_hPa_low_SLP_centroid	472	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-24_12_00_00	16.009
2024-05-24 18:00:00+00:00	47.484	148.829	1.5_hPa_low_SLP_centroid	7	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-24_18_00_00	16.429
2024-05-25 00:00:00+00:00	42.445	125.906	1.5_hPa_low_SLP_centroid	61	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-25_00_00_00	17.376
2024-05-25 03:00:00+00:00	45.805	32.096	1.5_hPa_low_SLP_centroid	38	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-25_03_00_00	17.462
2024-05-25 06:00:00+00:00	51.848	78.083	1.5_hPa_low_SLP_centroid	28	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-25_06_00_00	17.548
2024-05-25 12:00:00+00:00	52.856	74.402	1.5_hPa_low_SLP_centroid	74	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-25_12_00_00	18.289
2024-05-25 15:00:00+00:00	51.967	142.279	1.5_hPa_low_SLP_centroid	42	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-25_15_00_00	18.531
2024-05-25 18:00:00+00:00	72.186	86.174	1.5_hPa_low_SLP_centroid	59	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-25_18_00_00	18.774
2024-05-25 21:00:00+00:00	79.312	75.304	1.5_hPa_low_SLP_centroid	49	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-25_21_00_00	19.226
2024-05-26 00:00:00+00:00	90.721	86.054	1.5_hPa_low_SLP_centroid	24	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-26_00_00_00	19.678
2024-05-26 03:00:00+00:00	90.411	85.368	1.5_hPa_low_SLP_centroid	49	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-26_03_00_00	20.082
2024-05-26 06:00:00+00:00	77.003	71.132	1.5_hPa_low_SLP_centroid	61	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-26_06_00_00	20.485
2024-05-26 09:00:00+00:00	81.898	44.556	1.5_hPa_low_SLP_centroid	34	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-26_09_00_00	21.142
2024-05-26 12:00:00+00:00	70.257	67.504	1.5_hPa_low_SLP_centroid	38	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-26_12_00_00	21.798
2024-05-26 15:00:00+00:00	61.85	79.194	1.5_hPa_low_SLP_centroid	37	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-26_15_00_00	22.185
2024-05-26 18:00:00+00:00	65.484	90.747	1.5_hPa_low_SLP_centroid	41	/home/ushmo/Desktop/remaL_p	22.571

					ortfolio/gfs/wrfout_d02_2024-05-26_18_00_00	
2024-05-26 21:00:00+00:00	54.436	145.776	1.5_hPa_low_SLP_centroid	48	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-26_21_00_00	23.042
2024-05-27 00:00:00+00:00	48.002	119.288	1.5_hPa_low_SLP_centroid	51	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-27_00_00_00	23.512
2024-05-27 03:00:00+00:00	42.804	94.047	1.5_hPa_low_SLP_centroid	54	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-27_03_00_00	23.828
2024-05-27 06:00:00+00:00	39.422	80.564	1.5_hPa_low_SLP_centroid	59	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-27_06_00_00	24.144
2024-05-27 09:00:00+00:00	40.99	50.698	1.5_hPa_low_SLP_centroid	45	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-27_09_00_00	24.427
2024-05-27 12:00:00+00:00	35.19	67.42	1.5_hPa_low_SLP_centroid	34	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-27_12_00_00	24.711
2024-05-27 15:00:00+00:00	33.35	28.638	1.5_hPa_low_SLP_centroid	20	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-27_15_00_00	24.711
2024-05-27 18:00:00+00:00	33.325	148.433	1.5_hPa_low_SLP_centroid	154	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-27_18_00_00	24.711
2024-05-28 00:00:00+00:00	14.95	147.05	1.5_hPa_low_SLP_centroid	269	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-28_00_00_00	24.711
2024-05-28 03:00:00+00:00	17.264	137.295	1.5_hPa_low_SLP_centroid	708	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-28_03_00_00	24.711
2024-05-28 06:00:00+00:00	15.968	44.148	1.5_hPa_low_SLP_centroid	381	/home/ushmo/Desktop/remaL_portfolio/gfs/wrfout_d02_2024-05-28_06_00_00	24.711

Column block 3 of 3

time_utc	guide_longitude	search_radius_km	domain_min_slp_hpa	domain_min_outside_search
2024-05-24 00:00:00+00:00	89.261	300.0	994.595	False
2024-05-24 03:00:00+00:00	89.597	300.0	996.603	False
2024-05-24 06:00:00+00:00	89.933	300.0	994.715	False
2024-05-24 12:00:00+00:00	90.053	300.0	992.595	False
2024-05-24 18:00:00+00:00	90.328	300.0	989.55	False
2024-05-25 00:00:00+00:00	90.075	300.0	986.911	False
2024-05-25 03:00:00+00:00	90.099	300.0	985.81	False
2024-05-25 06:00:00+00:00	90.124	300.0	982.054	False
2024-05-25 12:00:00+00:00	89.835	300.0	979.108	False
2024-05-25 15:00:00+00:00	89.556	300.0	976.777	False
2024-05-25 18:00:00+00:00	89.278	300.0	974.391	False
2024-05-25 21:00:00+00:00	89.413	300.0	970.307	False
2024-05-26 00:00:00+00:00	89.549	300.0	966.838	False
2024-05-26 03:00:00+00:00	89.619	300.0	966.186	False
2024-05-26 06:00:00+00:00	89.69	300.0	968.131	False
2024-05-26 09:00:00+00:00	89.762	300.0	966.17	False
2024-05-26 12:00:00+00:00	89.834	300.0	967.224	False
2024-05-26 15:00:00+00:00	89.902	300.0	970.569	False
2024-05-26 18:00:00+00:00	89.97	300.0	973.094	False
2024-05-26 21:00:00+00:00	90.086	300.0	975.299	False
2024-05-27 00:00:00+00:00	90.203	300.0	977.045	False
2024-05-27 03:00:00+00:00	90.376	300.0	980.22	False

2024-05-27 06:00:00+00:00	90.549	300.0	982.225	False
2024-05-27 09:00:00+00:00	91.103	300.0	983.106	False
2024-05-27 12:00:00+00:00	91.656	300.0	984.181	False
2024-05-27 15:00:00+00:00	91.656	300.0	986.522	False
2024-05-27 18:00:00+00:00	91.656	300.0	990.562	True
2024-05-28 00:00:00+00:00	91.656	300.0	990.793	True
2024-05-28 03:00:00+00:00	91.656	300.0	992.558	True
2024-05-28 06:00:00+00:00	91.656	300.0	991.955	True

Remal_GFS_All_IMD_Wind_Matches.csv

Column block 1 of 2

time_utc	wind_imd	wind_model	wind_bias_kt	wind_centre_model	wind_centre_bias_kt	max_wind_distance_km
2024-05-24 00:00:00+00:00	25	35.808	10.808	18.84	-6.16	75.88
2024-05-24 03:00:00+00:00	25	42.155	17.155	6.694	-18.306	147.759
2024-05-24 06:00:00+00:00	25	38.007	13.007	6.433	-18.567	127.097
2024-05-24 12:00:00+00:00	25	37.204	12.204	9.664	-15.336	145.714
2024-05-24 18:00:00+00:00	25	47.484	22.484	7.187	-17.813	148.829
2024-05-25 00:00:00+00:00	30	42.445	12.445	2.915	-27.085	125.906
2024-05-25 03:00:00+00:00	30	45.805	15.805	8.398	-21.602	32.096
2024-05-25 06:00:00+00:00	30	51.848	21.848	17.104	-12.896	78.083
2024-05-25 12:00:00+00:00	35	52.856	17.856	3.093	-31.907	74.402
2024-05-25 15:00:00+00:00	35	51.967	16.967	11.839	-23.161	142.279
2024-05-25 18:00:00+00:00	40	72.186	32.186	2.716	-37.284	86.174
2024-05-25 21:00:00+00:00	45	79.312	34.312	10.053	-34.947	75.304
2024-05-26 00:00:00+00:00	50	90.721	40.721	6.005	-43.995	86.054
2024-05-26 03:00:00+00:00	50	90.411	40.411	3.161	-46.839	85.368
2024-05-26 06:00:00+00:00	50	77.003	27.003	5.61	-44.39	71.132
2024-05-26 09:00:00+00:00	55	81.898	26.898	12.385	-42.615	44.556
2024-05-26 12:00:00+00:00	60	70.257	10.257	4.189	-55.811	67.504
2024-05-26 15:00:00+00:00	60	61.85	1.85	6.514	-53.486	79.194
2024-05-26 18:00:00+00:00	60	65.484	5.484	7.783	-52.217	90.747
2024-05-26 21:00:00+00:00	55	54.436	-0.564	4.33	-50.67	145.776
2024-05-27 00:00:00+00:00	45	48.002	3.002	7.707	-37.293	119.288
2024-05-27 03:00:00+00:00	40	42.804	2.804	5.007	-34.993	94.047
2024-05-27 06:00:00+00:00	35	39.422	4.422	10.103	-24.897	80.564
2024-05-27 09:00:00+00:00	35	40.99	5.99	9.97	-25.03	50.698
2024-05-27 12:00:00+00:00	35	35.19	0.19	12.658	-22.342	67.42
2024-05-27 15:00:00+00:00	30	33.35	3.35	16.338	-13.662	28.638
2024-05-27 18:00:00+00:00	30	33.325	3.325	14.684	-15.316	148.433
2024-05-28 00:00:00+00:00	20	14.95	-5.05	3.226	-16.774	147.05
2024-05-28 03:00:00+00:00	20	17.264	-2.736	9.099	-10.901	137.295
2024-05-28 06:00:00+00:00	20	15.968	-4.032	10.815	-9.185	44.148

Column block 2 of 2

time_utc	lat_model_intensity	lon_model_intensity	centre_method	n_centroid_cells
2024-05-24 00:00:00+00:00	14.312	89.261	1.5_hPa_low_SLP_centroid	44
2024-05-24 03:00:00+00:00	15.094	89.525	1.5_hPa_low_SLP_centroid	520
2024-05-24 06:00:00+00:00	15.463	89.933	1.5_hPa_low_SLP_centroid	558
2024-05-24 12:00:00+00:00	16.009	90.053	1.5_hPa_low_SLP_centroid	472
2024-05-24 18:00:00+00:00	16.429	90.328	1.5_hPa_low_SLP_centroid	7
2024-05-25 00:00:00+00:00	17.376	90.075	1.5_hPa_low_SLP_centroid	61
2024-05-25 03:00:00+00:00	17.113	89.684	1.5_hPa_low_SLP_centroid	38
2024-05-25 06:00:00+00:00	17.548	90.124	1.5_hPa_low_SLP_centroid	28

2024-05-25 12:00:00+00:00	18.289	89.835	1.5_hPa_low_SLP_centroid	74
2024-05-25 15:00:00+00:00	18.848	89.489	1.5_hPa_low_SLP_centroid	42
2024-05-25 18:00:00+00:00	18.774	89.278	1.5_hPa_low_SLP_centroid	59
2024-05-25 21:00:00+00:00	19.174	89.513	1.5_hPa_low_SLP_centroid	49
2024-05-26 00:00:00+00:00	19.678	89.549	1.5_hPa_low_SLP_centroid	24
2024-05-26 03:00:00+00:00	19.988	89.625	1.5_hPa_low_SLP_centroid	49
2024-05-26 06:00:00+00:00	20.485	89.69	1.5_hPa_low_SLP_centroid	61
2024-05-26 09:00:00+00:00	21.092	89.788	1.5_hPa_low_SLP_centroid	34
2024-05-26 12:00:00+00:00	21.798	89.834	1.5_hPa_low_SLP_centroid	38
2024-05-26 15:00:00+00:00	22.274	89.745	1.5_hPa_low_SLP_centroid	37
2024-05-26 18:00:00+00:00	22.571	89.97	1.5_hPa_low_SLP_centroid	41
2024-05-26 21:00:00+00:00	23.111	90.094	1.5_hPa_low_SLP_centroid	48
2024-05-27 00:00:00+00:00	23.512	90.203	1.5_hPa_low_SLP_centroid	51
2024-05-27 03:00:00+00:00	23.811	90.393	1.5_hPa_low_SLP_centroid	54
2024-05-27 06:00:00+00:00	24.144	90.549	1.5_hPa_low_SLP_centroid	59
2024-05-27 09:00:00+00:00	24.367	91.004	1.5_hPa_low_SLP_centroid	45
2024-05-27 12:00:00+00:00	24.711	91.656	1.5_hPa_low_SLP_centroid	34
2024-05-27 15:00:00+00:00	24.929	92.343	1.5_hPa_low_SLP_centroid	20
2024-05-27 18:00:00+00:00	24.86	92.584	1.5_hPa_low_SLP_centroid	154
2024-05-28 00:00:00+00:00	24.56	90.886	1.5_hPa_low_SLP_centroid	269
2024-05-28 03:00:00+00:00	24.249	89.965	1.5_hPa_low_SLP_centroid	708
2024-05-28 06:00:00+00:00	24.449	89.34	1.5_hPa_low_SLP_centroid	381

S04_A05_Wind_Definition_Comparison_All_IMD_Matches.csv

model	centre_bias_kt	centre_rmse_kt	centre_corr	radius_max_bias_kt	radius_max_rmse_kt	radius_max_corr
FNL	-26.172	30.09	-0.037	13.04	18.623	0.789
GFS	-28.849	32.277	-0.301	13.013	18.224	0.798
CFSv2	-29.291	32.536	-0.286	4.735	17.414	0.133
ERA5	-28.14	31.525	-0.185	6.196	11.393	0.83

T03_A05_Wind_Stats_All_IMD_Matches.csv

Column block 1 of 2

model	n	mean_bias_kt	mae_kt	rmse_kt	correlation	time_weighted_bias_kt
FNL	30	13.04	13.905	18.623	0.789	13.555
GFS	30	13.013	13.839	18.224	0.798	13.336
CFSv2	30	4.735	13.735	17.414	0.133	6.635
ERA5	30	6.196	9.684	11.393	0.83	7.193

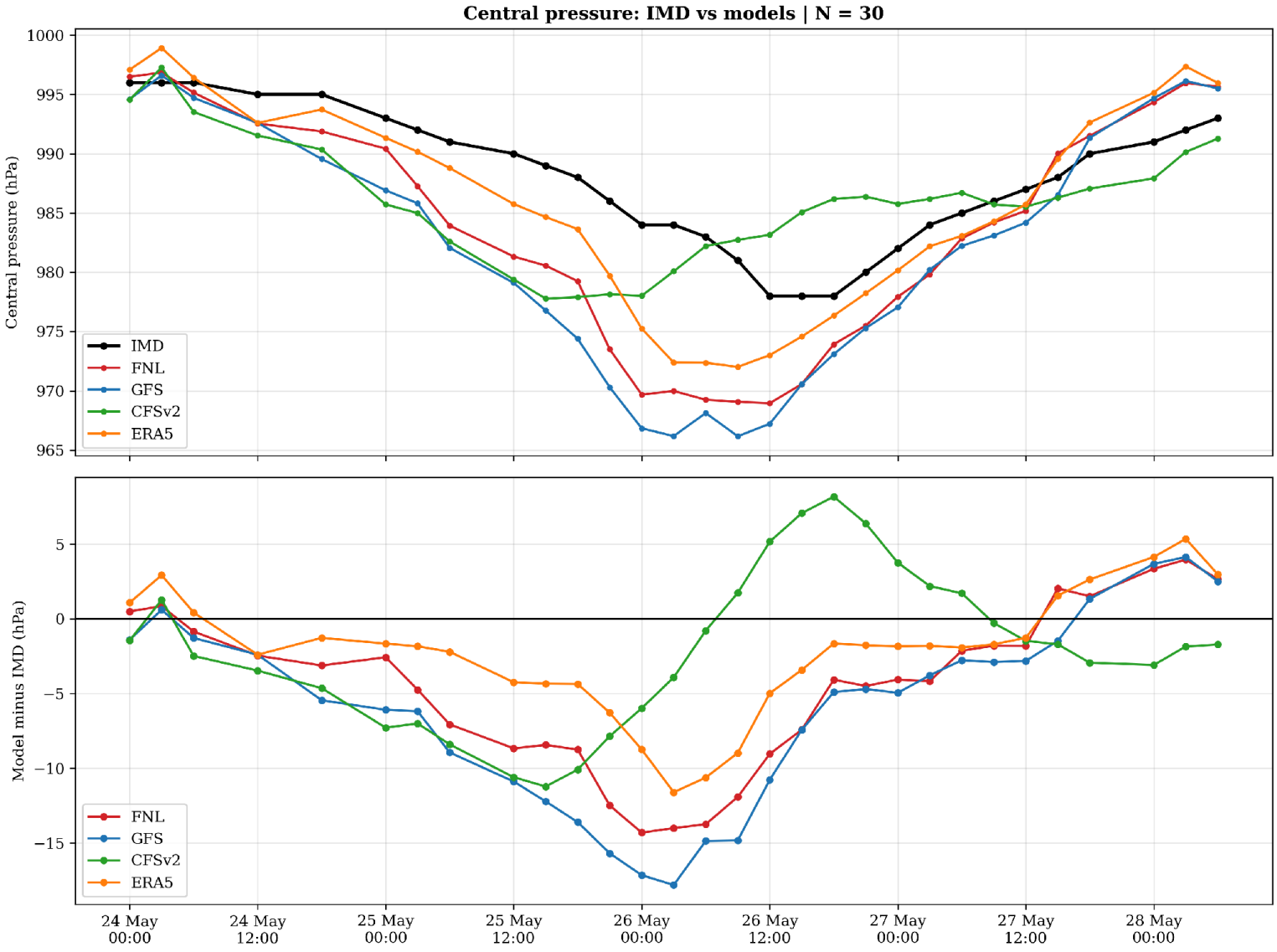
Column block 2 of 2

model	time_weighted_mae_kt	time_weighted_rmse_kt	time_weighted_correlation	time_weighted_duration_h
FNL	14.357	18.571	0.778	102.0
GFS	14.094	18.093	0.79	102.0
CFSv2	14.524	17.968	0.091	102.0
ERA5	10.261	11.86	0.803	102.0

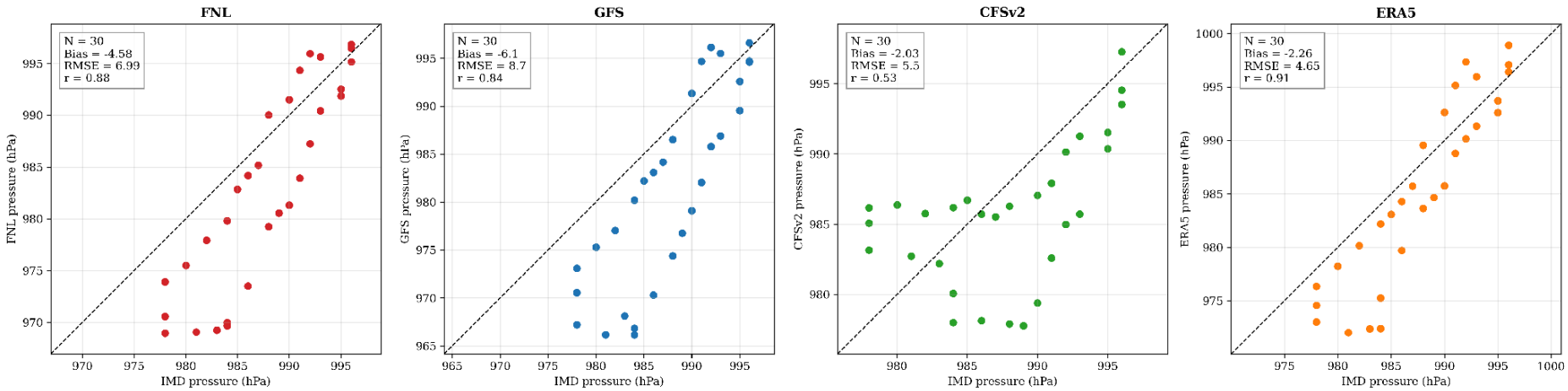
A06 - Central-pressure validation

Figures: 2 | CSV tables: 5 | NetCDF extracts: 0

F04_Pressure_Series_and_Bias.png



S03_Pressure_Scatter.png



Remal_CFSv2_All_IMD_Pressure_Matches.csv

Column block 1 of 2

time_utc	pres_imd	pres_model	pres_bias_hpa	lat_model_intensity	lon_model_intensity	centre_method
2024-05-24 00:00:00+00:00	996	994.543	-1.457	15.34	87.526	1.5_hPa_low_SLP_centroid
2024-05-24 03:00:00+00:00	996	997.265	1.265	15.39	87.516	1.5_hPa_low_SLP_centroid
2024-05-24 06:00:00+00:00	996	993.517	-2.483	14.652	87.715	1.5_hPa_low_SLP_centroid
2024-05-24 12:00:00+00:00	995	991.532	-3.468	16.156	87.448	1.5_hPa_low_SLP_centroid
2024-05-24 18:00:00+00:00	995	990.357	-4.643	16.983	87.904	1.5_hPa_low_SLP_centroid
2024-05-25 00:00:00+00:00	993	985.717	-7.283	17.443	87.535	1.5_hPa_low_SLP_centroid
2024-05-25 03:00:00+00:00	992	984.994	-7.006	17.841	87.929	1.5_hPa_low_SLP_centroid
2024-05-25 06:00:00+00:00	991	982.594	-8.406	17.813	87.904	1.5_hPa_low_SLP_centroid
2024-05-25 12:00:00+00:00	990	979.397	-10.603	19.227	87.703	1.5_hPa_low_SLP_centroid
2024-05-25 15:00:00+00:00	989	977.775	-11.225	19.671	87.201	1.5_hPa_low_SLP_centroid
2024-05-25 18:00:00+00:00	988	977.909	-10.091	19.97	86.767	1.5_hPa_low_SLP_centroid
2024-05-25 21:00:00+00:00	986	978.141	-7.859	20.193	86.529	1.5_hPa_low_SLP_centroid
2024-05-26 00:00:00+00:00	984	978.015	-5.985	20.445	86.314	1.5_hPa_low_SLP_centroid
2024-05-26 03:00:00+00:00	984	980.081	-3.919	20.629	86.088	1.5_hPa_low_SLP_centroid
2024-05-26 06:00:00+00:00	983	982.21	-0.79	20.796	86.065	1.5_hPa_low_SLP_centroid
2024-05-26 09:00:00+00:00	981	982.732	1.732	21.053	85.974	1.5_hPa_low_SLP_centroid
2024-05-26 12:00:00+00:00	978	983.162	5.162	21.493	85.758	1.5_hPa_low_SLP_centroid
2024-05-26 15:00:00+00:00	978	985.073	7.073	21.951	85.603	1.5_hPa_low_SLP_centroid
2024-05-26 18:00:00+00:00	978	986.179	8.179	22.043	85.716	1.5_hPa_low_SLP_centroid
2024-05-26 21:00:00+00:00	980	986.381	6.381	22.365	86.001	1.5_hPa_low_SLP_centroid
2024-05-27 00:00:00+00:00	982	985.759	3.759	22.765	86.122	1.5_hPa_low_SLP_centroid
2024-05-27 03:00:00+00:00	984	986.189	2.189	23.372	86.327	1.5_hPa_low_SLP_centroid
2024-05-27 06:00:00+00:00	985	986.709	1.709	23.663	86.288	1.5_hPa_low_SLP_centroid
2024-05-27 09:00:00+00:00	986	985.717	-0.283	23.819	86.447	1.5_hPa_low_SLP_centroid
2024-05-27 12:00:00+00:00	987	985.529	-1.471	24.252	86.774	1.5_hPa_low_SLP_centroid
2024-05-27 15:00:00+00:00	988	986.288	-1.712	24.639	86.91	1.5_hPa_low_SLP_centroid
2024-05-27 18:00:00+00:00	990	987.062	-2.938	24.597	87.108	1.5_hPa_low_SLP_centroid
2024-05-28 00:00:00+00:00	991	987.913	-3.087	24.713	87.847	1.5_hPa_low_SLP_centroid
2024-05-28 03:00:00+00:00	992	990.147	-1.853	24.829	88.337	1.5_hPa_low_SLP_centroid
2024-05-28 06:00:00+00:00	993	991.277	-1.723	24.969	89.611	1.5_hPa_low_SLP_centroid

Column block 2 of 2

time_utc	n_centroid_cells
2024-05-24 00:00:00+00:00	1506
2024-05-24 03:00:00+00:00	1340
2024-05-24 06:00:00+00:00	119
2024-05-24 12:00:00+00:00	734
2024-05-24 18:00:00+00:00	382
2024-05-25 00:00:00+00:00	63
2024-05-25 03:00:00+00:00	223
2024-05-25 06:00:00+00:00	23
2024-05-25 12:00:00+00:00	38
2024-05-25 15:00:00+00:00	72
2024-05-25 18:00:00+00:00	77
2024-05-25 21:00:00+00:00	79
2024-05-26 00:00:00+00:00	67
2024-05-26 03:00:00+00:00	82
2024-05-26 06:00:00+00:00	94
2024-05-26 09:00:00+00:00	78
2024-05-26 12:00:00+00:00	85
2024-05-26 15:00:00+00:00	114
2024-05-26 18:00:00+00:00	165
2024-05-26 21:00:00+00:00	176
2024-05-27 00:00:00+00:00	167
2024-05-27 03:00:00+00:00	172
2024-05-27 06:00:00+00:00	201
2024-05-27 09:00:00+00:00	181
2024-05-27 12:00:00+00:00	215
2024-05-27 15:00:00+00:00	205
2024-05-27 18:00:00+00:00	173
2024-05-28 00:00:00+00:00	191
2024-05-28 03:00:00+00:00	209
2024-05-28 06:00:00+00:00	51

Remal_ERA5_All_IMD_Pressure_Matches.csv

Column block 1 of 2

time_utc	pres_imd	pres_model	pres_bias_hpa	lat_model_intensity	lon_model_intensity	centre_method
2024-05-24 00:00:00+00:00	996	997.081	1.081	15.29	88.392	1.5_hPa_low_SLP_centroid
2024-05-24 03:00:00+00:00	996	998.919	2.919	15.831	88.395	1.5_hPa_low_SLP_centroid
2024-05-24 06:00:00+00:00	996	996.421	0.421	15.51	88.967	1.5_hPa_low_SLP_centroid
2024-05-24 12:00:00+00:00	995	992.596	-2.404	15.271	89.208	1.5_hPa_low_SLP_centroid
2024-05-24 18:00:00+00:00	995	993.724	-1.276	16.768	90.135	1.5_hPa_low_SLP_centroid
2024-05-25 00:00:00+00:00	993	991.339	-1.661	17.455	90.567	1.5_hPa_low_SLP_centroid
2024-05-25 03:00:00+00:00	992	990.159	-1.841	17.964	90.388	1.5_hPa_low_SLP_centroid
2024-05-25 06:00:00+00:00	991	988.791	-2.209	18.407	90.086	1.5_hPa_low_SLP_centroid
2024-05-25 12:00:00+00:00	990	985.755	-4.245	19.076	89.63	1.5_hPa_low_SLP_centroid
2024-05-25 15:00:00+00:00	989	984.658	-4.342	18.989	89.631	1.5_hPa_low_SLP_centroid
2024-05-25 18:00:00+00:00	988	983.638	-4.362	19.381	89.882	1.5_hPa_low_SLP_centroid
2024-05-25 21:00:00+00:00	986	979.713	-6.287	19.737	89.348	1.5_hPa_low_SLP_centroid
2024-05-26 00:00:00+00:00	984	975.25	-8.75	20.15	90.059	1.5_hPa_low_SLP_centroid
2024-05-26 03:00:00+00:00	984	972.388	-11.612	20.918	89.622	1.5_hPa_low_SLP_centroid
2024-05-26 06:00:00+00:00	983	972.373	-10.627	21.282	89.651	1.5_hPa_low_SLP_centroid
2024-05-26 09:00:00+00:00	981	972.022	-8.978	21.569	89.538	1.5_hPa_low_SLP_centroid

2024-05-26 12:00:00+00:00	978	973.012	-4.988	22.113	89.846	1.5_hPa_low_SLP_centroid
2024-05-26 15:00:00+00:00	978	974.579	-3.421	22.62	89.719	1.5_hPa_low_SLP_centroid
2024-05-26 18:00:00+00:00	978	976.344	-1.656	23.008	89.875	1.5_hPa_low_SLP_centroid
2024-05-26 21:00:00+00:00	980	978.227	-1.773	23.374	89.933	1.5_hPa_low_SLP_centroid
2024-05-27 00:00:00+00:00	982	980.161	-1.839	23.774	90.061	1.5_hPa_low_SLP_centroid
2024-05-27 03:00:00+00:00	984	982.191	-1.809	23.919	90.088	1.5_hPa_low_SLP_centroid
2024-05-27 06:00:00+00:00	985	983.073	-1.927	24.173	90.498	1.5_hPa_low_SLP_centroid
2024-05-27 09:00:00+00:00	986	984.295	-1.705	24.479	90.868	1.5_hPa_low_SLP_centroid
2024-05-27 12:00:00+00:00	987	985.724	-1.276	24.743	91.489	1.5_hPa_low_SLP_centroid
2024-05-27 15:00:00+00:00	988	989.553	1.553	25.082	92.277	1.5_hPa_low_SLP_centroid
2024-05-27 18:00:00+00:00	990	992.635	2.635	24.978	92.267	1.5_hPa_low_SLP_centroid
2024-05-28 00:00:00+00:00	991	995.14	4.14	24.313	94.265	1.5_hPa_low_SLP_centroid
2024-05-28 03:00:00+00:00	992	997.353	5.353	24.467	89.575	1.5_hPa_low_SLP_centroid
2024-05-28 06:00:00+00:00	993	995.971	2.971	24.46	89.102	1.5_hPa_low_SLP_centroid

Column block 2 of 2

time_utc	n_centroid_cells
2024-05-24 00:00:00+00:00	2479
2024-05-24 03:00:00+00:00	1999
2024-05-24 06:00:00+00:00	1661
2024-05-24 12:00:00+00:00	45
2024-05-24 18:00:00+00:00	173
2024-05-25 00:00:00+00:00	82
2024-05-25 03:00:00+00:00	82
2024-05-25 06:00:00+00:00	97
2024-05-25 12:00:00+00:00	52
2024-05-25 15:00:00+00:00	81
2024-05-25 18:00:00+00:00	90
2024-05-25 21:00:00+00:00	16
2024-05-26 00:00:00+00:00	45
2024-05-26 03:00:00+00:00	29
2024-05-26 06:00:00+00:00	41
2024-05-26 09:00:00+00:00	40
2024-05-26 12:00:00+00:00	48
2024-05-26 15:00:00+00:00	42
2024-05-26 18:00:00+00:00	37
2024-05-26 21:00:00+00:00	47
2024-05-27 00:00:00+00:00	49
2024-05-27 03:00:00+00:00	55
2024-05-27 06:00:00+00:00	51
2024-05-27 09:00:00+00:00	54
2024-05-27 12:00:00+00:00	42
2024-05-27 15:00:00+00:00	73
2024-05-27 18:00:00+00:00	230
2024-05-28 00:00:00+00:00	44
2024-05-28 03:00:00+00:00	407
2024-05-28 06:00:00+00:00	316

Remal_FNL_All_IMD_Pressure_Matches.csv

Column block 1 of 2

time_utc	pres_imd	pres_model	pres_bias_hpa	lat_model_intensity	lon_model_intensity	centre_method
2024-05-24 00:00:00+00:00	996	996.495	0.495	15.29	88.799	1.5_hPa_low_SLP_centroid
2024-05-24 03:00:00+00:00	996	996.843	0.843	14.956	89.836	1.5_hPa_low_SLP_centroid

2024-05-24 06:00:00+00:00	996	995.149	-0.851	15.377	89.978	1.5_hPa_low_SLP_centroid
2024-05-24 12:00:00+00:00	995	992.54	-2.46	16.07	90.13	1.5_hPa_low_SLP_centroid
2024-05-24 18:00:00+00:00	995	991.876	-3.124	16.271	89.618	1.5_hPa_low_SLP_centroid
2024-05-25 00:00:00+00:00	993	990.428	-2.572	17.226	90.325	1.5_hPa_low_SLP_centroid
2024-05-25 03:00:00+00:00	992	987.252	-4.748	18.106	90.653	1.5_hPa_low_SLP_centroid
2024-05-25 06:00:00+00:00	991	983.934	-7.066	18.59	89.573	1.5_hPa_low_SLP_centroid
2024-05-25 12:00:00+00:00	990	981.328	-8.672	18.177	89.73	1.5_hPa_low_SLP_centroid
2024-05-25 15:00:00+00:00	989	980.561	-8.439	18.408	89.695	1.5_hPa_low_SLP_centroid
2024-05-25 18:00:00+00:00	988	979.248	-8.752	18.854	89.459	1.5_hPa_low_SLP_centroid
2024-05-25 21:00:00+00:00	986	973.506	-12.494	18.79	89.718	1.5_hPa_low_SLP_centroid
2024-05-26 00:00:00+00:00	984	969.687	-14.313	19.639	89.448	1.5_hPa_low_SLP_centroid
2024-05-26 03:00:00+00:00	984	969.99	-14.01	20.104	89.828	1.5_hPa_low_SLP_centroid
2024-05-26 06:00:00+00:00	983	969.25	-13.75	20.522	89.601	1.5_hPa_low_SLP_centroid
2024-05-26 09:00:00+00:00	981	969.078	-11.922	21.174	89.95	1.5_hPa_low_SLP_centroid
2024-05-26 12:00:00+00:00	978	968.954	-9.046	21.715	90.086	1.5_hPa_low_SLP_centroid
2024-05-26 15:00:00+00:00	978	970.574	-7.426	22.176	89.98	1.5_hPa_low_SLP_centroid
2024-05-26 18:00:00+00:00	978	973.926	-4.074	22.518	90.244	1.5_hPa_low_SLP_centroid
2024-05-26 21:00:00+00:00	980	975.505	-4.495	23.108	90.418	1.5_hPa_low_SLP_centroid
2024-05-27 00:00:00+00:00	982	977.931	-4.069	23.528	90.696	1.5_hPa_low_SLP_centroid
2024-05-27 03:00:00+00:00	984	979.826	-4.174	23.877	90.686	1.5_hPa_low_SLP_centroid
2024-05-27 06:00:00+00:00	985	982.86	-2.14	24.197	91.109	1.5_hPa_low_SLP_centroid
2024-05-27 09:00:00+00:00	986	984.196	-1.804	24.57	91.423	1.5_hPa_low_SLP_centroid
2024-05-27 12:00:00+00:00	987	985.189	-1.811	24.844	92.035	1.5_hPa_low_SLP_centroid
2024-05-27 15:00:00+00:00	988	990.033	2.033	24.921	92.529	1.5_hPa_low_SLP_centroid
2024-05-27 18:00:00+00:00	990	991.512	1.512	25.156	93.164	1.5_hPa_low_SLP_centroid
2024-05-28 00:00:00+00:00	991	994.349	3.349	24.826	94.385	1.5_hPa_low_SLP_centroid
2024-05-28 03:00:00+00:00	992	995.956	3.956	24.394	90.264	1.5_hPa_low_SLP_centroid
2024-05-28 06:00:00+00:00	993	995.656	2.656	24.57	90.102	1.5_hPa_low_SLP_centroid

Column block 2 of 2

time_utc	n_centroid_cells
2024-05-24 00:00:00+00:00	1864
2024-05-24 03:00:00+00:00	523
2024-05-24 06:00:00+00:00	583
2024-05-24 12:00:00+00:00	259
2024-05-24 18:00:00+00:00	182
2024-05-25 00:00:00+00:00	441
2024-05-25 03:00:00+00:00	9
2024-05-25 06:00:00+00:00	4
2024-05-25 12:00:00+00:00	63
2024-05-25 15:00:00+00:00	103
2024-05-25 18:00:00+00:00	148
2024-05-25 21:00:00+00:00	18
2024-05-26 00:00:00+00:00	26
2024-05-26 03:00:00+00:00	30
2024-05-26 06:00:00+00:00	54
2024-05-26 09:00:00+00:00	44
2024-05-26 12:00:00+00:00	59
2024-05-26 15:00:00+00:00	49
2024-05-26 18:00:00+00:00	47
2024-05-26 21:00:00+00:00	41
2024-05-27 00:00:00+00:00	46
2024-05-27 03:00:00+00:00	43
2024-05-27 06:00:00+00:00	46
2024-05-27 09:00:00+00:00	43
2024-05-27 12:00:00+00:00	29
2024-05-27 15:00:00+00:00	160
2024-05-27 18:00:00+00:00	63

2024-05-28 00:00:00+00:00	60
2024-05-28 03:00:00+00:00	416
2024-05-28 06:00:00+00:00	344

Remal_GFS_All_IMD_Pressure_Matches.csv

Column block 1 of 2

time_utc	pres_imd	pres_model	pres_bias_hpa	lat_model_intensity	lon_model_intensity	centre_method
2024-05-24 00:00:00+00:00	996	994.595	-1.405	14.312	89.261	1.5_hPa_low_SLP_centroid
2024-05-24 03:00:00+00:00	996	996.603	0.603	15.094	89.525	1.5_hPa_low_SLP_centroid
2024-05-24 06:00:00+00:00	996	994.715	-1.285	15.463	89.933	1.5_hPa_low_SLP_centroid
2024-05-24 12:00:00+00:00	995	992.595	-2.405	16.009	90.053	1.5_hPa_low_SLP_centroid
2024-05-24 18:00:00+00:00	995	989.55	-5.45	16.429	90.328	1.5_hPa_low_SLP_centroid
2024-05-25 00:00:00+00:00	993	986.911	-6.089	17.376	90.075	1.5_hPa_low_SLP_centroid
2024-05-25 03:00:00+00:00	992	985.81	-6.19	17.113	89.684	1.5_hPa_low_SLP_centroid
2024-05-25 06:00:00+00:00	991	982.054	-8.946	17.548	90.124	1.5_hPa_low_SLP_centroid
2024-05-25 12:00:00+00:00	990	979.108	-10.892	18.289	89.835	1.5_hPa_low_SLP_centroid
2024-05-25 15:00:00+00:00	989	976.777	-12.223	18.848	89.489	1.5_hPa_low_SLP_centroid
2024-05-25 18:00:00+00:00	988	974.391	-13.609	18.774	89.278	1.5_hPa_low_SLP_centroid
2024-05-25 21:00:00+00:00	986	970.307	-15.693	19.174	89.513	1.5_hPa_low_SLP_centroid
2024-05-26 00:00:00+00:00	984	966.838	-17.162	19.678	89.549	1.5_hPa_low_SLP_centroid
2024-05-26 03:00:00+00:00	984	966.186	-17.814	19.988	89.625	1.5_hPa_low_SLP_centroid
2024-05-26 06:00:00+00:00	983	968.131	-14.869	20.485	89.69	1.5_hPa_low_SLP_centroid
2024-05-26 09:00:00+00:00	981	966.17	-14.83	21.092	89.788	1.5_hPa_low_SLP_centroid
2024-05-26 12:00:00+00:00	978	967.224	-10.776	21.798	89.834	1.5_hPa_low_SLP_centroid
2024-05-26 15:00:00+00:00	978	970.569	-7.431	22.274	89.745	1.5_hPa_low_SLP_centroid
2024-05-26 18:00:00+00:00	978	973.094	-4.906	22.571	89.97	1.5_hPa_low_SLP_centroid
2024-05-26 21:00:00+00:00	980	975.299	-4.701	23.111	90.094	1.5_hPa_low_SLP_centroid
2024-05-27 00:00:00+00:00	982	977.045	-4.955	23.512	90.203	1.5_hPa_low_SLP_centroid
2024-05-27 03:00:00+00:00	984	980.22	-3.78	23.811	90.393	1.5_hPa_low_SLP_centroid
2024-05-27 06:00:00+00:00	985	982.225	-2.775	24.144	90.549	1.5_hPa_low_SLP_centroid
2024-05-27 09:00:00+00:00	986	983.106	-2.894	24.367	91.004	1.5_hPa_low_SLP_centroid
2024-05-27 12:00:00+00:00	987	984.181	-2.819	24.711	91.656	1.5_hPa_low_SLP_centroid
2024-05-27 15:00:00+00:00	988	986.522	-1.478	24.929	92.343	1.5_hPa_low_SLP_centroid
2024-05-27 18:00:00+00:00	990	991.335	1.335	24.86	92.584	1.5_hPa_low_SLP_centroid
2024-05-28 00:00:00+00:00	991	994.677	3.677	24.56	90.886	1.5_hPa_low_SLP_centroid
2024-05-28 03:00:00+00:00	992	996.135	4.135	24.249	89.965	1.5_hPa_low_SLP_centroid
2024-05-28 06:00:00+00:00	993	995.502	2.502	24.449	89.34	1.5_hPa_low_SLP_centroid

Column block 2 of 2

time_utc	n_centroid_cells
2024-05-24 00:00:00+00:00	44
2024-05-24 03:00:00+00:00	520
2024-05-24 06:00:00+00:00	558
2024-05-24 12:00:00+00:00	472
2024-05-24 18:00:00+00:00	7
2024-05-25 00:00:00+00:00	61
2024-05-25 03:00:00+00:00	38
2024-05-25 06:00:00+00:00	28
2024-05-25 12:00:00+00:00	74
2024-05-25 15:00:00+00:00	42
2024-05-25 18:00:00+00:00	59
2024-05-25 21:00:00+00:00	49
2024-05-26 00:00:00+00:00	24

2024-05-26 03:00:00+00:00	49
2024-05-26 06:00:00+00:00	61
2024-05-26 09:00:00+00:00	34
2024-05-26 12:00:00+00:00	38
2024-05-26 15:00:00+00:00	37
2024-05-26 18:00:00+00:00	41
2024-05-26 21:00:00+00:00	48
2024-05-27 00:00:00+00:00	51
2024-05-27 03:00:00+00:00	54
2024-05-27 06:00:00+00:00	59
2024-05-27 09:00:00+00:00	45
2024-05-27 12:00:00+00:00	34
2024-05-27 15:00:00+00:00	20
2024-05-27 18:00:00+00:00	154
2024-05-28 00:00:00+00:00	269
2024-05-28 03:00:00+00:00	708
2024-05-28 06:00:00+00:00	381

T03_A06_Pressure_Stats_All_IMD_Matches.csv

Column block 1 of 2

model	n	mean_bias_hpa	mae_hpa	rmse_hpa	correlation	time_weighted_bias_hpa
FNL	30	-4.579	5.569	6.993	0.883	-4.461
GFS	30	-6.104	6.921	8.696	0.839	-5.96
CFSv2	30	-2.028	4.525	5.504	0.534	-2.493
ERA5	30	-2.264	3.669	4.655	0.914	-2.179

Column block 2 of 2

model	time_weighted_mae_hpa	time_weighted_rmse_hpa	time_weighted_correlation	time_weighted_duration_h
FNL	5.385	6.757	0.881	102.0
GFS	6.754	8.448	0.837	102.0
CFSv2	4.696	5.611	0.539	102.0
ERA5	3.511	4.463	0.911	102.0

A07 - Peak intensity and intensity-change rates

Figures: 0 | CSV tables: 8 | NetCDF extracts: 0

A07_exact_24h_changes.csv

model	start_time_utc	end_time_utc	model_wind_change_24h_kt	imd_wind_change_24h_kt	model_pressure_change_24h_hpa	imd_pressure_change_24h_hpa
FNL	2024-05-24 00:00:00+00:00	2024-05-25 00:00:00+00:00	17.131	5	-6.067	-3
FNL	2024-05-24 03:00:00+00:00	2024-05-25 03:00:00+00:00	19.308	5	-9.592	-4
FNL	2024-05-24 06:00:00+00:00	2024-05-25 06:00:00+00:00	18.752	5	-11.216	-5
FNL	2024-05-24 12:00:00+00:00	2024-05-25 12:00:00+00:00	20.871	10	-11.212	-5
FNL	2024-05-24 18:00:00+00:00	2024-05-25 18:00:00+00:00	21.589	15	-12.628	-7
FNL	2024-05-25 00:00:00+00:00	2024-05-26 00:00:00+00:00	43.563	20	-20.742	-9
FNL	2024-05-25 03:00:00+00:00	2024-05-26 03:00:00+00:00	31.617	20	-17.262	-8
FNL	2024-05-25 06:00:00+00:00	2024-05-26 06:00:00+00:00	27.063	20	-14.684	-8
FNL	2024-05-25 12:00:00+00:00	2024-05-26 12:00:00+00:00	14.599	25	-12.374	-12
FNL	2024-05-25 15:00:00+00:00	2024-05-26 15:00:00+00:00	3.954	25	-9.987	-11
FNL	2024-05-25 18:00:00+00:00	2024-05-26 18:00:00+00:00	-1.288	20	-5.322	-10
FNL	2024-05-25 21:00:00+00:00	2024-05-26 21:00:00+00:00	-31.149	10	1.998	-6
FNL	2024-05-26 00:00:00+00:00	2024-05-27 00:00:00+00:00	-37.516	-5	8.244	-2
FNL	2024-05-26 03:00:00+00:00	2024-05-27 03:00:00+00:00	-40.561	-10	9.835	0
FNL	2024-05-26 06:00:00+00:00	2024-05-27 06:00:00+00:00	-46.633	-15	13.609	2
FNL	2024-05-26 09:00:00+00:00	2024-05-27 09:00:00+00:00	-39.038	-20	15.118	5
FNL	2024-05-26 12:00:00+00:00	2024-05-27 12:00:00+00:00	-38.198	-25	16.234	9
FNL	2024-05-26 15:00:00+00:00	2024-05-27 15:00:00+00:00	-36.082	-30	19.46	10
FNL	2024-05-26 18:00:00+00:00	2024-05-27 18:00:00+00:00	-39.686	-30	17.586	12
FNL	2024-05-27 00:00:00+00:00	2024-05-28 00:00:00+00:00	-25.441	-25	16.418	9
FNL	2024-05-27 03:00:00+00:00	2024-05-28 03:00:00+00:00	-28.746	-20	16.131	8
FNL	2024-05-27 06:00:00+00:00	2024-05-28 06:00:00+00:00	-19.257	-15	12.796	8
GFS	2024-05-24 00:00:00+00:00	2024-05-25 00:00:00+00:00	6.637	5	-7.683	-3
GFS	2024-05-24 03:00:00+00:00	2024-05-25 03:00:00+00:00	3.65	5	-10.793	-4
GFS	2024-05-24 06:00:00+00:00	2024-05-25 06:00:00+00:00	13.842	5	-12.662	-5
GFS	2024-05-24 12:00:00+00:00	2024-05-25 12:00:00+00:00	15.652	10	-13.487	-5
GFS	2024-05-24 18:00:00+00:00	2024-05-25 18:00:00+00:00	24.701	15	-15.158	-7
GFS	2024-05-25 00:00:00+00:00	2024-05-26 00:00:00+00:00	48.276	20	-20.073	-9
GFS	2024-05-25 03:00:00+00:00	2024-05-26 03:00:00+00:00	44.607	20	-19.624	-8
GFS	2024-05-25 06:00:00+00:00	2024-05-26 06:00:00+00:00	25.154	20	-13.923	-8
GFS	2024-05-25 12:00:00+00:00	2024-05-26 12:00:00+00:00	17.4	25	-11.884	-12
GFS	2024-05-25 15:00:00+00:00	2024-05-26 15:00:00+00:00	9.883	25	-6.208	-11
GFS	2024-05-25 18:00:00+00:00	2024-05-26 18:00:00+00:00	-6.701	20	-1.298	-10
GFS	2024-05-25 21:00:00+00:00	2024-05-26 21:00:00+00:00	-24.875	10	4.992	-6
GFS	2024-05-26 00:00:00+00:00	2024-05-27 00:00:00+00:00	-42.719	-5	10.207	-2
GFS	2024-05-26 03:00:00+00:00	2024-05-27 03:00:00+00:00	-47.607	-10	14.034	0
GFS	2024-05-26 06:00:00+00:00	2024-05-27 06:00:00+00:00	-37.581	-15	14.095	2
GFS	2024-05-26 09:00:00+00:00	2024-05-27 09:00:00+00:00	-40.909	-20	16.935	5
GFS	2024-05-26 12:00:00+00:00	2024-05-27 12:00:00+00:00	-35.066	-25	16.958	9
GFS	2024-05-26 15:00:00+00:00	2024-05-27 15:00:00+00:00	-28.5	-30	15.953	10
GFS	2024-05-26 18:00:00+00:00	2024-05-27 18:00:00+00:00	-32.159	-30	18.241	12
GFS	2024-05-27 00:00:00+00:00	2024-05-28 00:00:00+00:00	-33.052	-25	17.632	9
GFS	2024-05-27 03:00:00+00:00	2024-05-28 03:00:00+00:00	-25.539	-20	15.915	8
GFS	2024-05-27 06:00:00+00:00	2024-05-28 06:00:00+00:00	-23.455	-15	13.276	8
CFSv2	2024-05-24 00:00:00+00:00	2024-05-25 00:00:00+00:00	27.996	5	-8.826	-3
CFSv2	2024-05-24 03:00:00+00:00	2024-05-25 03:00:00+00:00	29.537	5	-12.271	-4
CFSv2	2024-05-24 06:00:00+00:00	2024-05-25 06:00:00+00:00	24.004	5	-10.923	-5
CFSv2	2024-05-24 12:00:00+00:00	2024-05-25 12:00:00+00:00	18.802	10	-12.135	-5
CFSv2	2024-05-24 18:00:00+00:00	2024-05-25 18:00:00+00:00	11.662	15	-12.448	-7
CFSv2	2024-05-25 00:00:00+00:00	2024-05-26 00:00:00+00:00	6.224	20	-7.702	-9
CFSv2	2024-05-25 03:00:00+00:00	2024-05-26 03:00:00+00:00	3.073	20	-4.913	-8
CFSv2	2024-05-25 06:00:00+00:00	2024-05-26 06:00:00+00:00	-14.658	20	-0.384	-8

CFSv2	2024-05-25 12:00:00+00:00	2024-05-26 12:00:00+00:00	-23.818	25	3.765	-12
CFSv2	2024-05-25 15:00:00+00:00	2024-05-26 15:00:00+00:00	-37.658	25	7.298	-11
CFSv2	2024-05-25 18:00:00+00:00	2024-05-26 18:00:00+00:00	-28.551	20	8.27	-10
CFSv2	2024-05-25 21:00:00+00:00	2024-05-26 21:00:00+00:00	-30.056	10	8.24	-6
CFSv2	2024-05-26 00:00:00+00:00	2024-05-27 00:00:00+00:00	-23.569	-5	7.744	-2
CFSv2	2024-05-26 03:00:00+00:00	2024-05-27 03:00:00+00:00	-15.831	-10	6.108	0
CFSv2	2024-05-26 06:00:00+00:00	2024-05-27 06:00:00+00:00	-15.856	-15	4.499	2
CFSv2	2024-05-26 09:00:00+00:00	2024-05-27 09:00:00+00:00	-4.719	-20	2.986	5
CFSv2	2024-05-26 12:00:00+00:00	2024-05-27 12:00:00+00:00	-1.112	-25	2.367	9
CFSv2	2024-05-26 15:00:00+00:00	2024-05-27 15:00:00+00:00	1.056	-30	1.215	10
CFSv2	2024-05-26 18:00:00+00:00	2024-05-27 18:00:00+00:00	0.349	-30	0.883	12
CFSv2	2024-05-27 00:00:00+00:00	2024-05-28 00:00:00+00:00	-5.447	-25	2.155	9
CFSv2	2024-05-27 03:00:00+00:00	2024-05-28 03:00:00+00:00	-13.884	-20	3.958	8
CFSv2	2024-05-27 06:00:00+00:00	2024-05-28 06:00:00+00:00	-3.085	-15	4.569	8
ERA5	2024-05-24 00:00:00+00:00	2024-05-25 00:00:00+00:00	23.433	5	-5.742	-3
ERA5	2024-05-24 03:00:00+00:00	2024-05-25 03:00:00+00:00	25.262	5	-8.759	-4
ERA5	2024-05-24 06:00:00+00:00	2024-05-25 06:00:00+00:00	16.76	5	-7.63	-5
ERA5	2024-05-24 12:00:00+00:00	2024-05-25 12:00:00+00:00	0.949	10	-6.842	-5
ERA5	2024-05-24 18:00:00+00:00	2024-05-25 18:00:00+00:00	12.896	15	-10.086	-7
ERA5	2024-05-25 00:00:00+00:00	2024-05-26 00:00:00+00:00	27.375	20	-16.089	-9
ERA5	2024-05-25 03:00:00+00:00	2024-05-26 03:00:00+00:00	28.809	20	-17.771	-8
ERA5	2024-05-25 06:00:00+00:00	2024-05-26 06:00:00+00:00	21.421	20	-16.418	-8
ERA5	2024-05-25 12:00:00+00:00	2024-05-26 12:00:00+00:00	24.39	25	-12.742	-12
ERA5	2024-05-25 15:00:00+00:00	2024-05-26 15:00:00+00:00	18.657	25	-10.079	-11
ERA5	2024-05-25 18:00:00+00:00	2024-05-26 18:00:00+00:00	0.905	20	-7.295	-10
ERA5	2024-05-25 21:00:00+00:00	2024-05-26 21:00:00+00:00	-19.87	10	-1.486	-6
ERA5	2024-05-26 00:00:00+00:00	2024-05-27 00:00:00+00:00	-17.731	-5	4.911	-2
ERA5	2024-05-26 03:00:00+00:00	2024-05-27 03:00:00+00:00	-30.78	-10	9.803	0
ERA5	2024-05-26 06:00:00+00:00	2024-05-27 06:00:00+00:00	-32.311	-15	10.7	2
ERA5	2024-05-26 09:00:00+00:00	2024-05-27 09:00:00+00:00	-25.779	-20	12.273	5
ERA5	2024-05-26 12:00:00+00:00	2024-05-27 12:00:00+00:00	-32.888	-25	12.712	9
ERA5	2024-05-26 15:00:00+00:00	2024-05-27 15:00:00+00:00	-35.851	-30	14.973	10
ERA5	2024-05-26 18:00:00+00:00	2024-05-27 18:00:00+00:00	-31.187	-30	16.291	12
ERA5	2024-05-27 00:00:00+00:00	2024-05-28 00:00:00+00:00	-28.356	-25	14.979	9
ERA5	2024-05-27 03:00:00+00:00	2024-05-28 03:00:00+00:00	-27.302	-20	15.162	8
ERA5	2024-05-27 06:00:00+00:00	2024-05-28 06:00:00+00:00	-20.351	-15	12.898	8

A07_intensity_change_rates.csv

Column block 1 of 2

model	start_time_utc	end_time_utc	elapsed_hours	model_dwind_dt_kt_per_h	imd_dwind_dt_kt_per_h	model_dpressure_dt_hpa_per_h
FNL	2024-05-24 00:00:00+00:00	2024-05-24 03:00:00+00:00	3.0	2.628	0.0	0.116
FNL	2024-05-24 03:00:00+00:00	2024-05-24 06:00:00+00:00	3.0	0.89	0.0	-0.565
FNL	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	6.0	-0.28	0.0	-0.435
FNL	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	6.0	1.614	0.0	-0.111
FNL	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	6.0	-0.238	0.833	-0.241
FNL	2024-05-25 00:00:00+00:00	2024-05-25 03:00:00+00:00	3.0	3.354	0.0	-1.059
FNL	2024-05-25 03:00:00+00:00	2024-05-25 06:00:00+00:00	3.0	0.704	0.0	-1.106
FNL	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	6.0	0.073	0.833	-0.434
FNL	2024-05-25 12:00:00+00:00	2024-05-25 15:00:00+00:00	3.0	1.305	0.0	-0.256
FNL	2024-05-25 15:00:00+00:00	2024-05-25 18:00:00+00:00	3.0	2.162	1.667	-0.437
FNL	2024-05-25 18:00:00+00:00	2024-05-25 21:00:00+00:00	3.0	6.384	1.667	-1.914
FNL	2024-05-25 21:00:00+00:00	2024-05-26 00:00:00+00:00	3.0	0.466	1.667	-1.273
FNL	2024-05-26 00:00:00+00:00	2024-05-26 03:00:00+00:00	3.0	-0.628	0.0	0.101
FNL	2024-05-26 03:00:00+00:00	2024-05-26 06:00:00+00:00	3.0	-0.814	0.0	-0.247
FNL	2024-05-26 06:00:00+00:00	2024-05-26 09:00:00+00:00	3.0	-2.758	1.667	-0.057

FNL	2024-05-26 09:00:00+00:00	2024-05-26 12:00:00+00:00	3.0	-1.251	1.667	-0.041
FNL	2024-05-26 12:00:00+00:00	2024-05-26 15:00:00+00:00	3.0	-2.243	0.0	0.54
FNL	2024-05-26 15:00:00+00:00	2024-05-26 18:00:00+00:00	3.0	0.415	0.0	1.118
FNL	2024-05-26 18:00:00+00:00	2024-05-26 21:00:00+00:00	3.0	-3.57	-1.667	0.526
FNL	2024-05-26 21:00:00+00:00	2024-05-27 00:00:00+00:00	3.0	-1.657	-3.333	0.809
FNL	2024-05-27 00:00:00+00:00	2024-05-27 03:00:00+00:00	3.0	-1.643	-1.667	0.632
FNL	2024-05-27 03:00:00+00:00	2024-05-27 06:00:00+00:00	3.0	-2.837	-1.667	1.011
FNL	2024-05-27 06:00:00+00:00	2024-05-27 09:00:00+00:00	3.0	-0.226	0.0	0.445
FNL	2024-05-27 09:00:00+00:00	2024-05-27 12:00:00+00:00	3.0	-0.971	0.0	0.331
FNL	2024-05-27 12:00:00+00:00	2024-05-27 15:00:00+00:00	3.0	-1.537	-1.667	1.615
FNL	2024-05-27 15:00:00+00:00	2024-05-27 18:00:00+00:00	3.0	-0.787	0.0	0.493
FNL	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	6.0	-0.239	-1.667	0.473
FNL	2024-05-28 00:00:00+00:00	2024-05-28 03:00:00+00:00	3.0	-2.745	0.0	0.536
FNL	2024-05-28 03:00:00+00:00	2024-05-28 06:00:00+00:00	3.0	0.326	0.0	-0.1
GFS	2024-05-24 00:00:00+00:00	2024-05-24 03:00:00+00:00	3.0	2.116	0.0	0.669
GFS	2024-05-24 03:00:00+00:00	2024-05-24 06:00:00+00:00	3.0	-1.383	0.0	-0.629
GFS	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	6.0	-0.134	0.0	-0.353
GFS	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	6.0	1.713	0.0	-0.508
GFS	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	6.0	-0.84	0.833	-0.44
GFS	2024-05-25 00:00:00+00:00	2024-05-25 03:00:00+00:00	3.0	1.12	0.0	-0.367
GFS	2024-05-25 03:00:00+00:00	2024-05-25 06:00:00+00:00	3.0	2.015	0.0	-1.252
GFS	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	6.0	0.168	0.833	-0.491
GFS	2024-05-25 12:00:00+00:00	2024-05-25 15:00:00+00:00	3.0	-0.296	0.0	-0.777
GFS	2024-05-25 15:00:00+00:00	2024-05-25 18:00:00+00:00	3.0	6.739	1.667	-0.795
GFS	2024-05-25 18:00:00+00:00	2024-05-25 21:00:00+00:00	3.0	2.375	1.667	-1.362
GFS	2024-05-25 21:00:00+00:00	2024-05-26 00:00:00+00:00	3.0	3.803	1.667	-1.156
GFS	2024-05-26 00:00:00+00:00	2024-05-26 03:00:00+00:00	3.0	-0.103	0.0	-0.217
GFS	2024-05-26 03:00:00+00:00	2024-05-26 06:00:00+00:00	3.0	-4.469	0.0	0.648
GFS	2024-05-26 06:00:00+00:00	2024-05-26 09:00:00+00:00	3.0	1.632	1.667	-0.653
GFS	2024-05-26 09:00:00+00:00	2024-05-26 12:00:00+00:00	3.0	-3.881	1.667	0.351
GFS	2024-05-26 12:00:00+00:00	2024-05-26 15:00:00+00:00	3.0	-2.802	0.0	1.115
GFS	2024-05-26 15:00:00+00:00	2024-05-26 18:00:00+00:00	3.0	1.212	0.0	0.842
GFS	2024-05-26 18:00:00+00:00	2024-05-26 21:00:00+00:00	3.0	-3.683	-1.667	0.735
GFS	2024-05-26 21:00:00+00:00	2024-05-27 00:00:00+00:00	3.0	-2.145	-3.333	0.582
GFS	2024-05-27 00:00:00+00:00	2024-05-27 03:00:00+00:00	3.0	-1.733	-1.667	1.058
GFS	2024-05-27 03:00:00+00:00	2024-05-27 06:00:00+00:00	3.0	-1.127	-1.667	0.669
GFS	2024-05-27 06:00:00+00:00	2024-05-27 09:00:00+00:00	3.0	0.522	0.0	0.293
GFS	2024-05-27 09:00:00+00:00	2024-05-27 12:00:00+00:00	3.0	-1.933	0.0	0.359
GFS	2024-05-27 12:00:00+00:00	2024-05-27 15:00:00+00:00	3.0	-0.613	-1.667	0.78
GFS	2024-05-27 15:00:00+00:00	2024-05-27 18:00:00+00:00	3.0	-0.008	0.0	1.604
GFS	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	6.0	-3.062	-1.667	0.557
GFS	2024-05-28 00:00:00+00:00	2024-05-28 03:00:00+00:00	3.0	0.771	0.0	0.486
GFS	2024-05-28 03:00:00+00:00	2024-05-28 06:00:00+00:00	3.0	-0.432	0.0	-0.211
CFSv2	2024-05-24 00:00:00+00:00	2024-05-24 03:00:00+00:00	3.0	0.668	0.0	0.908
CFSv2	2024-05-24 03:00:00+00:00	2024-05-24 06:00:00+00:00	3.0	5.161	0.0	-1.25
CFSv2	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	6.0	-0.308	0.0	-0.331
CFSv2	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	6.0	1.427	0.0	-0.196
CFSv2	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	6.0	0.633	0.833	-0.773
CFSv2	2024-05-25 00:00:00+00:00	2024-05-25 03:00:00+00:00	3.0	1.182	0.0	-0.241
CFSv2	2024-05-25 03:00:00+00:00	2024-05-25 06:00:00+00:00	3.0	3.316	0.0	-0.8
CFSv2	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	6.0	-1.175	0.833	-0.533
CFSv2	2024-05-25 12:00:00+00:00	2024-05-25 15:00:00+00:00	3.0	3.506	0.0	-0.541
CFSv2	2024-05-25 15:00:00+00:00	2024-05-25 18:00:00+00:00	3.0	-3.032	1.667	0.045
CFSv2	2024-05-25 18:00:00+00:00	2024-05-25 21:00:00+00:00	3.0	0.713	1.667	0.078
CFSv2	2024-05-25 21:00:00+00:00	2024-05-26 00:00:00+00:00	3.0	-1.26	1.667	-0.042
CFSv2	2024-05-26 00:00:00+00:00	2024-05-26 03:00:00+00:00	3.0	0.132	0.0	0.689
CFSv2	2024-05-26 03:00:00+00:00	2024-05-26 06:00:00+00:00	3.0	-2.594	0.0	0.71
CFSv2	2024-05-26 06:00:00+00:00	2024-05-26 09:00:00+00:00	3.0	-1.793	1.667	0.174
CFSv2	2024-05-26 09:00:00+00:00	2024-05-26 12:00:00+00:00	3.0	-3.611	1.667	0.143
CFSv2	2024-05-26 12:00:00+00:00	2024-05-26 15:00:00+00:00	3.0	-1.108	0.0	0.637
CFSv2	2024-05-26 15:00:00+00:00	2024-05-26 18:00:00+00:00	3.0	0.004	0.0	0.369

CFSv2	2024-05-26 18:00:00+00:00	2024-05-26 21:00:00+00:00	3.0	0.211	-1.667	0.067
CFSv2	2024-05-26 21:00:00+00:00	2024-05-27 00:00:00+00:00	3.0	0.903	-3.333	-0.207
CFSv2	2024-05-27 00:00:00+00:00	2024-05-27 03:00:00+00:00	3.0	2.711	-1.667	0.143
CFSv2	2024-05-27 03:00:00+00:00	2024-05-27 06:00:00+00:00	3.0	-2.602	-1.667	0.173
CFSv2	2024-05-27 06:00:00+00:00	2024-05-27 09:00:00+00:00	3.0	1.919	0.0	-0.33
CFSv2	2024-05-27 09:00:00+00:00	2024-05-27 12:00:00+00:00	3.0	-2.408	0.0	-0.063
CFSv2	2024-05-27 12:00:00+00:00	2024-05-27 15:00:00+00:00	3.0	-0.385	-1.667	0.253
CFSv2	2024-05-27 15:00:00+00:00	2024-05-27 18:00:00+00:00	3.0	-0.232	0.0	0.258
CFSv2	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	6.0	-0.409	-1.667	0.142
CFSv2	2024-05-28 00:00:00+00:00	2024-05-28 03:00:00+00:00	3.0	-0.101	0.0	0.744
CFSv2	2024-05-28 03:00:00+00:00	2024-05-28 06:00:00+00:00	3.0	0.997	0.0	0.377
ERA5	2024-05-24 00:00:00+00:00	2024-05-24 03:00:00+00:00	3.0	-0.625	0.0	0.612
ERA5	2024-05-24 03:00:00+00:00	2024-05-24 06:00:00+00:00	3.0	4.352	0.0	-0.833
ERA5	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	6.0	2.141	0.0	-0.637
ERA5	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	6.0	-0.798	0.0	0.188
ERA5	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	6.0	0.699	0.833	-0.397
ERA5	2024-05-25 00:00:00+00:00	2024-05-25 03:00:00+00:00	3.0	-0.015	0.0	-0.393
ERA5	2024-05-25 03:00:00+00:00	2024-05-25 06:00:00+00:00	3.0	1.518	0.0	-0.456
ERA5	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	6.0	-0.494	0.833	-0.506
ERA5	2024-05-25 12:00:00+00:00	2024-05-25 15:00:00+00:00	3.0	-0.221	0.0	-0.366
ERA5	2024-05-25 15:00:00+00:00	2024-05-25 18:00:00+00:00	3.0	2.607	1.667	-0.34
ERA5	2024-05-25 18:00:00+00:00	2024-05-25 21:00:00+00:00	3.0	3.662	1.667	-1.309
ERA5	2024-05-25 21:00:00+00:00	2024-05-26 00:00:00+00:00	3.0	2.562	1.667	-1.488
ERA5	2024-05-26 00:00:00+00:00	2024-05-26 03:00:00+00:00	3.0	0.463	0.0	-0.954
ERA5	2024-05-26 03:00:00+00:00	2024-05-26 06:00:00+00:00	3.0	-0.945	0.0	-0.005
ERA5	2024-05-26 06:00:00+00:00	2024-05-26 09:00:00+00:00	3.0	-1.125	1.667	-0.117
ERA5	2024-05-26 09:00:00+00:00	2024-05-26 12:00:00+00:00	3.0	1.126	1.667	0.33
ERA5	2024-05-26 12:00:00+00:00	2024-05-26 15:00:00+00:00	3.0	-2.132	0.0	0.522
ERA5	2024-05-26 15:00:00+00:00	2024-05-26 18:00:00+00:00	3.0	-3.31	0.0	0.588
ERA5	2024-05-26 18:00:00+00:00	2024-05-26 21:00:00+00:00	3.0	-3.263	-1.667	0.628
ERA5	2024-05-26 21:00:00+00:00	2024-05-27 00:00:00+00:00	3.0	3.276	-3.333	0.644
ERA5	2024-05-27 00:00:00+00:00	2024-05-27 03:00:00+00:00	3.0	-3.887	-1.667	0.677
ERA5	2024-05-27 03:00:00+00:00	2024-05-27 06:00:00+00:00	3.0	-1.455	-1.667	0.294
ERA5	2024-05-27 06:00:00+00:00	2024-05-27 09:00:00+00:00	3.0	1.052	0.0	0.407
ERA5	2024-05-27 09:00:00+00:00	2024-05-27 12:00:00+00:00	3.0	-1.243	0.0	0.477
ERA5	2024-05-27 12:00:00+00:00	2024-05-27 15:00:00+00:00	3.0	-3.119	-1.667	1.276
ERA5	2024-05-27 15:00:00+00:00	2024-05-27 18:00:00+00:00	3.0	-1.755	0.0	1.027
ERA5	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	6.0	0.478	-1.667	0.418
ERA5	2024-05-28 00:00:00+00:00	2024-05-28 03:00:00+00:00	3.0	-3.536	0.0	0.738
ERA5	2024-05-28 03:00:00+00:00	2024-05-28 06:00:00+00:00	3.0	0.862	0.0	-0.461

Column block 2 of 2

model	**imd_dpressure_dt_hpa_per_h**
FNL	0.0
FNL	0.0
FNL	-0.167
FNL	0.0
FNL	-0.333
FNL	-0.333
FNL	-0.333
FNL	-0.167
FNL	-0.333
FNL	-0.333
FNL	-0.667
FNL	-0.667
FNL	0.0
FNL	-0.333
FNL	-0.667

FNL	-1.0
FNL	0.0
FNL	0.0
FNL	0.667
FNL	0.667
FNL	0.667
FNL	0.333
FNL	0.333
FNL	0.333
FNL	0.333
FNL	0.667
FNL	0.167
FNL	0.333
FNL	0.333
GFS	0.0
GFS	0.0
GFS	-0.167
GFS	0.0
GFS	-0.333
GFS	-0.333
GFS	-0.333
GFS	-0.167
GFS	-0.333
GFS	-0.333
GFS	-0.667
GFS	-0.667
GFS	0.0
GFS	-0.333
GFS	-0.667
GFS	-1.0
GFS	0.0
GFS	0.0
GFS	0.667
GFS	0.667
GFS	0.667
GFS	0.333
GFS	0.333
GFS	0.333
GFS	0.333
GFS	0.667
GFS	0.167
GFS	0.333
GFS	0.333
CFSv2	0.0
CFSv2	0.0
CFSv2	-0.167
CFSv2	0.0
CFSv2	-0.333
CFSv2	-0.333
CFSv2	-0.333
CFSv2	-0.167
CFSv2	-0.333
CFSv2	-0.333
CFSv2	-0.667
CFSv2	-0.667
CFSv2	0.0
CFSv2	-0.333
CFSv2	-0.667
CFSv2	-1.0
CFSv2	0.0
CFSv2	0.0

CFSv2	0.667
CFSv2	0.667
CFSv2	0.667
CFSv2	0.333
CFSv2	0.333
CFSv2	0.333
CFSv2	0.333
CFSv2	0.667
CFSv2	0.167
CFSv2	0.333
CFSv2	0.333
ERA5	0.0
ERA5	0.0
ERA5	-0.167
ERA5	0.0
ERA5	-0.333
ERA5	-0.333
ERA5	-0.333
ERA5	-0.167
ERA5	-0.333
ERA5	-0.333
ERA5	-0.667
ERA5	-0.667
ERA5	0.0
ERA5	-0.333
ERA5	-0.667
ERA5	-1.0
ERA5	0.0
ERA5	0.0
ERA5	0.667
ERA5	0.667
ERA5	0.667
ERA5	0.333
ERA5	0.333
ERA5	0.333
ERA5	0.333
ERA5	0.667
ERA5	0.167
ERA5	0.333
ERA5	0.333

Remal_CFSv2_All_IMD_Intensity_Matches.csv

Column block 1 of 3

time_utc	wind_imd	pres_imd	lat_model_intensity	lon_model_intensity	wind_model	wind_centre_model
2024-05-24 00:00:00+00:00	25	996	15.34	87.526	23.25	7.162
2024-05-24 03:00:00+00:00	25	996	15.39	87.516	25.254	6.693
2024-05-24 06:00:00+00:00	25	996	14.652	87.715	40.736	18.452
2024-05-24 12:00:00+00:00	25	995	16.156	87.448	38.887	6.666
2024-05-24 18:00:00+00:00	25	995	16.983	87.904	47.449	9.079
2024-05-25 00:00:00+00:00	30	993	17.443	87.535	51.246	4.023
2024-05-25 03:00:00+00:00	30	992	17.841	87.929	54.791	3.824
2024-05-25 06:00:00+00:00	30	991	17.813	87.904	64.74	14.121
2024-05-25 12:00:00+00:00	35	990	19.227	87.703	57.689	13.93
2024-05-25 15:00:00+00:00	35	989	19.671	87.201	68.206	17.427
2024-05-25 18:00:00+00:00	40	988	19.97	86.767	59.111	4.82
2024-05-25 21:00:00+00:00	45	986	20.193	86.529	61.249	4.293

2024-05-26 00:00:00+00:00	50	984	20.445	86.314	57.469	6.378
2024-05-26 03:00:00+00:00	50	984	20.629	86.088	57.865	9.977
2024-05-26 06:00:00+00:00	50	983	20.796	86.065	50.082	6.837
2024-05-26 09:00:00+00:00	55	981	21.053	85.974	44.703	7.883
2024-05-26 12:00:00+00:00	60	978	21.493	85.758	33.871	8.725
2024-05-26 15:00:00+00:00	60	978	21.951	85.603	30.548	2.649
2024-05-26 18:00:00+00:00	60	978	22.043	85.716	30.56	7.263
2024-05-26 21:00:00+00:00	55	980	22.365	86.001	31.192	5.483
2024-05-27 00:00:00+00:00	45	982	22.765	86.122	33.901	5.103
2024-05-27 03:00:00+00:00	40	984	23.372	86.327	42.034	8.699
2024-05-27 06:00:00+00:00	35	985	23.663	86.288	34.226	8.509
2024-05-27 09:00:00+00:00	35	986	23.819	86.447	39.984	8.088
2024-05-27 12:00:00+00:00	35	987	24.252	86.774	32.76	5.065
2024-05-27 15:00:00+00:00	30	988	24.639	86.91	31.605	6.302
2024-05-27 18:00:00+00:00	30	990	24.597	87.108	30.908	3.294
2024-05-28 00:00:00+00:00	20	991	24.713	87.847	28.454	7.075
2024-05-28 03:00:00+00:00	20	992	24.829	88.337	28.15	11.933
2024-05-28 06:00:00+00:00	20	993	24.969	89.611	31.141	11.513

Column block 2 of 3

time_utc	pres_model	max_wind_distance_km	centre_method	n_centroid_cells	wind_bias_kt	wind_centre_bias_kt
2024-05-24 00:00:00+00:00	994.543	148.879	1.5_hPa_low_SLP_centroid	1506	-1.75	-17.838
2024-05-24 03:00:00+00:00	997.265	144.136	1.5_hPa_low_SLP_centroid	1340	0.254	-18.307
2024-05-24 06:00:00+00:00	993.517	106.879	1.5_hPa_low_SLP_centroid	119	15.736	-6.548
2024-05-24 12:00:00+00:00	991.532	140.605	1.5_hPa_low_SLP_centroid	734	13.887	-18.334
2024-05-24 18:00:00+00:00	990.357	148.656	1.5_hPa_low_SLP_centroid	382	22.449	-15.921
2024-05-25 00:00:00+00:00	985.717	91.857	1.5_hPa_low_SLP_centroid	63	21.246	-25.977
2024-05-25 03:00:00+00:00	984.994	111.804	1.5_hPa_low_SLP_centroid	223	24.791	-26.176
2024-05-25 06:00:00+00:00	982.594	45.121	1.5_hPa_low_SLP_centroid	23	34.74	-15.879
2024-05-25 12:00:00+00:00	979.397	109.752	1.5_hPa_low_SLP_centroid	38	22.689	-21.07
2024-05-25 15:00:00+00:00	977.775	144.133	1.5_hPa_low_SLP_centroid	72	33.206	-17.573
2024-05-25 18:00:00+00:00	977.909	100.693	1.5_hPa_low_SLP_centroid	77	19.111	-35.18
2024-05-25 21:00:00+00:00	978.141	96.831	1.5_hPa_low_SLP_centroid	79	16.249	-40.707
2024-05-26 00:00:00+00:00	978.015	110.148	1.5_hPa_low_SLP_centroid	67	7.469	-43.622
2024-05-26 03:00:00+00:00	980.081	141.82	1.5_hPa_low_SLP_centroid	82	7.865	-40.023
2024-05-26 06:00:00+00:00	982.21	125.207	1.5_hPa_low_SLP_centroid	94	0.082	-43.163
2024-05-26 09:00:00+00:00	982.732	121.309	1.5_hPa_low_SLP_centroid	78	-10.297	-47.117
2024-05-26 12:00:00+00:00	983.162	149.032	1.5_hPa_low_SLP_centroid	85	-26.129	-51.275
2024-05-26 15:00:00+00:00	985.073	136.177	1.5_hPa_low_SLP_centroid	114	-29.452	-57.351
2024-05-26 18:00:00+00:00	986.179	114.247	1.5_hPa_low_SLP_centroid	165	-29.44	-52.737
2024-05-26 21:00:00+00:00	986.381	149.661	1.5_hPa_low_SLP_centroid	176	-23.808	-49.517
2024-05-27 00:00:00+00:00	985.759	129.598	1.5_hPa_low_SLP_centroid	167	-11.099	-39.897
2024-05-27 03:00:00+00:00	986.189	132.479	1.5_hPa_low_SLP_centroid	172	2.034	-31.301
2024-05-27 06:00:00+00:00	986.709	122.732	1.5_hPa_low_SLP_centroid	201	-0.774	-26.491
2024-05-27 09:00:00+00:00	985.717	98.083	1.5_hPa_low_SLP_centroid	181	4.984	-26.912
2024-05-27 12:00:00+00:00	985.529	97.479	1.5_hPa_low_SLP_centroid	215	-2.24	-29.935
2024-05-27 15:00:00+00:00	986.288	89.375	1.5_hPa_low_SLP_centroid	205	1.605	-23.698
2024-05-27 18:00:00+00:00	987.062	140.759	1.5_hPa_low_SLP_centroid	173	0.908	-26.706
2024-05-28 00:00:00+00:00	987.913	108.745	1.5_hPa_low_SLP_centroid	191	8.454	-12.925
2024-05-28 03:00:00+00:00	990.147	149.132	1.5_hPa_low_SLP_centroid	209	8.15	-8.067
2024-05-28 06:00:00+00:00	991.277	147.629	1.5_hPa_low_SLP_centroid	51	11.141	-8.487

Column block 3 of 3

time_utc	pres_bias_hpa
2024-05-24 00:00:00+00:00	-1.457

2024-05-24 03:00:00+00:00	1.265
2024-05-24 06:00:00+00:00	-2.483
2024-05-24 12:00:00+00:00	-3.468
2024-05-24 18:00:00+00:00	-4.643
2024-05-25 00:00:00+00:00	-7.283
2024-05-25 03:00:00+00:00	-7.006
2024-05-25 06:00:00+00:00	-8.406
2024-05-25 12:00:00+00:00	-10.603
2024-05-25 15:00:00+00:00	-11.225
2024-05-25 18:00:00+00:00	-10.091
2024-05-25 21:00:00+00:00	-7.859
2024-05-26 00:00:00+00:00	-5.985
2024-05-26 03:00:00+00:00	-3.919
2024-05-26 06:00:00+00:00	-0.79
2024-05-26 09:00:00+00:00	1.732
2024-05-26 12:00:00+00:00	5.162
2024-05-26 15:00:00+00:00	7.073
2024-05-26 18:00:00+00:00	8.179
2024-05-26 21:00:00+00:00	6.381
2024-05-27 00:00:00+00:00	3.759
2024-05-27 03:00:00+00:00	2.189
2024-05-27 06:00:00+00:00	1.709
2024-05-27 09:00:00+00:00	-0.283
2024-05-27 12:00:00+00:00	-1.471
2024-05-27 15:00:00+00:00	-1.712
2024-05-27 18:00:00+00:00	-2.938
2024-05-28 00:00:00+00:00	-3.087
2024-05-28 03:00:00+00:00	-1.853
2024-05-28 06:00:00+00:00	-1.723

Remal_ERA5_All_IMD_Intensity_Matches.csv

Column block 1 of 3

time_utc	wind_imd	pres_imd	lat_model_intensity	lon_model_intensity	wind_model	wind_centre_model
2024-05-24 00:00:00+00:00	25	996	15.29	88.392	19.81	6.473
2024-05-24 03:00:00+00:00	25	996	15.831	88.395	17.935	6.828
2024-05-24 06:00:00+00:00	25	996	15.51	88.967	30.99	9.911
2024-05-24 12:00:00+00:00	25	995	15.271	89.208	43.835	11.541
2024-05-24 18:00:00+00:00	25	995	16.768	90.135	39.048	10.362
2024-05-25 00:00:00+00:00	30	993	17.455	90.567	43.243	17.427
2024-05-25 03:00:00+00:00	30	992	17.964	90.388	43.197	5.0
2024-05-25 06:00:00+00:00	30	991	18.407	90.086	47.751	12.142
2024-05-25 12:00:00+00:00	35	990	19.076	89.63	44.784	9.418
2024-05-25 15:00:00+00:00	35	989	18.989	89.631	44.123	10.292
2024-05-25 18:00:00+00:00	40	988	19.381	89.882	51.944	17.011
2024-05-25 21:00:00+00:00	45	986	19.737	89.348	62.931	17.25
2024-05-26 00:00:00+00:00	50	984	20.15	90.059	70.618	8.197
2024-05-26 03:00:00+00:00	50	984	20.918	89.622	72.006	12.213
2024-05-26 06:00:00+00:00	50	983	21.282	89.651	69.171	8.396
2024-05-26 09:00:00+00:00	55	981	21.569	89.538	65.795	14.482
2024-05-26 12:00:00+00:00	60	978	22.113	89.846	69.174	1.686
2024-05-26 15:00:00+00:00	60	978	22.62	89.719	62.78	7.922
2024-05-26 18:00:00+00:00	60	978	23.008	89.875	52.848	5.1
2024-05-26 21:00:00+00:00	55	980	23.374	89.933	43.06	3.011
2024-05-27 00:00:00+00:00	45	982	23.774	90.061	52.887	2.518
2024-05-27 03:00:00+00:00	40	984	23.919	90.088	41.226	1.876
2024-05-27 06:00:00+00:00	35	985	24.173	90.498	36.86	7.026

2024-05-27 09:00:00+00:00	35	986	24.479	90.868	40.016	8.922
2024-05-27 12:00:00+00:00	35	987	24.743	91.489	36.286	6.778
2024-05-27 15:00:00+00:00	30	988	25.082	92.277	26.928	19.945
2024-05-27 18:00:00+00:00	30	990	24.978	92.267	21.662	10.886
2024-05-28 00:00:00+00:00	20	991	24.313	94.265	24.532	2.714
2024-05-28 03:00:00+00:00	20	992	24.467	89.575	13.924	9.297
2024-05-28 06:00:00+00:00	20	993	24.46	89.102	16.51	11.183

Column block 2 of 3

time_utc	pres_model	max_wind_distance_km	centre_method	n_centroid_cells	wind_bias_kt	wind_centre_bias_kt
2024-05-24 00:00:00+00:00	997.081	147.327	1.5_hPa_low_SLP_centroid	2479	-5.19	-18.527
2024-05-24 03:00:00+00:00	998.919	146.966	1.5_hPa_low_SLP_centroid	1999	-7.065	-18.172
2024-05-24 06:00:00+00:00	996.421	118.474	1.5_hPa_low_SLP_centroid	1661	5.99	-15.089
2024-05-24 12:00:00+00:00	992.596	38.422	1.5_hPa_low_SLP_centroid	45	18.835	-13.459
2024-05-24 18:00:00+00:00	993.724	147.916	1.5_hPa_low_SLP_centroid	173	14.048	-14.638
2024-05-25 00:00:00+00:00	991.339	127.212	1.5_hPa_low_SLP_centroid	82	13.243	-12.573
2024-05-25 03:00:00+00:00	990.159	134.641	1.5_hPa_low_SLP_centroid	82	13.197	-25.0
2024-05-25 06:00:00+00:00	988.791	146.929	1.5_hPa_low_SLP_centroid	97	17.751	-17.858
2024-05-25 12:00:00+00:00	985.755	122.001	1.5_hPa_low_SLP_centroid	52	9.784	-25.582
2024-05-25 15:00:00+00:00	984.658	146.356	1.5_hPa_low_SLP_centroid	81	9.123	-24.708
2024-05-25 18:00:00+00:00	983.638	145.374	1.5_hPa_low_SLP_centroid	90	11.944	-22.989
2024-05-25 21:00:00+00:00	979.713	145.283	1.5_hPa_low_SLP_centroid	16	17.931	-27.75
2024-05-26 00:00:00+00:00	975.25	60.035	1.5_hPa_low_SLP_centroid	45	20.618	-41.803
2024-05-26 03:00:00+00:00	972.388	94.502	1.5_hPa_low_SLP_centroid	29	22.006	-37.787
2024-05-26 06:00:00+00:00	972.373	70.32	1.5_hPa_low_SLP_centroid	41	19.171	-41.604
2024-05-26 09:00:00+00:00	972.022	88.318	1.5_hPa_low_SLP_centroid	40	10.795	-40.518
2024-05-26 12:00:00+00:00	973.012	91.829	1.5_hPa_low_SLP_centroid	48	9.174	-58.314
2024-05-26 15:00:00+00:00	974.579	137.213	1.5_hPa_low_SLP_centroid	42	2.78	-52.078
2024-05-26 18:00:00+00:00	976.344	139.351	1.5_hPa_low_SLP_centroid	37	-7.152	-54.9
2024-05-26 21:00:00+00:00	978.227	149.481	1.5_hPa_low_SLP_centroid	47	-11.94	-51.989
2024-05-27 00:00:00+00:00	980.161	116.395	1.5_hPa_low_SLP_centroid	49	7.887	-42.482
2024-05-27 03:00:00+00:00	982.191	55.136	1.5_hPa_low_SLP_centroid	55	1.226	-38.124
2024-05-27 06:00:00+00:00	983.073	44.67	1.5_hPa_low_SLP_centroid	51	1.86	-27.974
2024-05-27 09:00:00+00:00	984.295	32.216	1.5_hPa_low_SLP_centroid	54	5.016	-26.078
2024-05-27 12:00:00+00:00	985.724	47.408	1.5_hPa_low_SLP_centroid	42	1.286	-28.222
2024-05-27 15:00:00+00:00	989.553	19.205	1.5_hPa_low_SLP_centroid	73	-3.072	-10.055
2024-05-27 18:00:00+00:00	992.635	73.015	1.5_hPa_low_SLP_centroid	230	-8.338	-19.114
2024-05-28 00:00:00+00:00	995.14	78.639	1.5_hPa_low_SLP_centroid	44	4.532	-17.286
2024-05-28 03:00:00+00:00	997.353	47.126	1.5_hPa_low_SLP_centroid	407	-6.076	-10.703
2024-05-28 06:00:00+00:00	995.971	72.567	1.5_hPa_low_SLP_centroid	316	-3.49	-8.817

Column block 3 of 3

time_utc	pres_bias_hpa
2024-05-24 00:00:00+00:00	1.081
2024-05-24 03:00:00+00:00	2.919
2024-05-24 06:00:00+00:00	0.421
2024-05-24 12:00:00+00:00	-2.404
2024-05-24 18:00:00+00:00	-1.276
2024-05-25 00:00:00+00:00	-1.661
2024-05-25 03:00:00+00:00	-1.841
2024-05-25 06:00:00+00:00	-2.209
2024-05-25 12:00:00+00:00	-4.245
2024-05-25 15:00:00+00:00	-4.342
2024-05-25 18:00:00+00:00	-4.362
2024-05-25 21:00:00+00:00	-6.287

2024-05-26 00:00:00+00:00	-8.75
2024-05-26 03:00:00+00:00	-11.612
2024-05-26 06:00:00+00:00	-10.627
2024-05-26 09:00:00+00:00	-8.978
2024-05-26 12:00:00+00:00	-4.988
2024-05-26 15:00:00+00:00	-3.421
2024-05-26 18:00:00+00:00	-1.656
2024-05-26 21:00:00+00:00	-1.773
2024-05-27 00:00:00+00:00	-1.839
2024-05-27 03:00:00+00:00	-1.809
2024-05-27 06:00:00+00:00	-1.927
2024-05-27 09:00:00+00:00	-1.705
2024-05-27 12:00:00+00:00	-1.276
2024-05-27 15:00:00+00:00	1.553
2024-05-27 18:00:00+00:00	2.635
2024-05-28 00:00:00+00:00	4.14
2024-05-28 03:00:00+00:00	5.353
2024-05-28 06:00:00+00:00	2.971

Remal_FNL_All_IMD_Intensity_Matches.csv

Column block 1 of 3

time_utc	wind_imd	pres_imd	lat_model_intensity	lon_model_intensity	wind_model	wind_centre_model
2024-05-24 00:00:00+00:00	25	996	15.29	88.799	27.038	7.754
2024-05-24 03:00:00+00:00	25	996	14.956	89.836	34.923	13.693
2024-05-24 06:00:00+00:00	25	996	15.377	89.978	37.592	1.131
2024-05-24 12:00:00+00:00	25	995	16.07	90.13	35.911	10.485
2024-05-24 18:00:00+00:00	25	995	16.271	89.618	45.595	7.4
2024-05-25 00:00:00+00:00	30	993	17.226	90.325	44.169	1.746
2024-05-25 03:00:00+00:00	30	992	18.106	90.653	54.231	28.519
2024-05-25 06:00:00+00:00	30	991	18.59	89.573	56.344	34.842
2024-05-25 12:00:00+00:00	35	990	18.177	89.73	56.782	8.837
2024-05-25 15:00:00+00:00	35	989	18.408	89.695	60.698	9.777
2024-05-25 18:00:00+00:00	40	988	18.854	89.459	67.184	20.205
2024-05-25 21:00:00+00:00	45	986	18.79	89.718	86.335	19.839
2024-05-26 00:00:00+00:00	50	984	19.639	89.448	87.732	10.45
2024-05-26 03:00:00+00:00	50	984	20.104	89.828	85.848	14.312
2024-05-26 06:00:00+00:00	50	983	20.522	89.601	83.407	7.243
2024-05-26 09:00:00+00:00	55	981	21.174	89.95	75.134	8.753
2024-05-26 12:00:00+00:00	60	978	21.715	90.086	71.381	17.551
2024-05-26 15:00:00+00:00	60	978	22.176	89.98	64.652	6.929
2024-05-26 18:00:00+00:00	60	978	22.518	90.244	65.896	4.887
2024-05-26 21:00:00+00:00	55	980	23.108	90.418	55.187	6.914
2024-05-27 00:00:00+00:00	45	982	23.528	90.696	50.216	4.887
2024-05-27 03:00:00+00:00	40	984	23.877	90.686	45.286	7.628
2024-05-27 06:00:00+00:00	35	985	24.197	91.109	36.774	7.493
2024-05-27 09:00:00+00:00	35	986	24.57	91.423	36.096	4.559
2024-05-27 12:00:00+00:00	35	987	24.844	92.035	33.182	14.636
2024-05-27 15:00:00+00:00	30	988	24.921	92.529	28.571	19.435
2024-05-27 18:00:00+00:00	30	990	25.156	93.164	26.211	11.56
2024-05-28 00:00:00+00:00	20	991	24.826	94.385	24.774	1.954
2024-05-28 03:00:00+00:00	20	992	24.394	90.264	16.54	10.395
2024-05-28 06:00:00+00:00	20	993	24.57	90.102	17.518	11.038

Column block 2 of 3

time_utc	pres_model	max_wind_distance_km	centre_method	n_centroid_cells	wind_bias_kt	wind_centre_bias_kt
2024-05-24 00:00:00+00:00	996.495	149.615	1.5_hPa_low_SLP_centroid	1864	2.038	-17.246
2024-05-24 03:00:00+00:00	996.843	89.165	1.5_hPa_low_SLP_centroid	523	9.923	-11.307
2024-05-24 06:00:00+00:00	995.149	127.21	1.5_hPa_low_SLP_centroid	583	12.592	-23.869
2024-05-24 12:00:00+00:00	992.54	146.483	1.5_hPa_low_SLP_centroid	259	10.911	-14.515
2024-05-24 18:00:00+00:00	991.876	116.62	1.5_hPa_low_SLP_centroid	182	20.595	-17.6
2024-05-25 00:00:00+00:00	990.428	139.093	1.5_hPa_low_SLP_centroid	441	14.169	-28.254
2024-05-25 03:00:00+00:00	987.252	39.586	1.5_hPa_low_SLP_centroid	9	24.231	-1.481
2024-05-25 06:00:00+00:00	983.934	14.136	1.5_hPa_low_SLP_centroid	4	26.344	4.842
2024-05-25 12:00:00+00:00	981.328	74.966	1.5_hPa_low_SLP_centroid	63	21.782	-26.163
2024-05-25 15:00:00+00:00	980.561	126.794	1.5_hPa_low_SLP_centroid	103	25.698	-25.223
2024-05-25 18:00:00+00:00	979.248	94.754	1.5_hPa_low_SLP_centroid	148	27.184	-19.795
2024-05-25 21:00:00+00:00	973.506	40.925	1.5_hPa_low_SLP_centroid	18	41.335	-25.161
2024-05-26 00:00:00+00:00	969.687	110.946	1.5_hPa_low_SLP_centroid	26	37.732	-39.55
2024-05-26 03:00:00+00:00	969.99	71.378	1.5_hPa_low_SLP_centroid	30	35.848	-35.688
2024-05-26 06:00:00+00:00	969.25	98.637	1.5_hPa_low_SLP_centroid	54	33.407	-42.757
2024-05-26 09:00:00+00:00	969.078	91.983	1.5_hPa_low_SLP_centroid	44	20.134	-46.247
2024-05-26 12:00:00+00:00	968.954	58.178	1.5_hPa_low_SLP_centroid	59	11.381	-42.449
2024-05-26 15:00:00+00:00	970.574	83.473	1.5_hPa_low_SLP_centroid	49	4.652	-53.071
2024-05-26 18:00:00+00:00	973.926	102.523	1.5_hPa_low_SLP_centroid	47	5.896	-55.113
2024-05-26 21:00:00+00:00	975.505	137.336	1.5_hPa_low_SLP_centroid	41	0.187	-48.086
2024-05-27 00:00:00+00:00	977.931	83.324	1.5_hPa_low_SLP_centroid	46	5.216	-40.113
2024-05-27 03:00:00+00:00	979.826	68.569	1.5_hPa_low_SLP_centroid	43	5.286	-32.372
2024-05-27 06:00:00+00:00	982.86	61.054	1.5_hPa_low_SLP_centroid	46	1.774	-27.507
2024-05-27 09:00:00+00:00	984.196	46.467	1.5_hPa_low_SLP_centroid	43	1.096	-30.441
2024-05-27 12:00:00+00:00	985.189	37.624	1.5_hPa_low_SLP_centroid	29	-1.818	-20.364
2024-05-27 15:00:00+00:00	990.033	111.598	1.5_hPa_low_SLP_centroid	160	-1.429	-10.565
2024-05-27 18:00:00+00:00	991.512	146.968	1.5_hPa_low_SLP_centroid	63	-3.789	-18.44
2024-05-28 00:00:00+00:00	994.349	38.78	1.5_hPa_low_SLP_centroid	60	4.774	-18.046
2024-05-28 03:00:00+00:00	995.956	126.799	1.5_hPa_low_SLP_centroid	416	-3.46	-9.605
2024-05-28 06:00:00+00:00	995.656	147.856	1.5_hPa_low_SLP_centroid	344	-2.482	-8.962

Column block 3 of 3

time_utc	pres_bias_hpa
2024-05-24 00:00:00+00:00	0.495
2024-05-24 03:00:00+00:00	0.843
2024-05-24 06:00:00+00:00	-0.851
2024-05-24 12:00:00+00:00	-2.46
2024-05-24 18:00:00+00:00	-3.124
2024-05-25 00:00:00+00:00	-2.572
2024-05-25 03:00:00+00:00	-4.748
2024-05-25 06:00:00+00:00	-7.066
2024-05-25 12:00:00+00:00	-8.672
2024-05-25 15:00:00+00:00	-8.439
2024-05-25 18:00:00+00:00	-8.752
2024-05-25 21:00:00+00:00	-12.494
2024-05-26 00:00:00+00:00	-14.313
2024-05-26 03:00:00+00:00	-14.01
2024-05-26 06:00:00+00:00	-13.75
2024-05-26 09:00:00+00:00	-11.922
2024-05-26 12:00:00+00:00	-9.046
2024-05-26 15:00:00+00:00	-7.426
2024-05-26 18:00:00+00:00	-4.074
2024-05-26 21:00:00+00:00	-4.495
2024-05-27 00:00:00+00:00	-4.069
2024-05-27 03:00:00+00:00	-4.174
2024-05-27 06:00:00+00:00	-2.14
2024-05-27 09:00:00+00:00	-1.804

2024-05-27 12:00:00+00:00	-1.811
2024-05-27 15:00:00+00:00	2.033
2024-05-27 18:00:00+00:00	1.512
2024-05-28 00:00:00+00:00	3.349
2024-05-28 03:00:00+00:00	3.956
2024-05-28 06:00:00+00:00	2.656

Remal_GFS_All_IMD_Intensity_Matches.csv

Column block 1 of 3

time_utc	wind_imd	pres_imd	lat_model_intensity	lon_model_intensity	wind_model	wind_centre_model
2024-05-24 00:00:00+00:00	25	996	14.312	89.261	35.808	18.84
2024-05-24 03:00:00+00:00	25	996	15.094	89.525	42.155	6.694
2024-05-24 06:00:00+00:00	25	996	15.463	89.933	38.007	6.433
2024-05-24 12:00:00+00:00	25	995	16.009	90.053	37.204	9.664
2024-05-24 18:00:00+00:00	25	995	16.429	90.328	47.484	7.187
2024-05-25 00:00:00+00:00	30	993	17.376	90.075	42.445	2.915
2024-05-25 03:00:00+00:00	30	992	17.113	89.684	45.805	8.398
2024-05-25 06:00:00+00:00	30	991	17.548	90.124	51.848	17.104
2024-05-25 12:00:00+00:00	35	990	18.289	89.835	52.856	3.093
2024-05-25 15:00:00+00:00	35	989	18.848	89.489	51.967	11.839
2024-05-25 18:00:00+00:00	40	988	18.774	89.278	72.186	2.716
2024-05-25 21:00:00+00:00	45	986	19.174	89.513	79.312	10.053
2024-05-26 00:00:00+00:00	50	984	19.678	89.549	90.721	6.005
2024-05-26 03:00:00+00:00	50	984	19.988	89.625	90.411	3.161
2024-05-26 06:00:00+00:00	50	983	20.485	89.69	77.003	5.61
2024-05-26 09:00:00+00:00	55	981	21.092	89.788	81.898	12.385
2024-05-26 12:00:00+00:00	60	978	21.798	89.834	70.257	4.189
2024-05-26 15:00:00+00:00	60	978	22.274	89.745	61.85	6.514
2024-05-26 18:00:00+00:00	60	978	22.571	89.97	65.484	7.783
2024-05-26 21:00:00+00:00	55	980	23.111	90.094	54.436	4.33
2024-05-27 00:00:00+00:00	45	982	23.512	90.203	48.002	7.707
2024-05-27 03:00:00+00:00	40	984	23.811	90.393	42.804	5.007
2024-05-27 06:00:00+00:00	35	985	24.144	90.549	39.422	10.103
2024-05-27 09:00:00+00:00	35	986	24.367	91.004	40.99	9.97
2024-05-27 12:00:00+00:00	35	987	24.711	91.656	35.19	12.658
2024-05-27 15:00:00+00:00	30	988	24.929	92.343	33.35	16.338
2024-05-27 18:00:00+00:00	30	990	24.86	92.584	33.325	14.684
2024-05-28 00:00:00+00:00	20	991	24.56	90.886	14.95	3.226
2024-05-28 03:00:00+00:00	20	992	24.249	89.965	17.264	9.099
2024-05-28 06:00:00+00:00	20	993	24.449	89.34	15.968	10.815

Column block 2 of 3

time_utc	pres_model	max_wind_distance_km	centre_method	n_centroid_cells	wind_bias_kt	wind_centre_bias_kt
2024-05-24 00:00:00+00:00	994.595	75.88	1.5_hPa_low_SLP_centroid	44	10.808	-6.16
2024-05-24 03:00:00+00:00	996.603	147.759	1.5_hPa_low_SLP_centroid	520	17.155	-18.306
2024-05-24 06:00:00+00:00	994.715	127.097	1.5_hPa_low_SLP_centroid	558	13.007	-18.567
2024-05-24 12:00:00+00:00	992.595	145.714	1.5_hPa_low_SLP_centroid	472	12.204	-15.336
2024-05-24 18:00:00+00:00	989.55	148.829	1.5_hPa_low_SLP_centroid	7	22.484	-17.813
2024-05-25 00:00:00+00:00	986.911	125.906	1.5_hPa_low_SLP_centroid	61	12.445	-27.085
2024-05-25 03:00:00+00:00	985.81	32.096	1.5_hPa_low_SLP_centroid	38	15.805	-21.602
2024-05-25 06:00:00+00:00	982.054	78.083	1.5_hPa_low_SLP_centroid	28	21.848	-12.896
2024-05-25 12:00:00+00:00	979.108	74.402	1.5_hPa_low_SLP_centroid	74	17.856	-31.907
2024-05-25 15:00:00+00:00	976.777	142.279	1.5_hPa_low_SLP_centroid	42	16.967	-23.161

2024-05-25 18:00:00+00:00	974.391	86.174	1.5_hPa_low_SLP_centroid	59	32.186	-37.284
2024-05-25 21:00:00+00:00	970.307	75.304	1.5_hPa_low_SLP_centroid	49	34.312	-34.947
2024-05-26 00:00:00+00:00	966.838	86.054	1.5_hPa_low_SLP_centroid	24	40.721	-43.995
2024-05-26 03:00:00+00:00	966.186	85.368	1.5_hPa_low_SLP_centroid	49	40.411	-46.839
2024-05-26 06:00:00+00:00	968.131	71.132	1.5_hPa_low_SLP_centroid	61	27.003	-44.39
2024-05-26 09:00:00+00:00	966.17	44.556	1.5_hPa_low_SLP_centroid	34	26.898	-42.615
2024-05-26 12:00:00+00:00	967.224	67.504	1.5_hPa_low_SLP_centroid	38	10.257	-55.811
2024-05-26 15:00:00+00:00	970.569	79.194	1.5_hPa_low_SLP_centroid	37	1.85	-53.486
2024-05-26 18:00:00+00:00	973.094	90.747	1.5_hPa_low_SLP_centroid	41	5.484	-52.217
2024-05-26 21:00:00+00:00	975.299	145.776	1.5_hPa_low_SLP_centroid	48	-0.564	-50.67
2024-05-27 00:00:00+00:00	977.045	119.288	1.5_hPa_low_SLP_centroid	51	3.002	-37.293
2024-05-27 03:00:00+00:00	980.22	94.047	1.5_hPa_low_SLP_centroid	54	2.804	-34.993
2024-05-27 06:00:00+00:00	982.225	80.564	1.5_hPa_low_SLP_centroid	59	4.422	-24.897
2024-05-27 09:00:00+00:00	983.106	50.698	1.5_hPa_low_SLP_centroid	45	5.99	-25.03
2024-05-27 12:00:00+00:00	984.181	67.42	1.5_hPa_low_SLP_centroid	34	0.19	-22.342
2024-05-27 15:00:00+00:00	986.522	28.638	1.5_hPa_low_SLP_centroid	20	3.35	-13.662
2024-05-27 18:00:00+00:00	991.335	148.433	1.5_hPa_low_SLP_centroid	154	3.325	-15.316
2024-05-28 00:00:00+00:00	994.677	147.05	1.5_hPa_low_SLP_centroid	269	-5.05	-16.774
2024-05-28 03:00:00+00:00	996.135	137.295	1.5_hPa_low_SLP_centroid	708	-2.736	-10.901
2024-05-28 06:00:00+00:00	995.502	44.148	1.5_hPa_low_SLP_centroid	381	-4.032	-9.185

Column block 3 of 3

time_utc	pres_bias_hpa
2024-05-24 00:00:00+00:00	-1.405
2024-05-24 03:00:00+00:00	0.603
2024-05-24 06:00:00+00:00	-1.285
2024-05-24 12:00:00+00:00	-2.405
2024-05-24 18:00:00+00:00	-5.45
2024-05-25 00:00:00+00:00	-6.089
2024-05-25 03:00:00+00:00	-6.19
2024-05-25 06:00:00+00:00	-8.946
2024-05-25 12:00:00+00:00	-10.892
2024-05-25 15:00:00+00:00	-12.223
2024-05-25 18:00:00+00:00	-13.609
2024-05-25 21:00:00+00:00	-15.693
2024-05-26 00:00:00+00:00	-17.162
2024-05-26 03:00:00+00:00	-17.814
2024-05-26 06:00:00+00:00	-14.869
2024-05-26 09:00:00+00:00	-14.83
2024-05-26 12:00:00+00:00	-10.776
2024-05-26 15:00:00+00:00	-7.431
2024-05-26 18:00:00+00:00	-4.906
2024-05-26 21:00:00+00:00	-4.701
2024-05-27 00:00:00+00:00	-4.955
2024-05-27 03:00:00+00:00	-3.78
2024-05-27 06:00:00+00:00	-2.775
2024-05-27 09:00:00+00:00	-2.894
2024-05-27 12:00:00+00:00	-2.819
2024-05-27 15:00:00+00:00	-1.478
2024-05-27 18:00:00+00:00	1.335
2024-05-28 00:00:00+00:00	3.677
2024-05-28 03:00:00+00:00	4.135
2024-05-28 06:00:00+00:00	2.502

T04_A07_Peak_Intensity_Timing_All_IMD_Matches.csv

Column block 1 of 2

model	variable	imd_peak	imd_peak_times	n_tied_imd_times	model_peak	model_peak_time
FNL	Maximum wind (kt)	60	26 May 12:00; 26 May 15:00; 26 May 18:00	3	87.732	26 May 00:00
FNL	Minimum pressure (hPa)	978	26 May 12:00; 26 May 15:00; 26 May 18:00	3	968.954	26 May 12:00
GFS	Maximum wind (kt)	60	26 May 12:00; 26 May 15:00; 26 May 18:00	3	90.721	26 May 00:00
GFS	Minimum pressure (hPa)	978	26 May 12:00; 26 May 15:00; 26 May 18:00	3	966.17	26 May 09:00
CFSv2	Maximum wind (kt)	60	26 May 12:00; 26 May 15:00; 26 May 18:00	3	68.206	25 May 15:00
CFSv2	Minimum pressure (hPa)	978	26 May 12:00; 26 May 15:00; 26 May 18:00	3	977.775	25 May 15:00
ERA5	Maximum wind (kt)	60	26 May 12:00; 26 May 15:00; 26 May 18:00	3	72.006	26 May 03:00
ERA5	Minimum pressure (hPa)	978	26 May 12:00; 26 May 15:00; 26 May 18:00	3	972.022	26 May 09:00

Column block 2 of 2

model	value_error	timing_error_h_nearest_tie	timing_error_h_vs_first_tie	timing_error_h_vs_last_tie
FNL	27.732	-12.0	-12.0	-18.0
FNL	-9.046	0.0	0.0	-6.0
GFS	30.721	-12.0	-12.0	-18.0
GFS	-11.83	-3.0	-3.0	-9.0
CFSv2	8.206	-21.0	-21.0	-27.0
CFSv2	-0.225	-21.0	-21.0	-27.0
ERA5	12.006	-9.0	-9.0	-15.0
ERA5	-5.978	-3.0	-3.0	-9.0

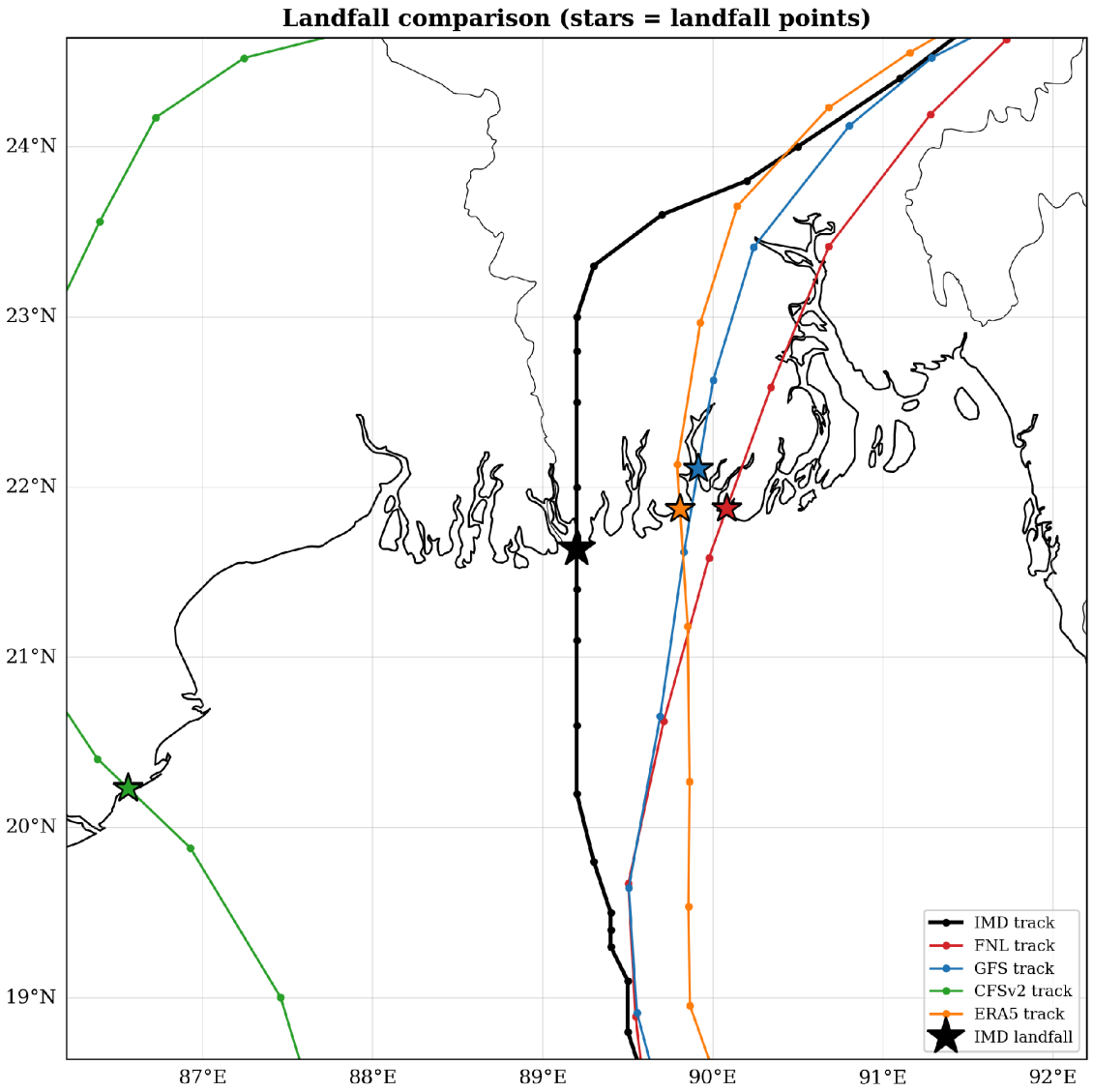
T04_intensity_change_summary.csv

model	max_model_wind_increase_kt_per_h	max_imd_wind_increase_kt_per_h	max_model_deepening_hpa_per_h	max_imd_deepening_hpa_per_h
FNL	6.384	1.667	1.914	1.0
GFS	6.739	1.667	1.362	1.0
CFSv2	5.161	1.667	1.25	1.0
ERA5	4.352	1.667	1.488	1.0

A08 - Landfall / coastal crossing

Figures: 1 | CSV tables: 2 | NetCDF extracts: 0

Remal_A6_Landfall_Map.png



Remal_A6_Landfall_Crossings.csv

source	landfall_time_utc	latitude	longitude	time_error_h	position_error_km	note
IMD	2024-05-26 17:24:00+00:00	21.64	89.2	0.0	0.0	derived from IMD track and WRF land mask
FNL	2024-05-26 13:44:00+00:00	21.876	90.083	-3.667	94.86	model minus IMD
GFS	2024-05-26 14:55:00+00:00	22.108	89.914	-2.483	90.182	model minus IMD
CFSv2	2024-05-25 22:01:00+00:00	20.231	86.562	-19.383	315.583	model minus IMD
ERA5	2024-05-26 10:21:00+00:00	21.872	89.807	-7.05	67.84	model minus IMD

Remal_A6_Landfall_Window.csv

Column block 1 of 2

model	time_utc	hours_from_imd_landfall	track_error_km	wind_imd	wind_model	wind_bias_kt
FNL	2024-05-26 06:00:00+00:00	-11.4	71.261	50	83.407	33.407
FNL	2024-05-26 12:00:00+00:00	-5.4	96.878	60	71.381	11.381
FNL	2024-05-26 18:00:00+00:00	0.6	153.475	60	65.896	5.896
FNL	2024-05-27 00:00:00+00:00	6.6	182.73	45	50.216	5.216
GFS	2024-05-26 06:00:00+00:00	-11.4	71.867	50	77.003	27.003
GFS	2024-05-26 12:00:00+00:00	-5.4	87.147	60	70.257	10.257
GFS	2024-05-26 18:00:00+00:00	0.6	132.099	60	65.484	5.484
GFS	2024-05-27 00:00:00+00:00	6.6	146.85	45	48.002	3.002
CFSv2	2024-05-26 06:00:00+00:00	-11.4	337.826	50	50.082	0.082
CFSv2	2024-05-26 12:00:00+00:00	-5.4	349.622	60	33.871	-26.129
CFSv2	2024-05-26 18:00:00+00:00	0.6	346.921	60	30.56	-29.44
CFSv2	2024-05-27 00:00:00+00:00	6.6	326.045	45	33.901	-11.099
ERA5	2024-05-26 06:00:00+00:00	-11.4	128.527	50	69.171	19.171
ERA5	2024-05-26 12:00:00+00:00	-5.4	130.208	60	69.174	9.174
ERA5	2024-05-26 18:00:00+00:00	0.6	159.307	60	52.848	-7.152
ERA5	2024-05-27 00:00:00+00:00	6.6	160.421	45	52.887	7.887

Column block 2 of 2

model	pres_imd	pres_model	pres_bias_hpa
FNL	983	969.25	-13.75
FNL	978	968.954	-9.046
FNL	978	973.926	-4.074
FNL	982	977.931	-4.069
GFS	983	968.131	-14.869
GFS	978	967.224	-10.776
GFS	978	973.094	-4.906
GFS	982	977.045	-4.955
CFSv2	983	982.21	-0.79
CFSv2	978	983.162	5.162
CFSv2	978	986.179	8.179
CFSv2	982	985.759	3.759
ERA5	983	972.373	-10.627
ERA5	978	973.012	-4.988
ERA5	978	976.344	-1.656
ERA5	982	980.161	-1.839

A09 - Common-period PDI

Figures: 0 | CSV tables: 1 | NetCDF extracts: 0

T05_A09_Common_Period_PDI.csv

Column block 1 of 2

source	start_utc	end_utc	n_timestamps	valid_intervals	integrated_hours	wind_threshold_kt
IMD	2024-05-24 00:00:00+00:00	2024-05-28 06:00:00+00:00	30	29	102.0	0
FNL	2024-05-24 00:00:00+00:00	2024-05-28 06:00:00+00:00	30	29	102.0	0
GFS	2024-05-24 00:00:00+00:00	2024-05-28 06:00:00+00:00	30	29	102.0	0
CFSv2	2024-05-24 00:00:00+00:00	2024-05-28 06:00:00+00:00	30	29	102.0	0
ERA5	2024-05-24 00:00:00+00:00	2024-05-28 06:00:00+00:00	30	29	102.0	0

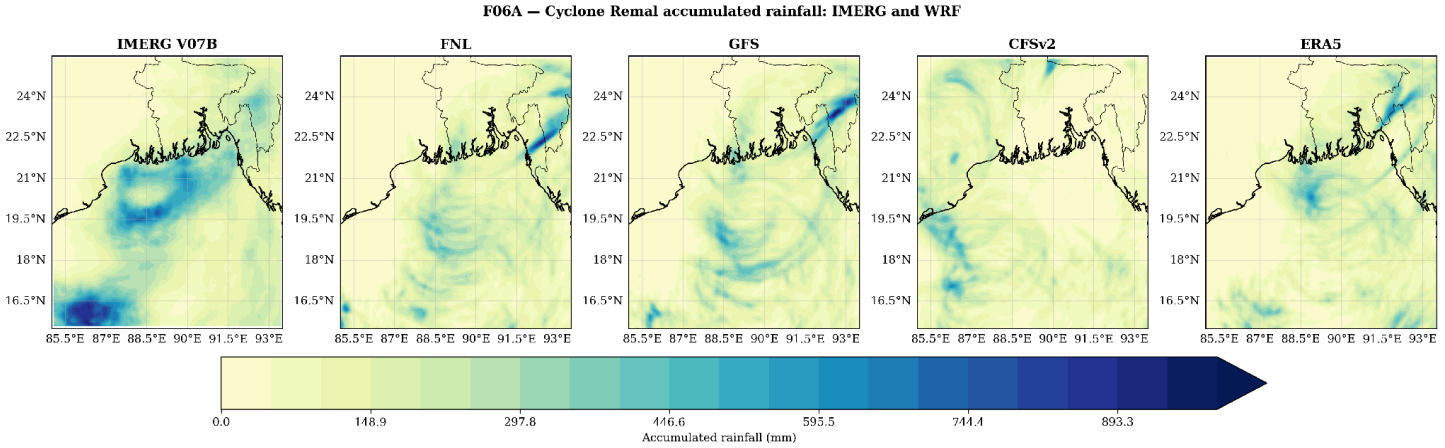
Column block 2 of 2

source	n_above_threshold	pdi_m3_s_minus2	difference_from_imd_pct	model_over_imd
IMD	30	3250622640.034	0.0	1.0
FNL	30	9190173751.993	182.72	2.827
GFS	30	9026507915.041	177.686	2.777
CFSv2	30	4992038093.35	53.572	1.536
ERA5	30	5743445568.125	76.688	1.767

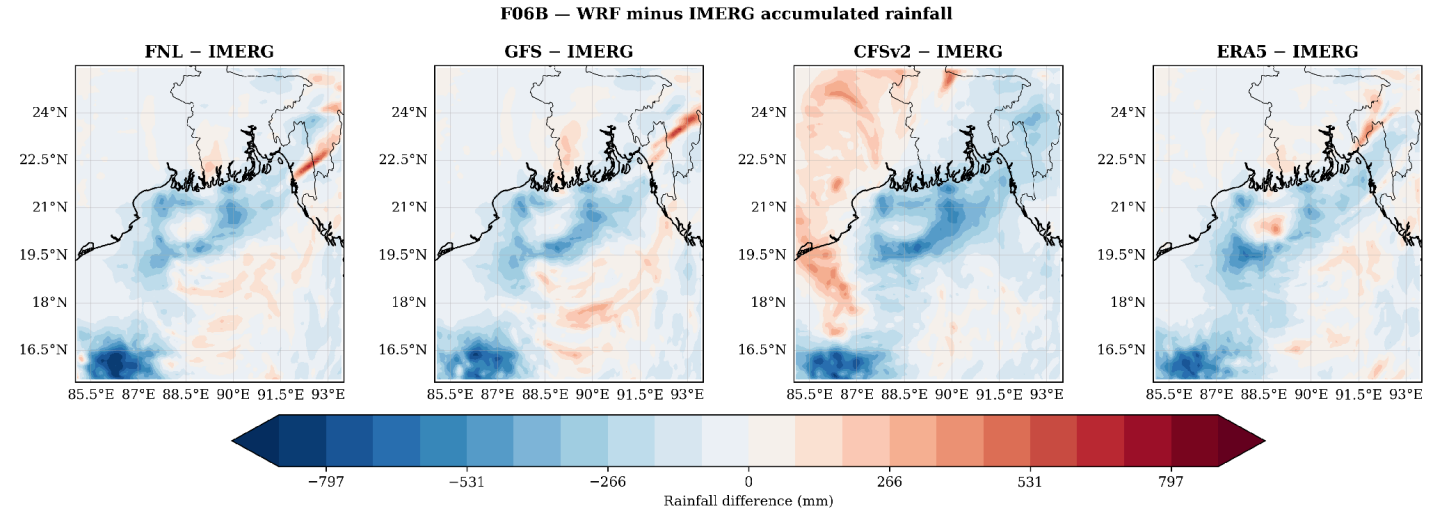
A10 - Rainfall validation against IMERG

Figures: 3 | CSV tables: 2 | NetCDF extracts: 1

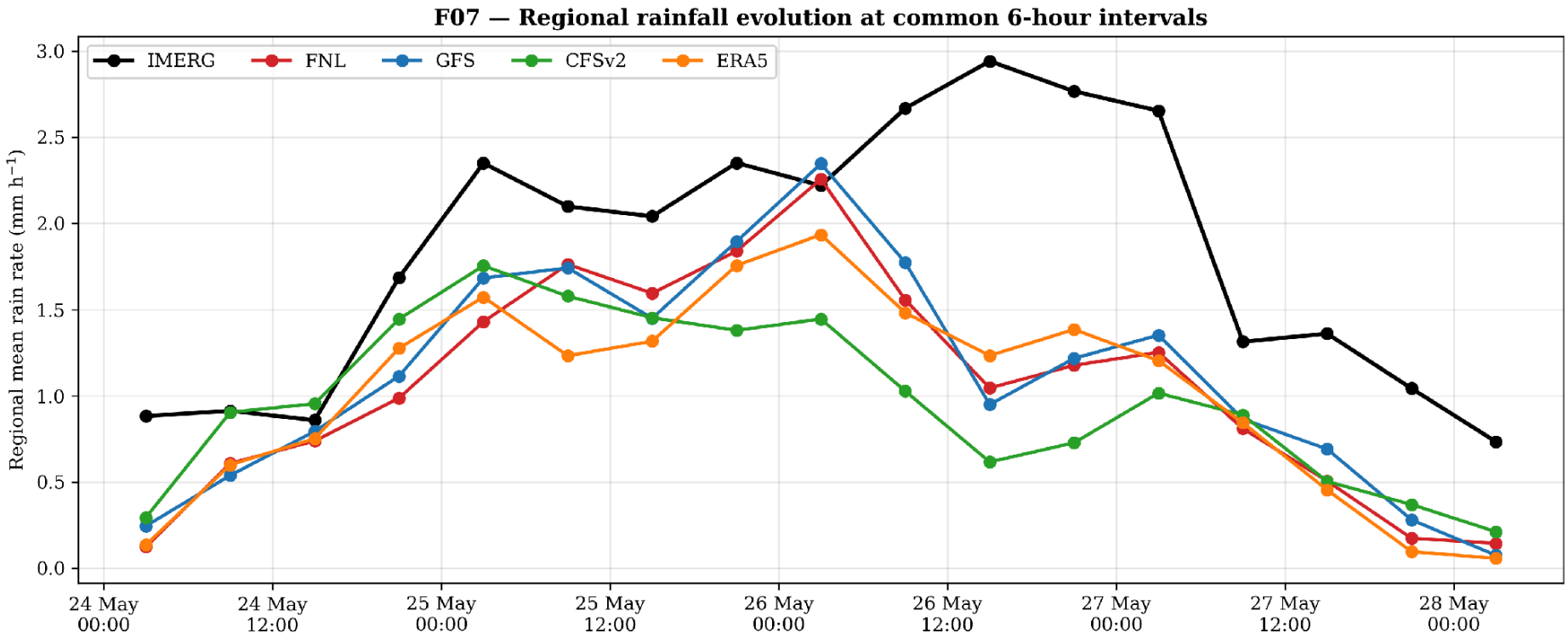
F06A_Rainfall_Accumulation_Maps.png



F06B_Rainfall_Difference_Maps.png



F07_Regional_6h_Rain_Rate.png



F07_Regional_6h_Rain_Rate.csv

source	start_time_utc	end_time_utc	mid_time_utc	interval_rain_mm	mean_rain_rate_mm_per_h
FNL	2024-05-24 00:00:00+00:00	2024-05-24 06:00:00+00:00	2024-05-24 03:00:00+00:00	0.739	0.123
FNL	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	2024-05-24 09:00:00+00:00	3.648	0.608
FNL	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	2024-05-24 15:00:00+00:00	4.425	0.738
FNL	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	2024-05-24 21:00:00+00:00	5.921	0.987
FNL	2024-05-25 00:00:00+00:00	2024-05-25 06:00:00+00:00	2024-05-25 03:00:00+00:00	8.577	1.43
FNL	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	2024-05-25 09:00:00+00:00	10.573	1.762
FNL	2024-05-25 12:00:00+00:00	2024-05-25 18:00:00+00:00	2024-05-25 15:00:00+00:00	9.569	1.595
FNL	2024-05-25 18:00:00+00:00	2024-05-26 00:00:00+00:00	2024-05-25 21:00:00+00:00	11.037	1.839
FNL	2024-05-26 00:00:00+00:00	2024-05-26 06:00:00+00:00	2024-05-26 03:00:00+00:00	13.541	2.257
FNL	2024-05-26 06:00:00+00:00	2024-05-26 12:00:00+00:00	2024-05-26 09:00:00+00:00	9.338	1.556
FNL	2024-05-26 12:00:00+00:00	2024-05-26 18:00:00+00:00	2024-05-26 15:00:00+00:00	6.268	1.045
FNL	2024-05-26 18:00:00+00:00	2024-05-27 00:00:00+00:00	2024-05-26 21:00:00+00:00	7.061	1.177
FNL	2024-05-27 00:00:00+00:00	2024-05-27 06:00:00+00:00	2024-05-27 03:00:00+00:00	7.509	1.252
FNL	2024-05-27 06:00:00+00:00	2024-05-27 12:00:00+00:00	2024-05-27 09:00:00+00:00	4.856	0.809
FNL	2024-05-27 12:00:00+00:00	2024-05-27 18:00:00+00:00	2024-05-27 15:00:00+00:00	3.029	0.505
FNL	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	2024-05-27 21:00:00+00:00	1.038	0.173
FNL	2024-05-28 00:00:00+00:00	2024-05-28 06:00:00+00:00	2024-05-28 03:00:00+00:00	0.861	0.144
GFS	2024-05-24 00:00:00+00:00	2024-05-24 06:00:00+00:00	2024-05-24 03:00:00+00:00	1.467	0.244
GFS	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	2024-05-24 09:00:00+00:00	3.23	0.538
GFS	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	2024-05-24 15:00:00+00:00	4.762	0.794

GFS	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	2024-05-24 21:00:00+00:00	6.686	1.114
GFS	2024-05-25 00:00:00+00:00	2024-05-25 06:00:00+00:00	2024-05-25 03:00:00+00:00	10.099	1.683
GFS	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	2024-05-25 09:00:00+00:00	10.452	1.742
GFS	2024-05-25 12:00:00+00:00	2024-05-25 18:00:00+00:00	2024-05-25 15:00:00+00:00	8.698	1.45
GFS	2024-05-25 18:00:00+00:00	2024-05-26 00:00:00+00:00	2024-05-25 21:00:00+00:00	11.373	1.896
GFS	2024-05-26 00:00:00+00:00	2024-05-26 06:00:00+00:00	2024-05-26 03:00:00+00:00	14.08	2.347
GFS	2024-05-26 06:00:00+00:00	2024-05-26 12:00:00+00:00	2024-05-26 09:00:00+00:00	10.634	1.772
GFS	2024-05-26 12:00:00+00:00	2024-05-26 18:00:00+00:00	2024-05-26 15:00:00+00:00	5.693	0.949
GFS	2024-05-26 18:00:00+00:00	2024-05-27 00:00:00+00:00	2024-05-26 21:00:00+00:00	7.301	1.217
GFS	2024-05-27 00:00:00+00:00	2024-05-27 06:00:00+00:00	2024-05-27 03:00:00+00:00	8.111	1.352
GFS	2024-05-27 06:00:00+00:00	2024-05-27 12:00:00+00:00	2024-05-27 09:00:00+00:00	5.188	0.865
GFS	2024-05-27 12:00:00+00:00	2024-05-27 18:00:00+00:00	2024-05-27 15:00:00+00:00	4.142	0.69
GFS	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	2024-05-27 21:00:00+00:00	1.679	0.28
GFS	2024-05-28 00:00:00+00:00	2024-05-28 06:00:00+00:00	2024-05-28 03:00:00+00:00	0.446	0.074
CFSv2	2024-05-24 00:00:00+00:00	2024-05-24 06:00:00+00:00	2024-05-24 03:00:00+00:00	1.762	0.294
CFSv2	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	2024-05-24 09:00:00+00:00	5.426	0.904
CFSv2	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	2024-05-24 15:00:00+00:00	5.725	0.954
CFSv2	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	2024-05-24 21:00:00+00:00	8.673	1.446
CFSv2	2024-05-25 00:00:00+00:00	2024-05-25 06:00:00+00:00	2024-05-25 03:00:00+00:00	10.525	1.754
CFSv2	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	2024-05-25 09:00:00+00:00	9.467	1.578
CFSv2	2024-05-25 12:00:00+00:00	2024-05-25 18:00:00+00:00	2024-05-25 15:00:00+00:00	8.716	1.453
CFSv2	2024-05-25 18:00:00+00:00	2024-05-26 00:00:00+00:00	2024-05-25 21:00:00+00:00	8.28	1.38
CFSv2	2024-05-26 00:00:00+00:00	2024-05-26 06:00:00+00:00	2024-05-26 03:00:00+00:00	8.672	1.445
CFSv2	2024-05-26 06:00:00+00:00	2024-05-26 12:00:00+00:00	2024-05-26 09:00:00+00:00	6.159	1.026
CFSv2	2024-05-26 12:00:00+00:00	2024-05-26 18:00:00+00:00	2024-05-26 15:00:00+00:00	3.695	0.616
CFSv2	2024-05-26 18:00:00+00:00	2024-05-27 00:00:00+00:00	2024-05-26 21:00:00+00:00	4.362	0.727
CFSv2	2024-05-27 00:00:00+00:00	2024-05-27 06:00:00+00:00	2024-05-27 03:00:00+00:00	6.091	1.015
CFSv2	2024-05-27 06:00:00+00:00	2024-05-27 12:00:00+00:00	2024-05-27 09:00:00+00:00	5.332	0.889
CFSv2	2024-05-27 12:00:00+00:00	2024-05-27 18:00:00+00:00	2024-05-27 15:00:00+00:00	3.016	0.503
CFSv2	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	2024-05-27 21:00:00+00:00	2.213	0.369
CFSv2	2024-05-28 00:00:00+00:00	2024-05-28 06:00:00+00:00	2024-05-28 03:00:00+00:00	1.265	0.211
ERA5	2024-05-24 00:00:00+00:00	2024-05-24 06:00:00+00:00	2024-05-24 03:00:00+00:00	0.812	0.135
ERA5	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	2024-05-24 09:00:00+00:00	3.595	0.599
ERA5	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	2024-05-24 15:00:00+00:00	4.508	0.751
ERA5	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	2024-05-24 21:00:00+00:00	7.657	1.276
ERA5	2024-05-25 00:00:00+00:00	2024-05-25 06:00:00+00:00	2024-05-25 03:00:00+00:00	9.435	1.572
ERA5	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	2024-05-25 09:00:00+00:00	7.392	1.232
ERA5	2024-05-25 12:00:00+00:00	2024-05-25 18:00:00+00:00	2024-05-25 15:00:00+00:00	7.9	1.317
ERA5	2024-05-25 18:00:00+00:00	2024-05-26 00:00:00+00:00	2024-05-25 21:00:00+00:00	10.538	1.756
ERA5	2024-05-26 00:00:00+00:00	2024-05-26 06:00:00+00:00	2024-05-26 03:00:00+00:00	11.608	1.935
ERA5	2024-05-26 06:00:00+00:00	2024-05-26 12:00:00+00:00	2024-05-26 09:00:00+00:00	8.887	1.481
ERA5	2024-05-26 12:00:00+00:00	2024-05-26 18:00:00+00:00	2024-05-26 15:00:00+00:00	7.399	1.233
ERA5	2024-05-26 18:00:00+00:00	2024-05-27 00:00:00+00:00	2024-05-26 21:00:00+00:00	8.308	1.385
ERA5	2024-05-27 00:00:00+00:00	2024-05-27 06:00:00+00:00	2024-05-27 03:00:00+00:00	7.224	1.204
ERA5	2024-05-27 06:00:00+00:00	2024-05-27 12:00:00+00:00	2024-05-27 09:00:00+00:00	5.057	0.843
ERA5	2024-05-27 12:00:00+00:00	2024-05-27 18:00:00+00:00	2024-05-27 15:00:00+00:00	2.72	0.453
ERA5	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	2024-05-27 21:00:00+00:00	0.574	0.096
ERA5	2024-05-28 00:00:00+00:00	2024-05-28 06:00:00+00:00	2024-05-28 03:00:00+00:00	0.336	0.056
IMERG	2024-05-24 00:00:00+00:00	2024-05-24 06:00:00+00:00	2024-05-24 03:00:00+00:00	5.295	0.883
IMERG	2024-05-24 06:00:00+00:00	2024-05-24 12:00:00+00:00	2024-05-24 09:00:00+00:00	5.475	0.913
IMERG	2024-05-24 12:00:00+00:00	2024-05-24 18:00:00+00:00	2024-05-24 15:00:00+00:00	5.149	0.858
IMERG	2024-05-24 18:00:00+00:00	2024-05-25 00:00:00+00:00	2024-05-24 21:00:00+00:00	10.118	1.686
IMERG	2024-05-25 00:00:00+00:00	2024-05-25 06:00:00+00:00	2024-05-25 03:00:00+00:00	14.1	2.35
IMERG	2024-05-25 06:00:00+00:00	2024-05-25 12:00:00+00:00	2024-05-25 09:00:00+00:00	12.593	2.099
IMERG	2024-05-25 12:00:00+00:00	2024-05-25 18:00:00+00:00	2024-05-25 15:00:00+00:00	12.247	2.041
IMERG	2024-05-25 18:00:00+00:00	2024-05-26 00:00:00+00:00	2024-05-25 21:00:00+00:00	14.109	2.351
IMERG	2024-05-26 00:00:00+00:00	2024-05-26 06:00:00+00:00	2024-05-26 03:00:00+00:00	13.318	2.22
IMERG	2024-05-26 06:00:00+00:00	2024-05-26 12:00:00+00:00	2024-05-26 09:00:00+00:00	16.01	2.668
IMERG	2024-05-26 12:00:00+00:00	2024-05-26 18:00:00+00:00	2024-05-26 15:00:00+00:00	17.649	2.941
IMERG	2024-05-26 18:00:00+00:00	2024-05-27 00:00:00+00:00	2024-05-26 21:00:00+00:00	16.601	2.767
IMERG	2024-05-27 00:00:00+00:00	2024-05-27 06:00:00+00:00	2024-05-27 03:00:00+00:00	15.923	2.654

IMERG	2024-05-27 06:00:00+00:00	2024-05-27 12:00:00+00:00	2024-05-27 09:00:00+00:00	7.884	1.314
IMERG	2024-05-27 12:00:00+00:00	2024-05-27 18:00:00+00:00	2024-05-27 15:00:00+00:00	8.163	1.361
IMERG	2024-05-27 18:00:00+00:00	2024-05-28 00:00:00+00:00	2024-05-27 21:00:00+00:00	6.253	1.042
IMERG	2024-05-28 00:00:00+00:00	2024-05-28 06:00:00+00:00	2024-05-28 03:00:00+00:00	4.4	0.733

T06_Rainfall_Spatial_Scores.csv

model	n_valid_cells	bias	mae	rmse	corr
FNL	13924	-76.979	107.033	168.234	0.412
GFS	13924	-70.793	107.217	163.856	0.44
CFSv2	13924	-85.926	159.963	207.234	-0.12
ERA5	13924	-81.209	107.603	163.93	0.47

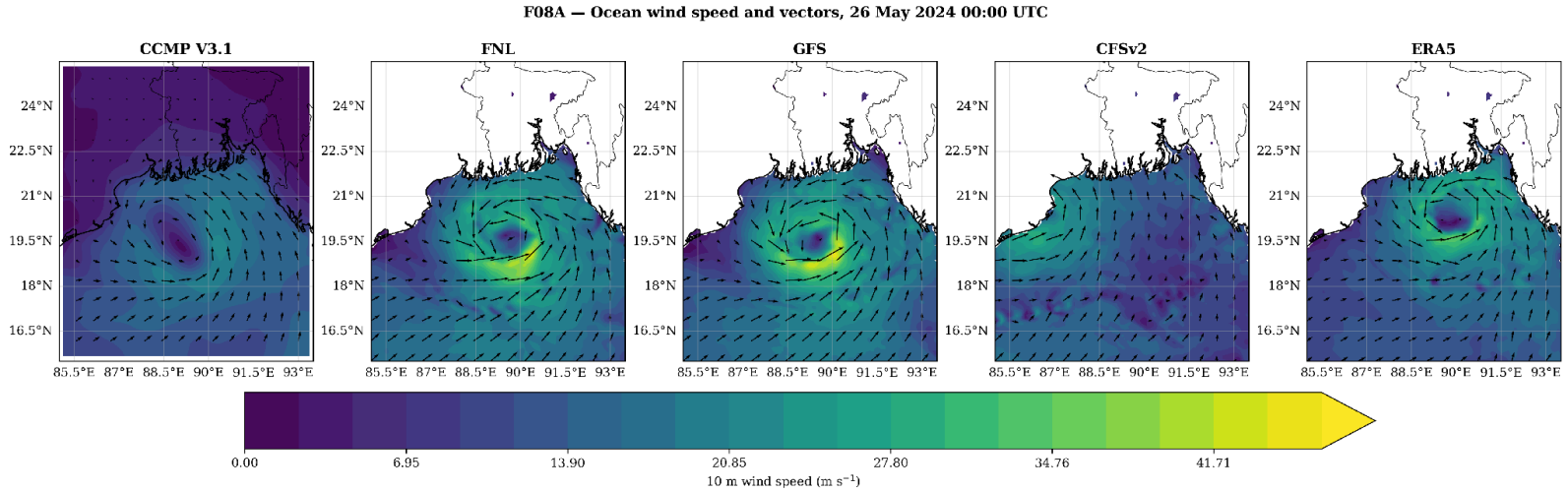
NetCDF diagnostic extracts

- F06_common_grid_rainfall_fields.nc

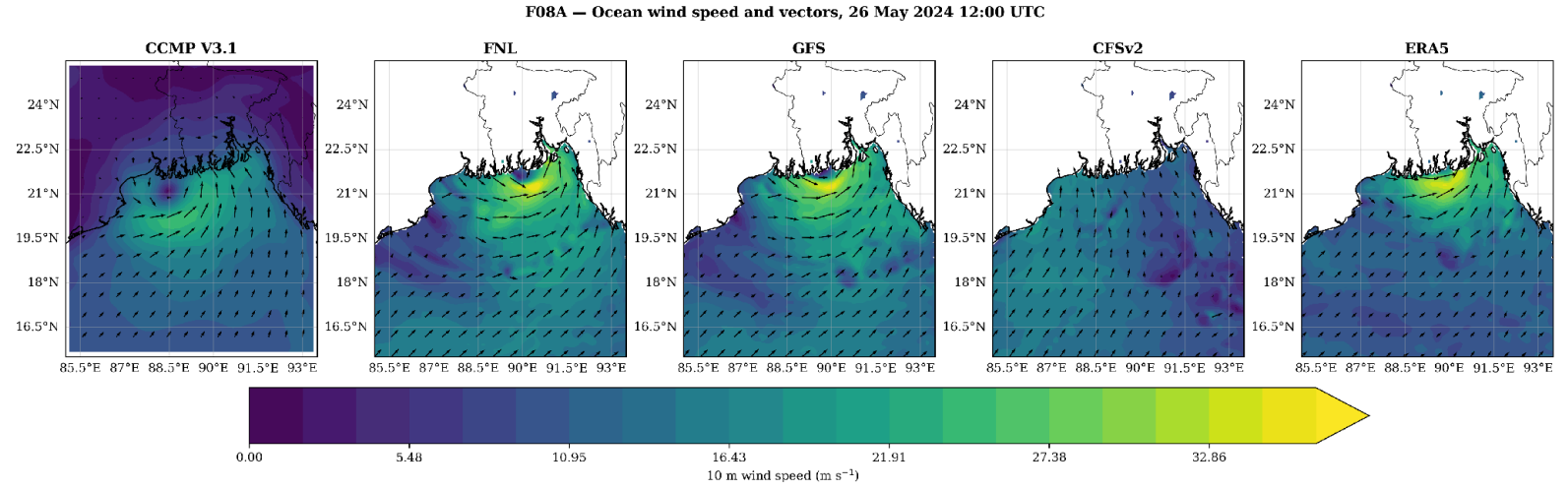
A11 - Ocean-wind validation against CCMP

Figures: 4 | CSV tables: 2 | NetCDF extracts: 0

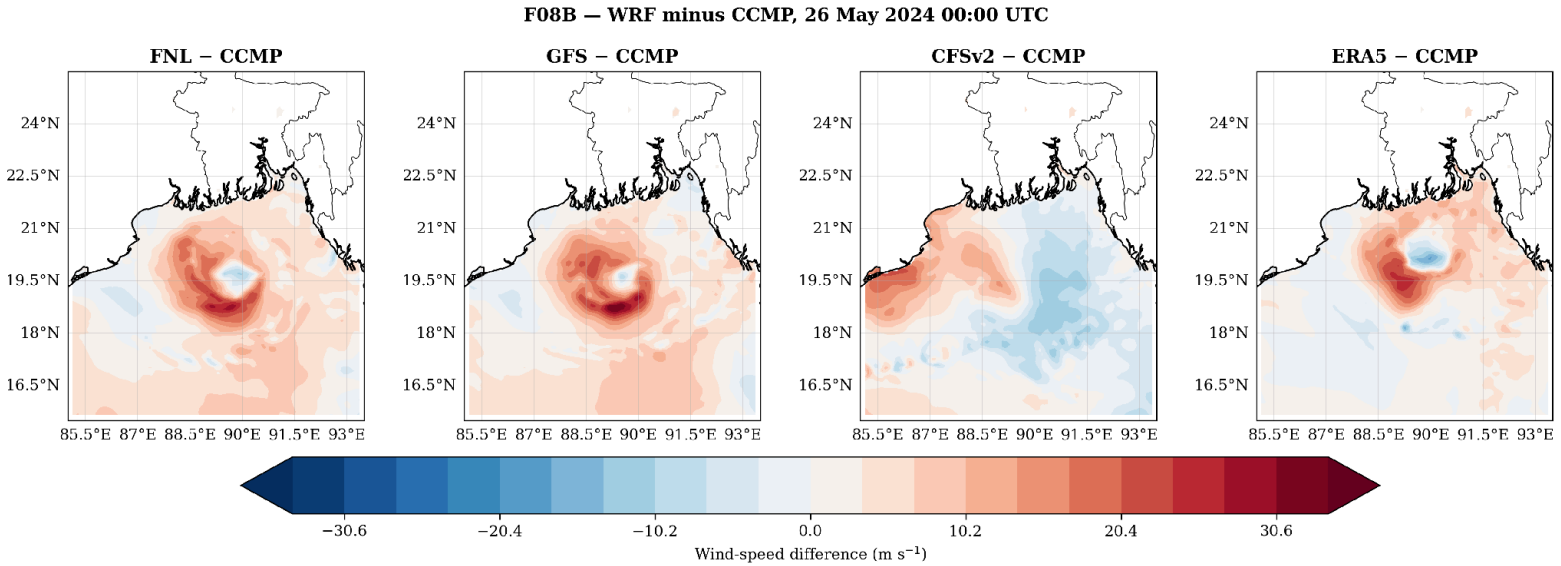
F08A_CCMP_WRF_Wind_20240526_0000.png



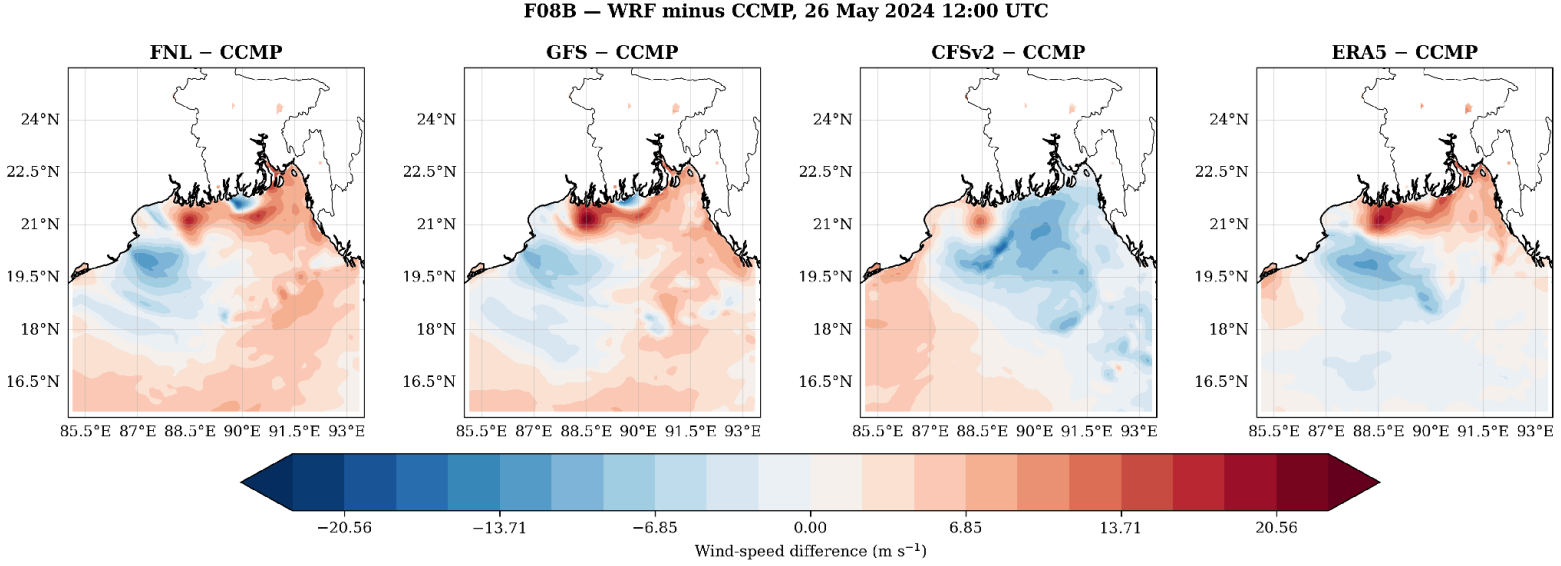
F08A_CCMP_WRF_Wind_20240526_1200.png



F08B_CCMP_Differences_20240526_0000.png



F08B_CCMP_Differences_20240526_1200.png



T06_CCMP_Wind_Score_Summary.csv

model	speed_bias_ms	speed_mae_ms	speed_rmse_ms	speed_spatial_corr	vector_rmse_ms
CFSv2	-0.499	4.599	5.731	-0.153	11.261
ERA5	1.258	3.312	5.024	0.481	10.444

FNL	3.871	5.046	6.2	0.535	11.093
GFS	3.66	4.778	6.11	0.558	10.45

T06_CCMP_Wind_Scores_By_Time.csv

Column block 1 of 2

time_utc	model	n_valid_cells	speed_bias_ms	speed_mae_ms	speed_rmse_ms	speed_spatial_corr
2024-05-26 00:00:00+00:00	FNL	7826	4.995	5.554	7.128	0.557
2024-05-26 00:00:00+00:00	GFS	7826	5.049	5.556	7.365	0.549
2024-05-26 00:00:00+00:00	CFSv2	7826	-0.142	5.046	6.439	-0.33
2024-05-26 00:00:00+00:00	ERA5	7826	2.464	3.703	5.693	0.416
2024-05-26 12:00:00+00:00	FNL	7826	2.747	4.537	5.272	0.513
2024-05-26 12:00:00+00:00	GFS	7826	2.271	3.999	4.855	0.566
2024-05-26 12:00:00+00:00	CFSv2	7826	-0.855	4.151	5.024	0.024
2024-05-26 12:00:00+00:00	ERA5	7826	0.051	2.92	4.355	0.546

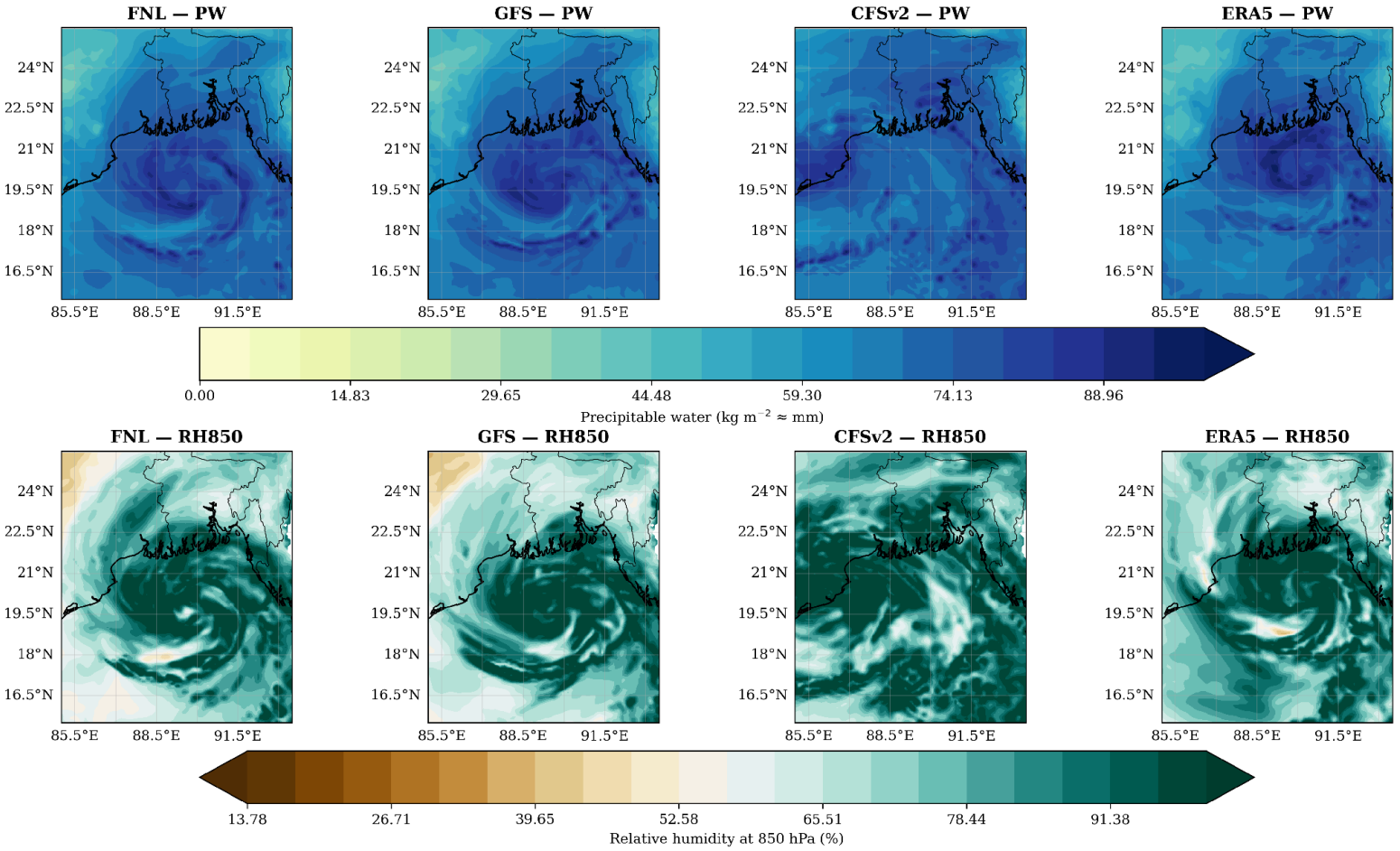
Column block 2 of 2

time_utc	u_bias_ms	v_bias_ms	vector_rmse_ms
2024-05-26 00:00:00+00:00	3.063	-0.284	10.242
2024-05-26 00:00:00+00:00	3.236	-0.237	9.883
2024-05-26 00:00:00+00:00	1.521	3.556	13.012
2024-05-26 00:00:00+00:00	3.591	-0.899	11.17
2024-05-26 12:00:00+00:00	5.31	-1.274	11.944
2024-05-26 12:00:00+00:00	5.337	-1.041	11.016
2024-05-26 12:00:00+00:00	-1.844	1.188	9.509
2024-05-26 12:00:00+00:00	3.006	-0.924	9.718

A12 - Moisture and latent heat flux

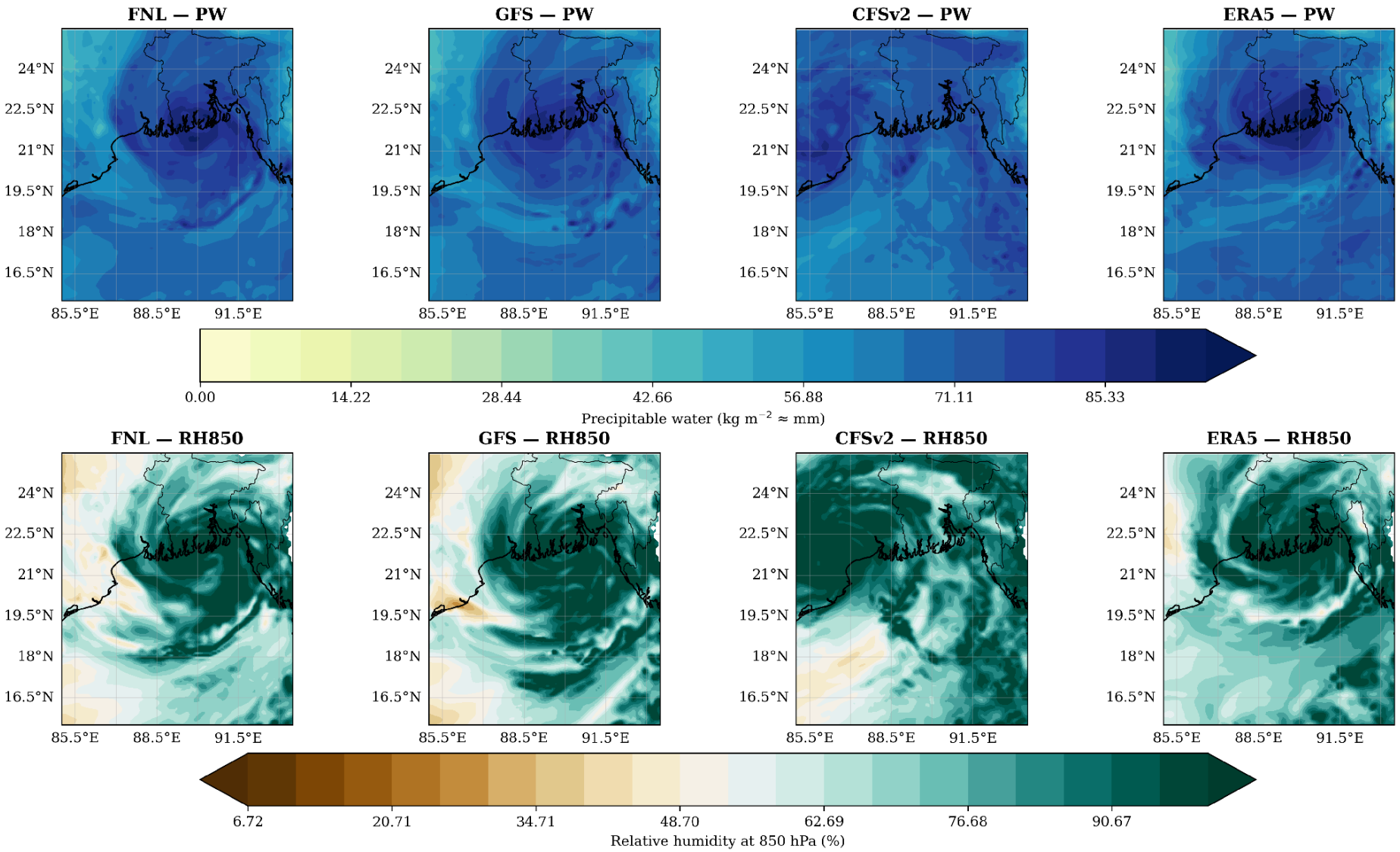
Figures: 4 | CSV tables: 1 | NetCDF extracts: 12

F09_PW_RH850_20240526_0000.png



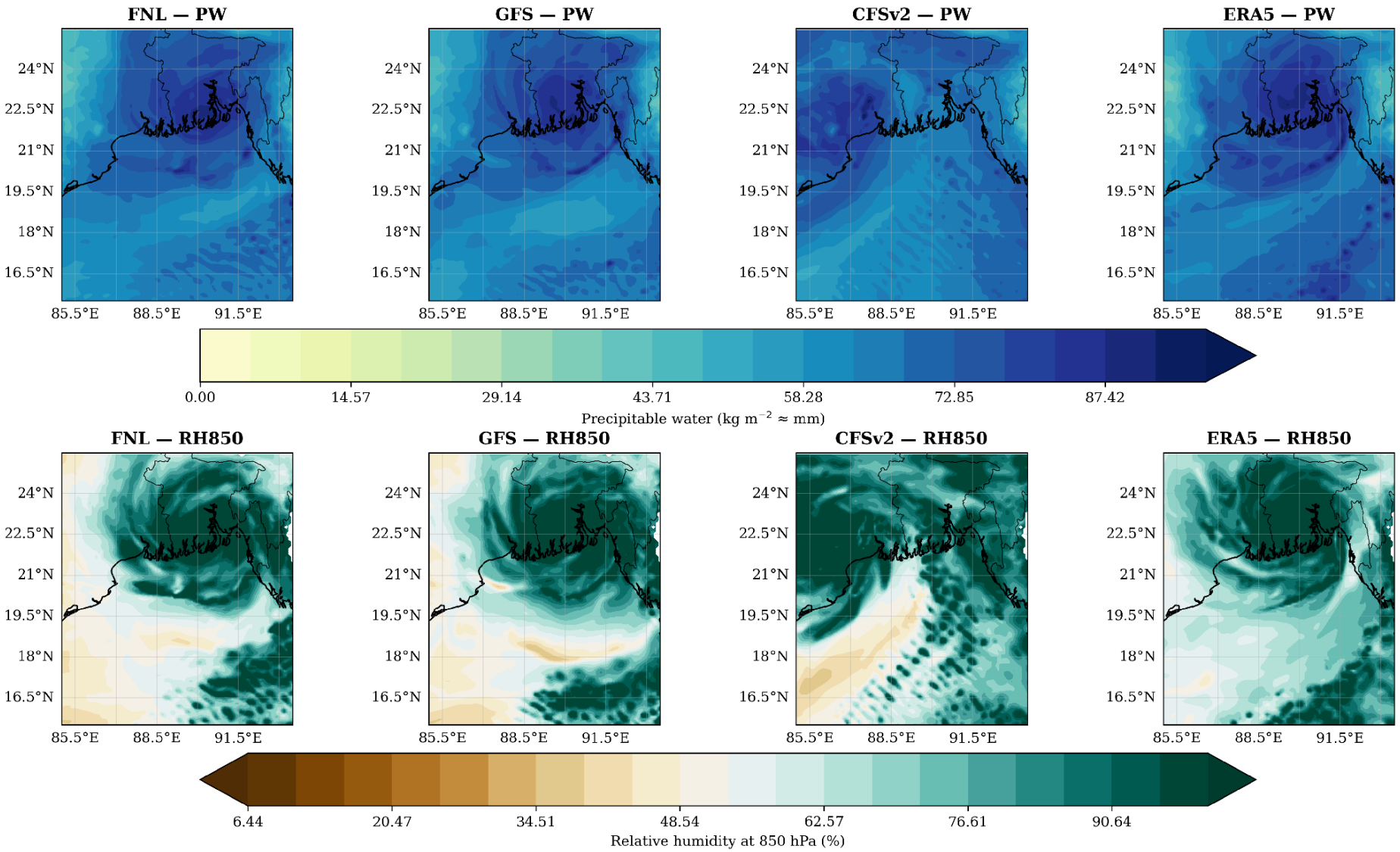
F09_PW_RH850_20240526_1200.png

F09 — Moisture diagnostics, 26 May 2024 12:00 UTC

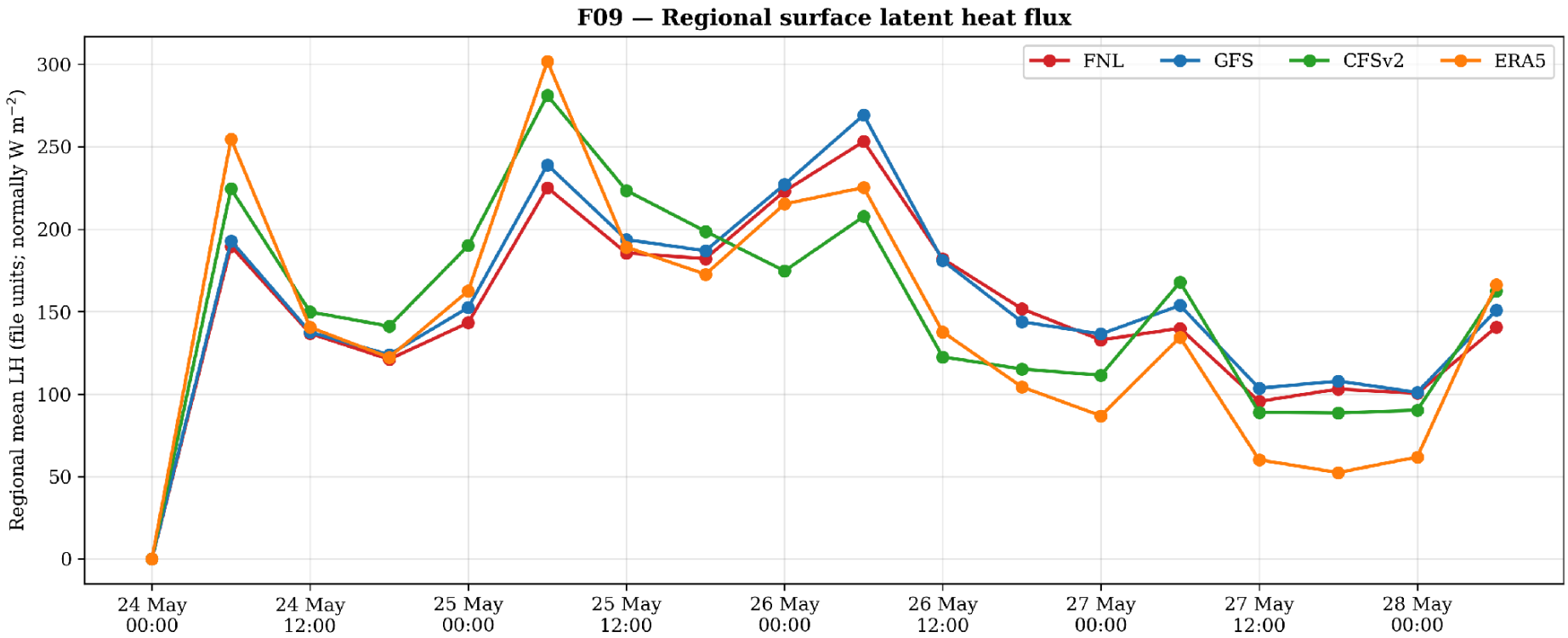


F09_PW_RH850_20240526_1800.png

F09 — Moisture diagnostics, 26 May 2024 18:00 UTC



F09_Regional_Latent_Heat_Flux.png



A12_regional_latent_heat_flux.csv

model	time_utc	regional_mean_LH	units_from_file	description_from_file
FNL	2024-05-24 00:00:00+00:00	0.0	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-24 06:00:00+00:00	189.722	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-24 12:00:00+00:00	136.802	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-24 18:00:00+00:00	121.11	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-25 00:00:00+00:00	143.29	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-25 06:00:00+00:00	225.261	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-25 12:00:00+00:00	185.553	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-25 18:00:00+00:00	182.172	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-26 00:00:00+00:00	223.073	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-26 06:00:00+00:00	253.201	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-26 12:00:00+00:00	182.074	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-26 18:00:00+00:00	151.758	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-27 00:00:00+00:00	132.862	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-27 06:00:00+00:00	139.932	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-27 12:00:00+00:00	95.586	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-27 18:00:00+00:00	103.104	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-28 00:00:00+00:00	100.453	W m-2	LATENT HEAT FLUX AT THE SURFACE
FNL	2024-05-28 06:00:00+00:00	140.68	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-24 00:00:00+00:00	0.0	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-24 06:00:00+00:00	193.001	W m-2	LATENT HEAT FLUX AT THE SURFACE

GFS	2024-05-24 12:00:00+00:00	137.62	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-24 18:00:00+00:00	123.916	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-25 00:00:00+00:00	152.52	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-25 06:00:00+00:00	239.062	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-25 12:00:00+00:00	193.69	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-25 18:00:00+00:00	186.951	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-26 00:00:00+00:00	227.188	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-26 06:00:00+00:00	269.274	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-26 12:00:00+00:00	180.945	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-26 18:00:00+00:00	143.894	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-27 00:00:00+00:00	136.458	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-27 06:00:00+00:00	153.789	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-27 12:00:00+00:00	103.591	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-27 18:00:00+00:00	107.903	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-28 00:00:00+00:00	101.099	W m-2	LATENT HEAT FLUX AT THE SURFACE
GFS	2024-05-28 06:00:00+00:00	150.857	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-24 00:00:00+00:00	0.0	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-24 06:00:00+00:00	224.531	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-24 12:00:00+00:00	149.853	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-24 18:00:00+00:00	141.179	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-25 00:00:00+00:00	190.1	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-25 06:00:00+00:00	281.181	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-25 12:00:00+00:00	223.483	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-25 18:00:00+00:00	198.711	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-26 00:00:00+00:00	174.745	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-26 06:00:00+00:00	207.665	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-26 12:00:00+00:00	122.599	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-26 18:00:00+00:00	115.23	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-27 00:00:00+00:00	111.454	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-27 06:00:00+00:00	167.921	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-27 12:00:00+00:00	89.03	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-27 18:00:00+00:00	88.564	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-28 00:00:00+00:00	90.264	W m-2	LATENT HEAT FLUX AT THE SURFACE
CFSv2	2024-05-28 06:00:00+00:00	162.691	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-24 00:00:00+00:00	0.0	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-24 06:00:00+00:00	254.753	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-24 12:00:00+00:00	140.51	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-24 18:00:00+00:00	122.27	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-25 00:00:00+00:00	162.582	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-25 06:00:00+00:00	301.856	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-25 12:00:00+00:00	189.156	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-25 18:00:00+00:00	172.669	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-26 00:00:00+00:00	215.317	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-26 06:00:00+00:00	225.319	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-26 12:00:00+00:00	137.746	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-26 18:00:00+00:00	104.393	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-27 00:00:00+00:00	86.865	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-27 06:00:00+00:00	134.496	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-27 12:00:00+00:00	60.165	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-27 18:00:00+00:00	52.437	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-28 00:00:00+00:00	61.88	W m-2	LATENT HEAT FLUX AT THE SURFACE
ERA5	2024-05-28 06:00:00+00:00	166.307	W m-2	LATENT HEAT FLUX AT THE SURFACE

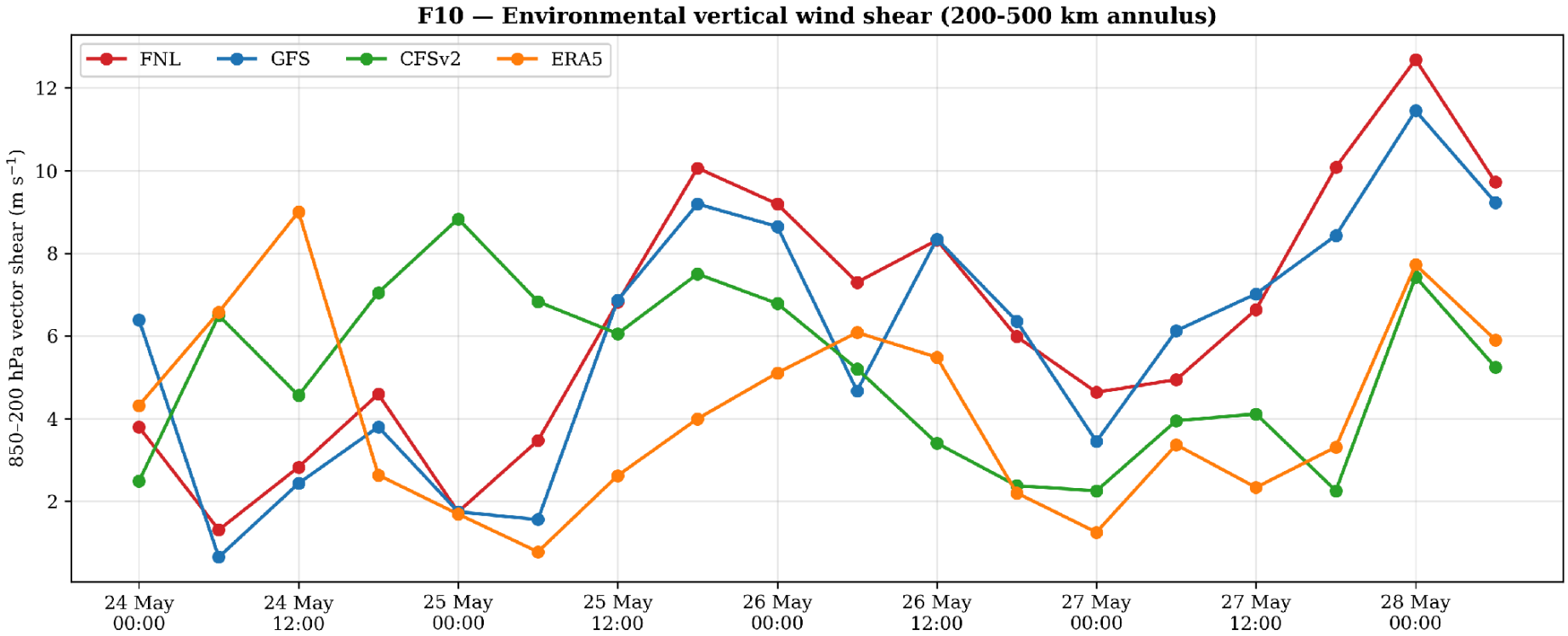
NetCDF diagnostic extracts

- A12_CFSv2_20240526_0000_PW_RH850.nc
- A12_CFSv2_20240526_1200_PW_RH850.nc
- A12_CFSv2_20240526_1800_PW_RH850.nc
- A12_ERA5_20240526_0000_PW_RH850.nc
- A12_ERA5_20240526_1200_PW_RH850.nc
- A12_ERA5_20240526_1800_PW_RH850.nc
- A12_FNL_20240526_0000_PW_RH850.nc
- A12_FNL_20240526_1200_PW_RH850.nc
- A12_FNL_20240526_1800_PW_RH850.nc
- A12_GFS_20240526_0000_PW_RH850.nc
- A12_GFS_20240526_1200_PW_RH850.nc
- A12_GFS_20240526_1800_PW_RH850.nc

A13 - Environmental 850-200 hPa shear

Figures: 1 | CSV tables: 1 | NetCDF extracts: 0

F10_Environmental_850_200hPa_Shear.png



A13_850_200hPa_environmental_shear.csv

Column block 1 of 3

model	time_utc	centre_latitude	centre_longitude	inner_radius_km	outer_radius_km	coverage_fraction
FNL	2024-05-24 00:00:00+00:00	15.271	88.835	200	500	1.0
FNL	2024-05-24 06:00:00+00:00	15.377	89.978	200	500	1.0
FNL	2024-05-24 12:00:00+00:00	16.07	90.13	200	500	1.0
FNL	2024-05-24 18:00:00+00:00	16.271	89.618	200	500	1.0
FNL	2024-05-25 00:00:00+00:00	17.226	90.325	200	500	1.0
FNL	2024-05-25 06:00:00+00:00	18.59	89.573	200	500	1.0
FNL	2024-05-25 12:00:00+00:00	18.177	89.73	200	500	1.0
FNL	2024-05-25 18:00:00+00:00	18.854	89.459	200	500	1.0
FNL	2024-05-26 00:00:00+00:00	19.639	89.448	200	500	0.996
FNL	2024-05-26 06:00:00+00:00	20.522	89.601	200	500	0.99
FNL	2024-05-26 12:00:00+00:00	21.715	90.086	200	500	0.987
FNL	2024-05-26 18:00:00+00:00	22.518	90.244	200	500	0.985
FNL	2024-05-27 00:00:00+00:00	23.528	90.696	200	500	0.974

FNL	2024-05-27 06:00:00+00:00	24.197	91.109	200	500	0.972
FNL	2024-05-27 12:00:00+00:00	24.844	92.035	200	500	0.968
FNL	2024-05-27 18:00:00+00:00	24.844	92.035	200	500	0.968
FNL	2024-05-28 00:00:00+00:00	24.844	92.035	200	500	0.969
FNL	2024-05-28 06:00:00+00:00	24.844	92.035	200	500	0.971
GFS	2024-05-24 00:00:00+00:00	14.312	89.261	200	500	1.0
GFS	2024-05-24 06:00:00+00:00	15.463	89.933	200	500	1.0
GFS	2024-05-24 12:00:00+00:00	16.009	90.053	200	500	1.0
GFS	2024-05-24 18:00:00+00:00	16.429	90.328	200	500	1.0
GFS	2024-05-25 00:00:00+00:00	17.376	90.075	200	500	1.0
GFS	2024-05-25 06:00:00+00:00	17.548	90.124	200	500	1.0
GFS	2024-05-25 12:00:00+00:00	18.289	89.835	200	500	1.0
GFS	2024-05-25 18:00:00+00:00	18.774	89.278	200	500	1.0
GFS	2024-05-26 00:00:00+00:00	19.678	89.549	200	500	0.994
GFS	2024-05-26 06:00:00+00:00	20.485	89.69	200	500	0.99
GFS	2024-05-26 12:00:00+00:00	21.798	89.834	200	500	0.986
GFS	2024-05-26 18:00:00+00:00	22.571	89.97	200	500	0.985
GFS	2024-05-27 00:00:00+00:00	23.512	90.203	200	500	0.975
GFS	2024-05-27 06:00:00+00:00	24.144	90.549	200	500	0.972
GFS	2024-05-27 12:00:00+00:00	24.711	91.656	200	500	0.967
GFS	2024-05-27 18:00:00+00:00	24.711	91.656	200	500	0.967
GFS	2024-05-28 00:00:00+00:00	24.711	91.656	200	500	0.967
GFS	2024-05-28 06:00:00+00:00	24.711	91.656	200	500	0.97
CFSv2	2024-05-24 00:00:00+00:00	15.34	87.526	200	500	1.0
CFSv2	2024-05-24 06:00:00+00:00	14.652	87.715	200	500	1.0
CFSv2	2024-05-24 12:00:00+00:00	16.156	87.448	200	500	1.0
CFSv2	2024-05-24 18:00:00+00:00	16.983	87.904	200	500	1.0
CFSv2	2024-05-25 00:00:00+00:00	17.443	87.535	200	500	1.0
CFSv2	2024-05-25 06:00:00+00:00	17.813	87.904	200	500	1.0
CFSv2	2024-05-25 12:00:00+00:00	19.227	87.703	200	500	1.0
CFSv2	2024-05-25 18:00:00+00:00	19.97	86.767	200	500	1.0
CFSv2	2024-05-26 00:00:00+00:00	20.445	86.314	200	500	1.0
CFSv2	2024-05-26 06:00:00+00:00	20.796	86.065	200	500	1.0
CFSv2	2024-05-26 12:00:00+00:00	21.493	85.758	200	500	1.0
CFSv2	2024-05-26 18:00:00+00:00	22.043	85.716	200	500	1.0
CFSv2	2024-05-27 00:00:00+00:00	22.765	86.122	200	500	1.0
CFSv2	2024-05-27 06:00:00+00:00	23.663	86.288	200	500	1.0
CFSv2	2024-05-27 12:00:00+00:00	24.252	86.774	200	500	1.0
CFSv2	2024-05-27 18:00:00+00:00	24.597	87.108	200	500	1.0
CFSv2	2024-05-28 00:00:00+00:00	24.713	87.847	200	500	1.0
CFSv2	2024-05-28 06:00:00+00:00	24.969	89.611	200	500	0.983
ERA5	2024-05-24 00:00:00+00:00	15.353	88.286	200	500	1.0
ERA5	2024-05-24 06:00:00+00:00	15.465	88.999	200	500	1.0
ERA5	2024-05-24 12:00:00+00:00	15.271	89.208	200	500	1.0
ERA5	2024-05-24 18:00:00+00:00	16.768	90.135	200	500	1.0
ERA5	2024-05-25 00:00:00+00:00	17.455	90.567	200	500	1.0
ERA5	2024-05-25 06:00:00+00:00	18.407	90.086	200	500	0.999
ERA5	2024-05-25 12:00:00+00:00	19.076	89.63	200	500	0.999
ERA5	2024-05-25 18:00:00+00:00	19.381	89.882	200	500	0.994
ERA5	2024-05-26 00:00:00+00:00	20.15	90.059	200	500	0.99
ERA5	2024-05-26 06:00:00+00:00	21.282	89.651	200	500	0.989
ERA5	2024-05-26 12:00:00+00:00	22.113	89.846	200	500	0.987
ERA5	2024-05-26 18:00:00+00:00	23.008	89.875	200	500	0.984
ERA5	2024-05-27 00:00:00+00:00	23.774	90.061	200	500	0.976
ERA5	2024-05-27 06:00:00+00:00	24.173	90.498	200	500	0.973
ERA5	2024-05-27 12:00:00+00:00	24.743	91.489	200	500	0.967
ERA5	2024-05-27 18:00:00+00:00	24.743	91.489	200	500	0.968
ERA5	2024-05-28 00:00:00+00:00	24.743	91.489	200	500	0.968
ERA5	2024-05-28 06:00:00+00:00	24.743	91.489	200	500	0.971

Column block 2 of 3

model	coverage_ok	u850_mean_ms	v850_mean_ms	u200_mean_ms	v200_mean_ms	delta_u_ms
FNL	True	3.123	2.231	-0.414	3.603	-3.538
FNL	True	2.548	3.832	1.948	2.671	-0.6
FNL	True	1.328	4.435	2.002	1.689	0.673
FNL	True	2.025	3.466	-1.425	0.434	-3.45
FNL	True	-0.642	2.512	-1.217	4.157	-0.575
FNL	True	-2.963	1.836	-2.051	5.181	0.913
FNL	True	0.189	3.282	-6.478	1.867	-6.667
FNL	True	2.493	3.09	-7.569	3.101	-10.062
FNL	True	3.415	4.016	-5.619	2.309	-9.034
FNL	True	4.502	3.188	-2.795	3.056	-7.297
FNL	True	4.702	2.185	-3.57	3.06	-8.271
FNL	True	4.162	4.043	-1.727	2.979	-5.889
FNL	True	5.576	7.192	4.744	2.631	-0.832
FNL	True	6.58	6.865	7.279	1.973	0.699
FNL	True	7.255	6.196	8.133	-0.384	0.878
FNL	True	5.768	8.79	7.058	-1.206	1.289
FNL	True	8.231	10.96	4.6	-1.197	-3.631
FNL	True	8.032	7.972	1.845	0.475	-6.188
GFS	True	5.805	3.501	-0.512	2.53	-6.317
GFS	True	2.259	3.398	2.148	2.759	-0.112
GFS	True	1.419	4.156	0.779	1.809	-0.64
GFS	True	1.294	4.292	-0.864	1.166	-2.158
GFS	True	-0.868	2.333	-1.691	3.872	-0.823
GFS	True	-1.458	3.27	-3.014	3.234	-1.556
GFS	True	-0.1	3.594	-6.76	1.918	-6.661
GFS	True	1.772	3.063	-7.412	3.484	-9.184
GFS	True	3.387	4.276	-4.419	0.556	-7.806
GFS	True	3.704	2.009	-0.743	3.441	-4.447
GFS	True	4.235	2.425	-4.106	2.663	-8.341
GFS	True	4.391	3.375	-1.966	3.213	-6.357
GFS	True	5.883	7.065	4.981	3.74	-0.901
GFS	True	6.875	7.384	8.218	1.407	1.342
GFS	True	6.473	6.145	7.79	-0.748	1.317
GFS	True	5.629	8.01	7.556	-0.195	1.927
GFS	True	8.071	11.112	5.089	0.058	-2.983
GFS	True	9.054	7.139	2.082	1.089	-6.972
CFSv2	True	1.441	0.916	-0.626	2.296	-2.067
CFSv2	True	3.307	4.298	-2.089	0.683	-5.396
CFSv2	True	1.966	2.56	-1.964	0.252	-3.93
CFSv2	True	1.134	0.935	-5.713	2.613	-6.846
CFSv2	True	-0.494	1.144	-9.049	3.345	-8.555
CFSv2	True	-0.319	2.152	-7.01	3.517	-6.69
CFSv2	True	-0.869	2.829	-6.907	3.278	-6.037
CFSv2	True	0.118	2.039	-7.329	2.922	-7.448
CFSv2	True	1.704	2.023	-5.076	2.271	-6.78
CFSv2	True	1.649	1.391	-3.527	0.866	-5.176
CFSv2	True	0.578	2.023	-2.578	0.745	-3.156
CFSv2	True	0.766	3.37	-0.935	1.709	-1.701
CFSv2	True	1.078	4.605	1.093	2.353	0.015
CFSv2	True	3.06	5.712	3.821	1.839	0.76
CFSv2	True	5.232	5.182	3.453	1.473	-1.778
CFSv2	True	5.835	3.811	4.45	2.03	-1.385
CFSv2	True	6.971	6.734	5.072	-0.448	-1.899
CFSv2	True	6.43	5.579	3.615	1.155	-2.815
ERA5	True	3.008	3.021	-1.3	2.716	-4.307
ERA5	True	3.781	4.636	-1.813	1.184	-5.594
ERA5	True	6.122	5.13	-1.442	0.247	-7.564

ERA5	True	2.842	5.005	1.259	2.901	-1.583
ERA5	True	1.069	5.055	0.328	3.538	-0.741
ERA5	True	-0.865	3.777	-0.825	4.553	0.04
ERA5	True	-0.861	3.945	-2.997	5.466	-2.136
ERA5	True	0.581	3.866	-3.037	5.56	-3.618
ERA5	True	0.403	2.106	-2.801	6.079	-3.204
ERA5	True	2.276	2.821	-3.155	5.564	-5.431
ERA5	True	3.18	4.122	-2.272	4.705	-5.452
ERA5	True	3.771	4.06	1.615	3.614	-2.156
ERA5	True	4.357	4.223	5.368	3.49	1.011
ERA5	True	4.792	5.426	5.169	2.079	0.377
ERA5	True	6.405	4.05	4.52	2.675	-1.884
ERA5	True	7.454	5.566	6.821	2.316	-0.633
ERA5	True	8.114	7.753	5.642	0.436	-2.472
ERA5	True	7.614	5.682	3.872	1.115	-3.741

Column block 3 of 3

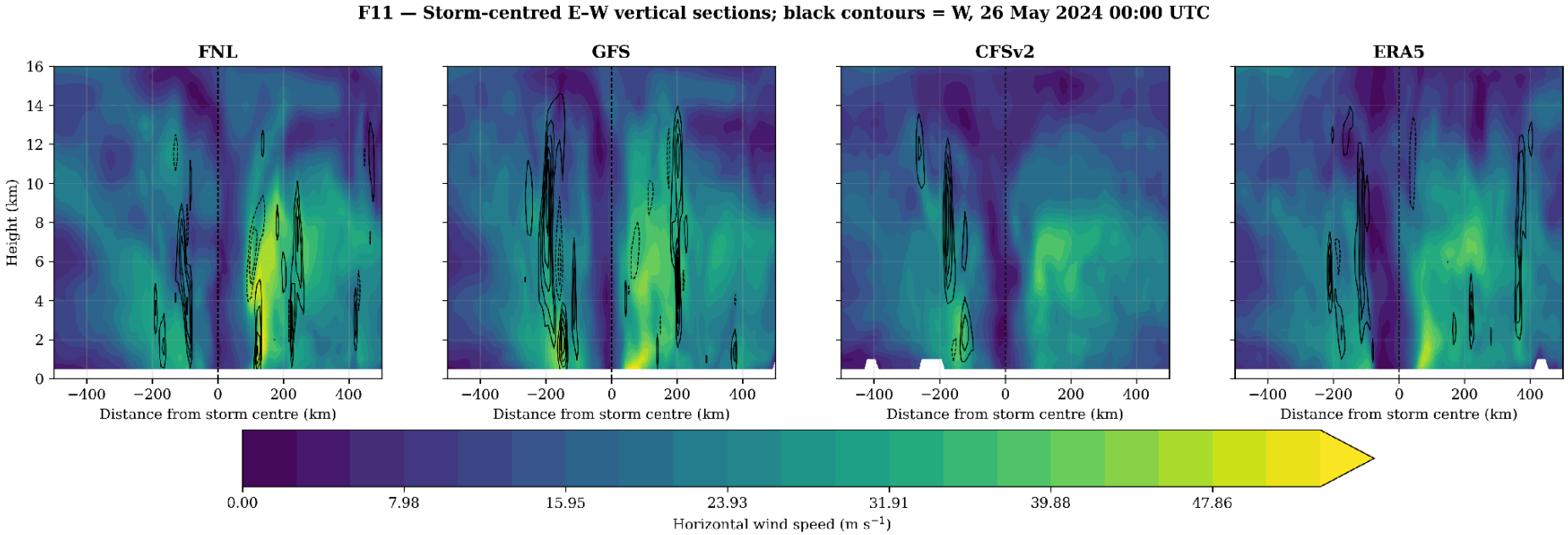
model	**delta_v_ms**	**shear_850_200_ms**
FNL	1.372	3.794
FNL	-1.161	1.307
FNL	-2.746	2.827
FNL	-3.032	4.592
FNL	1.644	1.742
FNL	3.345	3.467
FNL	-1.415	6.815
FNL	0.012	10.062
FNL	-1.707	9.194
FNL	-0.131	7.299
FNL	0.874	8.318
FNL	-1.065	5.984
FNL	-4.561	4.636
FNL	-4.892	4.942
FNL	-6.58	6.638
FNL	-9.996	10.079
FNL	-12.157	12.688
FNL	-7.497	9.72
GFS	-0.97	6.391
GFS	-0.639	0.649
GFS	-2.347	2.432
GFS	-3.126	3.799
GFS	1.538	1.745
GFS	-0.036	1.557
GFS	-1.677	6.868
GFS	0.42	9.194
GFS	-3.72	8.647
GFS	1.431	4.672
GFS	0.238	8.345
GFS	-0.162	6.359
GFS	-3.325	3.445
GFS	-5.977	6.126
GFS	-6.893	7.017
GFS	-8.205	8.429
GFS	-11.053	11.449
GFS	-6.05	9.231
CFSv2	1.38	2.485
CFSv2	-3.616	6.495
CFSv2	-2.308	4.558
CFSv2	1.678	7.049

CFSv2	2.201	8.834
CFSv2	1.366	6.828
CFSv2	0.449	6.054
CFSv2	0.883	7.5
CFSv2	0.248	6.785
CFSv2	-0.525	5.202
CFSv2	-1.278	3.405
CFSv2	-1.661	2.377
CFSv2	-2.252	2.252
CFSv2	-3.873	3.947
CFSv2	-3.709	4.113
CFSv2	-1.781	2.256
CFSv2	-7.182	7.429
CFSv2	-4.424	5.244
ERA5	-0.305	4.318
ERA5	-3.452	6.573
ERA5	-4.882	9.003
ERA5	-2.105	2.633
ERA5	-1.517	1.688
ERA5	0.776	0.777
ERA5	1.521	2.622
ERA5	1.693	3.995
ERA5	3.973	5.104
ERA5	2.742	6.084
ERA5	0.583	5.483
ERA5	-0.446	2.202
ERA5	-0.733	1.249
ERA5	-3.347	3.368
ERA5	-1.376	2.333
ERA5	-3.25	3.311
ERA5	-7.316	7.723
ERA5	-4.567	5.904

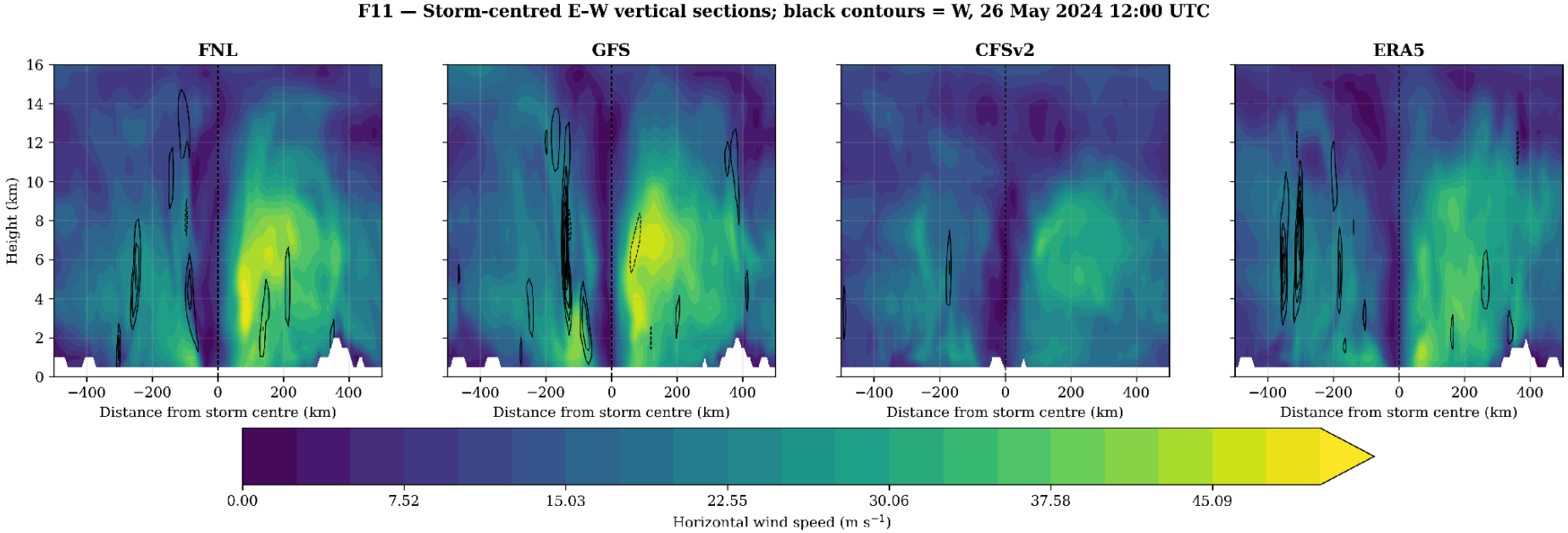
A14 - Storm-centred vertical sections

Figures: 3 | CSV tables: 0 | NetCDF extracts: 12

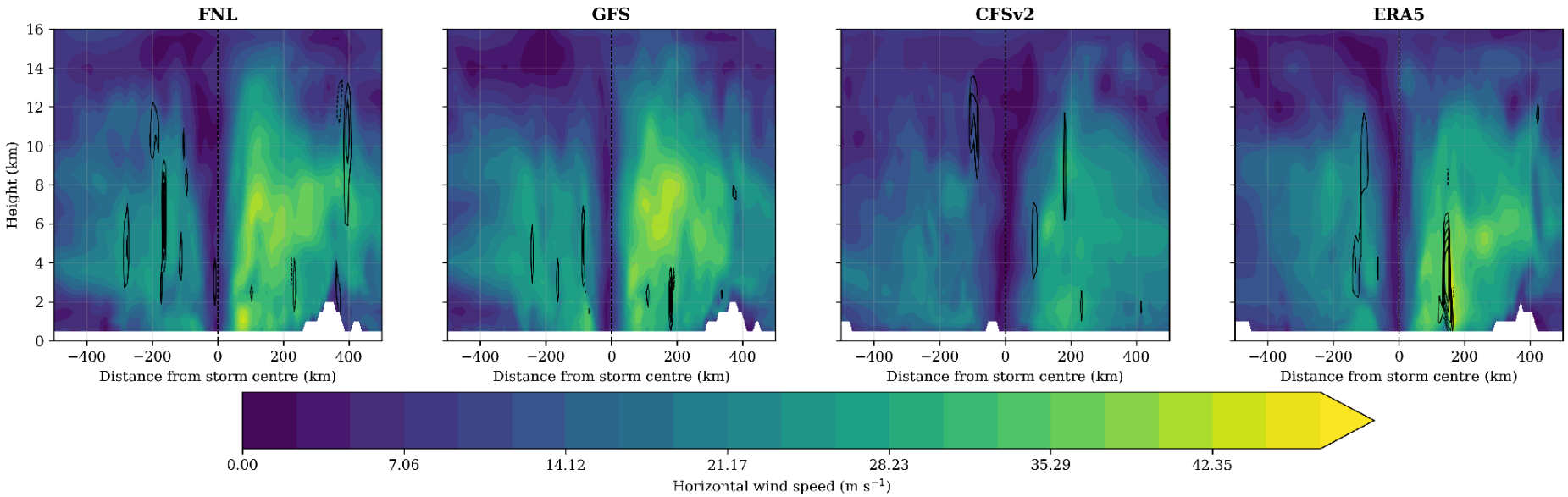
F11_Vertical_Section_20240526_0000.png



F11_Vertical_Section_20240526_1200.png



F11_Vertical_Section_20240526_1800.png



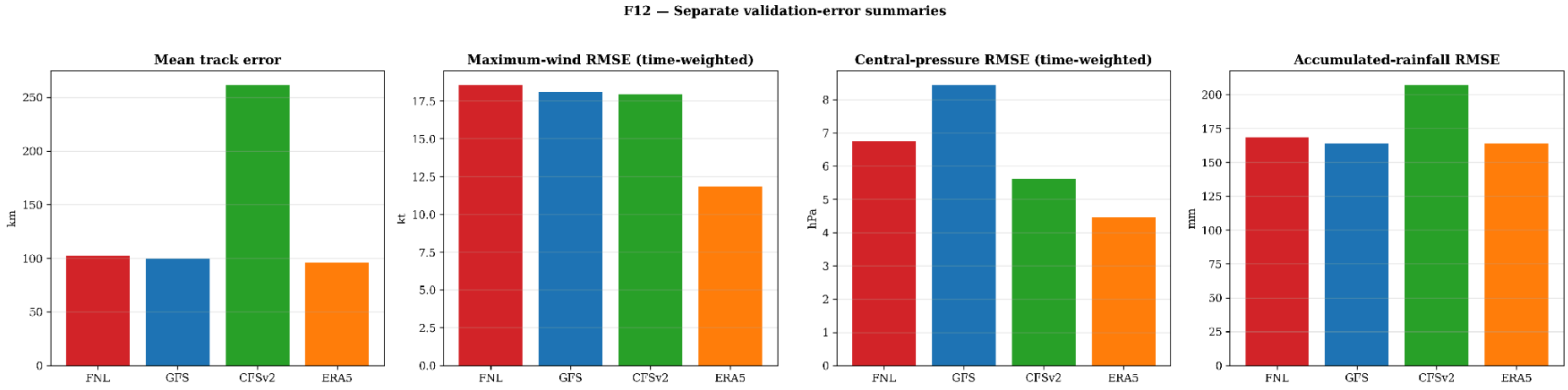
NetCDF diagnostic extracts

- A14_CFSv2_20240526_0000_vertical_section.nc
- A14_CFSv2_20240526_1200_vertical_section.nc
- A14_CFSv2_20240526_1800_vertical_section.nc
- A14_ERA5_20240526_0000_vertical_section.nc
- A14_ERA5_20240526_1200_vertical_section.nc
- A14_ERA5_20240526_1800_vertical_section.nc
- A14_FNL_20240526_0000_vertical_section.nc
- A14_FNL_20240526_1200_vertical_section.nc
- A14_FNL_20240526_1800_vertical_section.nc
- A14_GFS_20240526_0000_vertical_section.nc
- A14_GFS_20240526_1200_vertical_section.nc
- A14_GFS_20240526_1800_vertical_section.nc

A15 - Consolidated performance and checks

Figures: 1 | CSV tables: 2 | NetCDF extracts: 0

F12_Separate_Validation_Error_Bars.png



A15_Final_Checks.csv

check	result	detail
Same number of shared 6-hourly track timestamps for all models	PASS	counts = [18, 18, 18, 18]
Same number of full-time intensity timestamps for all models	PASS	counts = [30, 30, 30, 30]
Every supplied IMD intensity timestamp has an exact WRF match in al...	PASS	used 30 of 30 IMD timestamps
FNL: no missing full-time matched wind/pressure values	PASS	missing wind = 0, missing pressure = 0
FNL: no duplicate 6-hourly model-track times	PASS	0 duplicates
FNL: no duplicate full-time intensity times	PASS	0 duplicates
FNL: full-time model wind/pressure broadly plausible	PASS	wind 16.5-87.7 kt; pressure 969.0-996.8 hPa
FNL: 6-hourly tracking candidates requiring review	WARN	3 rejected/carry-forward timestamps; see A02
GFS: no missing full-time matched wind/pressure values	PASS	missing wind = 0, missing pressure = 0
GFS: no duplicate 6-hourly model-track times	PASS	0 duplicates
GFS: no duplicate full-time intensity times	PASS	0 duplicates
GFS: full-time model wind/pressure broadly plausible	PASS	wind 15.0-90.7 kt; pressure 966.2-996.6 hPa
GFS: 6-hourly tracking candidates requiring review	WARN	3 rejected/carry-forward timestamps; see A02
CFSv2: no missing full-time matched wind/pressure values	PASS	missing wind = 0, missing pressure = 0
CFSv2: no duplicate 6-hourly model-track times	PASS	0 duplicates
CFSv2: no duplicate full-time intensity times	PASS	0 duplicates
CFSv2: full-time model wind/pressure broadly plausible	PASS	wind 23.2-68.2 kt; pressure 977.8-997.3 hPa
CFSv2: 6-hourly tracking candidates requiring review	PASS	0 rejected/carry-forward timestamps; see A02
ERA5: no missing full-time matched wind/pressure values	PASS	missing wind = 0, missing pressure = 0
ERA5: no duplicate 6-hourly model-track times	PASS	0 duplicates
ERA5: no duplicate full-time intensity times	PASS	0 duplicates
ERA5: full-time model wind/pressure broadly plausible	PASS	wind 13.9-72.0 kt; pressure 972.0-998.9 hPa
ERA5: 6-hourly tracking candidates requiring review	WARN	3 rejected/carry-forward timestamps; see A02
FNL: landfall crossing found	PASS	IMD source: derived from IMD track and WRF land mask
GFS: landfall crossing found	PASS	IMD source: derived from IMD track and WRF land mask
CFSv2: landfall crossing found	PASS	IMD source: derived from IMD track and WRF land mask
ERA5: landfall crossing found	PASS	IMD source: derived from IMD track and WRF land mask
A01 domain/configuration output created	PASS	A01_configuration

A04 translation speed created	PASS	A04_translation_speed
A05/A06 full-time intensity coverage table created	PASS	A05_wind_validation / A06_pressure_validation
A07 intensity-change rates created	PASS	A07_intensity_change
A10 IMERG rainfall validation ran	PASS	A10_rainfall
A11 CCMP wind validation ran	PASS	A11_ccmp_wind
A12 PW/RH850 diagnostics ran	PASS	A12_moisture_flux
A12 latent-heat-flux diagnostic ran	PASS	A12_moisture_flux
A13 environmental shear diagnostic ran	PASS	A13_shear
A16 sampling-sensitivity analysis ran	PASS	A16_sampling_sensitivity
A16 regular-subset timestamps are exactly 6 hours apart	PASS	regular subset N = 18; gaps = [np.float64(6.0), np.float64(6.0), np...
A14 vertical sections ran	PASS	12 gridded section files created

T07_Consolidated_Performance_and_Diagnostics.csv

Column block 1 of 6

model	n_shared_track_times	n_shared_intensity_times	n_carried_forward_in_sample	mean_position_error_km	track_rmse_km	translation_speed_bias_kmh
FNL	18	30	3	102.128	123.464	-1.241
GFS	18	30	3	99.46	110.864	-0.987
CFSv2	18	30	0	261.708	273.257	0.303
ERA5	18	30	3	95.789	109.202	-0.931

Column block 2 of 6

model	translation_speed_rmse_kmh	wind_bias_kt	wind_rmse_kt	wind_corr	wind_bias_kt_unweighted_all_matches	wind_rmse_kt_unweighted_all_matches
FNL	7.544	13.555	18.571	0.778	13.04	18.623
GFS	7.596	13.336	18.093	0.79	13.013	18.224
CFSv2	5.716	6.635	17.968	0.091	4.735	17.414
ERA5	7.608	7.193	11.86	0.803	6.196	11.393

Column block 3 of 6

model	wind_corr_unweighted_all_matches	pressure_bias_hpa	pressure_rmse_hpa	pressure_corr	pressure_bias_hpa_unweighted_all_matches	pressure_rmse_hpa_unweighted_all_matches
FNL	0.789	-4.461	6.757	0.881	-4.579	6.993
GFS	0.798	-5.96	8.448	0.837	-6.104	8.696
CFSv2	0.133	-2.493	5.611	0.539	-2.028	5.504
ERA5	0.83	-2.179	4.463	0.911	-2.264	4.655

Column block 4 of 6

model	pressure_corr_unweighted_all_matches	peak_wind_error_kt	peak_wind_timing_error_h	min_pressure_error_hpa	min_pressure_timing_error_h	landfall_time_error_h
FNL	0.883	27.732	-12.0	-9.046	0.0	-3.667
GFS	0.839	30.721	-12.0	-11.83	-3.0	-2.483
CFSv2	0.534	8.206	-21.0	-0.225	-21.0	-19.383
ERA5	0.914	12.006	-9.0	-5.978	-3.0	-7.05

Column block 5 of 6

model	landfall_position_error_km	pdi_model_over_imd	rainfall_bias_mm	rainfall_rmse_mm	rainfall_spatial_corr	ccmp_speed_bias_ms_mean_selected_times
FNL	94.86	2.827	-76.979	168.234	0.412	3.871
GFS	90.182	2.777	-70.793	163.856	0.44	3.66
CFSv2	315.583	1.536	-85.926	207.234	-0.12	-0.499
ERA5	67.84	1.767	-81.209	163.93	0.47	1.258

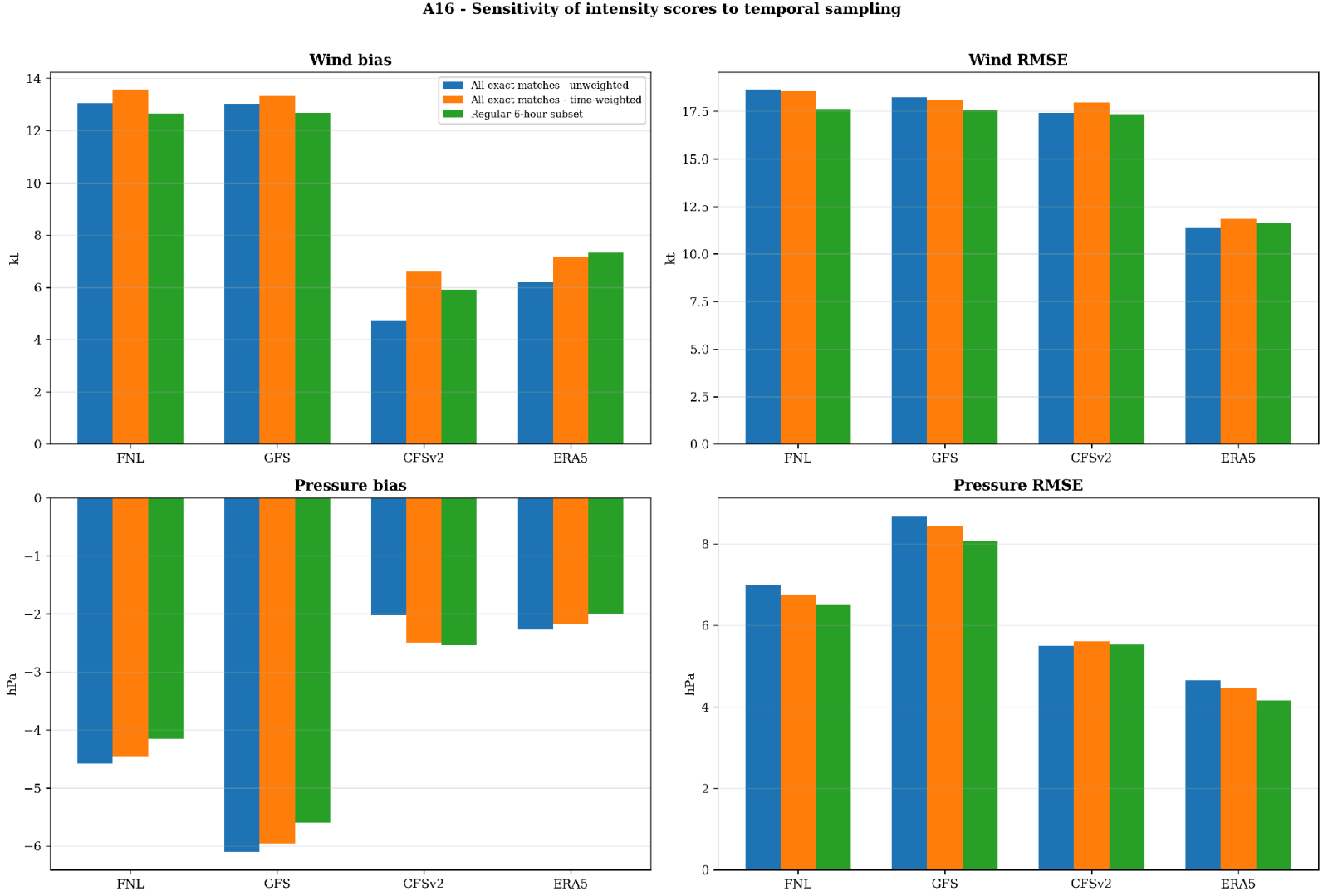
Column block 6 of 6

model	ccmp_speed_rmse_ms_mean_selected_times	ccmp_speed_spatial_corr_mean_selected_times	ccmp_vector_rmse_ms_mean_selected_times	mean_environmental_shear_850_200_ms
FNL	6.2	0.535	11.093	6.339
GFS	6.11	0.558	10.45	5.909
CFSv2	5.731	-0.153	11.261	5.156
ERA5	5.024	0.481	10.444	4.132

A16 - Temporal-sampling sensitivity

Figures: 1 | CSV tables: 3 | NetCDF extracts: 0

A16_Sampling_Sensitivity_Bias_RMSE.png



A16_PDI_Full_vs_6hour_Sensitivity.csv

Column block 1 of 2

source	full_n_timestamps	full_valid_intervals	full_integrated_hours	full_pdi_m3_s_minus2	six_hour_n_timestamps	six_hour_valid_intervals
IMD	30	29	102.0	3250622640.034	18	17
FNL	30	29	102.0	9190173751.993	18	17
GFS	30	29	102.0	9026507915.041	18	17
CFSv2	30	29	102.0	4992038093.35	18	17
ERA5	30	29	102.0	5743445568.125	18	17

Column block 2 of 2

source	six_hour_integrated_hours	six_hour_pdi_m3_s_minus2	six_hour_minus_full_pct
IMD	102.0	3274424943.157	0.732
FNL	102.0	9144822927.689	-0.493
GFS	102.0	9012925855.956	-0.15
CFSv2	102.0	4730777756.36	-5.234
ERA5	102.0	5975226940.887	4.036

A16_Timestamp_Support.csv

time_utc	full_exact_match_support	regular_6h_subset	elapsed_from_previous_full_match_h
2024-05-24 00:00:00+00:00	True	True	
2024-05-24 03:00:00+00:00	True	False	3.0
2024-05-24 06:00:00+00:00	True	True	3.0
2024-05-24 12:00:00+00:00	True	True	6.0
2024-05-24 18:00:00+00:00	True	True	6.0
2024-05-25 00:00:00+00:00	True	True	6.0
2024-05-25 03:00:00+00:00	True	False	3.0
2024-05-25 06:00:00+00:00	True	True	3.0
2024-05-25 12:00:00+00:00	True	True	6.0
2024-05-25 15:00:00+00:00	True	False	3.0
2024-05-25 18:00:00+00:00	True	True	3.0
2024-05-25 21:00:00+00:00	True	False	3.0
2024-05-26 00:00:00+00:00	True	True	3.0
2024-05-26 03:00:00+00:00	True	False	3.0
2024-05-26 06:00:00+00:00	True	True	3.0
2024-05-26 09:00:00+00:00	True	False	3.0
2024-05-26 12:00:00+00:00	True	True	3.0
2024-05-26 15:00:00+00:00	True	False	3.0
2024-05-26 18:00:00+00:00	True	True	3.0
2024-05-26 21:00:00+00:00	True	False	3.0
2024-05-27 00:00:00+00:00	True	True	3.0
2024-05-27 03:00:00+00:00	True	False	3.0
2024-05-27 06:00:00+00:00	True	True	3.0
2024-05-27 09:00:00+00:00	True	False	3.0
2024-05-27 12:00:00+00:00	True	True	3.0
2024-05-27 15:00:00+00:00	True	False	3.0
2024-05-27 18:00:00+00:00	True	True	3.0
2024-05-28 00:00:00+00:00	True	True	6.0
2024-05-28 03:00:00+00:00	True	False	3.0
2024-05-28 06:00:00+00:00	True	True	3.0

A16_Wind_Pressure_Sampling_Sensitivity.csv

Column block 1 of 2

model	variable	unit	sampling_scheme	n	bias	mae
FNL	Wind	kt	All exact matches - unweighted	30	13.04	13.905
FNL	Wind	kt	All exact matches - time-weighted	30	13.555	14.357
FNL	Wind	kt	Regular 6-hour subset	18	12.65	13.549
FNL	Pressure	hPa	All exact matches - unweighted	30	-4.579	5.569
FNL	Pressure	hPa	All exact matches - time-weighted	30	-4.461	5.385
FNL	Pressure	hPa	Regular 6-hour subset	18	-4.149	5.039
GFS	Wind	kt	All exact matches - unweighted	30	13.013	13.839
GFS	Wind	kt	All exact matches - time-weighted	30	13.336	14.094
GFS	Wind	kt	Regular 6-hour subset	18	12.676	13.685
GFS	Pressure	hPa	All exact matches - unweighted	30	-6.104	6.921
GFS	Pressure	hPa	All exact matches - time-weighted	30	-5.96	6.754
GFS	Pressure	hPa	Regular 6-hour subset	18	-5.602	6.436
CFSv2	Wind	kt	All exact matches - unweighted	30	4.735	13.735
CFSv2	Wind	kt	All exact matches - time-weighted	30	6.635	14.524
CFSv2	Wind	kt	Regular 6-hour subset	18	5.916	13.852
CFSv2	Pressure	hPa	All exact matches - unweighted	30	-2.028	4.525
CFSv2	Pressure	hPa	All exact matches - time-weighted	30	-2.493	4.696
CFSv2	Pressure	hPa	Regular 6-hour subset	18	-2.535	4.624
ERA5	Wind	kt	All exact matches - unweighted	30	6.196	9.684
ERA5	Wind	kt	All exact matches - time-weighted	30	7.193	10.261
ERA5	Wind	kt	Regular 6-hour subset	18	7.331	10.016
ERA5	Pressure	hPa	All exact matches - unweighted	30	-2.264	3.669
ERA5	Pressure	hPa	All exact matches - time-weighted	30	-2.179	3.511
ERA5	Pressure	hPa	Regular 6-hour subset	18	-1.998	3.248

Column block 2 of 2

model	rmse	correlation
FNL	18.623	0.789
FNL	18.571	0.778
FNL	17.631	0.814
FNL	6.993	0.883
FNL	6.757	0.881
FNL	6.508	0.889
GFS	18.224	0.798
GFS	18.093	0.79
GFS	17.53	0.814
GFS	8.696	0.839
GFS	8.448	0.837
GFS	8.089	0.849
CFSv2	17.414	0.133
CFSv2	17.968	0.091
CFSv2	17.346	0.115
CFSv2	5.504	0.534
CFSv2	5.611	0.539

CFSv2	5.529	0.574
ERA5	11.393	0.83
ERA5	11.86	0.803
ERA5	11.646	0.832
ERA5	4.655	0.914
ERA5	4.463	0.911
ERA5	4.165	0.923

Report build summary

Figures are reproduced from the PNG/JPG outputs saved by the notebook. Tables are reproduced from CSV outputs; wide tables are split into column blocks. NetCDF files remain available in their original analysis folders and the _ALL_NETCDF mirror.

No figure or CSV embedding errors were recorded.